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(54) **SEE-THROUGH COMPUTER DISPLAY SYSTEMS**

(71) Applicant: **Mentor Acquisition One, LLC**,
Plantation, FL (US)

(72) Inventors: **Ralph F. Osterhout**, San Francisco, CA
(US); **John N. Border**, Campbell, CA
(US); **Robert Michael Lohse**, Palo
Alto, CA (US); **John D. Haddick**,
Corte Madera, CA (US); **Nima L.
Shams**, San Jose, CA (US)

(73) Assignee: **Mentor Acquisition One, LLC**,
Plantation, FL (US)

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(56) **References Cited**

U.S. PATENT DOCUMENTS

1,897,833 A 2/1933 Benway
2,064,604 A 12/1936 Paul
(Continued)

FOREIGN PATENT DOCUMENTS

CA 2316473 A1 1/2001
CA 2362895 A1 12/2002
(Continued)

OTHER PUBLICATIONS

US 8,792,178 B2, 07/2014, Totani (withdrawn)
(Continued)

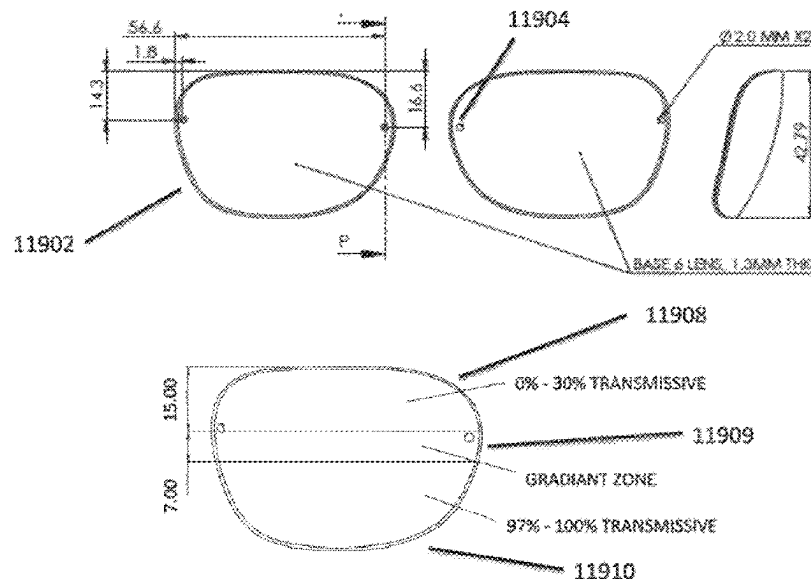
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(74) *Attorney, Agent, or Firm* — Morrison & Foerster
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(57) **ABSTRACT**

Aspects of the present invention relate to providing see-
through computer display optics.

20 Claims, 155 Drawing Sheets



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continuation of application No. 16/538,720, filed on Aug. 12, 2019, now Pat. No. 11,126,003, which is a continuation of application No. 15/488,916, filed on Apr. 17, 2017, now Pat. No. 10,379,365, which is a continuation of application No. 14/742,249, filed on Jun. 17, 2015, now Pat. No. 9,746,676, which is a continuation of application No. 14/589,713, filed on Jan. 5, 2015, now Pat. No. 9,529,195, which is a continuation-in-part of application No. 14/561,146, filed on Dec. 4, 2014, now Pat. No. 9,594,246, and a continuation-in-part of application No. 14/559,126, filed on Dec. 3, 2014, now Pat. No. 10,684,687, and a continuation-in-part of application No. 14/554,044, filed on Nov. 26, 2014, now Pat. No. 9,448,409, and a continuation-in-part of application No. 14/504,723, filed on Oct. 2, 2014, now Pat. No. 9,671,613, and a continuation-in-part of application No. 14/498,765, filed on Sep. 26, 2014, now Pat. No. 9,366,868, and a continuation-in-part of application No. 14/489,706, filed on Sep. 18, 2014, now abandoned, and a continuation-in-part of application No. 14/325,991, filed on Jul. 8, 2014, now Pat. No. 9,366,867, and a continuation-in-part of application No. 14/296,699, filed on Jun. 5, 2014, now Pat. No. 9,841,599, and a continuation-in-part of application No. 14/254,253, filed on Apr. 16, 2014, now Pat. No. 9,952,664, and a continuation-in-part of application No. 14/216,175, filed on Mar. 17, 2014, now Pat. No. 9,298,007, and a continuation-in-part of application No. 14/181,459, filed on Feb. 14, 2014, now Pat. No. 9,715,112, which is a continuation-in-part of application No. 14/178,047, filed on Feb. 11, 2014, now Pat. No. 9,229,233, said application No. 14/589,713 is a continuation-in-part of application No. 14/172,901, filed on Feb. 4, 2014, now abandoned, and a continuation-in-part of application No. 14/163,646, filed on Jan. 24, 2014, now Pat. No. 9,400,390, and a continuation-in-part of application No. 14/160,377, filed on Jan. 21, 2014, now abandoned.

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 See application file for complete search history.

(56)**References Cited****U.S. PATENT DOCUMENTS**

3,305,294	A	2/1967	Alvarez
3,531,190	A	9/1970	Leblanc
3,671,111	A	6/1972	Okner
4,034,401	A	7/1977	Mann
4,145,125	A	3/1979	Chika
4,513,812	A	4/1985	Papst
4,668,155	A	5/1987	Kaufmann
4,695,129	A	9/1987	Faessen
4,811,739	A	3/1989	Silver
4,852,988	A	8/1989	Velez
4,928,301	A	5/1990	Smoot
5,028,127	A	7/1991	Spitzberg
D327,674	S	7/1992	Kuo
5,151,722	A	9/1992	Massof
5,257,094	A	10/1993	Larussa
D352,930	S	11/1994	Tsuji
5,483,307	A	1/1996	Anderson
D375,748	S	11/1996	Hartman
D376,790	S	12/1996	Taylor
5,596,451	A	1/1997	Handschy
5,621,424	A	4/1997	Shimada
5,625,372	A	4/1997	Hildebrand
D383,148	S	9/1997	Lee
5,699,057	A	12/1997	Ikeda
5,699,194	A	12/1997	Takahashi
5,717,422	A	2/1998	Ferguson
D392,959	S	3/1998	Edwards
5,729,242	A	3/1998	Margerum
5,767,841	A	6/1998	Hartman
5,788,195	A	8/1998	Rice
5,808,589	A	9/1998	Ferguson
5,808,800	A	9/1998	Handschy
5,808,802	A	9/1998	Hur
D410,638	S	6/1999	Loughnane
5,914,818	A	6/1999	Tejada
5,949,583	A	9/1999	Rallison
5,954,642	A	9/1999	Johnson
5,971,538	A	10/1999	Heffner
5,991,084	A	11/1999	Hildebrand
6,028,608	A	2/2000	Jenkins
6,034,653	A	3/2000	Robertson
6,046,712	A	4/2000	Beller

(56)

References Cited

U.S. PATENT DOCUMENTS

6,076,927	A	6/2000	Owens	7,777,690	B2	8/2010	Winsor
6,137,675	A	10/2000	Perkins	7,777,723	B2	8/2010	Namiki
6,147,805	A	11/2000	Ferguson	7,777,960	B2	8/2010	Freeman
6,157,291	A	12/2000	Kuenster	7,791,889	B2	9/2010	Belady
6,160,552	A	12/2000	Wilsher	7,792,552	B2	9/2010	Thomas
6,160,666	A	12/2000	Rallison	7,806,525	B2	10/2010	Howell
6,195,136	B1	2/2001	Handschy	7,812,842	B2	10/2010	Gordon
6,204,974	B1	3/2001	Spitzer	7,813,743	B1	10/2010	Loeb
6,222,677	B1	4/2001	Budd	7,830,370	B2	11/2010	Yamazaki
6,297,749	B1	10/2001	Smith	D628,616	S	12/2010	Yuan
D451,892	S	12/2001	Carrere	7,850,301	B2	12/2010	Dichiara
6,347,764	B1	2/2002	Brandon	7,855,743	B2	12/2010	Sako
6,359,723	B1	3/2002	Handschy	D631,881	S	2/2011	Quinn
6,369,952	B1	4/2002	Rallison	D631,882	S	2/2011	Odgers
6,379,009	B1	4/2002	Ferguson	7,928,926	B2	4/2011	Yamamoto
6,384,982	B1	5/2002	Spitzer	8,004,765	B2	8/2011	Amitai
6,392,656	B1	5/2002	Someya	D645,492	S	9/2011	Zhao
D460,071	S	7/2002	Sheehan	D645,493	S	9/2011	Zhao
6,421,031	B1	7/2002	Ronzani	8,018,579	B1	9/2011	Krah
6,433,760	B1	8/2002	Vaissie	D646,316	S	10/2011	Zhao
6,456,438	B1	9/2002	Lee	D647,947	S	11/2011	Yu
6,461,000	B1	10/2002	Magarill	8,079,713	B2	12/2011	Ashkenazi
6,478,429	B1	11/2002	Aritake	8,089,568	B1	1/2012	Brown
6,480,174	B1	11/2002	Kaufmann	8,092,007	B2	1/2012	Dichiara
6,491,389	B2	12/2002	Yaguchi	8,166,421	B2	4/2012	Magal
6,491,391	B1	12/2002	Blum et al.	8,212,859	B2	7/2012	Tang
D470,144	S	2/2003	Li	8,228,315	B1	7/2012	Starner
6,535,182	B2	3/2003	Stanton	D665,838	S	8/2012	Kim
D473,871	S	4/2003	Santos	8,235,529	B1	8/2012	Raffle
6,563,626	B1	5/2003	Iwasaki	8,246,170	B2	8/2012	Yamamoto
D478,052	S	8/2003	Thomas, Jr.	D667,482	S	9/2012	Healy
6,642,945	B1	11/2003	Sharpe	D667,483	S	9/2012	Krsmanovic
6,747,611	B1	6/2004	Budd	D669,066	S	10/2012	Olsson
6,771,294	B1	8/2004	Pulli	D671,590	S	11/2012	Klinar
6,795,041	B2	9/2004	Ogawa	8,337,013	B2	12/2012	Howell
6,824,265	B1	11/2004	Harper	8,376,548	B2	2/2013	Schultz
6,847,336	B1	1/2005	Lemelson	8,378,924	B2	2/2013	Jacobsen
6,906,836	B2	6/2005	Parker	8,384,999	B1	2/2013	Crosby
6,943,754	B2	9/2005	Aughey	D680,112	S	4/2013	Monahan
D512,027	S	11/2005	Sarasjoki	D680,152	S	4/2013	Olsson
D513,233	S	12/2005	Stauffer	8,427,396	B1	4/2013	Kim
6,977,776	B2	12/2005	Volkenandt et al.	8,430,507	B2	4/2013	Howell
6,987,787	B1	1/2006	Mick	8,434,863	B2	5/2013	Howell
D514,525	S	2/2006	Stauffer	D685,019	S	6/2013	Li
7,003,308	B1	2/2006	Fuoss	8,473,241	B2	6/2013	Foxlin
7,016,116	B2	3/2006	Dolgoff	8,487,838	B2	7/2013	Lewis
7,030,925	B1	4/2006	Tsunekawa	8,489,326	B1	7/2013	Na
D521,493	S	5/2006	Wai	8,494,215	B2	7/2013	Kimchi
7,088,234	B2	8/2006	Naito	8,505,430	B2	8/2013	Miralles
D529,467	S	10/2006	Rose	D689,862	S	9/2013	Liu
D541,226	S	4/2007	Wakisaka	8,531,394	B2	9/2013	Maltz
7,199,934	B2	4/2007	Yamasaki	D690,684	S	10/2013	Lee
7,206,134	B2	4/2007	Weissman	D692,047	S	10/2013	Shin
D559,793	S	1/2008	Fan	8,553,910	B1	10/2013	Dong
7,347,551	B2	3/2008	Ferguson et al.	8,564,883	B2	10/2013	Totani
D571,816	S	6/2008	Corcoran	8,570,273	B1	10/2013	Smith
7,380,936	B2	6/2008	Howell	8,570,656	B1	10/2013	Weissman
7,401,918	B2	7/2008	Howell	D693,398	S	11/2013	Rubin
7,414,791	B2	8/2008	Urakawa	8,576,276	B2	11/2013	Bar-zeev
7,417,617	B2	8/2008	Eichenlaub	8,576,491	B2	11/2013	Takagi
7,425,065	B2	9/2008	Wang	8,587,869	B2	11/2013	Totani
7,457,040	B2	11/2008	Amitai	8,593,795	B1	11/2013	Chi
7,481,531	B2	1/2009	Howell	8,594,467	B2	11/2013	Lu
7,488,294	B2	2/2009	Torch	D696,668	S	12/2013	Chen
7,500,747	B2	3/2009	Howell	8,611,015	B2	12/2013	Wheeler
7,522,344	B1	4/2009	Curatu	8,638,498	B2	1/2014	Bohn et al.
7,542,210	B2	6/2009	Chirieleison, Sr.	8,662,686	B2	3/2014	Takagi
7,543,943	B1	6/2009	Hubby, Jr.	8,665,214	B2	3/2014	Forutanpour
7,582,828	B2	9/2009	Ryan	8,670,183	B2	3/2014	Clavin
7,646,540	B2	1/2010	Dolgoff	8,678,581	B2	3/2014	Blum
7,677,723	B2	3/2010	Howell	8,696,113	B2	4/2014	Lewis
7,690,799	B2	4/2010	Nestorovic	8,698,157	B2	4/2014	Hanamura
7,728,799	B2	6/2010	Kerr	8,711,487	B2	4/2014	Takeda
7,733,571	B1	6/2010	Li	D704,764	S	5/2014	Markovitz
7,771,046	B2	8/2010	Howell	8,730,129	B2	5/2014	Solomon
				8,743,052	B1	6/2014	Keller
				8,745,058	B1	6/2014	Garcia-Barrio
				8,750,541	B1	6/2014	Dong
				8,752,963	B2	6/2014	Meculloch

(56)

References Cited

U.S. PATENT DOCUMENTS

8,760,765 B2	6/2014	Gupta	D751,551 S	3/2016	Ho
8,767,306 B1	7/2014	Miao	D751,552 S	3/2016	Osterhout
8,770,742 B2	7/2014	Howell	9,274,338 B2	3/2016	Robbins et al.
8,786,675 B2	7/2014	Deering	9,286,728 B2	3/2016	Osterhout
8,786,686 B1	7/2014	Amirparviz	9,292,973 B2	3/2016	Bar-Zeev et al.
8,787,006 B2	7/2014	Golko	9,298,001 B2	3/2016	Border
8,803,867 B2	8/2014	Oikawa	9,298,002 B2	3/2016	Border
8,814,691 B2	8/2014	Haddick	9,298,007 B2	3/2016	Border
8,823,071 B2	9/2014	Oyamada	9,299,194 B2	3/2016	Border
8,824,779 B1	9/2014	Smyth	D753,114 S	4/2016	Osterhout
8,832,557 B2	9/2014	Fadell	9,310,610 B2	4/2016	Border
8,836,768 B1	9/2014	Rafii	9,316,833 B2	4/2016	Border
8,837,880 B2	9/2014	Takeda	9,323,325 B2	4/2016	Perez et al.
8,854,433 B1	10/2014	Rafii	D756,363 S	5/2016	Mathis
8,854,735 B2	10/2014	Totani	D757,006 S	5/2016	Cazalet
8,866,702 B1	10/2014	Wong	9,329,387 B2	5/2016	Border
8,866,849 B1	10/2014	Cho	9,354,445 B1	5/2016	Weaver
8,867,139 B2	10/2014	Gupta	9,355,867 B2	5/2016	Stockum
D716,808 S	11/2014	Yeom	9,366,868 B2	6/2016	Border
D716,813 S	11/2014	Deng	9,377,625 B2	6/2016	Border
8,878,749 B1	11/2014	Wu	D761,796 S	7/2016	Heinrich
D719,568 S	12/2014	Heinrich	9,400,390 B2	7/2016	Osterhout
D719,569 S	12/2014	Heinrich	9,401,540 B2	7/2016	Osterhout
D719,570 S	12/2014	Heinrich	D765,076 S	8/2016	Rochat
8,922,530 B2	12/2014	Pance	9,423,612 B2	8/2016	Border
8,929,589 B2	1/2015	Publicover et al.	9,423,842 B2	8/2016	Osterhout
D723,092 S	2/2015	Markovitz	9,436,006 B2	9/2016	Border
D723,093 S	2/2015	Li	9,448,409 B2	9/2016	Border
8,947,323 B1	2/2015	Raffle	D768,759 S	10/2016	Markovitz
8,948,935 B1	2/2015	Peeters	D769,873 S	10/2016	Cazalet
8,955,973 B2	2/2015	Raffle	9,482,880 B1	11/2016	Chandrasekhar
8,957,835 B2	2/2015	Hoellwarth	9,523,856 B2	12/2016	Osterhout
8,964,298 B2	2/2015	Haddick	9,529,195 B2	12/2016	Osterhout
D724,083 S	3/2015	Olsson	9,529,199 B2	12/2016	Osterhout
8,970,495 B1	3/2015	Biffle	9,651,787 B2	5/2017	Haddick
8,971,023 B2	3/2015	Olsson	9,651,788 B2	5/2017	Osterhout
8,982,014 B2	3/2015	Evans	9,651,789 B2	5/2017	Osterhout
8,982,471 B1	3/2015	Starner	9,672,210 B2	6/2017	Osterhout
D727,317 S	4/2015	Olsson	9,684,172 B2	6/2017	Border
9,010,929 B2	4/2015	Lewis	D792,400 S	7/2017	Osterhout
9,020,832 B2	4/2015	Fisher	D793,391 S	8/2017	Nakagawa
D728,573 S	5/2015	Deng	D793,467 S	8/2017	Krause
9,024,842 B1	5/2015	Prada Gomez	D794,022 S	8/2017	Limaye
9,031,273 B2	5/2015	Dong	D795,865 S	8/2017	Porter
9,033,502 B2	5/2015	Nistico	9,720,505 B2	8/2017	Gribetz et al.
D730,975 S	6/2015	Stables	9,746,676 B2	8/2017	Osterhout
D732,025 S	6/2015	Heinrich	D796,504 S	9/2017	Natsume
9,046,686 B2	6/2015	Saito	D796,506 S	9/2017	Natsume
9,046,999 B1	6/2015	Teller	D800,118 S	10/2017	Xing
9,063,563 B1	6/2015	Gray	D803,832 S	11/2017	Lin
D733,709 S	7/2015	Kawai	9,846,308 B2	12/2017	Osterhout
9,076,368 B2	7/2015	Evans	9,897,822 B2	2/2018	Osterhout
9,096,920 B1	8/2015	Gomez	10,013,053 B2	7/2018	Cederlund et al.
9,105,261 B2	8/2015	Horii	10,025,379 B2	7/2018	Drake et al.
9,107,622 B2	8/2015	Nistico	10,379,365 B2	8/2019	Osterhout
9,116,337 B1	8/2015	Miao	11,126,003 B2	9/2021	Osterhout et al.
D738,373 S	9/2015	Davies	11,892,644 B2 *	2/2024	Osterhout G06F 3/03542
9,122,054 B2	9/2015	Osterhout	2001/0019240 A1	9/2001	Takahashi
9,128,281 B2	9/2015	Osterhout	2001/0050817 A1	12/2001	Travers
9,129,157 B2	9/2015	Chao	2002/0005108 A1	1/2002	Ludwig
9,129,295 B2	9/2015	Border	2002/0021498 A1	2/2002	Ohtaka
9,143,693 B1	9/2015	Zhou	2002/0054272 A1	5/2002	Ebata
D741,398 S	10/2015	Echeverri	2002/0109903 A1	8/2002	Kaeriyama
9,158,115 B1	10/2015	Worley	2002/0148655 A1	10/2002	Cho
9,158,116 B1	10/2015	Osterhout	2002/0149545 A1	10/2002	Hanayama
D743,963 S	11/2015	Osterhout	2002/0183101 A1	12/2002	Oh
9,176,582 B1	11/2015	Johnson	2002/0191297 A1	12/2002	Gleckman
D744,581 S	12/2015	Votel	2003/0030597 A1	2/2003	Geist
D745,007 S	12/2015	Cazalet	2003/0030912 A1	2/2003	Gleckman
9,202,233 B1	12/2015	Siegel	2003/0151834 A1	8/2003	Penn
9,225,934 B2	12/2015	Cho	2003/0209953 A1	11/2003	Park
9,229,233 B2	1/2016	Osterhout	2003/0234823 A1	12/2003	Sato
9,229,234 B2	1/2016	Osterhout	2004/0008158 A1	1/2004	Chi
9,235,051 B2	1/2016	Salter	2004/0024287 A1	2/2004	Patton
9,269,193 B2	2/2016	Saito	2004/0027312 A1	2/2004	Owada
			2004/0030448 A1	2/2004	Solomon
			2004/0032392 A1	2/2004	Chi
			2004/0066363 A1	4/2004	Yamano
			2004/0066547 A1	4/2004	Parker

(56)

References Cited

U.S. PATENT DOCUMENTS

2004/0080541	A1	4/2004	Saiga	2009/0108837	A1	4/2009	Johansson
2004/0130522	A1	7/2004	Lin	2009/0110241	A1	4/2009	Takemoto
2004/0132509	A1	7/2004	Glezerman	2009/0147331	A1	6/2009	Ashkenazi
2004/0150631	A1	8/2004	Fleck	2009/0183929	A1	7/2009	Zhang
2004/0194880	A1	10/2004	Jiang	2009/0251441	A1	10/2009	Edgecomb
2004/0227994	A1	11/2004	Bruzzzone	2009/0279180	A1	11/2009	Amitai
2005/0010091	A1	1/2005	Woods	2010/0001572	A1	1/2010	Masunaga
2005/0010563	A1	1/2005	Gross	2010/0007852	A1	1/2010	Bietry
2005/0041289	A1	2/2005	Berman	2010/0045928	A1	2/2010	Levy
2005/0122319	A1	6/2005	Sakurai	2010/0046075	A1	2/2010	Powell
2005/0156915	A1	7/2005	Fisher	2010/0056274	A1	3/2010	Usitalo
2005/0157949	A1	7/2005	Aiso	2010/0060713	A1	3/2010	Snyder
2005/0212980	A1	9/2005	Miyazaki	2010/0079356	A1	4/2010	Hoellwarth
2005/0264752	A1	12/2005	Howell	2010/0079508	A1	4/2010	Hodge
2006/0023158	A1	2/2006	Howell et al.	2010/0079733	A1	4/2010	Lu
2006/0047386	A1	3/2006	Kanevsky	2010/0082368	A1	4/2010	Gecelter
2006/0050146	A1	3/2006	Richardson	2010/0085325	A1	4/2010	King-smith
2006/0061542	A1	3/2006	Stokic	2010/0094161	A1	4/2010	Kiderman
2006/0092131	A1	5/2006	Kuroki	2010/0097580	A1	4/2010	Yamamoto
2006/0098293	A1	5/2006	Garoutte	2010/0103075	A1	4/2010	Kalaboukis
2006/0119794	A1	6/2006	Hillis	2010/0113062	A1	5/2010	Lee
2006/0132457	A1	6/2006	Rimas-ribikauskas	2010/0130140	A1	5/2010	Waku
2006/0132924	A1	6/2006	Mimran	2010/0149073	A1	6/2010	Chaum
2006/0152686	A1	7/2006	Yeralan	2010/0178101	A1	7/2010	Day
2006/0170652	A1	8/2006	Bannai	2010/0194682	A1	8/2010	Orr
2006/0173351	A1	8/2006	Marcotte	2010/0240988	A1	9/2010	Varga
2006/0178827	A1	8/2006	Aoyama	2010/0241450	A1	9/2010	Gierhart
2006/0215111	A1	9/2006	Mihashi	2010/0254017	A1	10/2010	Martins
2006/0224238	A1	10/2006	Azar	2010/0280904	A1	11/2010	Ahuja
2006/0238550	A1	10/2006	Page	2010/0283774	A1	11/2010	Bovet
2006/0239629	A1	10/2006	Qi	2010/0290127	A1	11/2010	Kessler
2006/0250322	A1	11/2006	Hall	2010/0309426	A1	12/2010	Howell
2006/0250696	A1	11/2006	Mcguire	2010/0329301	A1	12/2010	Pang
2006/0285315	A1	12/2006	Tufenkjian	2011/0006982	A1	1/2011	Rhee
2006/0288233	A1	12/2006	Kozlay	2011/0007081	A1	1/2011	Gordon
2007/0003168	A1	1/2007	Oliver	2011/0012874	A1	1/2011	Kurozuka
2007/0004451	A1	1/2007	C. Anderson	2011/0089325	A1	4/2011	Ottney
2007/0024750	A1	2/2007	Wing	2011/0096100	A1	4/2011	Sprague
2007/0024763	A1	2/2007	Chung	2011/0102234	A1	5/2011	Adams
2007/0024764	A1	2/2007	Chung	2011/0130958	A1	6/2011	Stahl
2007/0024820	A1	2/2007	Chung	2011/0131495	A1	6/2011	Bull
2007/0024823	A1	2/2007	Chung	2011/0157236	A1	6/2011	Inoue
2007/0025273	A1	2/2007	Chung	2011/0159931	A1	6/2011	Boss
2007/0030243	A1	2/2007	Ishii	2011/0164047	A1	7/2011	Pance
2007/0030456	A1	2/2007	Duncan	2011/0164163	A1	7/2011	Bilbrey
2007/0035563	A1	2/2007	Biocca	2011/0164221	A1	7/2011	Tilleman
2007/0038960	A1	2/2007	Rekimoto	2011/0176106	A1	7/2011	Lewkowski
2007/0058868	A1	3/2007	Seino	2011/0196610	A1	8/2011	Waldman
2007/0069976	A1	3/2007	Willins	2011/0199171	A1	8/2011	Prest
2007/0070859	A1	3/2007	Hirayama	2011/0201213	A1	8/2011	Dabov
2007/0100637	A1	5/2007	Mccune	2011/0202823	A1	8/2011	Berger
2007/0120806	A1	5/2007	Schmidt	2011/0205209	A1	8/2011	Kurokawa
2007/0120836	A1	5/2007	Yamaguchi	2011/0211056	A1	9/2011	Publicover et al.
2007/0132662	A1	6/2007	Morita	2011/0213664	A1	9/2011	Osterhout
2007/0178950	A1	8/2007	Lewis	2011/0221672	A1	9/2011	Osterhout
2007/0233376	A1	10/2007	Gershony	2011/0221896	A1	9/2011	Haddick
2007/0263174	A1	11/2007	Cheng	2011/0227820	A1	9/2011	Haddick
2007/0273611	A1	11/2007	Torch	2011/0234475	A1	9/2011	Endo
2007/0282682	A1	12/2007	Dietz	2011/0234631	A1	9/2011	Kim
2007/0296684	A1	12/2007	Thomas	2011/0241975	A1	10/2011	Mukawa
2008/0005702	A1	1/2008	Skourup	2011/0248963	A1	10/2011	Lawrence
2008/0066973	A1	3/2008	Furuki	2011/0285638	A1	11/2011	Harris
2008/0121441	A1	5/2008	Sheets	2011/0285764	A1	11/2011	Kimura
2008/0122736	A1	5/2008	Ronzani	2012/0021806	A1	1/2012	Maltz
2008/0125288	A1	5/2008	Case	2012/0026088	A1	2/2012	Goran
2008/0143954	A1	6/2008	Abreu	2012/0026455	A1	2/2012	Takahashi
2008/0186255	A1	8/2008	Cohen	2012/0032874	A1	2/2012	Mukawa
2008/0191965	A1	8/2008	Pandozy	2012/0035934	A1	2/2012	Cunningham
2008/0219025	A1	9/2008	Spitzer	2012/0050140	A1	3/2012	Border
2008/0266645	A1	10/2008	Dharmatilleke	2012/0050493	A1	3/2012	Ernst
2008/0291277	A1	11/2008	Jacobsen	2012/0056093	A1	3/2012	Poteet
2008/0298639	A1	12/2008	Tsunekawa	2012/0062444	A1	3/2012	Cok
2009/0013204	A1	1/2009	Kobayashi	2012/0062594	A1	3/2012	Campbell
2009/0015735	A1	1/2009	Simmonds	2012/0062850	A1	3/2012	Travis
2009/0040296	A1	2/2009	Moscato	2012/0062998	A1	3/2012	Schultz
				2012/0068913	A1	3/2012	Bar-zeev
				2012/0069413	A1	3/2012	Schultz
				2012/0075168	A1	3/2012	Osterhout
				2012/0078628	A1	3/2012	Ghulman

(56)

References Cited

U.S. PATENT DOCUMENTS

2012/0081800	A1	4/2012	Cheng	2013/0196757	A1	8/2013	Latta
2012/0092328	A1	4/2012	Flaks	2013/0201080	A1	8/2013	Evans
2012/0092329	A1	4/2012	Koo	2013/0201081	A1	8/2013	Evans
2012/0096095	A1	4/2012	Bhargava	2013/0207887	A1	8/2013	Raffle
2012/0113514	A1	5/2012	Rodman	2013/0207970	A1	8/2013	Shpunt
2012/0119978	A1	5/2012	Border	2013/0214909	A1	8/2013	Meijers
2012/0120103	A1	5/2012	Border	2013/0215149	A1	8/2013	Hayashi
2012/0120498	A1	5/2012	Harrison	2013/0222919	A1	8/2013	Komatsu
2012/0127062	A1	5/2012	Bar-zeev	2013/0230215	A1	9/2013	Gurman
2012/0127284	A1	5/2012	Bar-zeev	2013/0234914	A1	9/2013	Fujimaki
2012/0133885	A1	5/2012	Howell	2013/0235331	A1	9/2013	Heinrich
2012/0154920	A1	6/2012	Harrison	2013/0241805	A1	9/2013	Gomez
2012/0162270	A1	6/2012	Fleck	2013/0241948	A1	9/2013	Kimura
2012/0163013	A1	6/2012	Buelow, II	2013/0242405	A1	9/2013	Gupta
2012/0169608	A1	7/2012	Forutanpour	2013/0248691	A1	9/2013	Mirov
2012/0176682	A1	7/2012	Dejong	2013/0249778	A1	9/2013	Morimoto
2012/0188245	A1	7/2012	Hyatt	2013/0249787	A1	9/2013	Morimoto
2012/0194550	A1	8/2012	Osterhout	2013/0250207	A1	9/2013	Bohn
2012/0194553	A1	8/2012	Osterhout	2013/0250430	A1	9/2013	Robbins
2012/0194784	A1	8/2012	Shih	2013/0250503	A1	9/2013	Olsson
2012/0200935	A1	8/2012	Miyao	2013/0257622	A1	10/2013	Davalos
2012/0206443	A1	8/2012	Kimura	2013/0257709	A1	10/2013	Raffle
2012/0206817	A1	8/2012	Totani	2013/0258111	A1	10/2013	Frank
2012/0212398	A1	8/2012	Border	2013/0265212	A1	10/2013	Kato
2012/0212484	A1	8/2012	Haddick	2013/0265227	A1	10/2013	Julian
2012/0212499	A1	8/2012	Haddick	2013/0278631	A1	10/2013	Border
2012/0212593	A1	8/2012	Na	2013/0293530	A1	11/2013	Perez
2012/0218301	A1	8/2012	Miller	2013/0293580	A1	11/2013	Spivack
2012/0223885	A1	9/2012	Perez	2013/0300637	A1	11/2013	Smits
2012/0229367	A1	9/2012	Magyari	2013/0300652	A1	11/2013	Raffle
2012/0233000	A1	9/2012	Fisher	2013/0321265	A1	12/2013	Bychkov
2012/0235885	A1	9/2012	Miller	2013/0321271	A1	12/2013	Bychkov
2012/0237085	A1	9/2012	Meier	2013/0321932	A1	12/2013	Hsu
2012/0242251	A1	9/2012	Kwisthout	2013/0335301	A1	12/2013	Wong
2012/0242570	A1	9/2012	Kobayashi	2013/0335435	A1	12/2013	Ambrus
2012/0242697	A1	9/2012	Border	2013/0335461	A1	12/2013	Rekimoto
2012/0242698	A1	9/2012	Haddick	2013/0336528	A1	12/2013	Itani
2012/0249741	A1	10/2012	Maciocci	2013/0336629	A1	12/2013	Mulholland
2012/0249797	A1	10/2012	Haddick	2013/0342564	A1	12/2013	Kinnebrew
2012/0250152	A1	10/2012	Larson	2013/0342571	A1	12/2013	Kinnebrew
2012/0264510	A1	10/2012	Wigdor	2013/0342981	A1	12/2013	Cox
2012/0268449	A1	10/2012	Choi	2013/0346245	A1	12/2013	Desore
2012/0287398	A1	11/2012	Baker	2014/0028704	A1	1/2014	Wu
2012/0293548	A1	11/2012	Perez	2014/0029498	A1	1/2014	Kim
2012/0294478	A1	11/2012	Publicover	2014/0043682	A1	2/2014	Hussey
2012/0306850	A1	12/2012	Balan	2014/0062854	A1	3/2014	Cho
2012/0307198	A1	12/2012	Ifergan	2014/0063054	A1	3/2014	Osterhout
2012/0326948	A1	12/2012	Crocco	2014/0063055	A1	3/2014	Osterhout
2012/0327040	A1	12/2012	Simon et al.	2014/0063473	A1	3/2014	Pasolini
2012/0327116	A1	12/2012	Liu	2014/0078043	A1	3/2014	Kim
2013/0009366	A1	1/2013	Hannegan	2014/0078282	A1	3/2014	Aoki
2013/0009907	A1	1/2013	Rosenberg	2014/0091984	A1	4/2014	Ashbrook
2013/0044042	A1	2/2013	Olsson	2014/0101608	A1	4/2014	Ryskamp
2013/0063695	A1	3/2013	Hsieh	2014/0104142	A1	4/2014	Bickerstaff
2013/0069985	A1	3/2013	Wong	2014/0104692	A1	4/2014	Bickerstaff
2013/0070344	A1	3/2013	Takeda	2014/0111864	A1	4/2014	Margulis
2013/0077049	A1	3/2013	Bohn	2014/0125668	A1	5/2014	Steed
2013/0077147	A1	3/2013	Efimov	2014/0125785	A1	5/2014	Na
2013/0083009	A1	4/2013	Geisner	2014/0129328	A1	5/2014	Mathew
2013/0083055	A1	4/2013	Piemonte	2014/0139655	A1	5/2014	Mimar
2013/0088413	A1	4/2013	Raffle	2014/0146394	A1	5/2014	Tout
2013/0100259	A1	4/2013	Ramaswamy	2014/0147829	A1	5/2014	Jerauld
2013/0106674	A1	5/2013	Wheeler	2014/0152530	A1	6/2014	Venkatesha
2013/0120224	A1	5/2013	Cajigas	2014/0152558	A1	6/2014	Salter
2013/0120841	A1	5/2013	Shpunt	2014/0152676	A1	6/2014	Rohn
2013/0121562	A1	5/2013	Barnum	2014/0153173	A1	6/2014	Pombo
2013/0127906	A1	5/2013	Sugita	2014/0159995	A1	6/2014	Adams
2013/0127980	A1	5/2013	Haddick	2014/0160055	A1	6/2014	Margolis
2013/0135198	A1	5/2013	Hodge	2014/0160137	A1	6/2014	Martin
2013/0141434	A1	6/2013	Sugden	2014/0160157	A1	6/2014	Poulos
2013/0154913	A1	6/2013	Genc	2014/0160170	A1	6/2014	Lyons
2013/0162632	A1	6/2013	Varga	2014/0168056	A1	6/2014	Swaminathan
2013/0176533	A1	7/2013	Raffle	2014/0168266	A1	6/2014	Kimura
2013/0185052	A1	7/2013	Boyd	2014/0168716	A1	6/2014	King
2013/0194389	A1	8/2013	Vaught	2014/0168735	A1	6/2014	Yuan
				2014/0176591	A1	6/2014	Klein
				2014/0176603	A1	6/2014	Kumar
				2014/0177023	A1	6/2014	Gao
				2014/0183269	A1	7/2014	Glaser

(56)

References Cited

U.S. PATENT DOCUMENTS

2014/0195918	A1	7/2014	Friedlander	2015/0205127	A1	7/2015	Border
2014/0204759	A1	7/2014	Guo et al.	2015/0205128	A1	7/2015	Border
2014/0206416	A1	7/2014	Aurongzeb	2015/0205129	A1	7/2015	Border
2014/0213280	A1	7/2014	Sandel	2015/0205130	A1	7/2015	Border
2014/0222929	A1	8/2014	Grossman	2015/0205131	A1	7/2015	Border
2014/0225814	A1	8/2014	English	2015/0205132	A1	7/2015	Osterhout
2014/0232651	A1	8/2014	Kress	2015/0205135	A1	7/2015	Border
2014/0240313	A1	8/2014	Varga	2015/0205346	A1	7/2015	Border
2014/0247286	A1	9/2014	Chi	2015/0205347	A1	7/2015	Border
2014/0253588	A1	9/2014	Mandala	2015/0205348	A1	7/2015	Nortrup
2014/0253605	A1	9/2014	Border	2015/0205349	A1	7/2015	Nortrup
2014/0267010	A1	9/2014	Pasquero	2015/0205351	A1	7/2015	Osterhout
2014/0285631	A1	9/2014	Janky	2015/0205373	A1	7/2015	Osterhout
2014/0306866	A1	10/2014	Miller	2015/0205378	A1	7/2015	Osterhout
2014/0310075	A1	10/2014	Ricci	2015/0205384	A1	7/2015	Osterhout
2014/0320389	A1	10/2014	Scavezze	2015/0205385	A1	7/2015	Osterhout
2014/0320971	A1	10/2014	Gupta	2015/0205387	A1	7/2015	Osterhout
2014/0341441	A1	11/2014	Slaby	2015/0205388	A1	7/2015	Osterhout
2014/0347572	A1	11/2014	Liu	2015/0205401	A1	7/2015	Osterhout
2014/0361957	A1	12/2014	Hua	2015/0205402	A1	7/2015	Osterhout
2014/0361976	A1	12/2014	Osman	2015/0205494	A1	7/2015	Scott
2014/0363797	A1	12/2014	Hu	2015/0205566	A1	7/2015	Osterhout
2014/0372957	A1	12/2014	Keane	2015/0206008	A1	7/2015	Border
2014/0375542	A1	12/2014	Robbins	2015/0206173	A1	7/2015	Nortrup
2014/0375545	A1	12/2014	Ackerman	2015/0212324	A1	7/2015	Osterhout
2014/0375680	A1	12/2014	Ackerman	2015/0212327	A1	7/2015	Osterhout
2014/0375683	A1	12/2014	Salter	2015/0213584	A1	7/2015	Ishikawa
2015/0002371	A1	1/2015	Burgess	2015/0213650	A1	7/2015	Barzuza
2015/0022542	A1	1/2015	Baba	2015/0226966	A1	8/2015	Osterhout
2015/0029088	A1	1/2015	Kim	2015/0226967	A1	8/2015	Osterhout
2015/0035744	A1	2/2015	Robbins	2015/0228099	A1	8/2015	Osterhout
2015/0042544	A1	2/2015	Sugihara	2015/0228119	A1	8/2015	Osterhout
2015/0084862	A1	3/2015	Sugihara	2015/0228120	A1	8/2015	Osterhout
2015/0097719	A1	4/2015	Balachandreswaran	2015/0229019	A1	8/2015	Osterhout
2015/0134143	A1	5/2015	Willenborg	2015/0234508	A1	8/2015	Cho
2015/0143297	A1	5/2015	Wheeler	2015/0235422	A1	8/2015	Lohse
2015/0145839	A1	5/2015	Hack	2015/0235429	A1	8/2015	Miller
2015/0146004	A1	5/2015	Kritt	2015/0235622	A1	8/2015	Border
2015/0161913	A1	6/2015	Dominguez	2015/0241963	A1	8/2015	Nortrup
2015/0168730	A1	6/2015	Ashkenazi	2015/0241964	A1	8/2015	Nortrup
2015/0168731	A1	6/2015	Robbins	2015/0241965	A1	8/2015	Nortrup
2015/0169953	A1	6/2015	Border	2015/0241966	A1	8/2015	Nortrup
2015/0178932	A1	6/2015	Wyatt	2015/0243039	A1	8/2015	Holz
2015/0181383	A1	6/2015	Schulz	2015/0245131	A1	8/2015	Facteau
2015/0186636	A1	7/2015	Tharappel	2015/0253573	A1	9/2015	Sako
2015/0198807	A1	7/2015	Hirai	2015/0260986	A1	9/2015	Nortrup
2015/0201834	A1	7/2015	Border	2015/0261015	A1	9/2015	Han
2015/0201835	A1	7/2015	Border	2015/0277113	A1	10/2015	Border
2015/0201836	A1	7/2015	Border	2015/0277116	A1	10/2015	Richards
2015/0201836	A1	7/2015	Border	2015/0277118	A1	10/2015	Border
2015/0202962	A1	7/2015	Habashima	2015/0277120	A1	10/2015	Border
2015/0205035	A1	7/2015	Border	2015/0277122	A1	10/2015	Border
2015/0205100	A1	7/2015	Border	2015/0277549	A1	10/2015	Border
2015/0205101	A1	7/2015	Border	2015/0277559	A1	10/2015	Vescovi
2015/0205102	A1	7/2015	Border	2015/0279104	A1	10/2015	Border
2015/0205103	A1	7/2015	Border	2015/0279107	A1	10/2015	Border
2015/0205104	A1	7/2015	Border	2015/0279108	A1	10/2015	Border
2015/0205105	A1	7/2015	Border	2015/0287048	A1	10/2015	Nortrup
2015/0205107	A1	7/2015	Border	2015/0293587	A1	10/2015	Wilairat
2015/0205108	A1	7/2015	Border	2015/0294156	A1	10/2015	Border
2015/0205111	A1	7/2015	Border	2015/0294627	A1	10/2015	Yoo
2015/0205112	A1	7/2015	Border	2015/0301593	A1	10/2015	Border
2015/0205113	A1	7/2015	Border	2015/0302646	A1	10/2015	Osterhout
2015/0205114	A1	7/2015	Border	2015/0302647	A1	10/2015	Osterhout
2015/0205115	A1	7/2015	Border	2015/0304368	A1	10/2015	Vaccari
2015/0205116	A1	7/2015	Border	2015/0309313	A1	10/2015	Border
2015/0205117	A1	7/2015	Border	2015/0309314	A1	10/2015	Border
2015/0205118	A1	7/2015	Border	2015/0309317	A1	10/2015	Osterhout
2015/0205119	A1	7/2015	Osterhout	2015/0309534	A1	10/2015	Osterhout
2015/0205120	A1	7/2015	Border	2015/0309562	A1	10/2015	Shams
2015/0205121	A1	7/2015	Border	2015/0309995	A1	10/2015	Osterhout
2015/0205122	A1	7/2015	Border	2015/0316766	A1	11/2015	Weaver
2015/0205123	A1	7/2015	Border	2015/0316769	A1	11/2015	Border
2015/0205124	A1	7/2015	Border	2015/0316770	A1	11/2015	Border
2015/0205125	A1	7/2015	Border	2015/0316771	A1	11/2015	Border
2015/0205126	A1	7/2015	Schowengerdt	2015/0316772	A1	11/2015	Border
				2015/0331241	A1	11/2015	Haddick
				2015/0332032	A1	11/2015	Alameh
				2015/0338661	A1	11/2015	Osterhout

(56)

References Cited

U.S. PATENT DOCUMENTS

2015/0346496	A1	12/2015	Haddick
2015/0346511	A1	12/2015	Osterhout
2015/0347823	A1	12/2015	Monnerat
2015/0355466	A1	12/2015	Border
2015/0355468	A1	12/2015	Osterhout
2015/0356772	A1	12/2015	Osterhout
2015/0356775	A1	12/2015	Osterhout
2015/0356776	A1	12/2015	Osterhout
2015/0356777	A1	12/2015	Osterhout
2015/0356778	A1	12/2015	Osterhout
2015/0356779	A1	12/2015	Osterhout
2015/0363975	A1	12/2015	Osterhout
2015/0382305	A1	12/2015	Drincic
2016/0005003	A1	1/2016	Norris
2016/0011417	A1	1/2016	Border
2016/0015470	A1	1/2016	Border
2016/0018640	A1	1/2016	Haddick
2016/0018641	A1	1/2016	Haddick
2016/0018642	A1	1/2016	Haddick
2016/0018644	A1	1/2016	Border
2016/0018645	A1	1/2016	Haddick
2016/0018646	A1	1/2016	Osterhout
2016/0018647	A1	1/2016	Osterhout
2016/0018648	A1	1/2016	Osterhout
2016/0018649	A1	1/2016	Osterhout
2016/0018650	A1	1/2016	Haddick
2016/0018651	A1	1/2016	Haddick
2016/0018652	A1	1/2016	Haddick
2016/0018653	A1	1/2016	Haddick
2016/0018654	A1	1/2016	Haddick
2016/0019715	A1	1/2016	Haddick
2016/0019719	A1	1/2016	Osterhout
2016/0021304	A1	1/2016	Osterhout
2016/0025974	A1	1/2016	Osterhout
2016/0025977	A1	1/2016	Osterhout
2016/0025979	A1	1/2016	Border
2016/0025980	A1	1/2016	Osterhout
2016/0026239	A1	1/2016	Border
2016/0027211	A1	1/2016	Osterhout
2016/0027414	A1	1/2016	Osterhout
2016/0045810	A1	2/2016	Minkovitch
2016/0048019	A1	2/2016	Haddick
2016/0048021	A1	2/2016	Border
2016/0048023	A1	2/2016	Haddick
2016/0048025	A1	2/2016	Cazalet
2016/0048160	A1	2/2016	Haddick
2016/0049008	A1	2/2016	Haddick
2016/0054566	A1	2/2016	Osterhout
2016/0062118	A1	3/2016	Osterhout
2016/0062121	A1	3/2016	Border
2016/0062122	A1	3/2016	Border
2016/0077342	A1	3/2016	Osterhout
2016/0085071	A1	3/2016	Border
2016/0085072	A1	3/2016	Haddick
2016/0085278	A1	3/2016	Osterhout
2016/0091718	A1	3/2016	Border
2016/0091719	A1	3/2016	Border
2016/0103325	A1	4/2016	Mirza
2016/0109709	A1	4/2016	Osterhout
2016/0109711	A1	4/2016	Border
2016/0109713	A1	4/2016	Osterhout
2016/0116738	A1	4/2016	Osterhout
2016/0116745	A1	4/2016	Osterhout
2016/0116979	A1	4/2016	Border
2016/0131904	A1	5/2016	Border
2016/0131911	A1	5/2016	Border
2016/0131912	A1	5/2016	Border
2016/0132082	A1	5/2016	Border
2016/0133201	A1	5/2016	Border
2016/0137312	A1	5/2016	Osterhout
2016/0147063	A1	5/2016	Border
2016/0147064	A1	5/2016	Border
2016/0147065	A1	5/2016	Border
2016/0147070	A1	5/2016	Border
2016/0154242	A1	6/2016	Border
2016/0154244	A1	6/2016	Border
2016/0161743	A1	6/2016	Osterhout
2016/0161747	A1	6/2016	Osterhout
2016/0170207	A1	6/2016	Haddick
2016/0170208	A1	6/2016	Border
2016/0170209	A1	6/2016	Border
2016/0170699	A1	6/2016	Border
2016/0171769	A1	6/2016	Haddick
2016/0171846	A1	6/2016	Brav
2016/0178904	A1	6/2016	Deleeuw
2016/0187651	A1	6/2016	Border
2016/0187658	A1	6/2016	Osterhout
2016/0202946	A1	7/2016	Osterhout
2016/0207457	A1	7/2016	Border
2016/0209674	A1	7/2016	Montalban
2016/0216516	A1	7/2016	Border
2016/0216517	A1	7/2016	Border
2016/0231571	A1	8/2016	Border
2016/0239985	A1	8/2016	Haddick
2016/0240008	A1	8/2016	Haddick
2016/0246055	A1	8/2016	Border
2016/0252731	A1	9/2016	Border
2016/0259166	A1	9/2016	Border
2016/0274361	A1	9/2016	Border
2016/0282626	A1	9/2016	Border
2016/0286177	A1	9/2016	Border
2016/0286203	A1	9/2016	Border
2016/0286210	A1	9/2016	Border
2016/0370606	A1	12/2016	Huynh
2017/0031395	A1	2/2017	Osterhout
2017/0219831	A1	8/2017	Haddick
2017/0220865	A1	8/2017	Osterhout
2017/0227778	A1	8/2017	Osterhout
2017/0227793	A1	8/2017	Abreu
2017/0235133	A1	8/2017	Border
2017/0235134	A1	8/2017	Border
2017/0337187	A1	11/2017	Osterhout
2017/0351098	A1	12/2017	Osterhout
2018/0003988	A1	1/2018	Osterhout
2018/0059434	A1	3/2018	Heisey
2022/0019085	A1	1/2022	Osterhout

FOREIGN PATENT DOCUMENTS

CA	2388766	A1	12/2003
CN	107250882	A	10/2017
EP	368898	A1	5/1990
EP	777867	A1	6/1997
EP	1326121	A2	7/2003
EP	2207164	A2	7/2010
EP	2486450	A1	8/2012
EP	2490130	A1	8/2012
EP	2502410	A1	9/2012
EP	2674834	A2	12/2013
GB	2491984	A	12/2012
JP	07110735	A	4/1995
JP	200102036	A	4/2000
JP	2005138755	A	6/2005
JP	2009171505	A	7/2009
JP	5017989	B2	9/2012
JP	2012212990	A	11/2012
KR	1020110101944		9/2011
WO	2011143655	A1	11/2011
WO	2012040030	A2	3/2012
WO	2012058175	A1	5/2012
WO	2012064546	A1	5/2012
WO	2012082807	A2	6/2012
WO	2012118573	A1	9/2012
WO	2012118575	A2	9/2012
WO	2013043288	A2	3/2013
WO	2013049248	A2	4/2013
WO	2013050650	A1	4/2013
WO	2013103825	A1	7/2013
WO	2013110846	A1	8/2013
WO	2013170073	A1	11/2013
WO	2013176079	A1	11/2013
WO	2015109145	A1	7/2015
WO	2015164276	A1	10/2015
WO	2015179877	A2	11/2015

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	2015195444	A1	12/2015
WO	2016044035	A1	3/2016
WO	2016073734	A1	5/2016
WO	2016133886	A1	8/2016
WO	2016205601	A1	12/2016
WO	2017100074	A1	6/2017
WO	2018044537	A1	3/2018

OTHER PUBLICATIONS

Jacob, R. "Eye Tracking in Advanced Interface Design", Virtual Environments and Advanced Interface Design, Oxford University Press, Inc. (Jun. 1995).

Rolland, J. et al., "High-resolution inset head-mounted display", Optical Society of America, vol. 37, No. 19, Applied Optics, (Jul. 1, 1998).

Tanriverdi, V. et al. (Apr. 2000). "Interacting With Eye Movements in Virtual Environments," Department of Electrical Engineering and Computer Science, Tufts University, Medford, MA 02155, USA, Proceedings of the SIGCHI conference on Human Factors in Computing Systems, eight pages.

Yoshida, A. et al., "Design and Applications of a High Resolution Insert Head Mounted Display", (Jun. 1994).

Non-Final Office Action mailed Jan. 20, 2023, for U.S. Appl. No. 17/389,174, filed Jul. 29, 2021, eight pages.

Non-Final Office Action mailed Jul. 7, 2023, for U.S. Appl. No. 17/389,174, filed Jul. 29, 2021, eight pages.

Notice of Allowance mailed Oct. 23, 2023, for U.S. Appl. No. 17/389,174, filed Jul. 29, 2021, seven pages.

PCT/US2015/059264, "International Application Serial No. PCT/US2015/059264, International Search Report and Written Opinion mailed Feb. 19, 2016", Osterhout Group, Inc., eleven Pages.

15857713.0 European Application Serial No. 15857713.0, Extended European Search Report Received mailed Oct. 16, 2017, Osterhout Group, Inc., seven pages.

Allison, R. S. et al. "Tolerance of Temporal Delay in Virtual Environments," VR '01 Proceedings of the Virtual Reality 2001 Conference (VR'01), Centre for Vision Research and Departments of Computer Science and Psychology, Mar. 2001, 1-8.

Bezryadin, et al. "Brightness Calculation in Digital Image Processing," Technologies for Digital Fulfillment 2007, Las Vegas, NV, 2007, pp. 1-6.

Bimber, Oliver et al. (2005). "Spatial Augmented Reality: Merging Real and Virtual Worlds," A. K. Peters, Ltd., Wellesley, MA.

Clements-Cortes, et al. "Short-Term Effects of Rhythmic Sensory Stimulation in Alzheimer's Disease: An Exploratory Pilot Study," Journal of Alzheimer's Disease 52 (2016), IOS Press Feb. 9, 2016, pp. 651-660.

Fathi, A. et al. (2012). "Social interactions: A First-Person Perspective," Computer Vision and Pattern Recognition (CVPR), IEEE Conference on. IEEE, 2012, eight Pages.

Final Office Action mailed Aug. 7, 2018, for U.S. Appl. No. 15/488,916, filed Apr. 17, 2017, eight pages.

Huang, "Image Completion Using Planar Structure Guidelines," ACM Transactions on Graphics, 33(4):129, Jul. 14, 2011, 1-10.

Janin, Adam L. et al. (1993). "Calibration of Head-Mounted Displays for Augmented Reality Applications", Research and Technology Boeing Computer Services MS 7L-48 P.O. Box 24346 Seattle, WA 98124-0346 Virtual Reality Annual International Symposium, 1993., 1993 IEEE, ten Pages.

Lang, Manuel et al. "Nonlinear Disparity Mapping For Stereoscopic 3D", Jul. 2010, pp. 1-10.

Logbar Inc., "Ring: Shortcut Everything", <https://www.kickstarter.com/projects/1761670738/ring-shortcut-everything>, Dec. 2014, twenty-two pages.

Mastandrea. "Mycestro, The Next Generation 3D Mouse," <https://www.kickstarter.com/projects/mycestro/mycestrotm-the-next-generation-3d-mouse>. Dec. 2014, twenty-two pages.

Non-Final Office Action mailed Oct. 5, 2017, for U.S. Appl. No. 15/488,916, filed Apr. 17, 2017, nine pages.

Notice of Allowance mailed Mar. 25, 2019, for U.S. Appl. No. 15/488,916, filed Apr. 17, 2017, eight pages.

Osterhout, R. (Oct. 29, 2015). "Commercial and Social Implications", accessed on Aug. 26, 2017, interviewed on Oct. 14, 2015, twelve pages.

Pamplona, Vitor R. et al., "Photorealistic Models for Pupil Light Reflex and Iridal Pattern Deformation", ACM Transactions on Graphics, vol. 28, No. 4, Article 106, Publication date: Aug. 2009, pp. 1-12.

PCT/2016/064441, Application Serial No. PCT/US2016/064441, International Search Report and Written Opinion mailed Feb. 7, 2017, Osterhout Group, Inc., sixteen pages.

PCT/US2015/011697, "International Application Serial No. PCT/US2015/011697, International Preliminary Report on Patentability and Written Opinion mailed Jul. 28, 2016," Osterhout Group, Inc., ten pages.

PCT/US2015/011697, "International Application Serial No. PCT/US2015/011697, International Search Report and Written Opinion mailed Apr. 13, 2015," Osterhout Group, Inc., fourteen pages.

PCT/US2015/026704, "International Search Report and Written Opinion" mailed Aug. 21, 2015, fifteen pages.

PCT/US2015/035192, "International Applications Serial No. PCT/US2015/035192, International Search Report and Written Opinion mailed Sep. 3, 2015," Osterhout Group, Inc., eleven pages.

PCT/US2015/059264, International Application Serial No. PCT/US2015/059264, International Preliminary report on Patentability and Written Opinion mailed May 18, 2017, Osterhout Group, Inc., eight pages.

PCT/US2015033379, International Application Serial No. PCT/US2015033379, International Search Report Patentability and Written Opinion, mailed Nov. 30, 2015, Osterhout Group, Inc. twelve pages.

PCT/US2016/0042440, filed Jul. 15, 2016. (copy not attached.).

PCT/US2016/018040, filed Feb. 16, 2016. (copy not attached.).

PCT/US2016/018040, "International Application Serial No. PCT/US2016/018040, International Search Report and Written Opinion mailed Jul. 6, 2016," Osterhout Group, Inc. ten pages.

PCT/US2016/038008, filed Jun. 17, 2016. (copy not attached.).

PCT/US2016/038008, International Application Serial No. PCT/US2016/038008, International Preliminary Report on Patentability mailed on Dec. 28, 2017, Osterhout Group, Inc. six pages.

PCT/US2016/038008, International Application Serial No. PCT/US2016/038008, International Search Report and Written Opinion mailed on Oct. 27, 2016, Osterhout Group, Inc. eight pages.

PCT/US2016/042440, "Application Serial No. PCT/US2016/042440, The International Search Report and Written Opinion mailed Oct. 13, 2016," Osterhout Group, Inc. seven pages.

PCT/US2016/058203, filed Oct. 21, 2016. (copy not attached.).

PCTUS2017046701, "Application Serial No. PCTUS2017046701, International Search Report and the Written Opinion mailed Nov. 6, 2017", seven pages.

Plainis, et al., "The Physiologic Mechanism of Accommodation", Cataract & Refractive Surgery Today Europe, Apr. 2014, pp. 23-29.

Schedwill, "Bidirectional OLED Microdisplay", Fraunhofer Research Institution for Organics, Materials and Electronic Device COMEDD, Apr. 11, 2014, two pages.

U.S. Appl. No. 14/966,586, filed Dec. 11, 2015. (copy not attached.).

U.S. Appl. No. 14/970,647, filed Dec. 16, 2015. (copy not attached.).

U.S. Appl. No. 14/181,459, filed Feb. 14, 2016. (Copy not attached).

U.S. Appl. No. 14/623,932, filed Feb. 17, 2015. (Copy not attached).

U.S. Appl. No. 14/659,781, filed Mar. 17, 2015. (Copy not attached).

U.S. Appl. No. 14/670,677, filed Mar. 27, 2015. (Copy not attached).

U.S. Appl. No. 14/671,885, filed Mar. 27, 2015. (Copy not attached).

U.S. Appl. No. 14/671,899, filed Mar. 27, 2015. (Copy not attached).

U.S. Appl. No. 14/671,906, filed Mar. 27, 2015. (Copy not attached).

U.S. Appl. No. 14/743,047, filed Jun. 18, 2015. (Copy not attached).

U.S. Appl. No. 14/802,878, filed Jul. 17, 2015. (Copy not attached).

U.S. Appl. No. 14/806,385, filed Jul. 22, 2015. (Copy not attached).

U.S. Appl. No. 14/806,410, filed Jul. 22, 2015. (Copy not attached).

(56)

References Cited**OTHER PUBLICATIONS**

U.S. Appl. No. 14/880,809, filed Oct. 12, 2015. (Copy not attached).
 U.S. Appl. No. 14/919,981, filed Oct. 22, 2015. (Copy not attached).
 U.S. Appl. No. 14/970,653, filed Dec. 16, 2015. (Copy not attached).
 U.S. Appl. No. 15/051,365, filed Feb. 23, 2016. (Copy not attached).
 U.S. Appl. No. 15/053,054, filed Feb. 25, 2016. (Copy not attached).
 U.S. Appl. No. 15/053,110, filed Feb. 25, 2016. (Copy not attached).
 U.S. Appl. No. 15/056,573, filed Feb. 29, 2016. (Copy not attached).
 U.S. Appl. No. 15/058,383, filed Mar. 2, 2016. (Copy not attached).
 U.S. Appl. No. 15/058,835, filed Mar. 2, 2016. (Copy not attached).
 U.S. Appl. No. 15/063,667, filed Mar. 8, 2016. (Copy not attached).
 U.S. Appl. No. 15/063,682, filed Mar. 8, 2016. (Copy not attached).
 U.S. Appl. No. 15/063,691, filed Mar. 8, 2016. (Copy not attached).
 U.S. Appl. No. 15/063,702, filed Mar. 8, 2016. (Copy not attached).
 U.S. Appl. No. 15/063,714, filed Mar. 8, 2016. (Copy not attached).
 U.S. Appl. No. 15/094,039, filed Apr. 8, 2016. (Copy not attached).
 U.S. Appl. No. 15/149,456, filed May 9, 2016. (Copy not attached).
 U.S. Appl. No. 15/155,139, filed May 16, 2016. (Copy not attached).
 U.S. Appl. No. 15/155,476, filed May 16, 2016. (Copy not attached).
 U.S. Appl. No. 15/157,573, filed May 18, 2016. (Copy not attached).
 U.S. Appl. No. 15/162,737, filed May 24, 2016. (Copy not attached).
 U.S. Appl. No. 15/167,621, filed May 27, 2016. (Copy not attached).
 U.S. Appl. No. 15/167,648, filed May 27, 2016. (Copy not attached).
 U.S. Appl. No. 15/167,665, filed May 27, 2016. (Copy not attached).
 U.S. Appl. No. 15/167,679, filed May 27, 2016. (Copy not attached).
 U.S. Appl. No. 15/167,695, filed May 27, 2016. (Copy not attached).
 U.S. Appl. No. 15/167,708, filed May 27, 2016. (Copy not attached).
 U.S. Appl. No. 15/167,720, filed May 27, 2016. (Copy not attached).
 U.S. Appl. No. 15/170,256, filed Jun. 1, 2016. (Copy not attached).
 U.S. Appl. No. 15/210,957, filed Jul. 15, 2016. (Copy not attached).
 U.S. Appl. No. 15/214,591, filed Jul. 20, 2016. (Copy not attached).
 U.S. Appl. No. 15/223,423, filed Jul. 29, 2016. (Copy not attached).
 U.S. Appl. No. 15/241,314, filed Aug. 19, 2016. (Copy not attached).
 U.S. Appl. No. 15/242,757, filed Aug. 22, 2016. (Copy not attached).
 U.S. Appl. No. 15/242,893, filed Aug. 22, 2016. (Copy not attached).
 U.S. Appl. No. 15/249,637, filed Aug. 29, 2016. (Copy not attached).
 U.S. Appl. No. 15/259,465, filed Sep. 8, 2016. (Copy not attached).
 U.S. Appl. No. 15/259,473, filed Sep. 8, 2016. (Copy not attached).

U.S. Appl. No. 15/281,504, filed Sep. 30, 2016. (Copy not attached).
 U.S. Appl. No. 15/334,412, filed Oct. 26, 2016. (Copy not attached).
 U.S. Appl. No. 29/553,028, filed Jan. 28, 2016. (Copy not attached).
 U.S. Appl. No. 29/555,129, filed Feb. 18, 2016. (Copy not attached).
 U.S. Appl. No. 29/575,093, filed Aug. 22, 2016. (Copy not attached).
 U.S. Appl. No. 29/581,145, filed Oct. 17, 2016. (Copy not attached).
 Vogel, et al., "Data glasses controlled by eye movements", Information and communication, Fraunhofer-Gesellschaft, Sep. 22, 2013, two pages.
 Walton, Z. "Wear This Smartphone Controller on Your Finger," <http://www.webpronews.com/wear-this-smartphone-controller-on-your-finger-2012-06>, five pages.
 Ye, Hui et al., "High Quality Voice Morphing", Cambridge University Engineering Department Trumpington Street, Cambridge, England, CB2 1PZ, 2004, I-9-I-12.
 "Audio Spotlight," by Holosonics, <http://www.holosonics.com>, accessed Jul. 3, 2014, three pages.
 "Genius Ring Mice," <http://www.geniusnet.com/Genius/wSite/productCompare/compare.jsp>, Dec. 23, 2014, one page.
 "Help Requested! Comments and input needed for new coaxial UAS-DIY Drones," <http://diydrone.com/profiles/blogs/help-requested-comments-and-input-needed-for-new-coaxial-uas>, Mar. 5, 2015, 1-3.
 "How Ascent AeroSystems is looking to add to your outdoor adventure," <http://droneblog.com/2015/03/23/how-ascent-aerosystems-is-looking-to-add-to-your-outdoor-adventure/>!prettyPhoto, Mar. 23, 2015, 1-10.
 "Lightberry," <https://web.archive.org/web/20131201194408/http://lightberry.eu/>, Dec. 1, 2013, eleven pages.
 "Meet Nod, the Bluetooth Ring That Wants to Replace your Mouse," <http://www.pcmag.com/article2/0,2817,2457238,00.asp>, Apr. 29, 2014, six pages.
 "Sound from Ultrasound," Wikipedia entry, http://en.m.wikipedia.org/wiki/Sound_from_ultrasound, accessed Jul. 3, 2014, thirteen pages.
 Non-Final Office Action mailed Dec. 16, 2020, for U.S. Appl. No. 16/538,720, filed Aug. 12, 2019, nine pages.
 Notice of Allowance mailed Apr. 30, 2021, for U.S. Appl. No. 16/538,720, filed Aug. 12, 2019, seven pages.

* cited by examiner

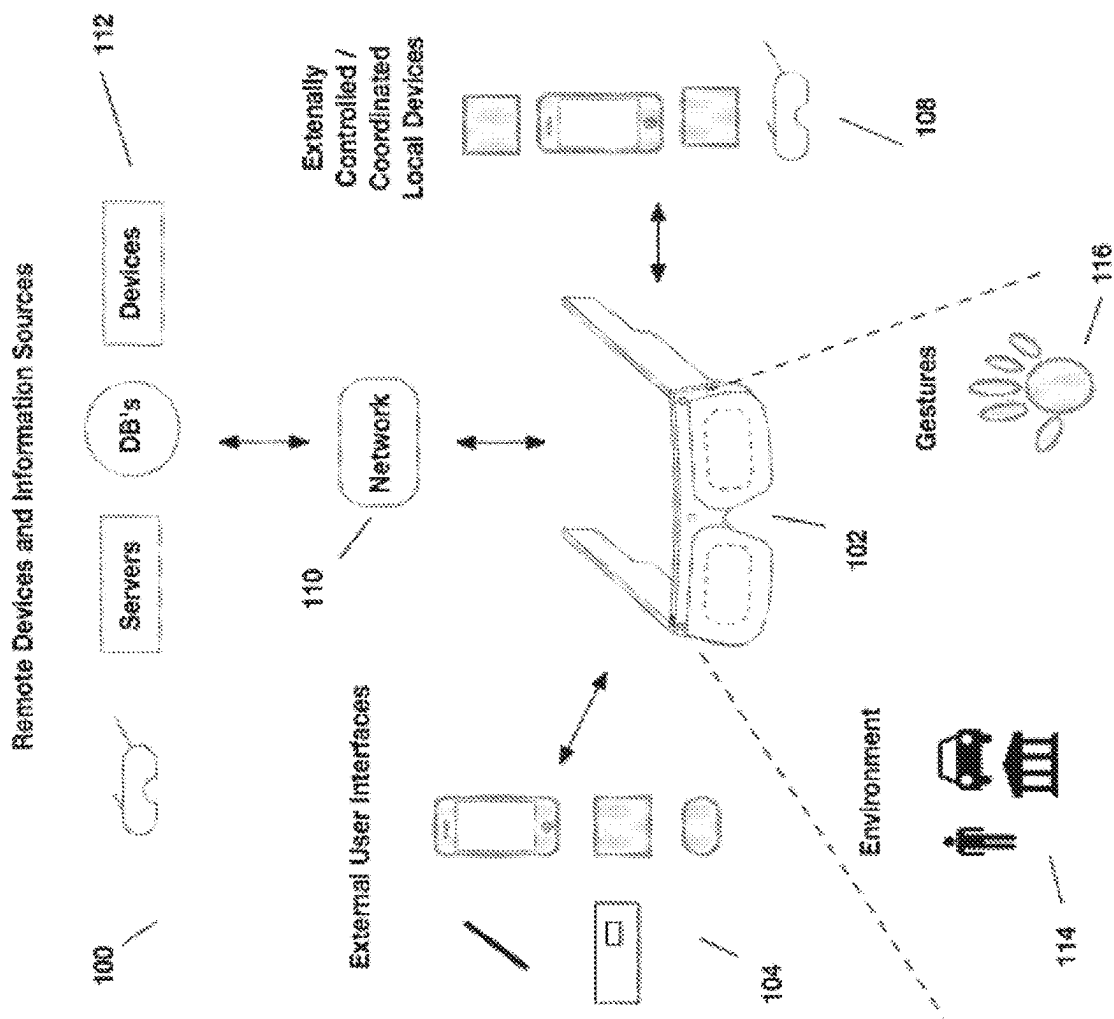


Figure 1

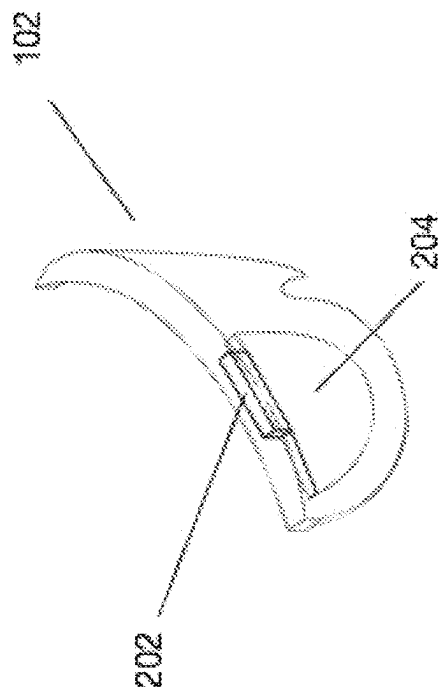
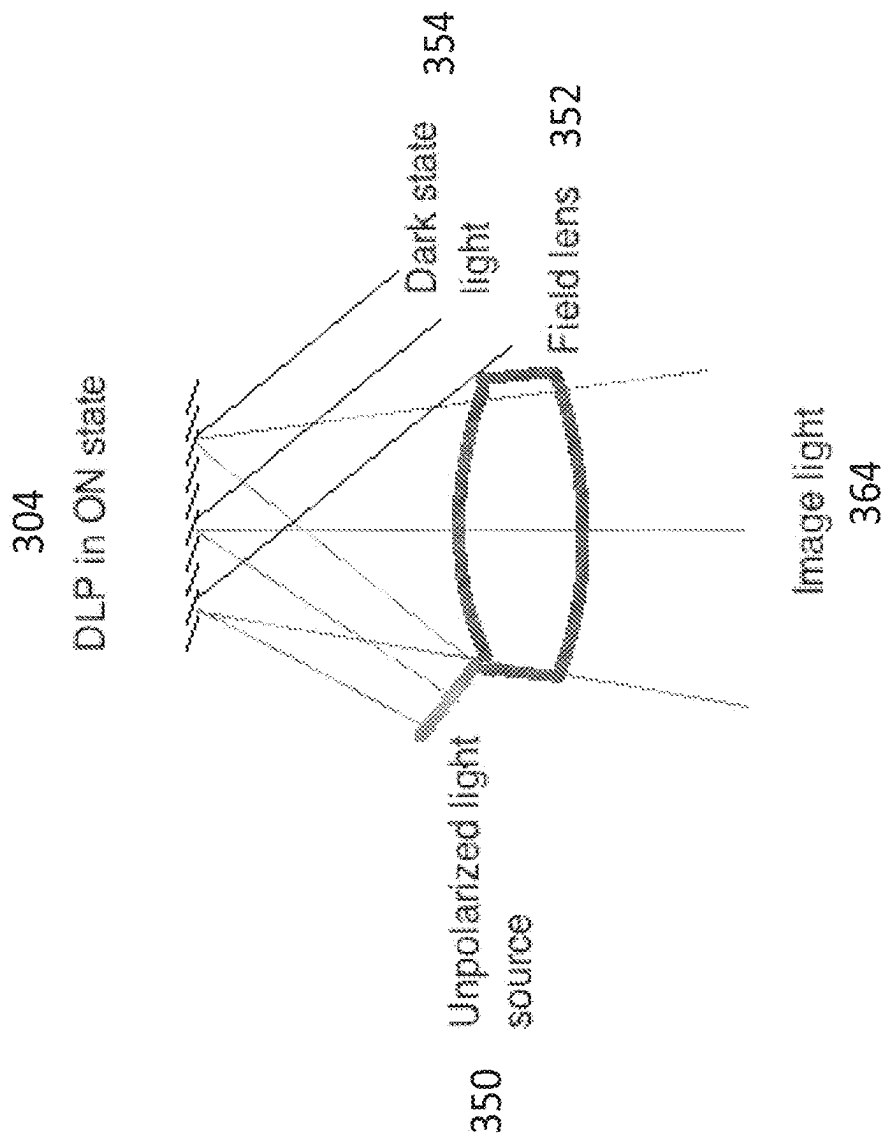


Figure 2



PRIOR ART

Figure 3a

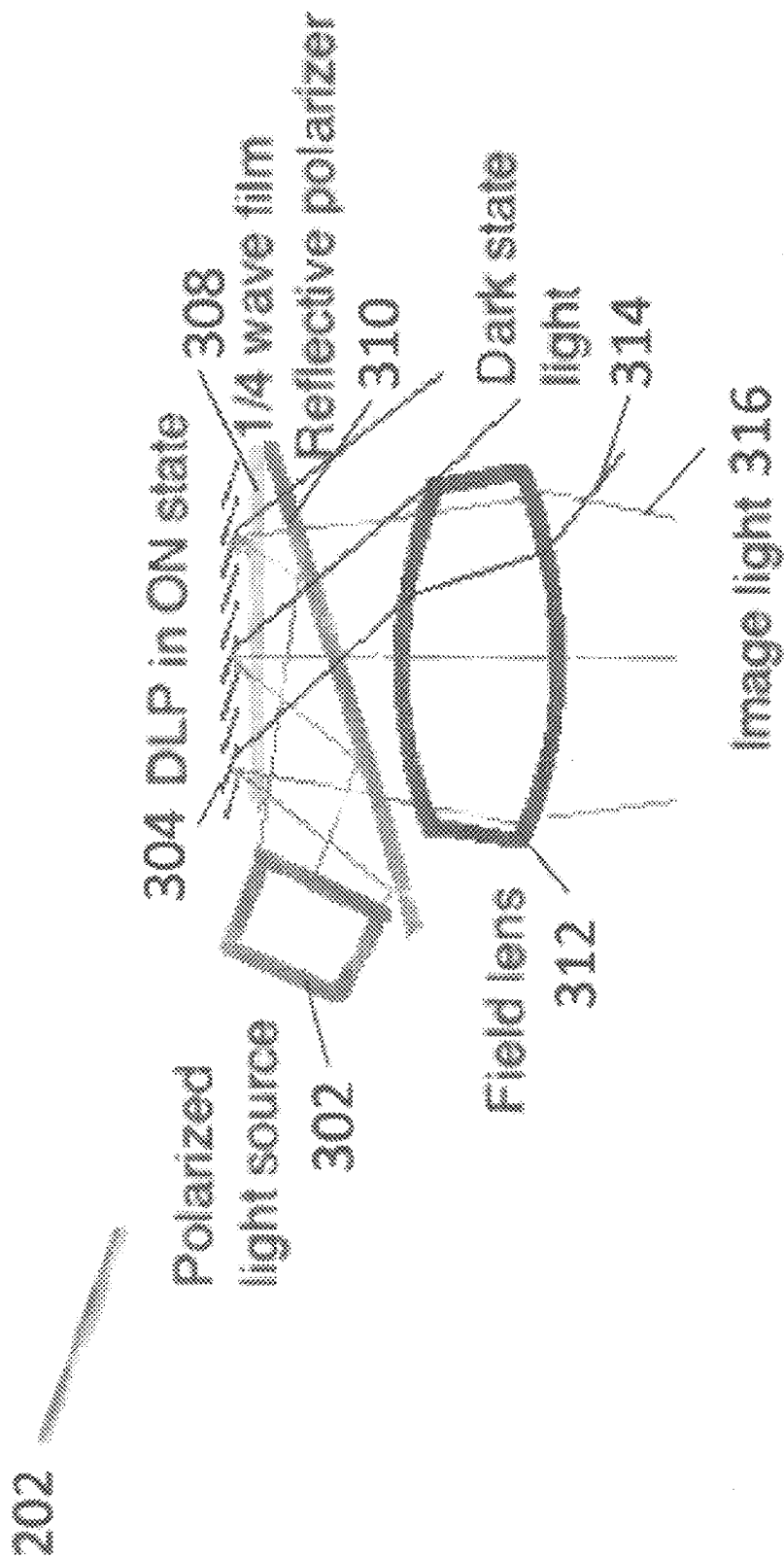


Figure 3b

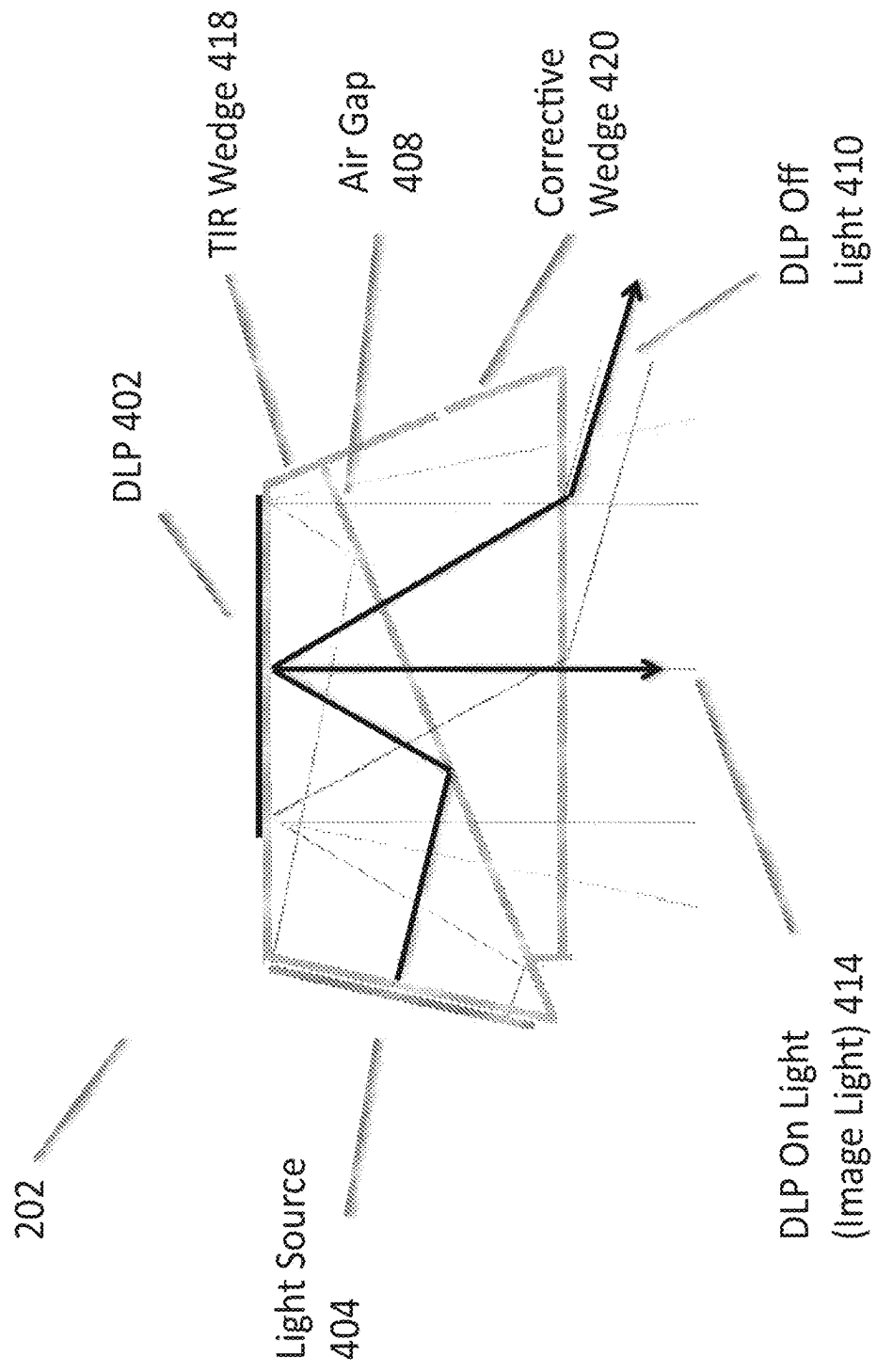


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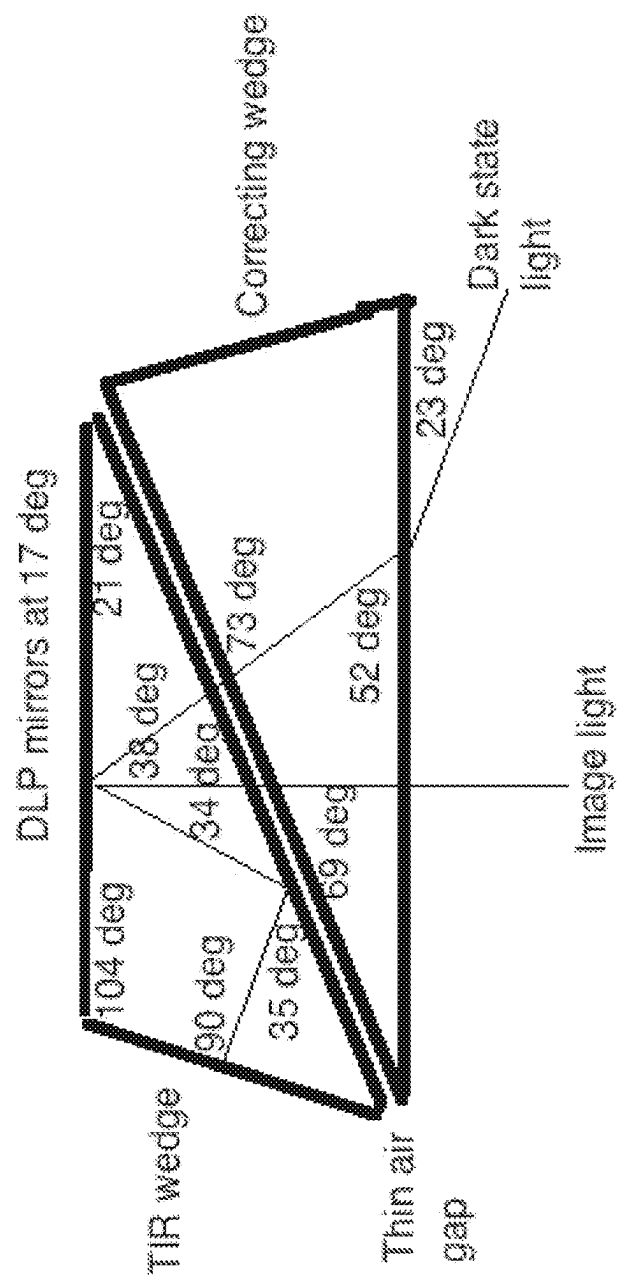


Figure 4a

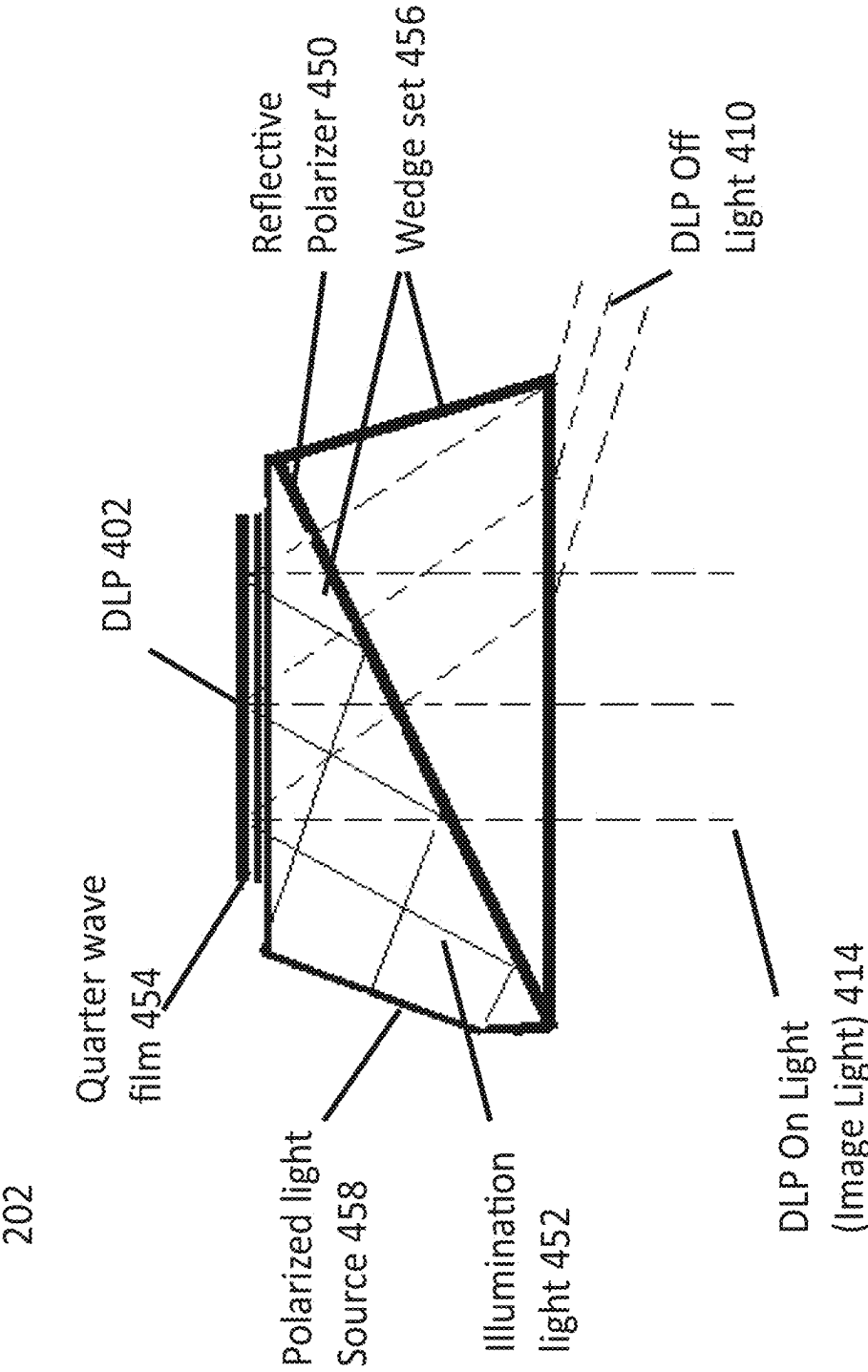


Figure 4b

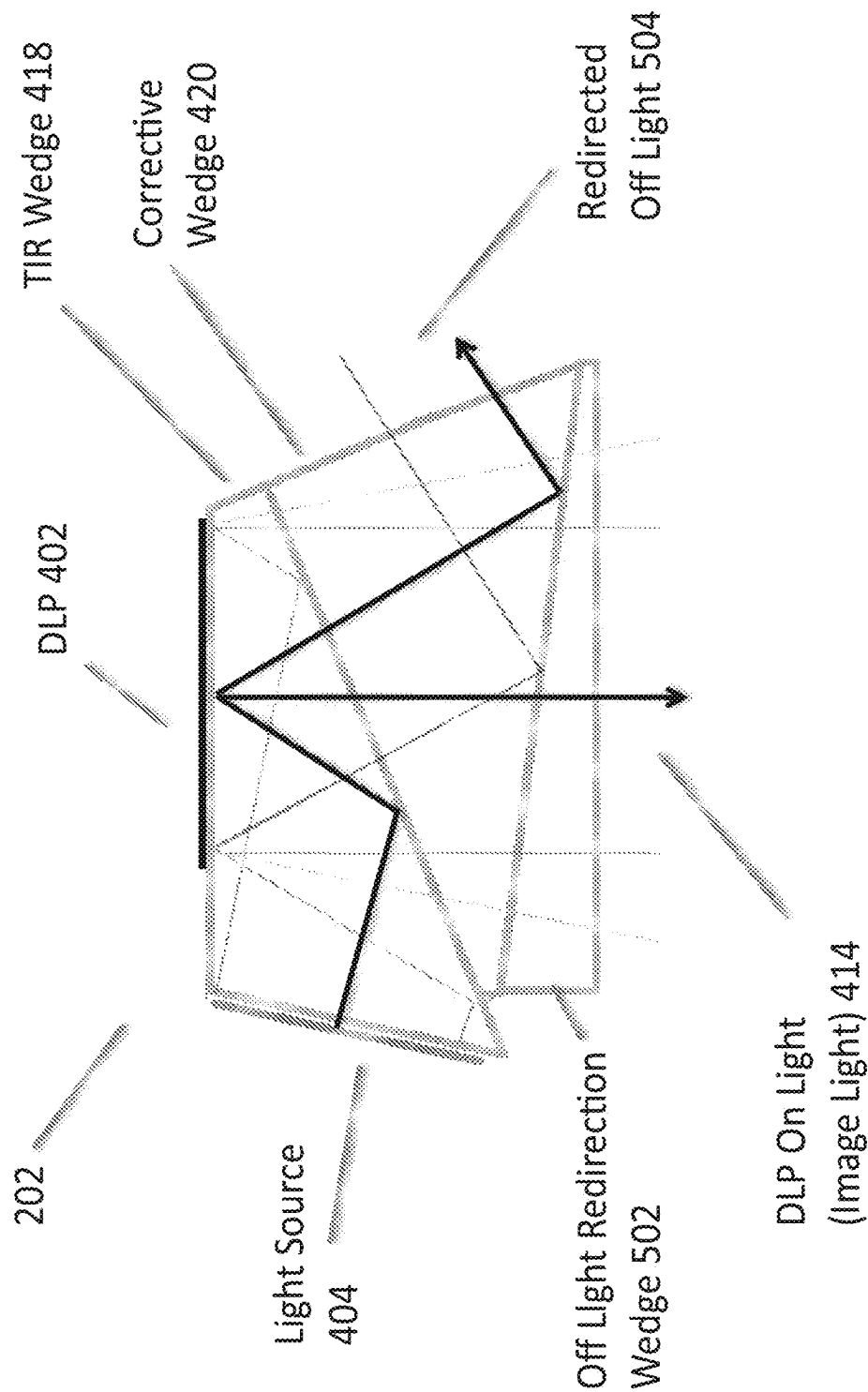


Figure 5

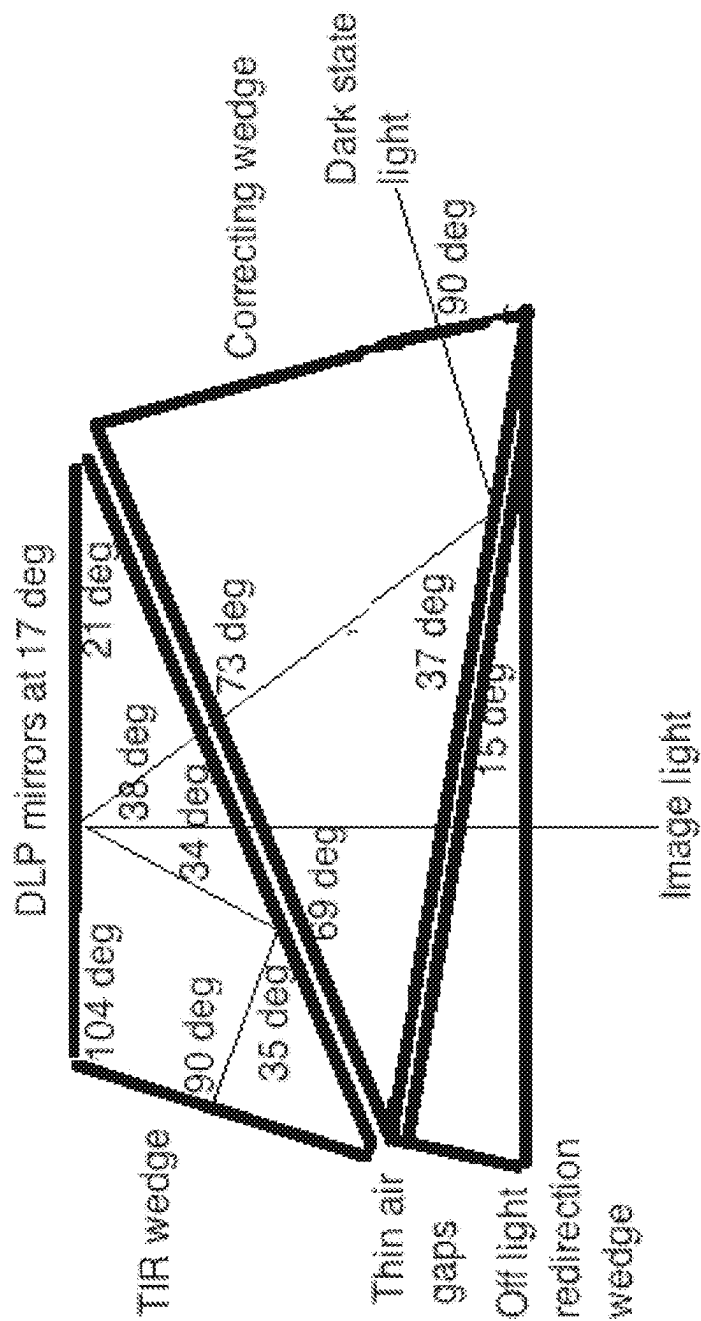


Figure 5a

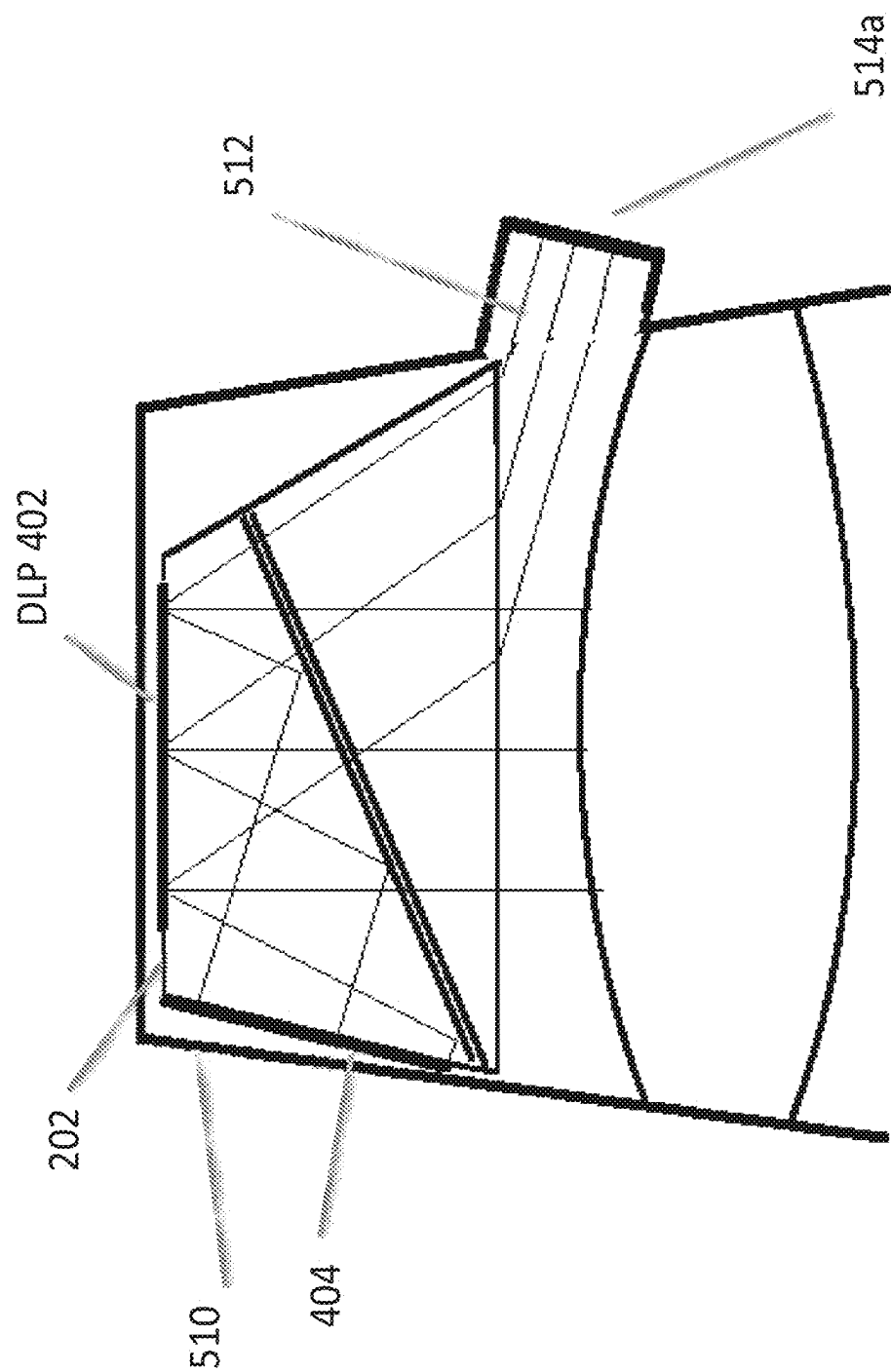


Figure 5b

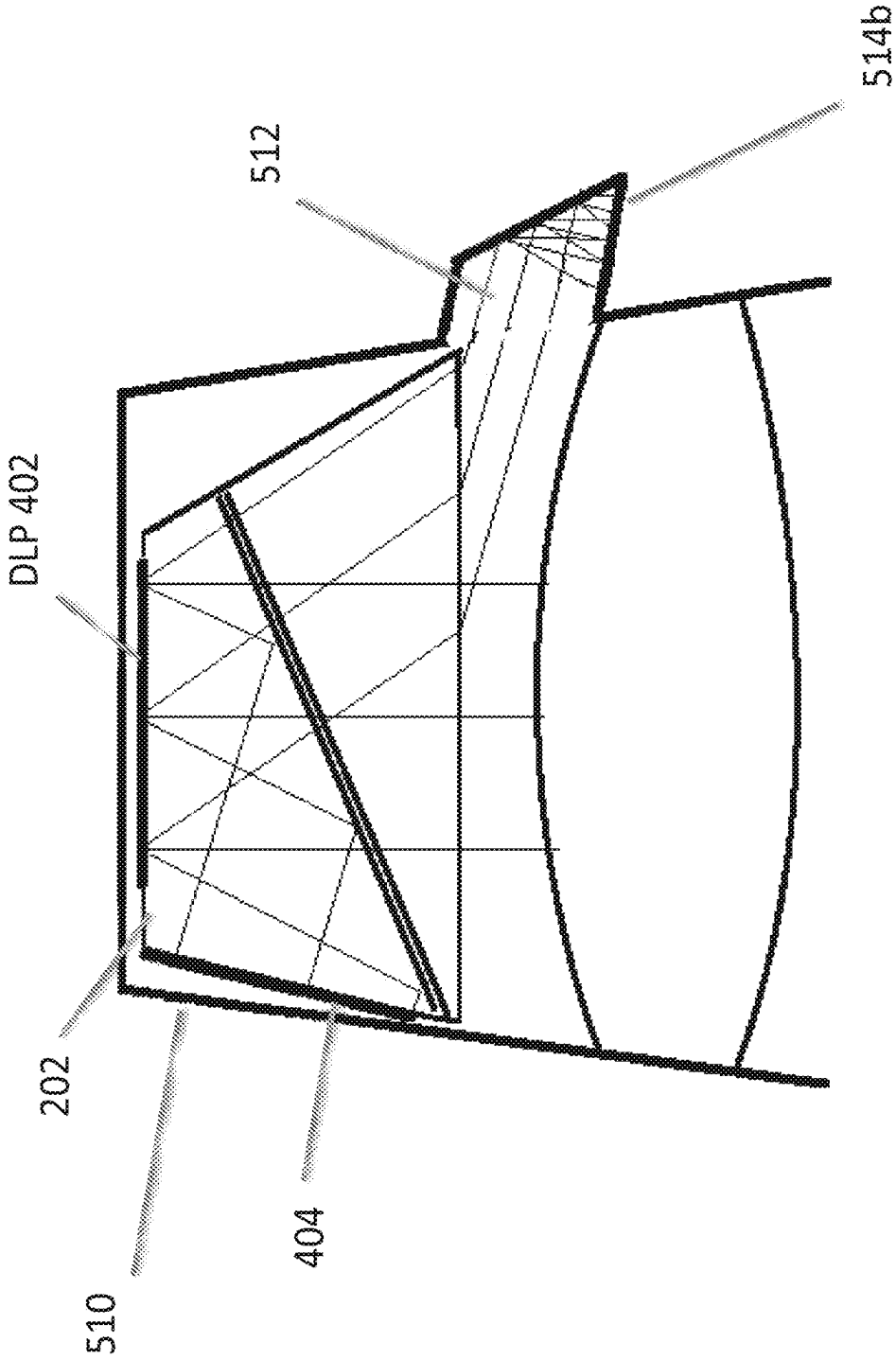


Figure 5c

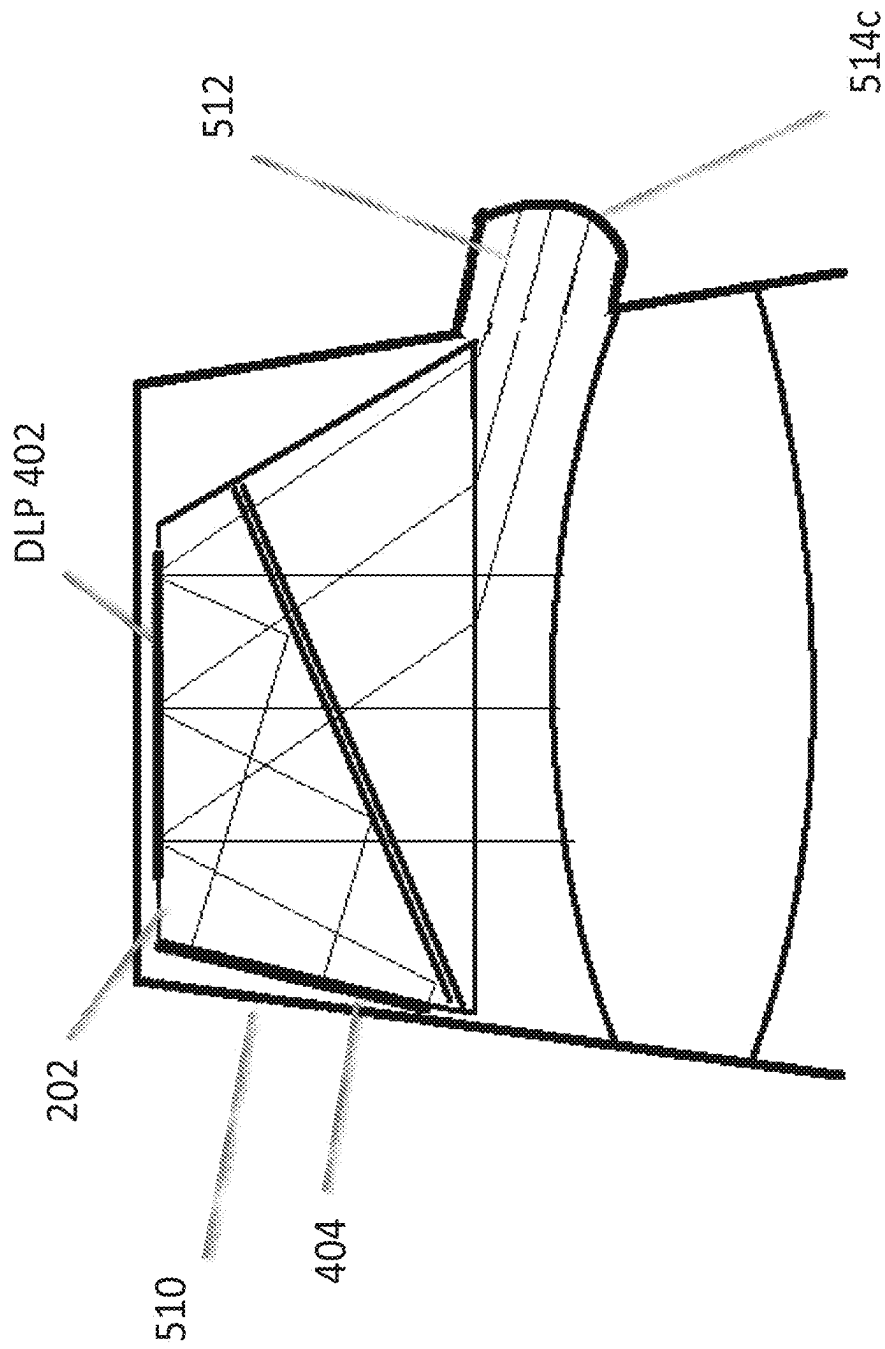


Figure 5d

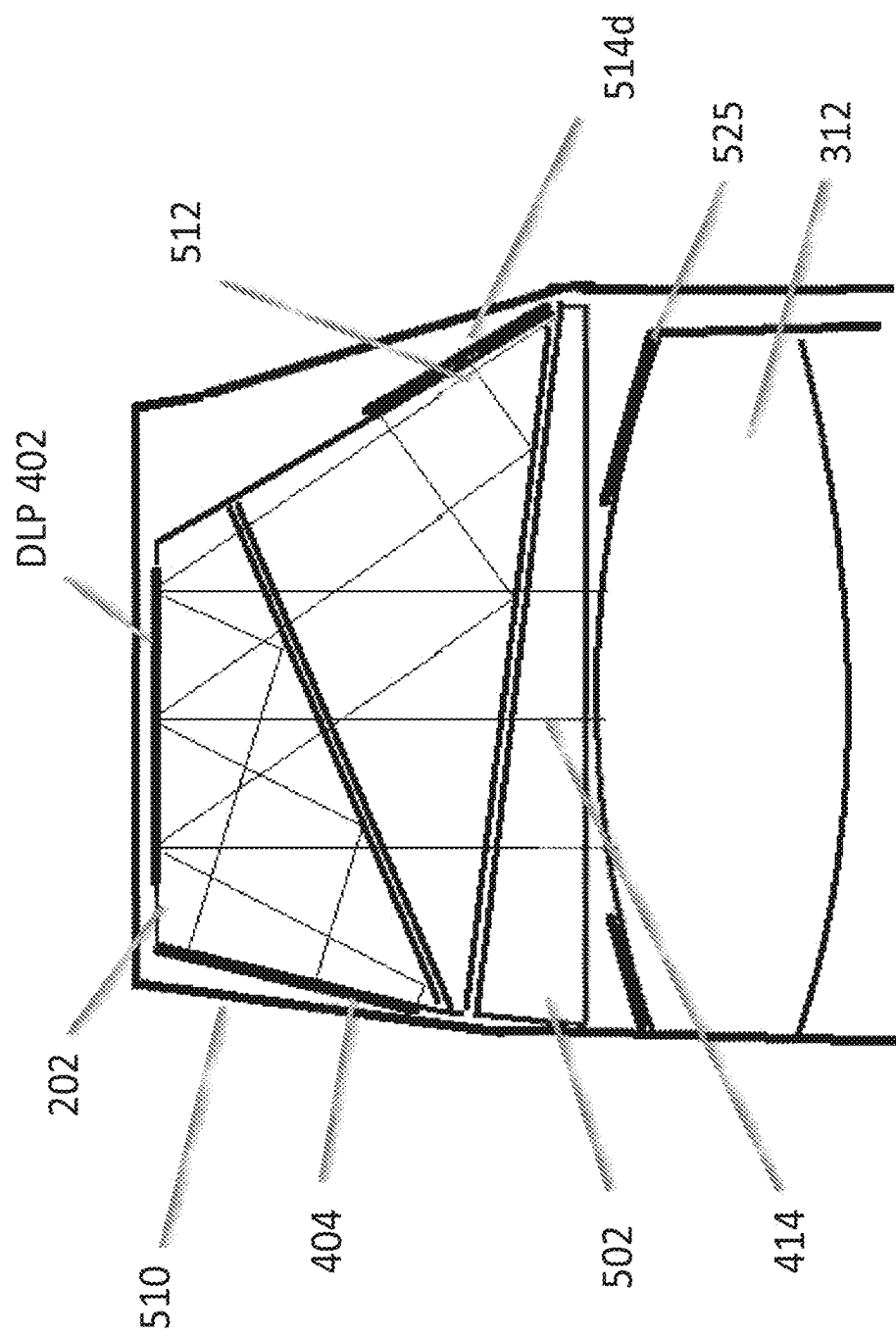


Figure 5e

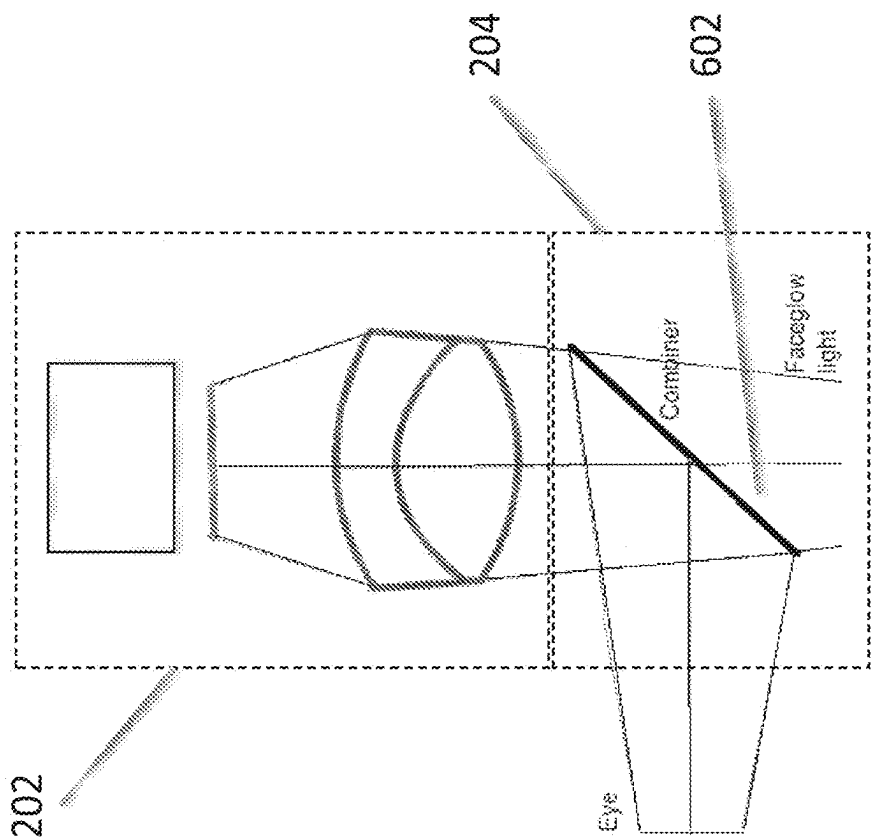


Figure 6

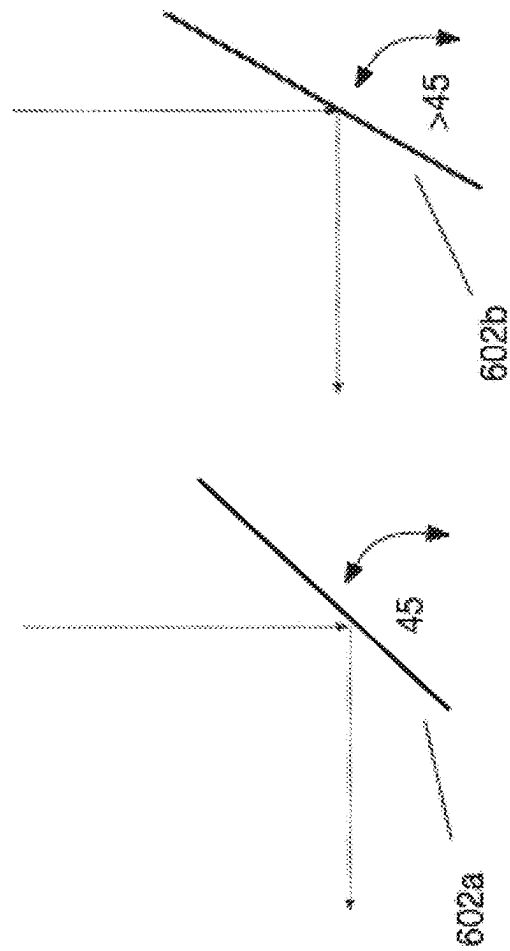


Figure 7

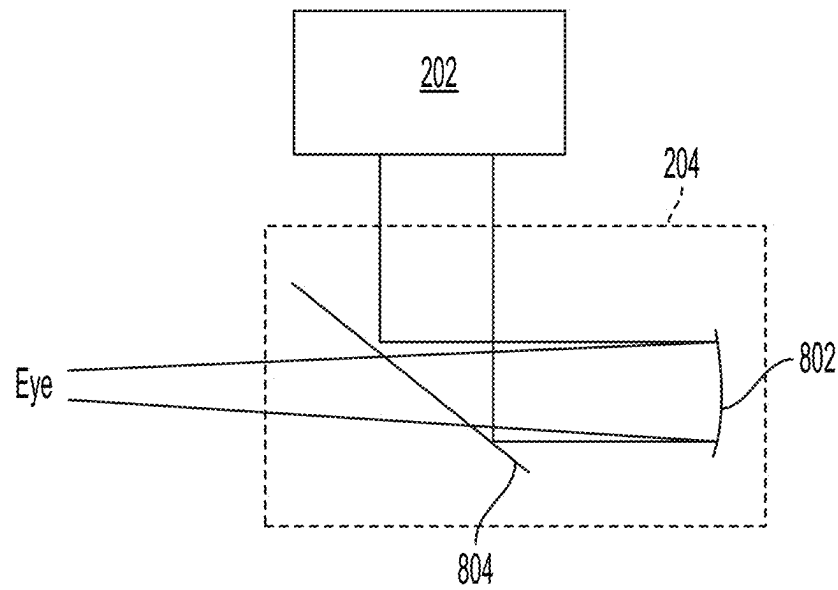


FIG. 8

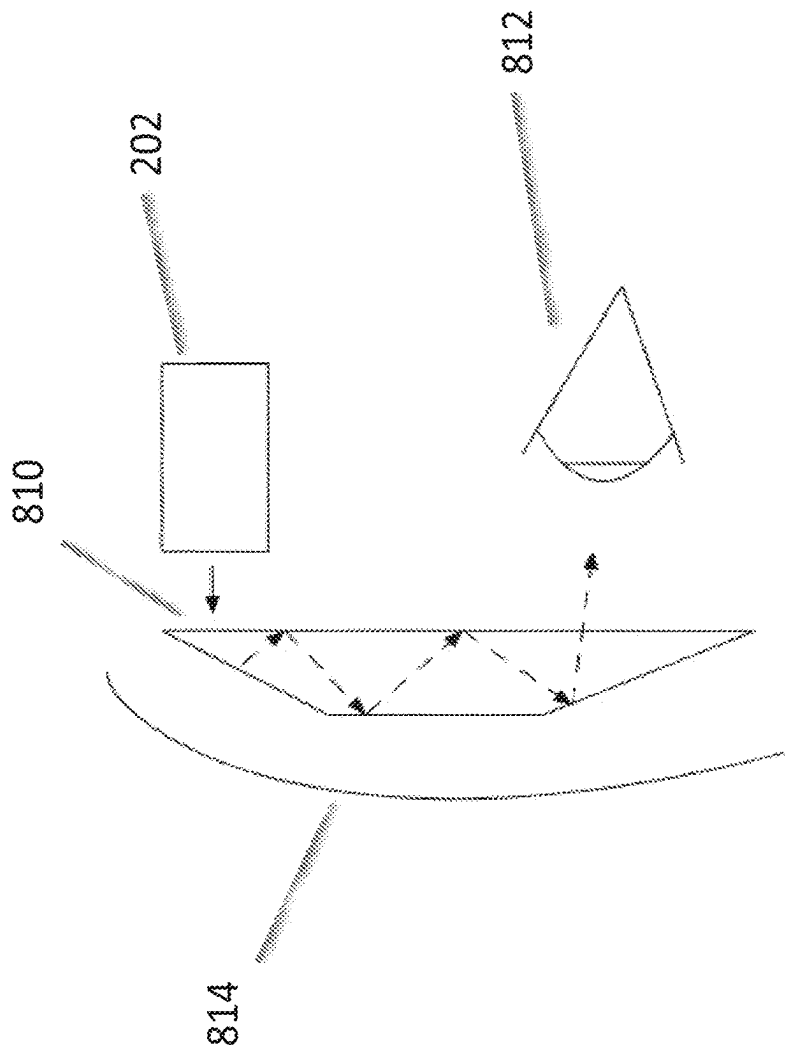


Figure 8a

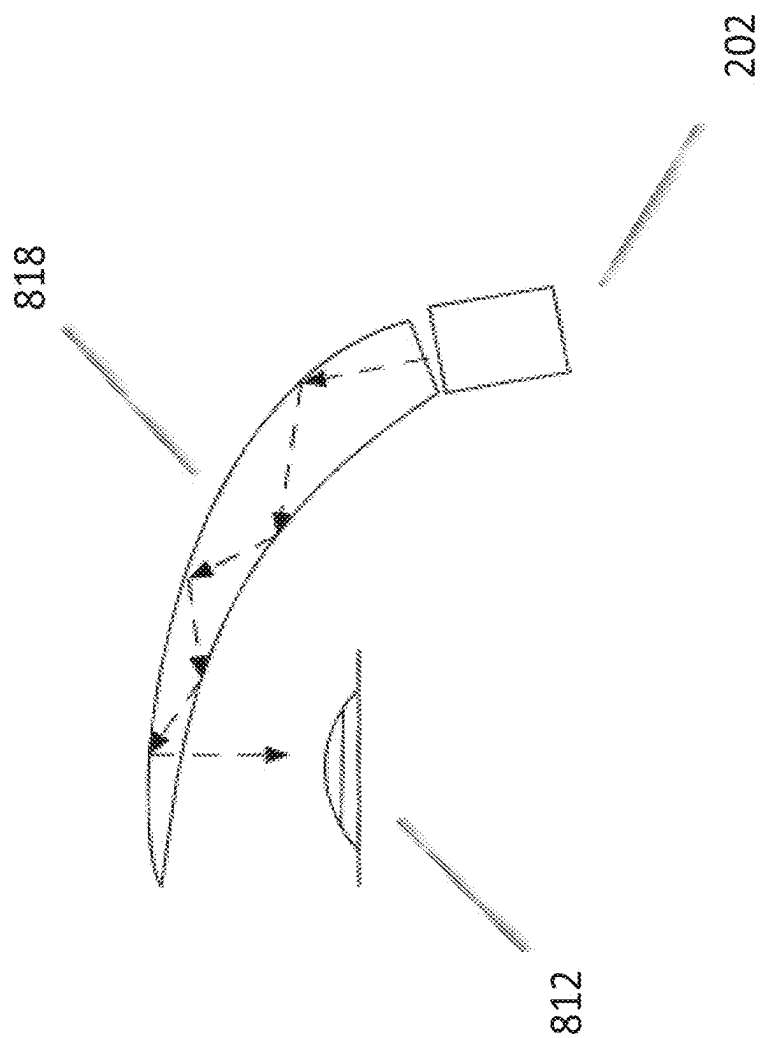


Figure 8b

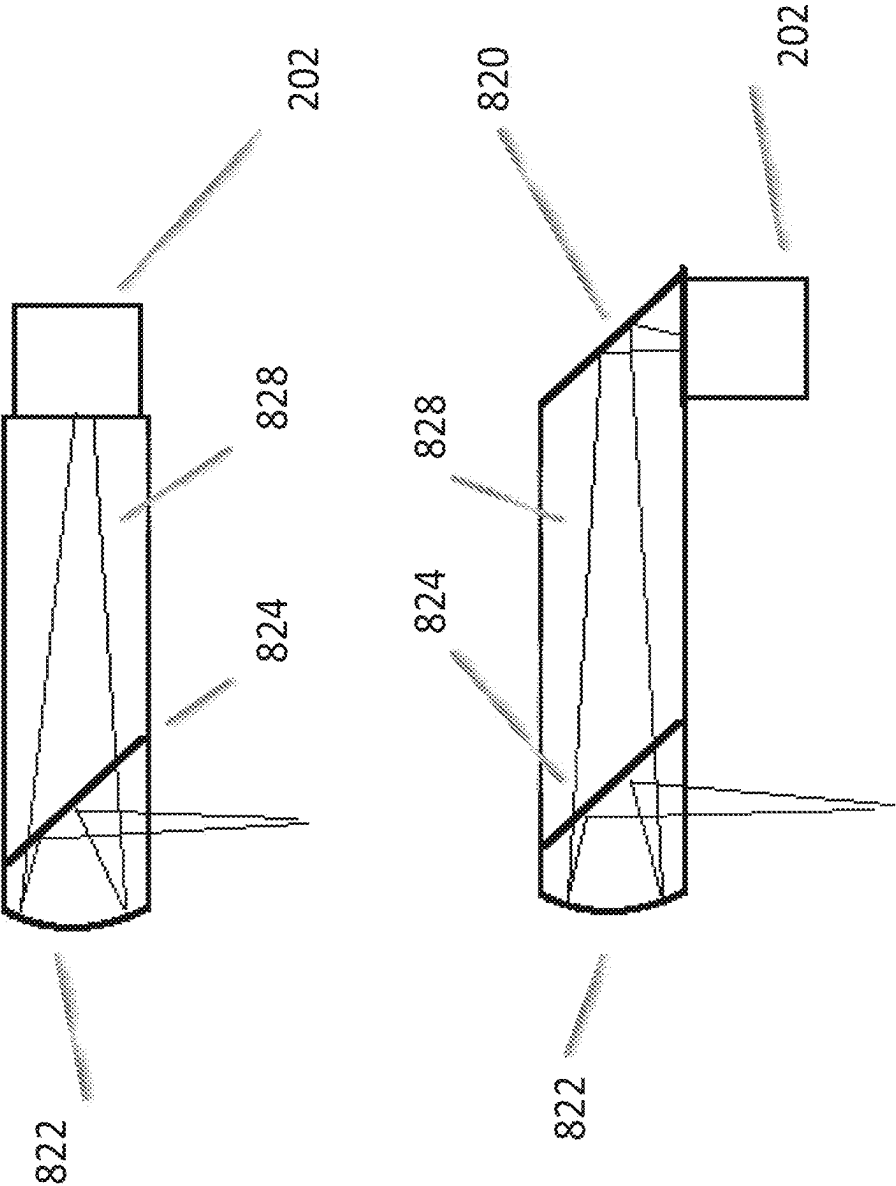


Figure 8c

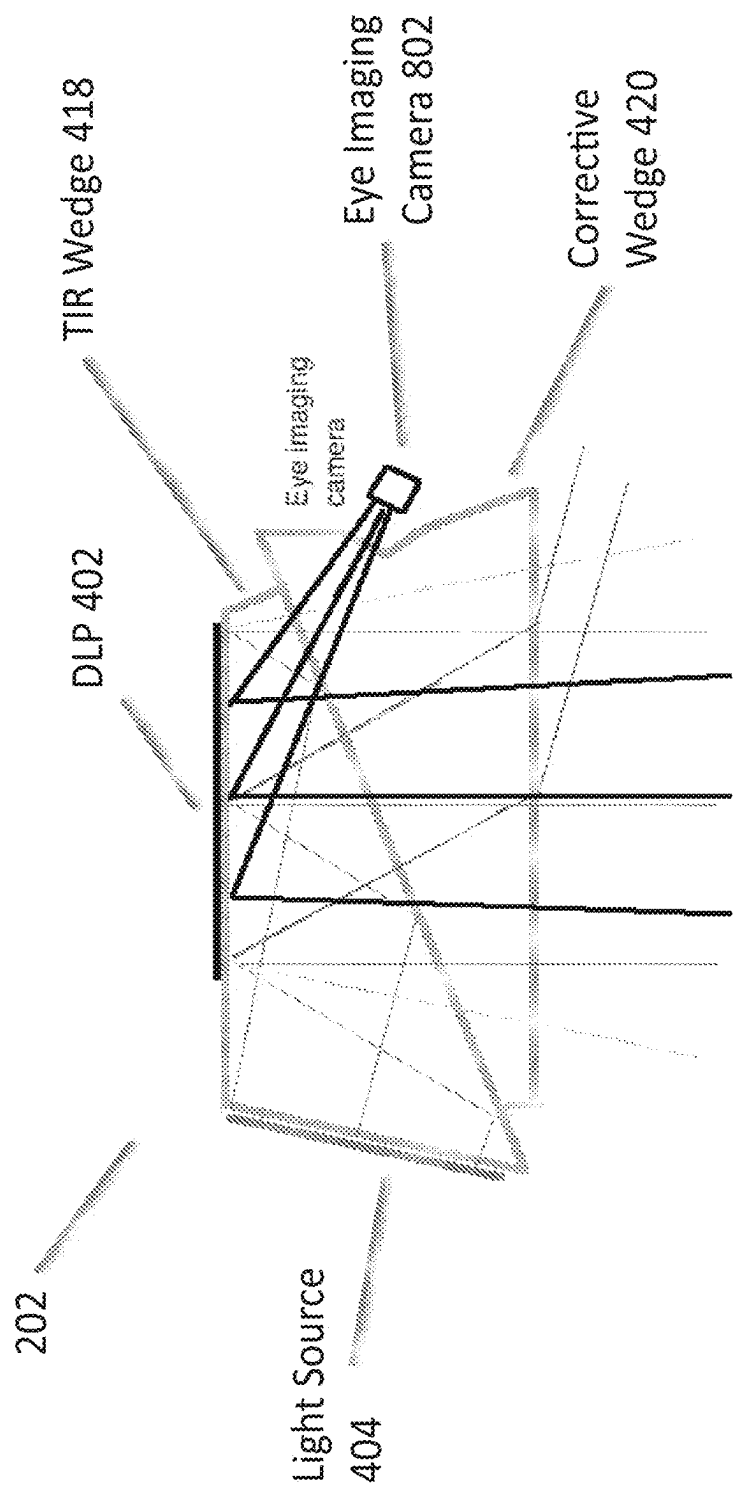


Figure 9

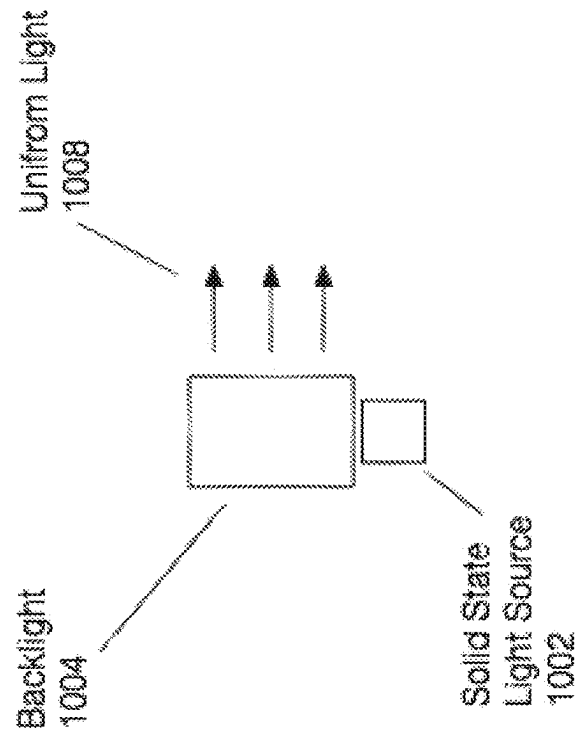


Figure 10

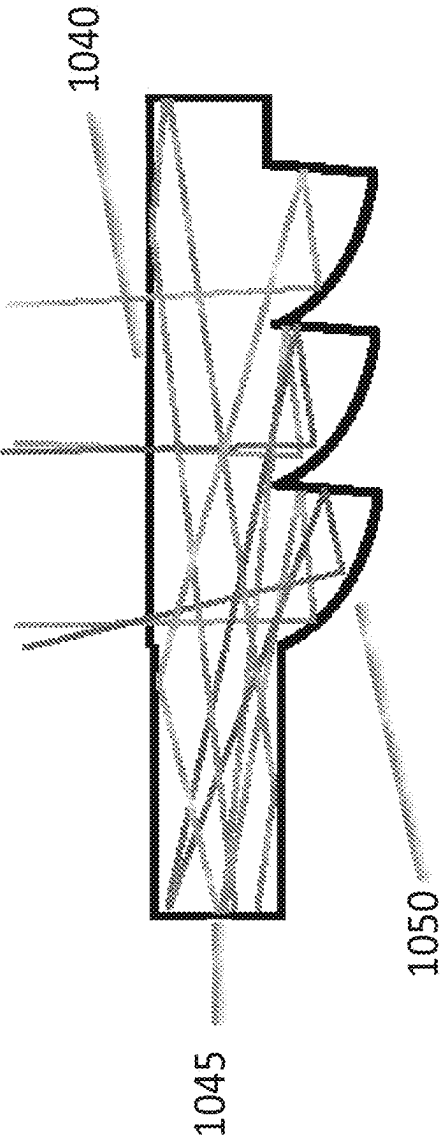


Figure 10a

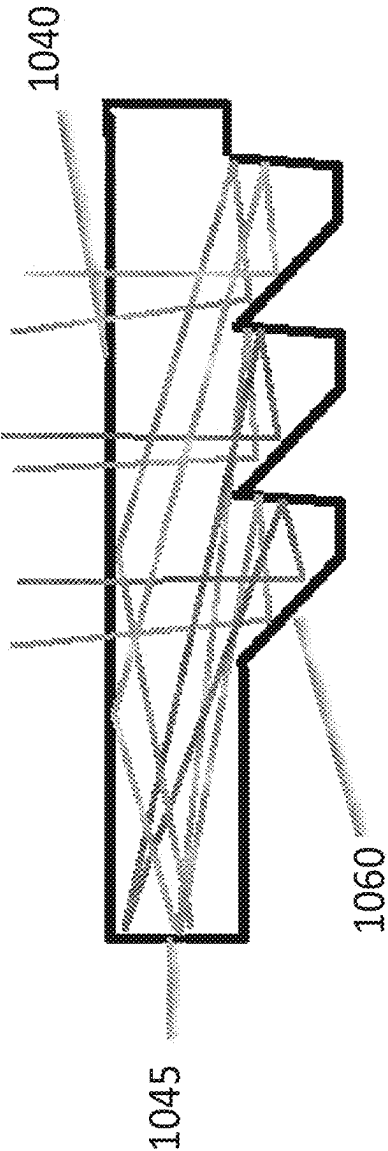


Figure 10b

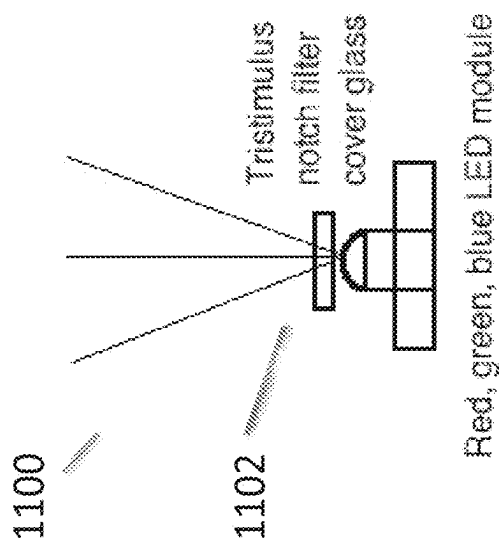


Figure 11a

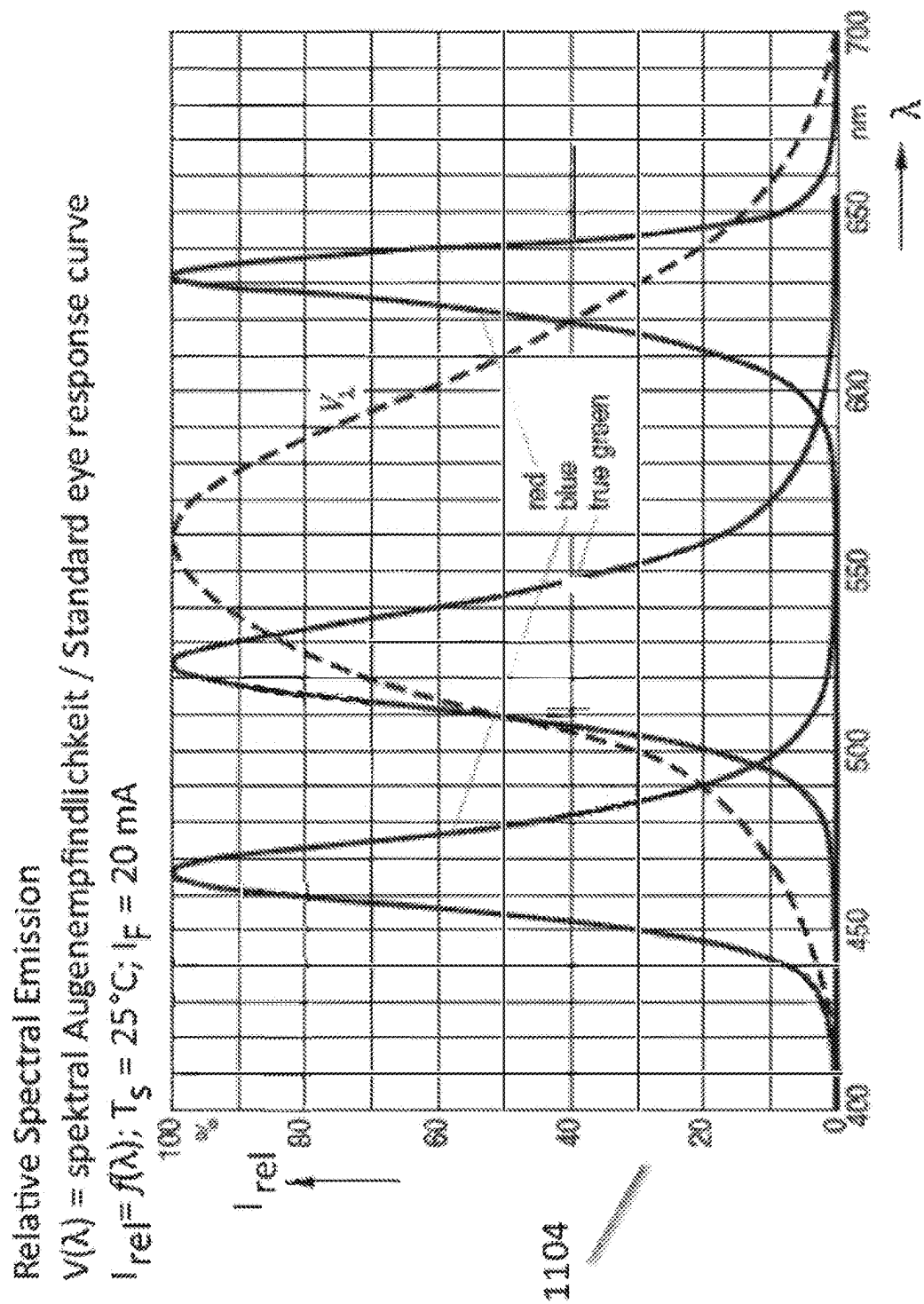


Fig. 11b

1108

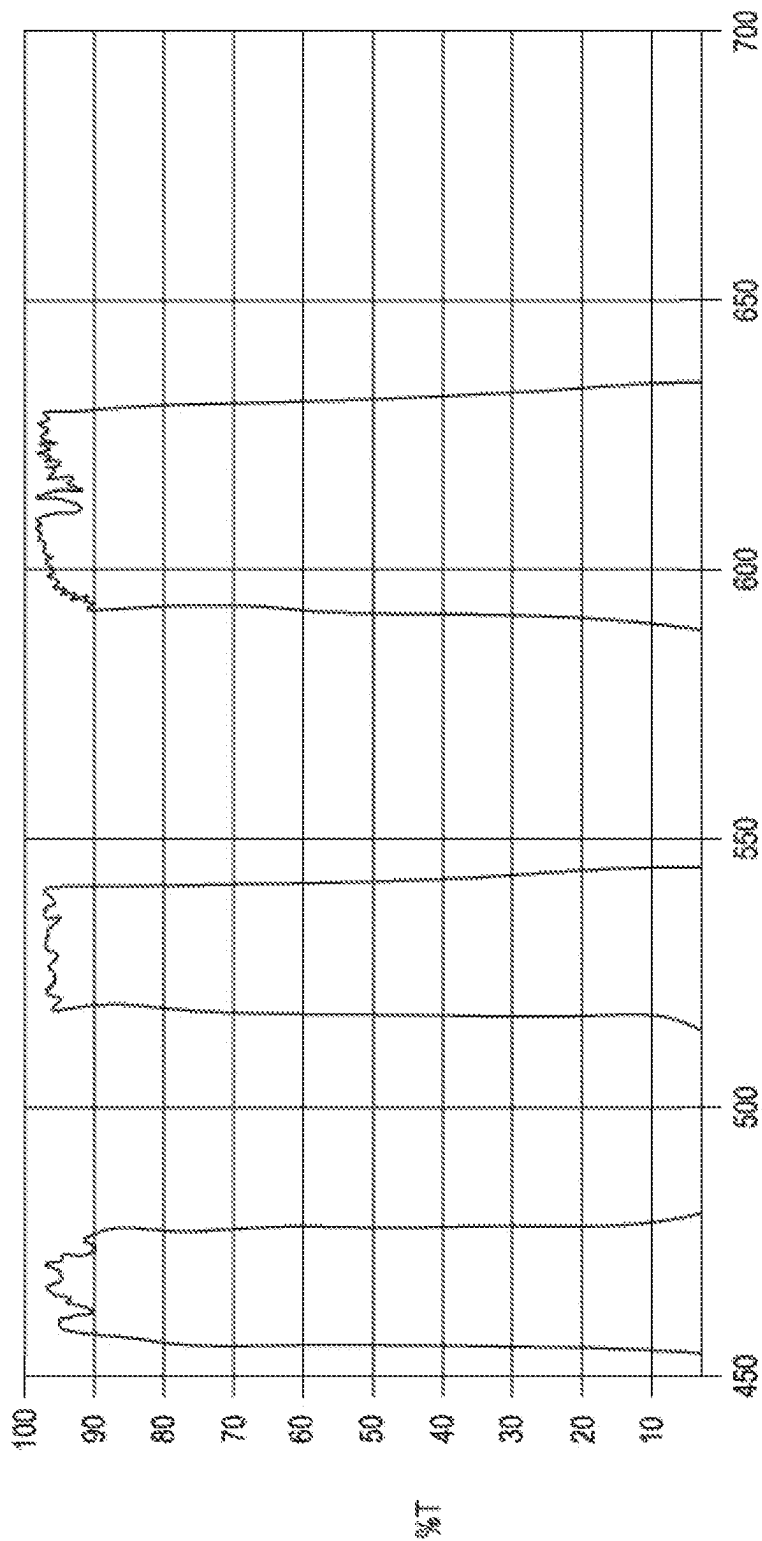


Fig. 11c

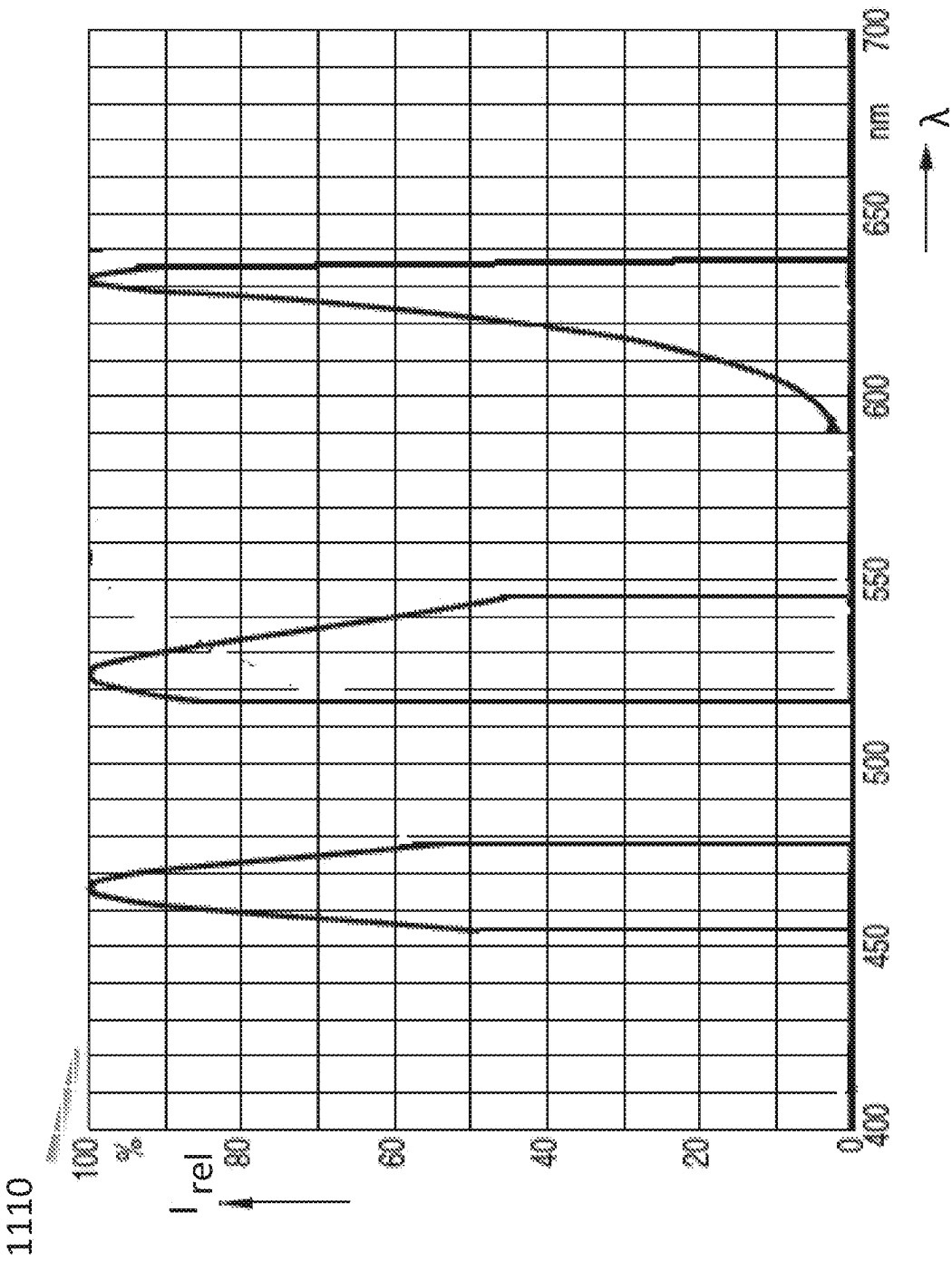


Fig. 11d

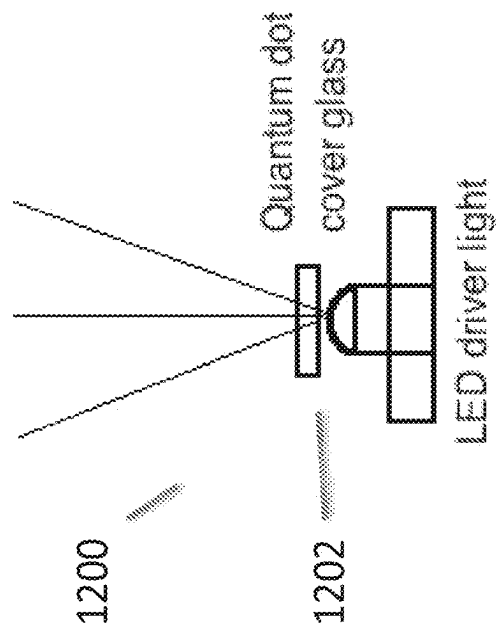


Figure 12a

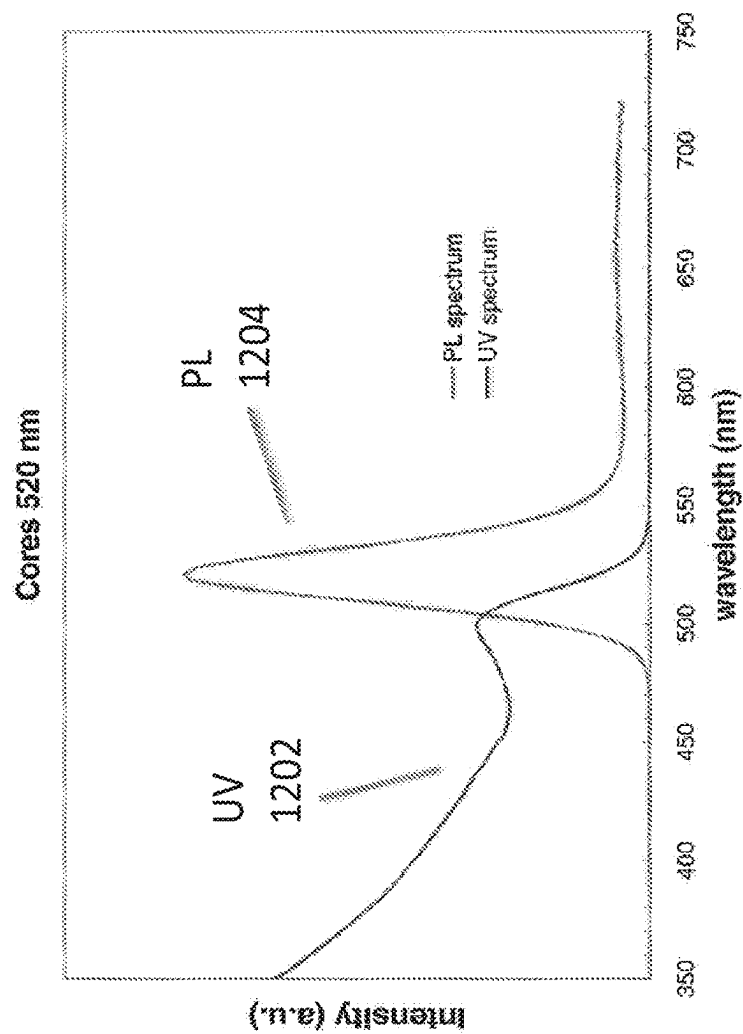


Figure 12b

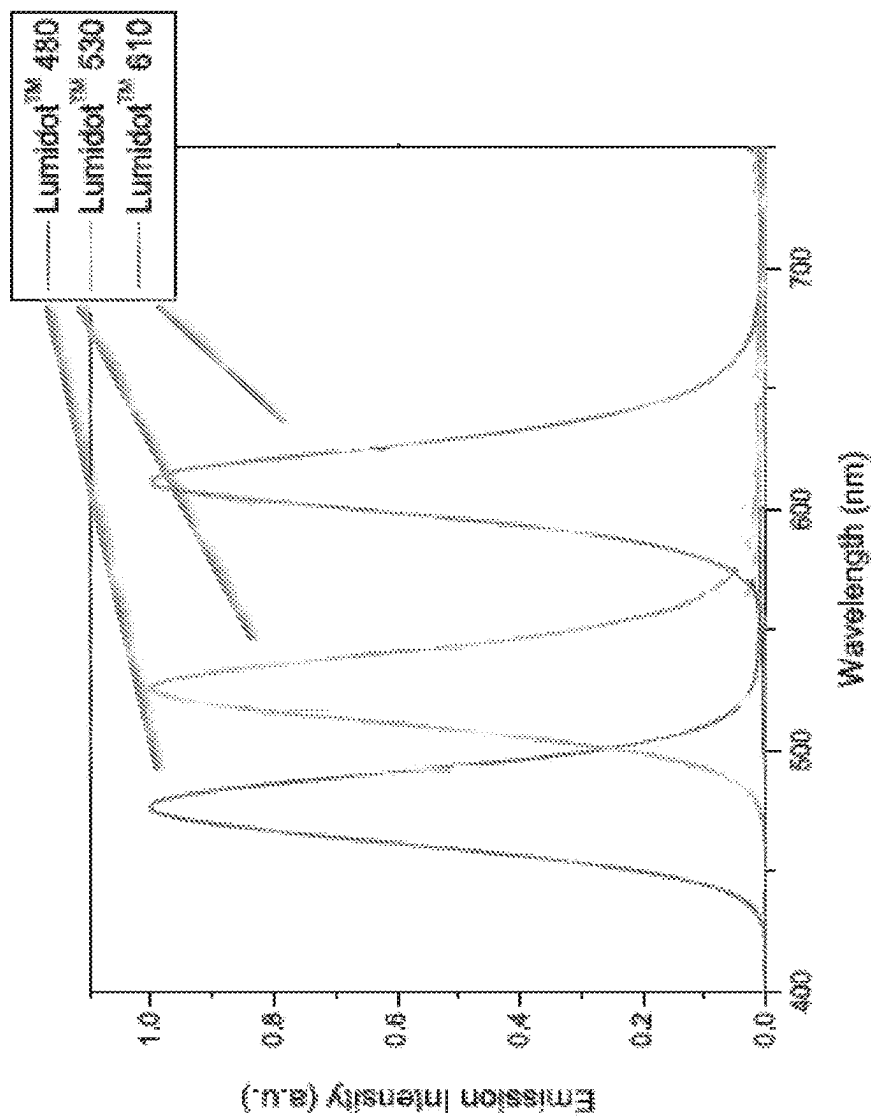


Figure 12c

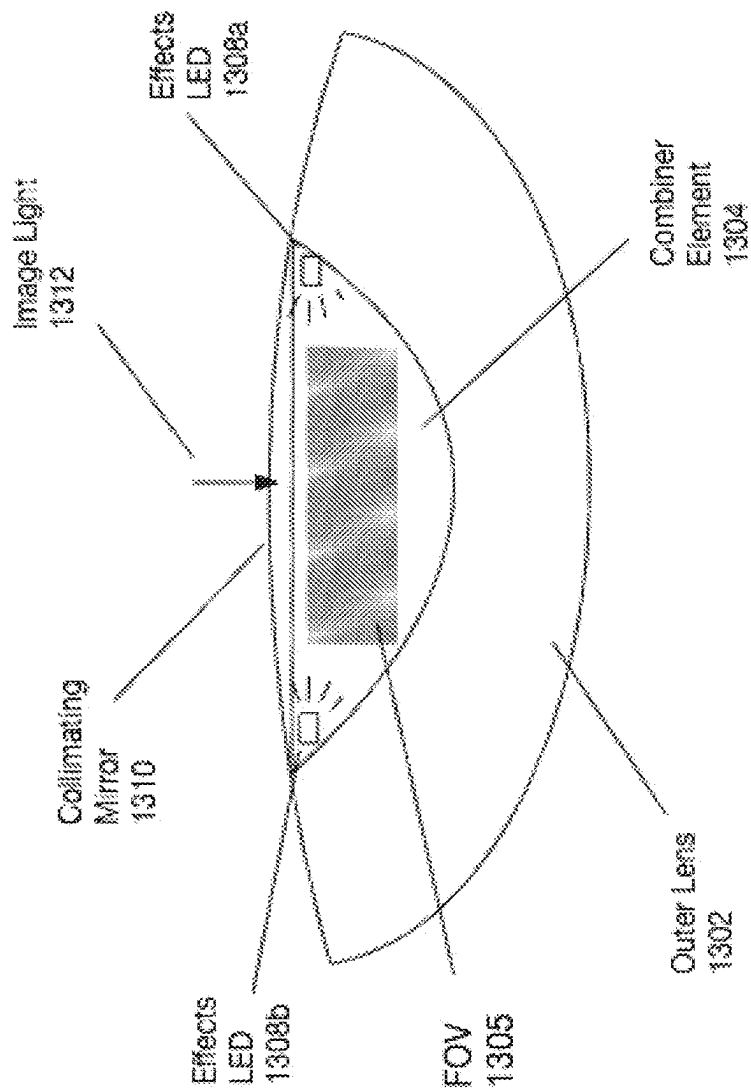


Figure 13a

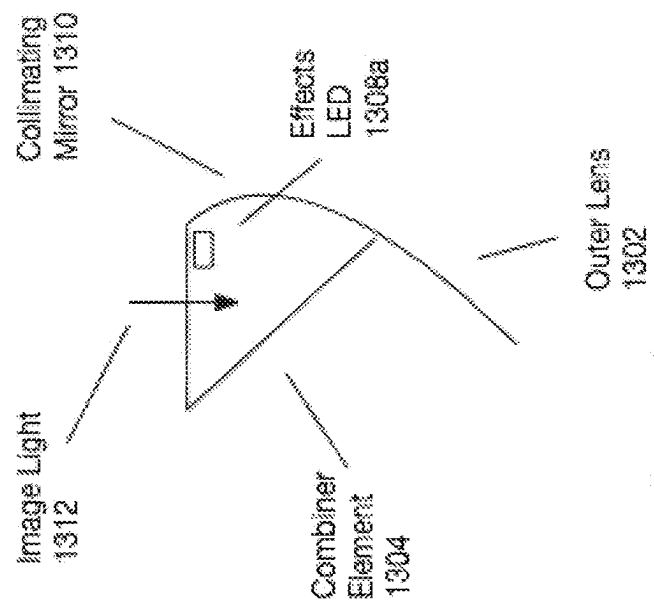


Figure 13b

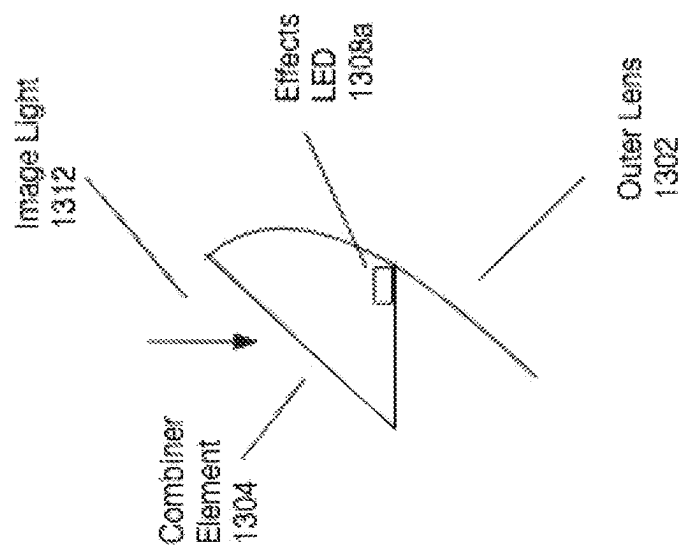


Figure 13c

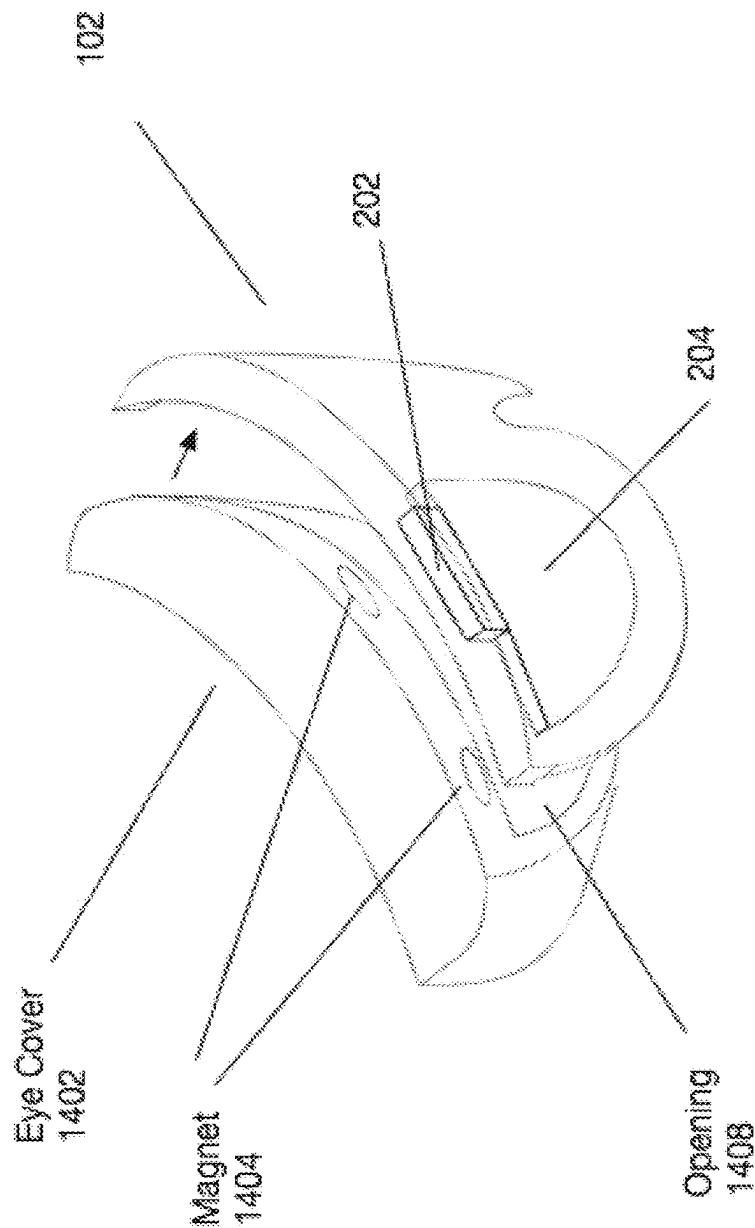


Figure 14a

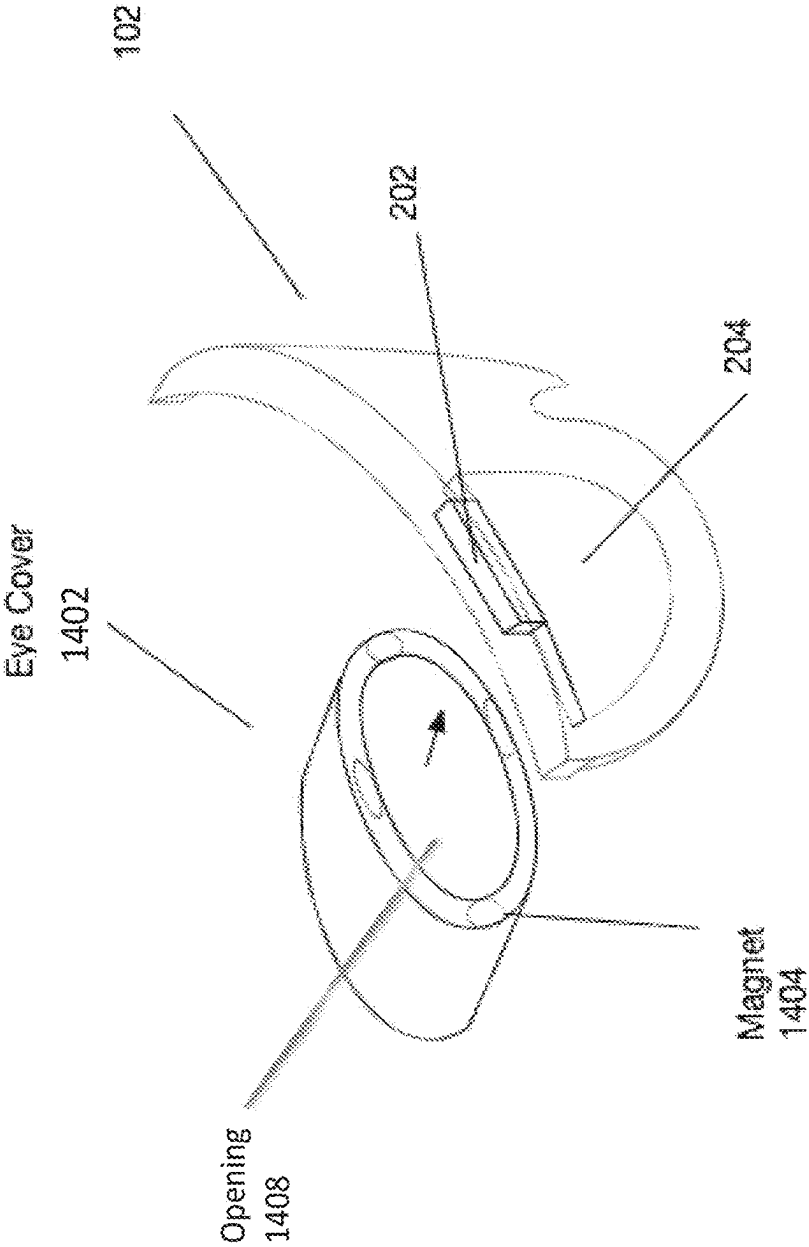


Figure 14b

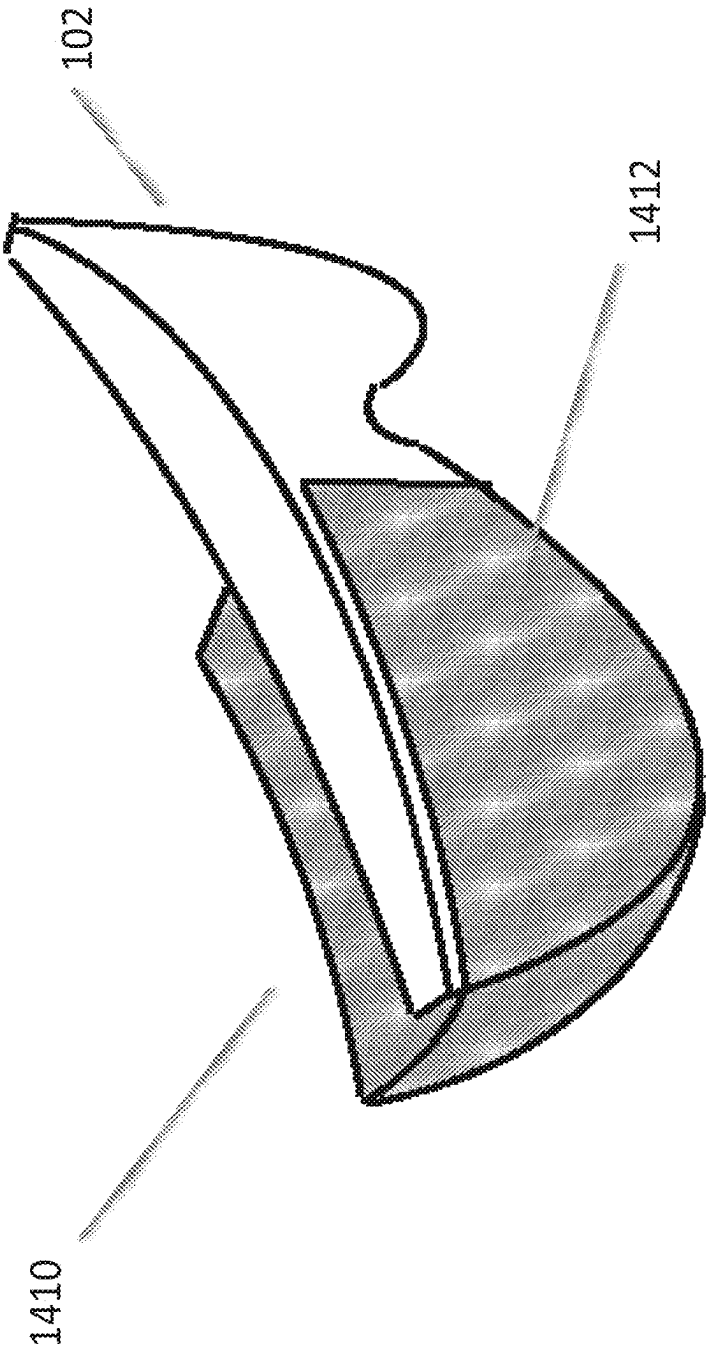


Figure 14c

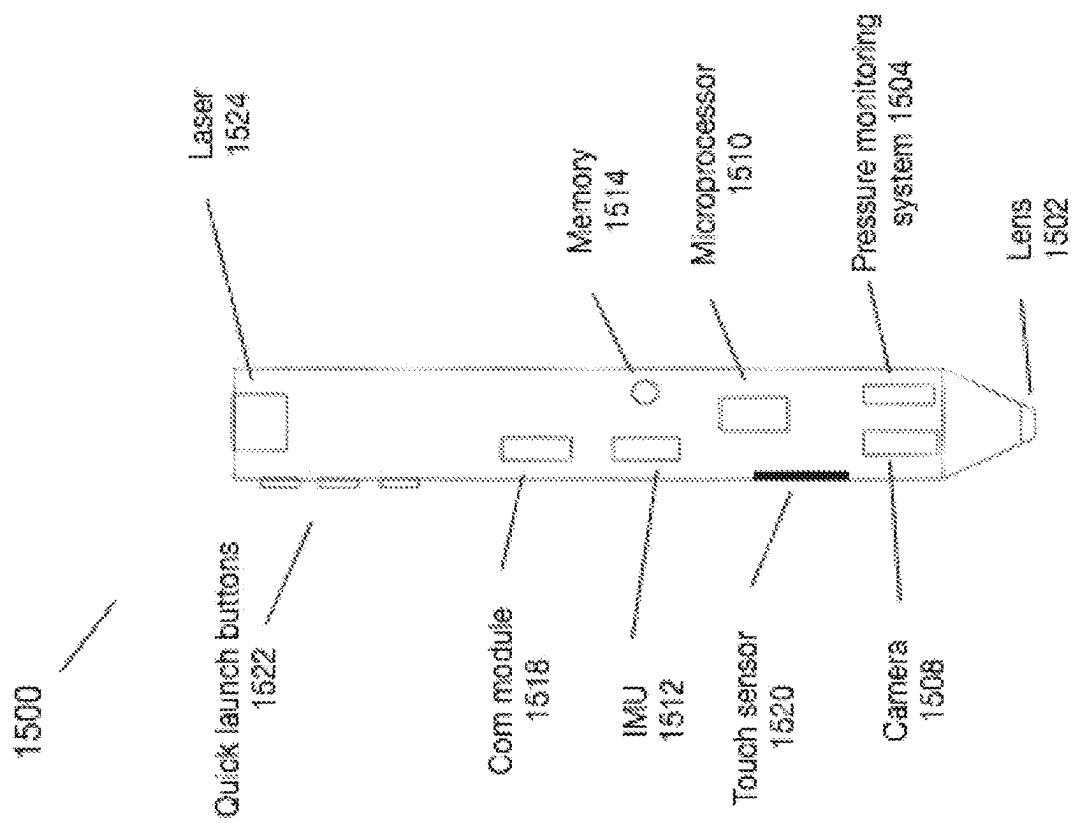


Figure 15

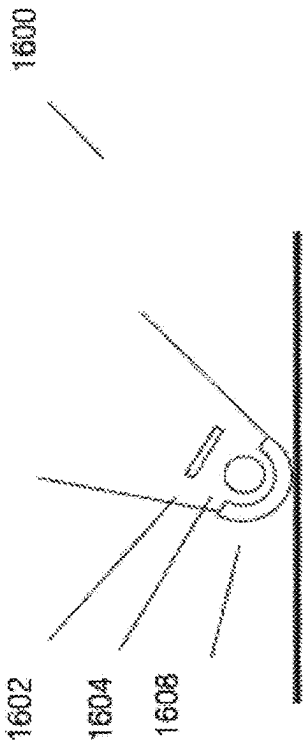


Fig.
16A

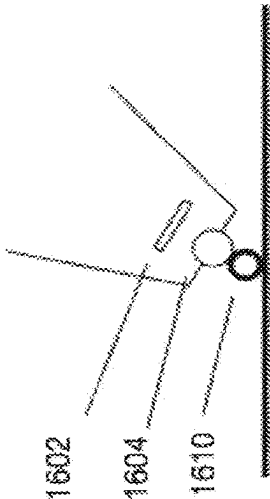


Fig.
16B

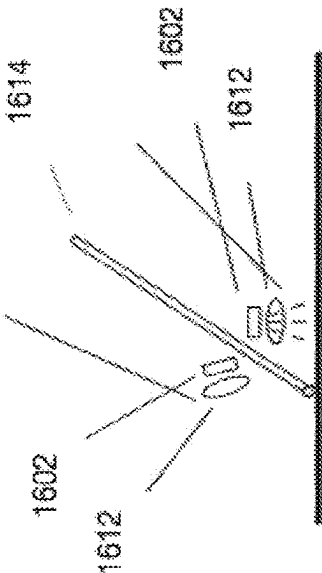


Fig.
16C

Fig. 17A

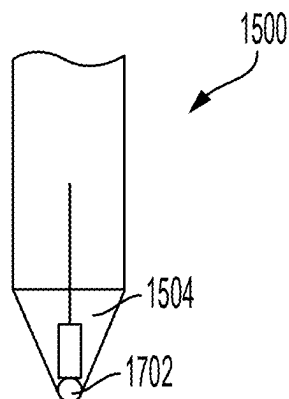


Fig. 17B

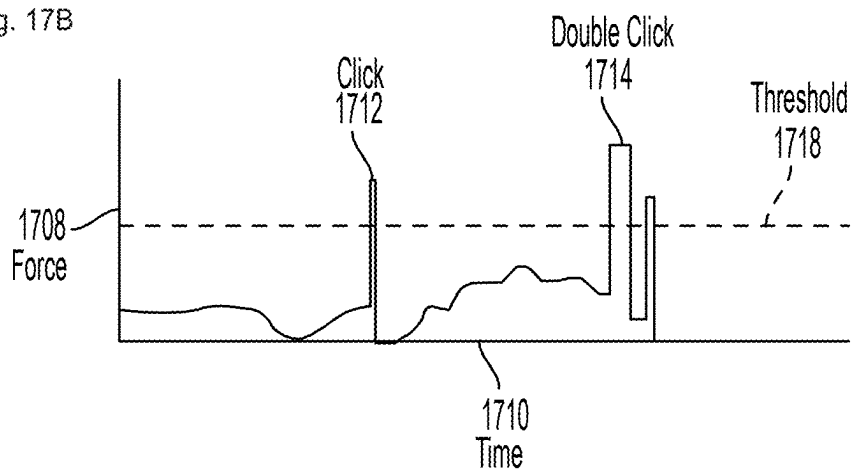


Fig. 17C

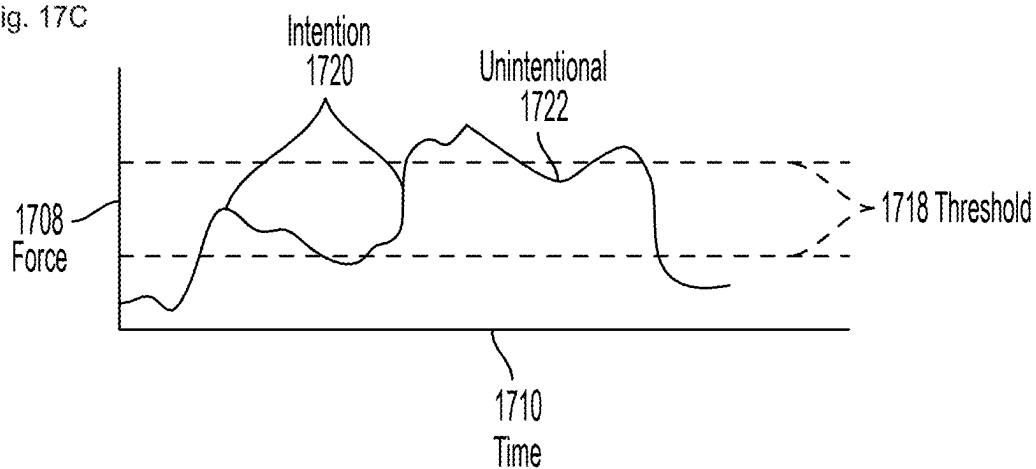
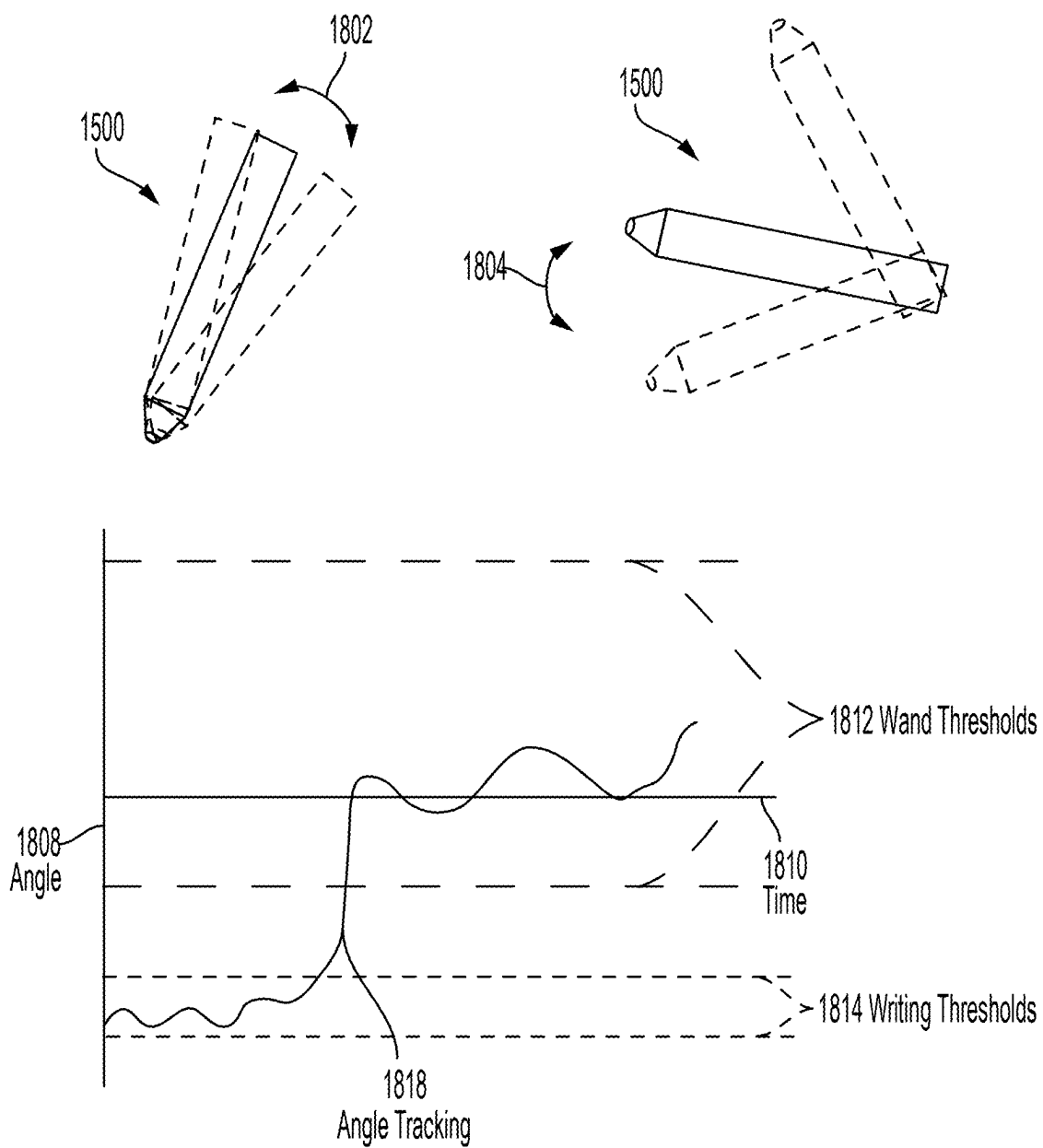
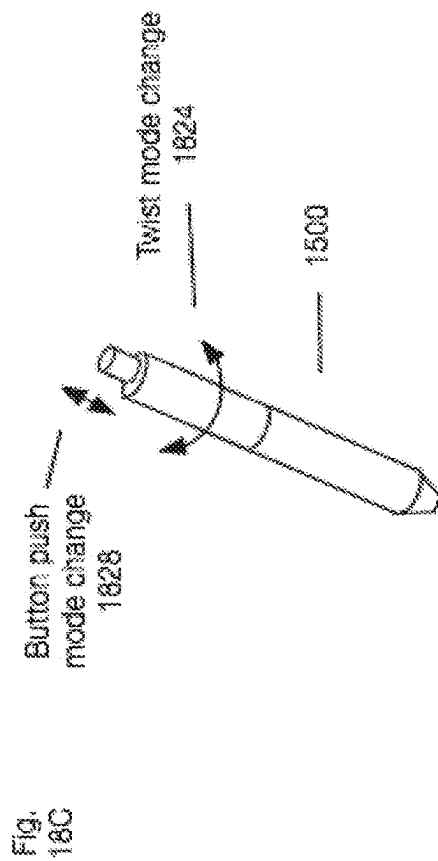
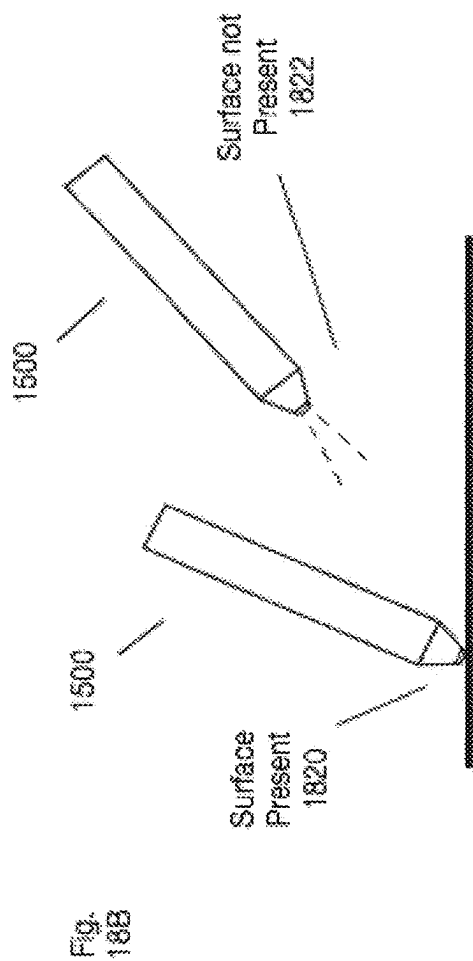
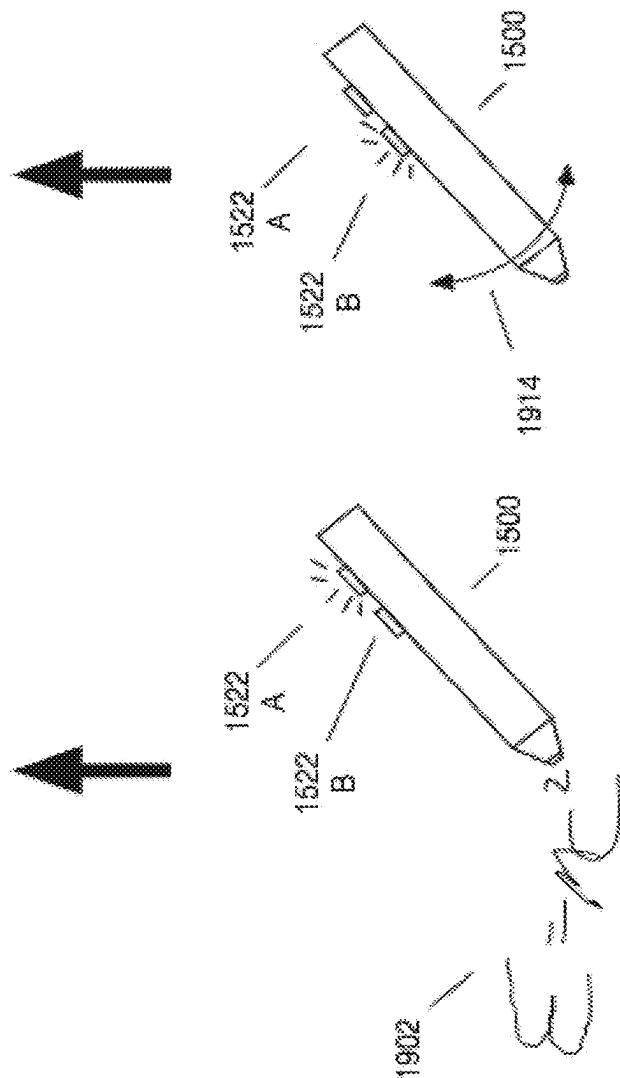
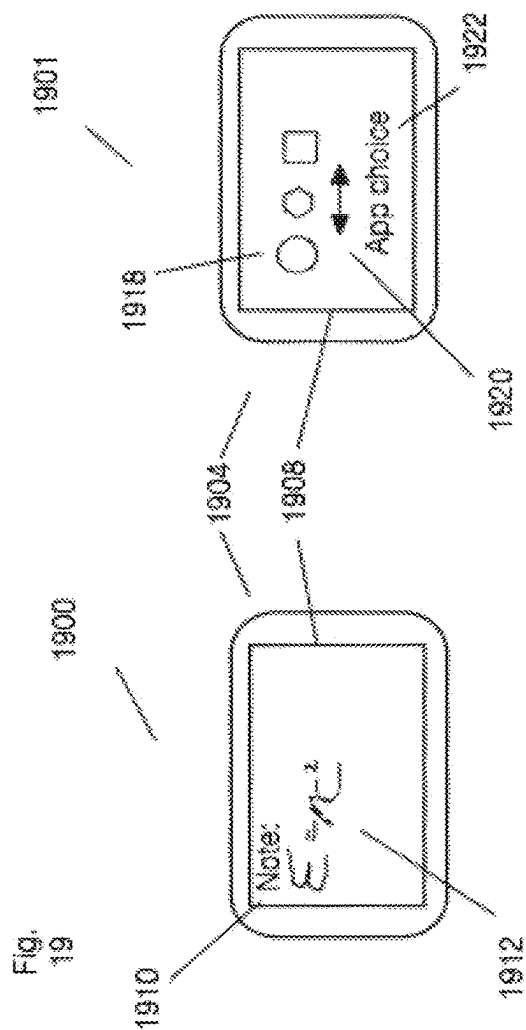


Fig. 18A







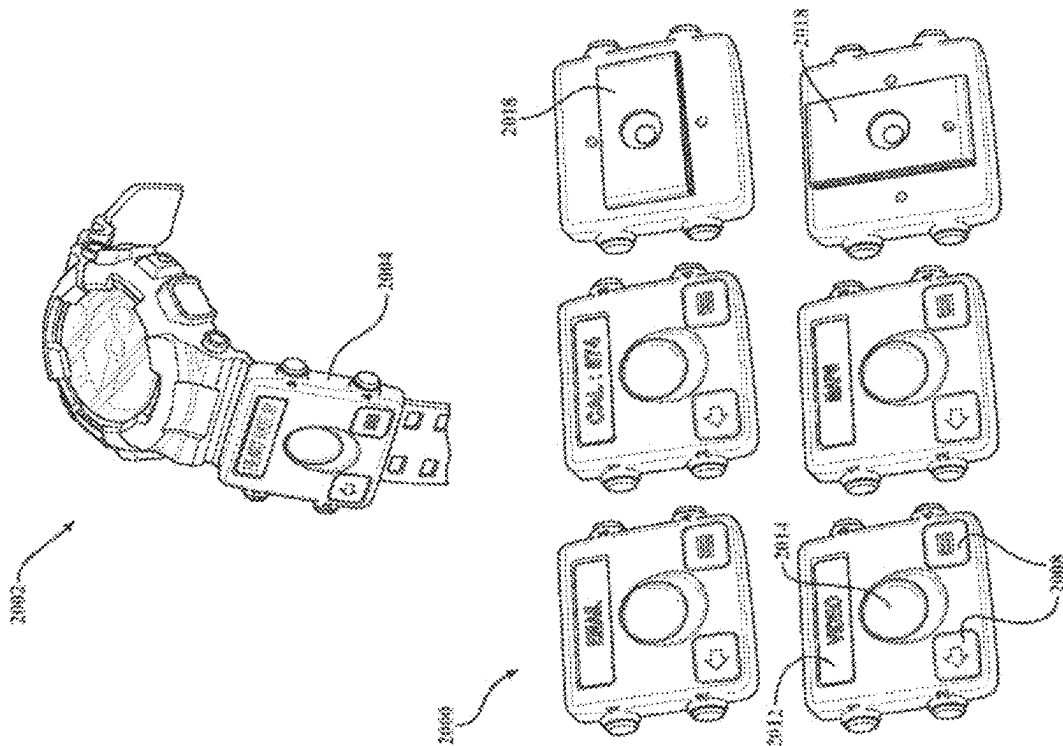


FIG. 20

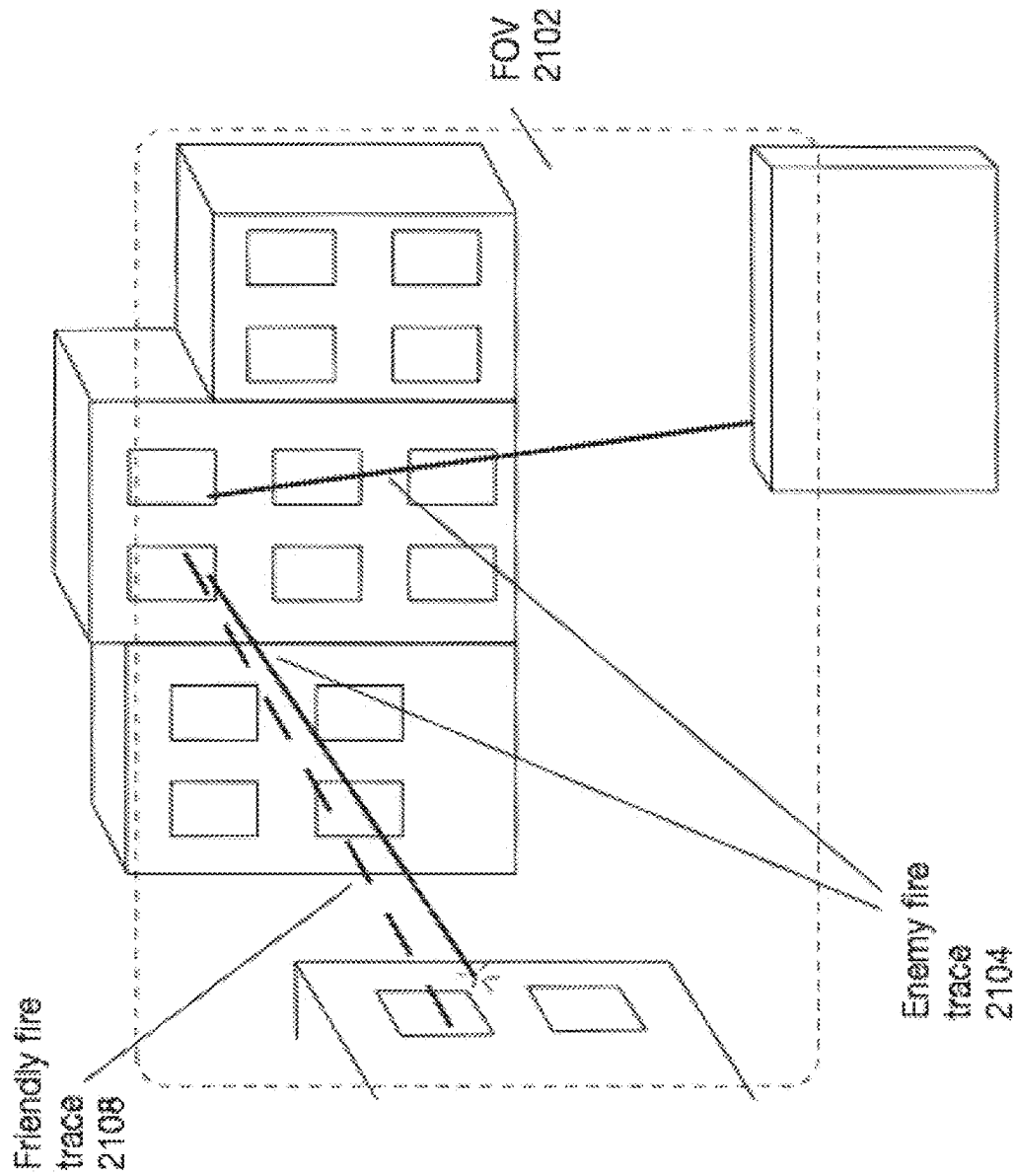


Figure 21

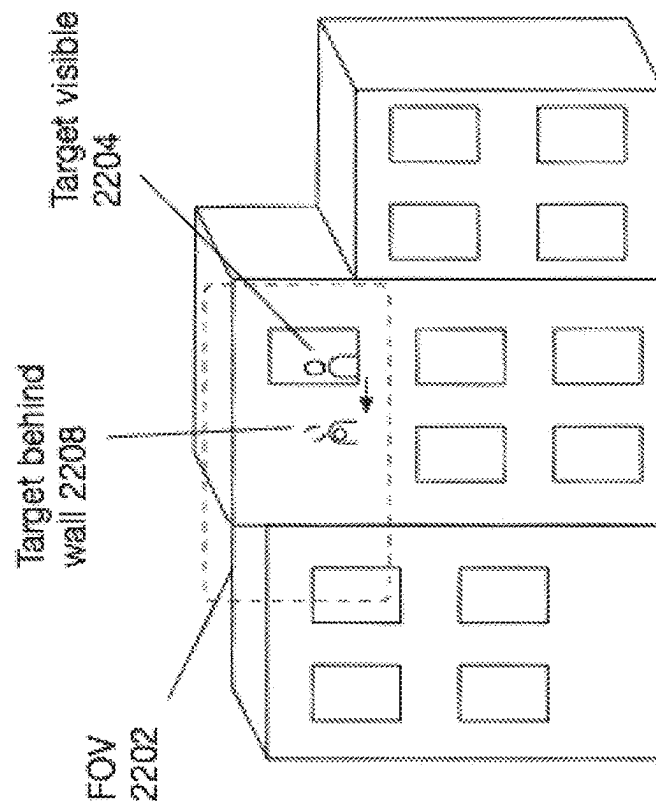


Figure 22

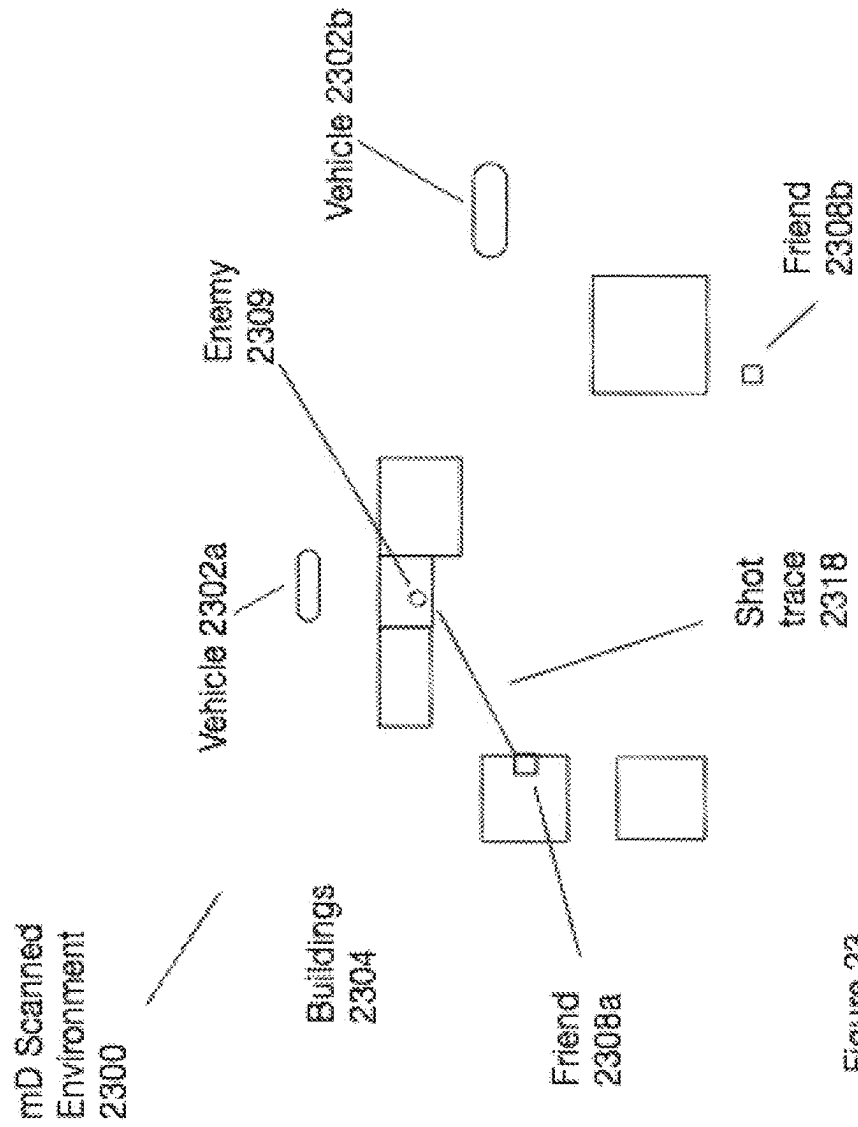


Figure 23

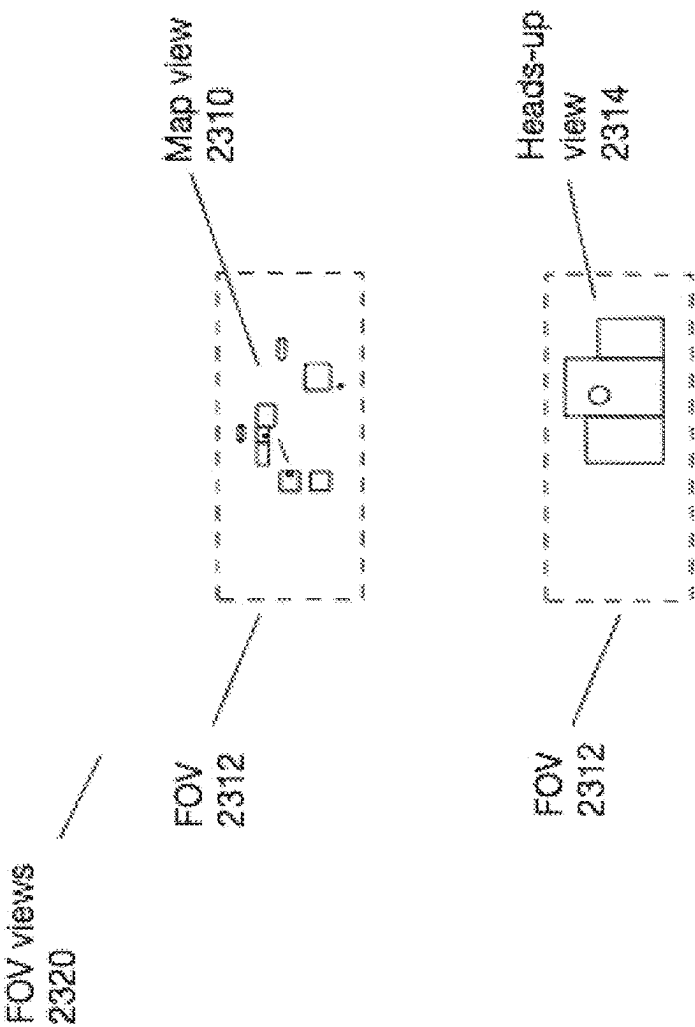


Figure 23a

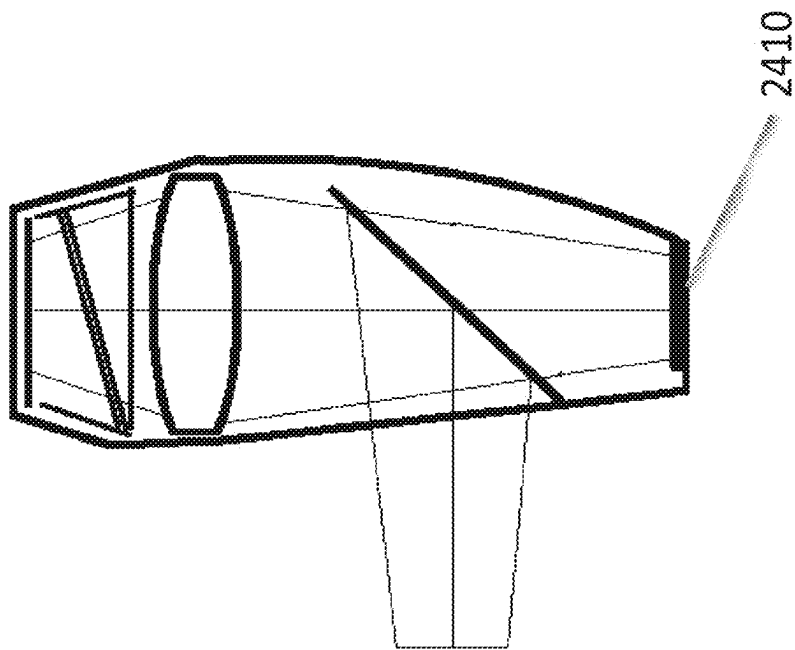


Figure 24

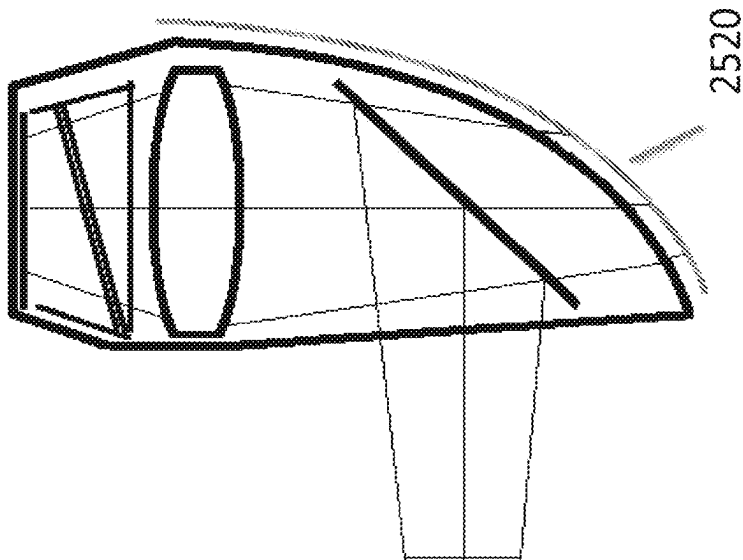


Figure 25

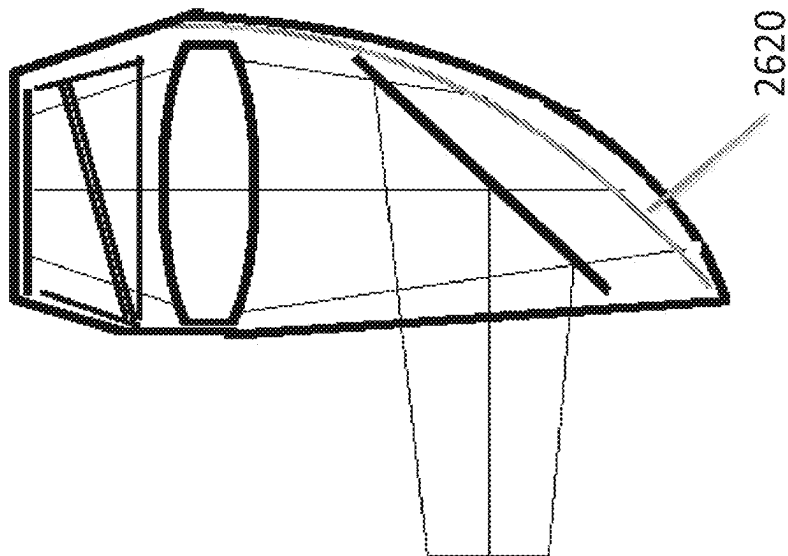


Figure 26

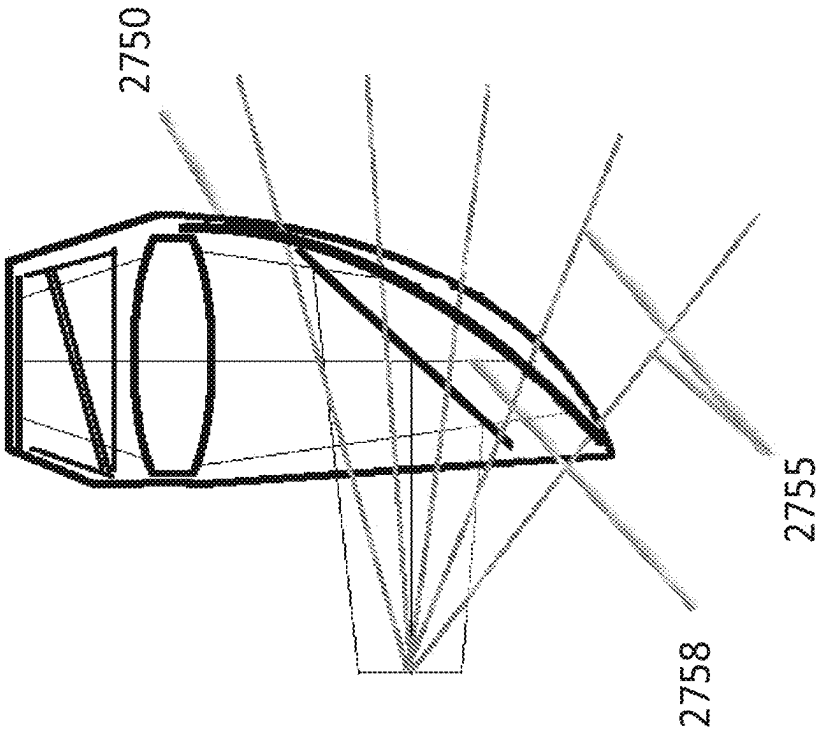


Figure 27

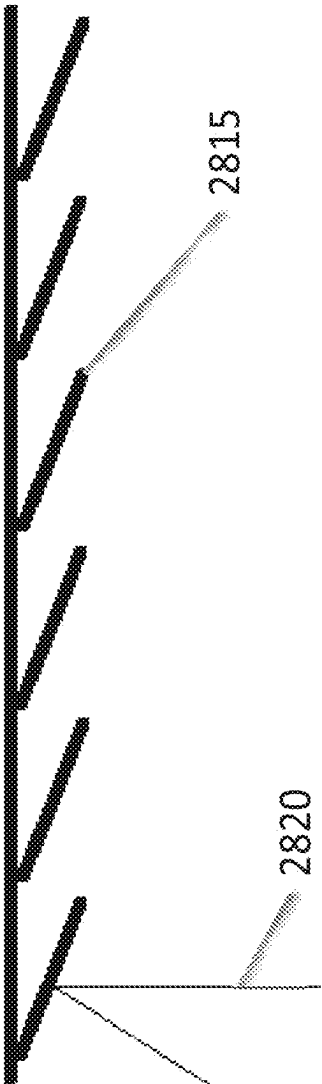


Figure 28a

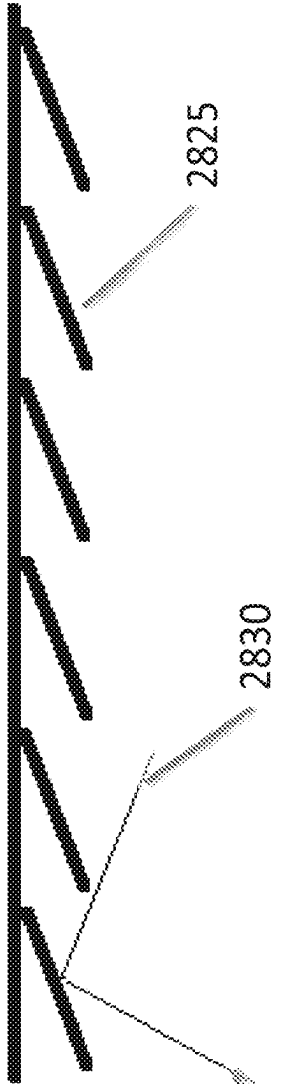


Figure 28b

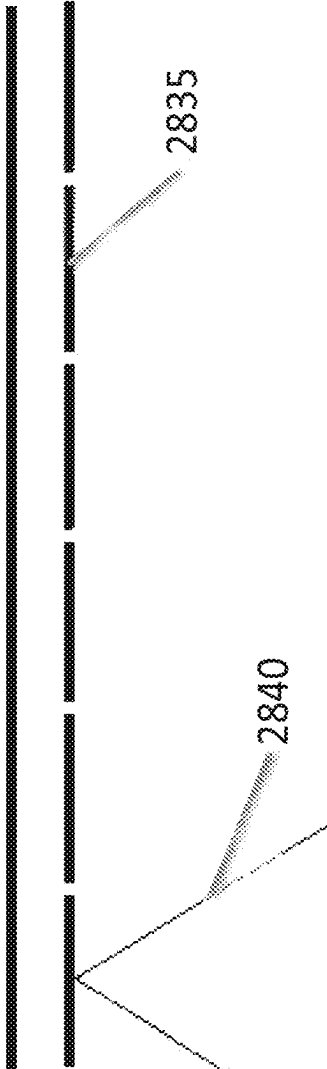


Figure 28c

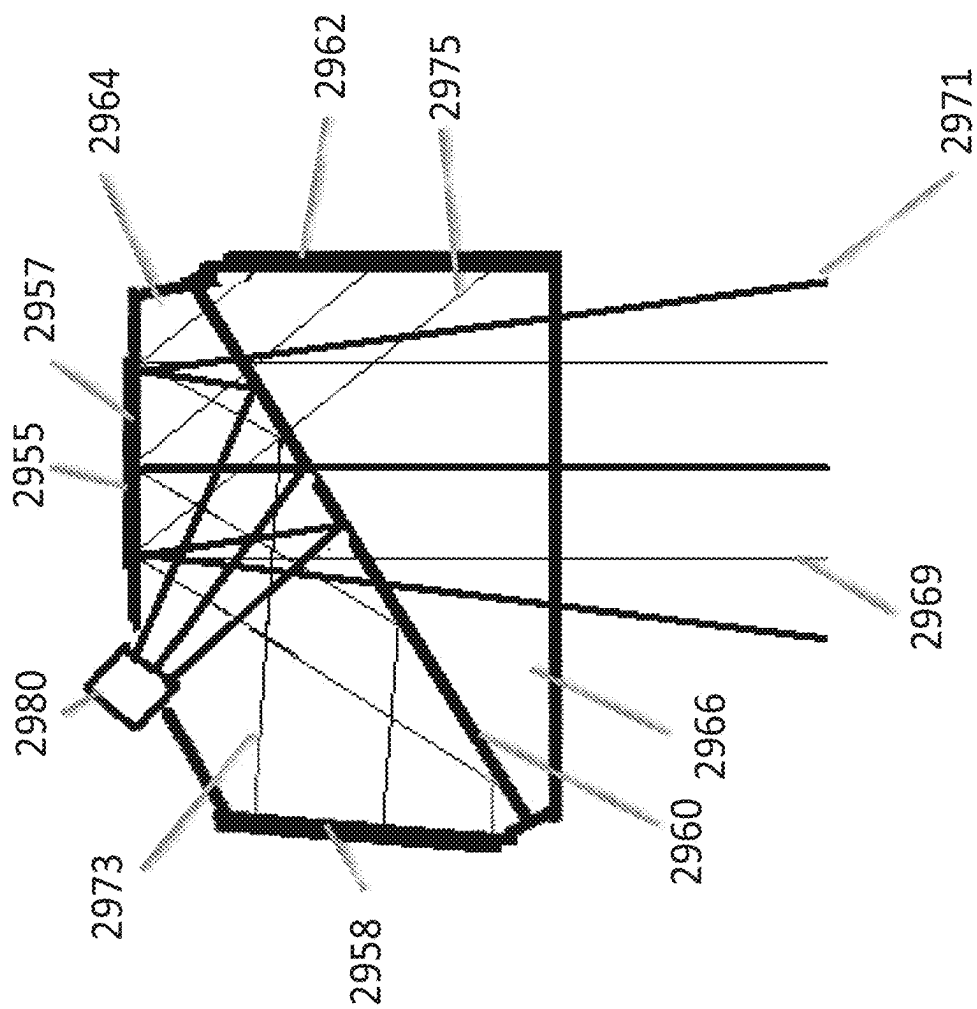


Figure 29

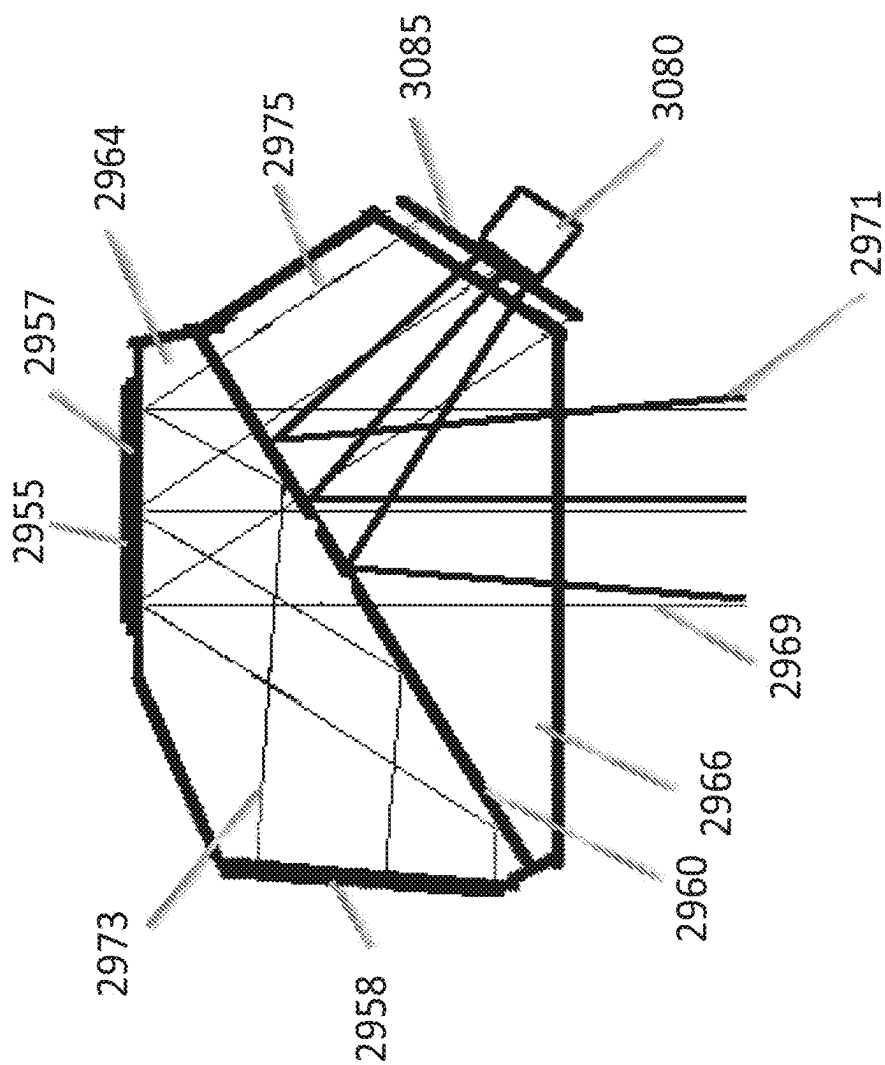


Figure 30

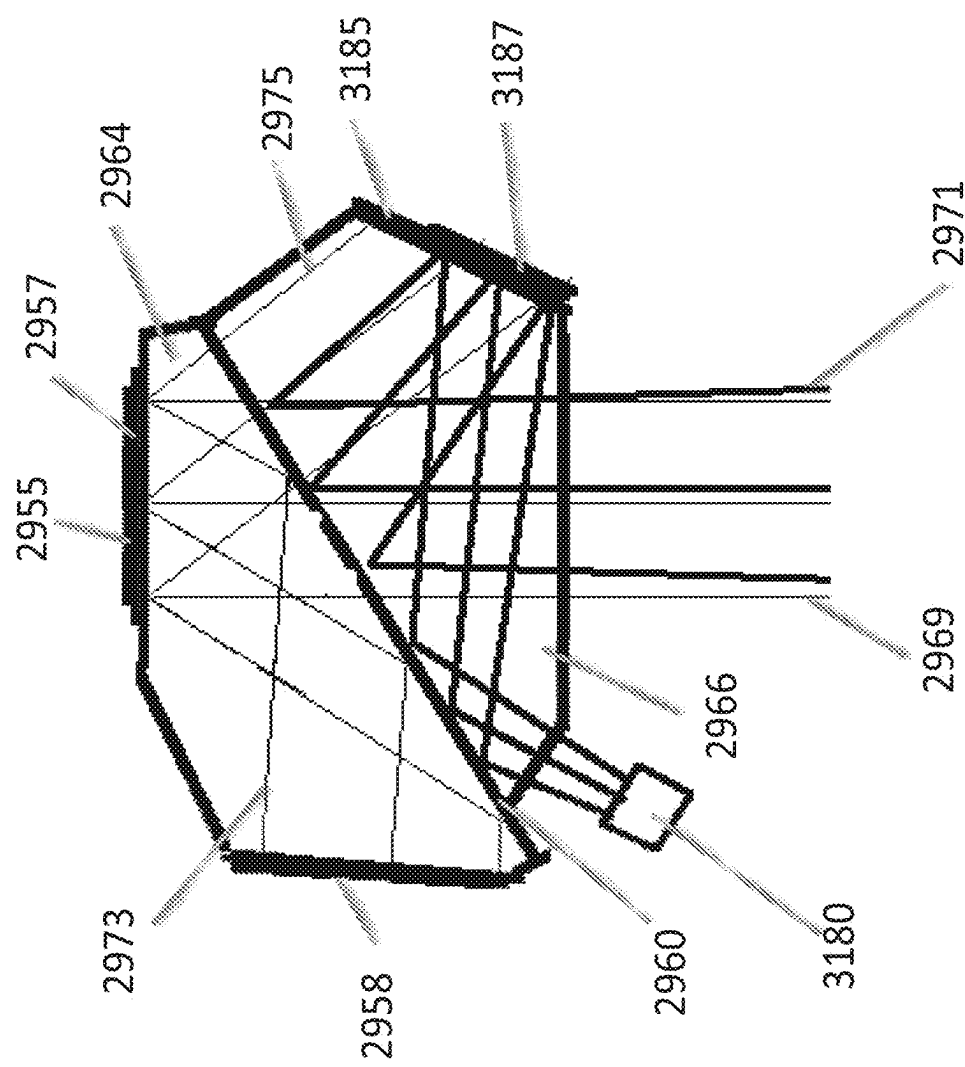


Figure 31

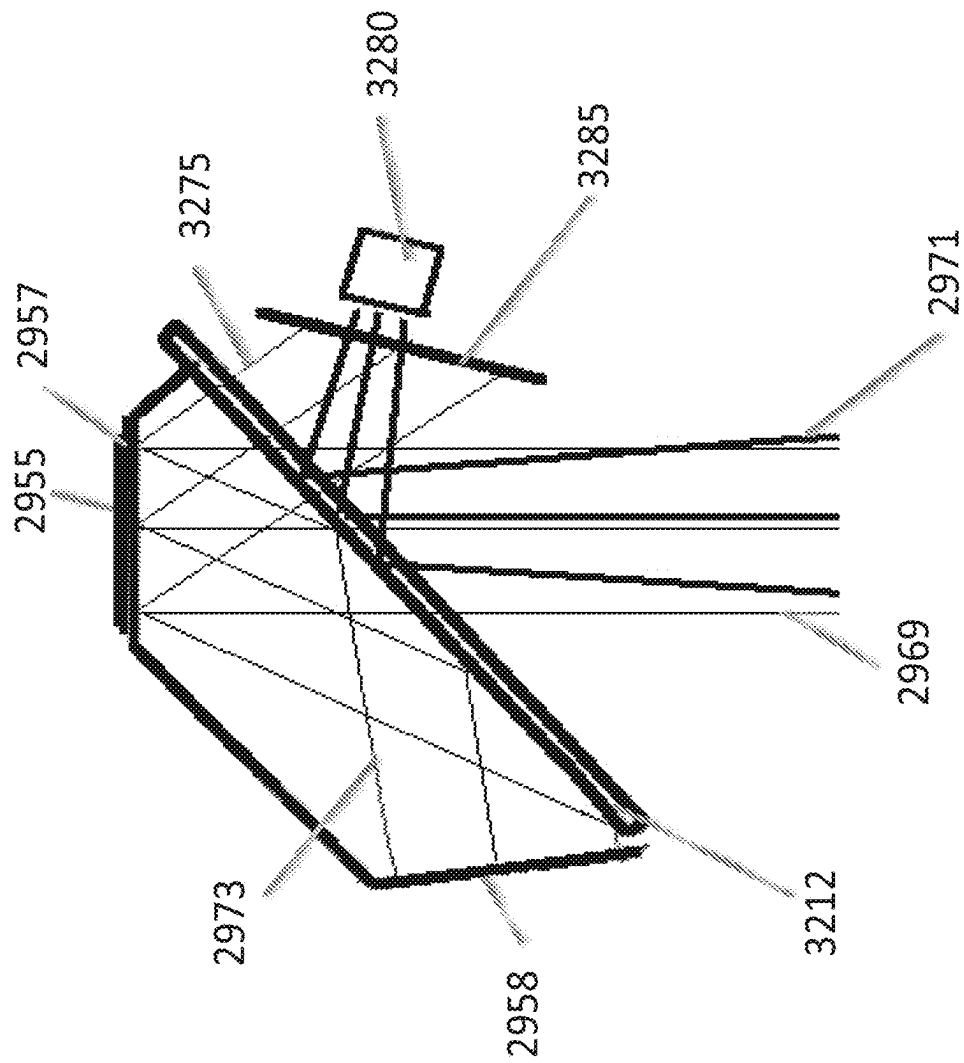


Figure 32

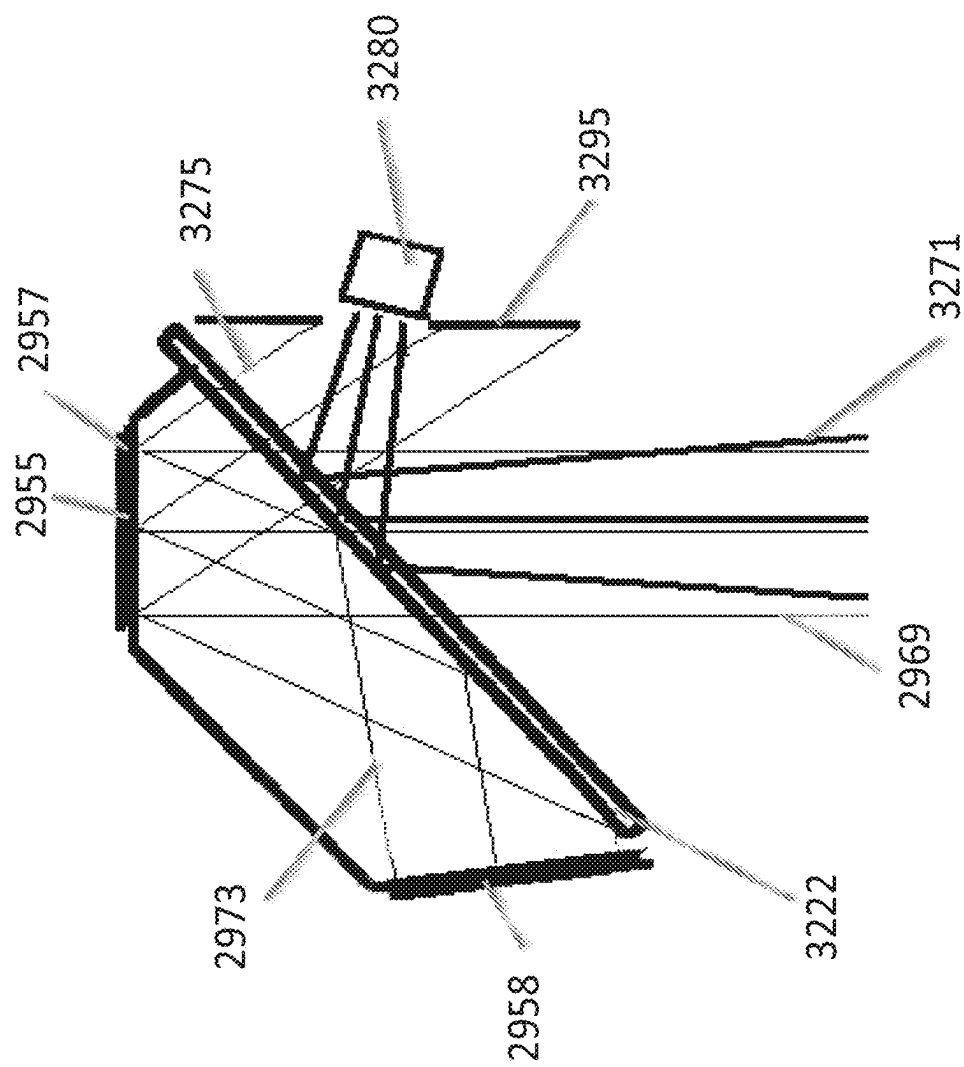


Figure 32a

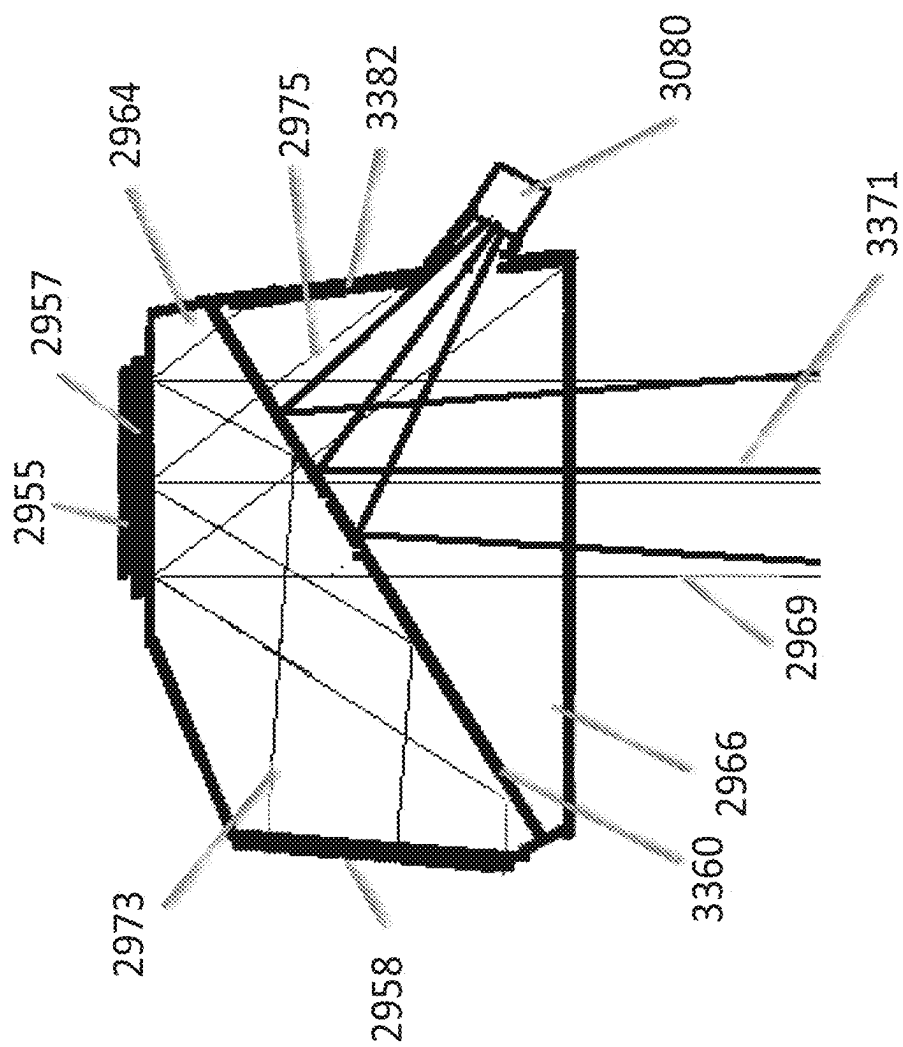


Figure 33

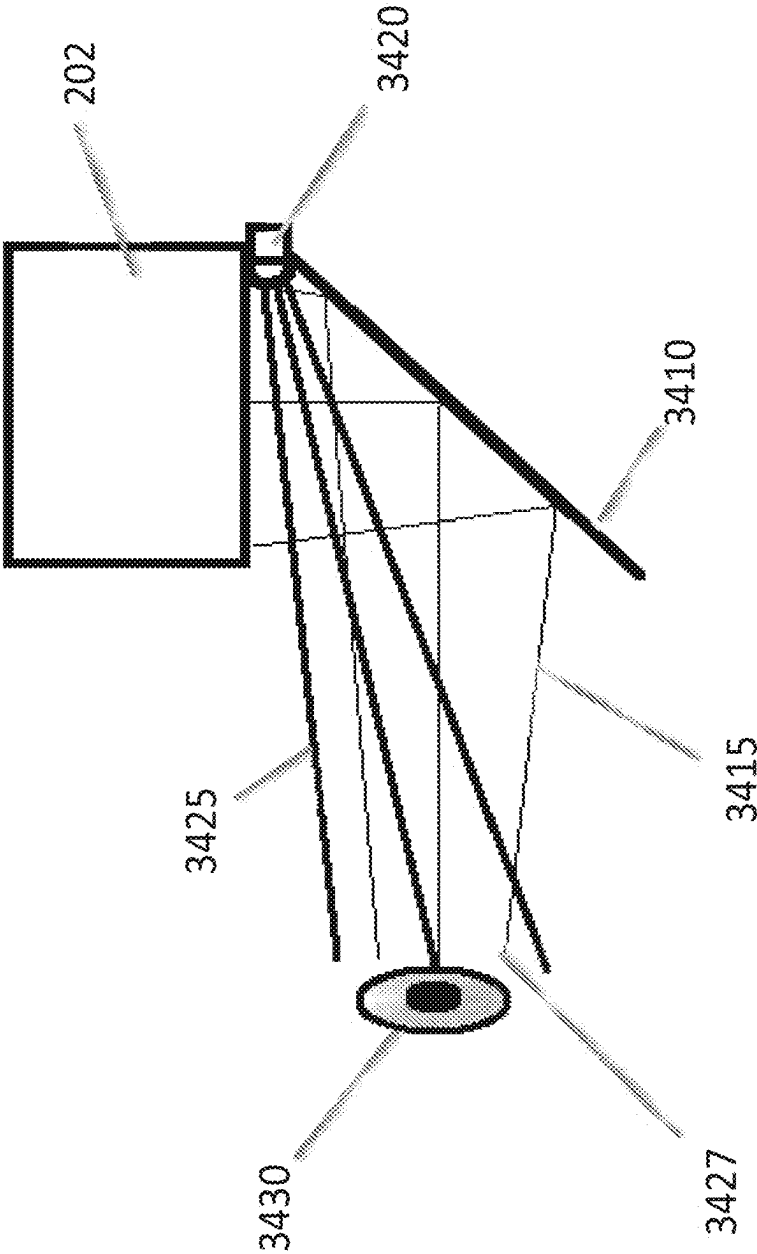


Figure 34

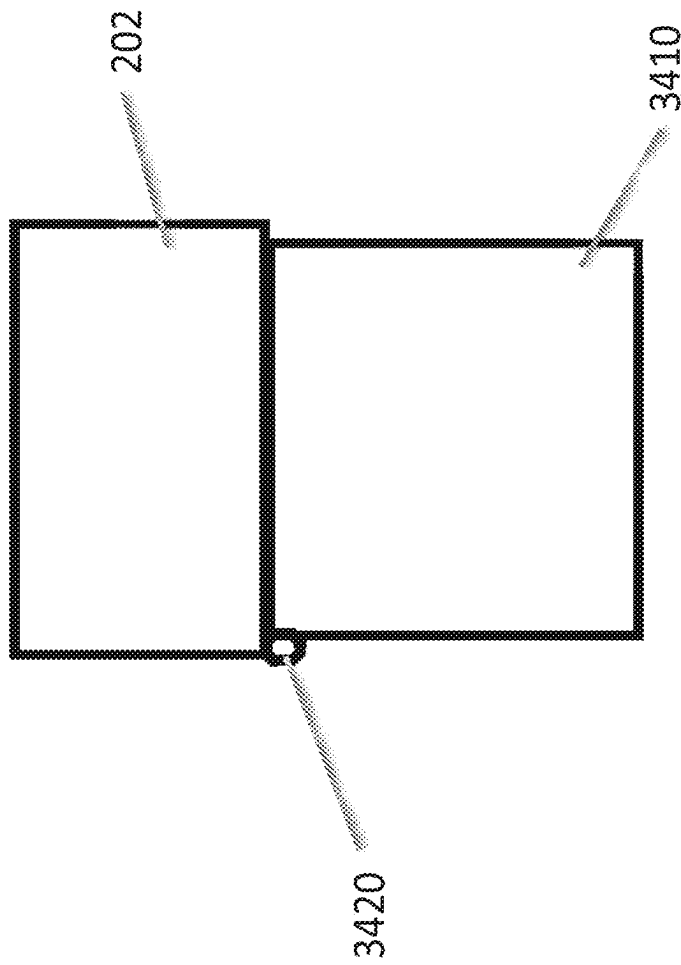


Figure 34a

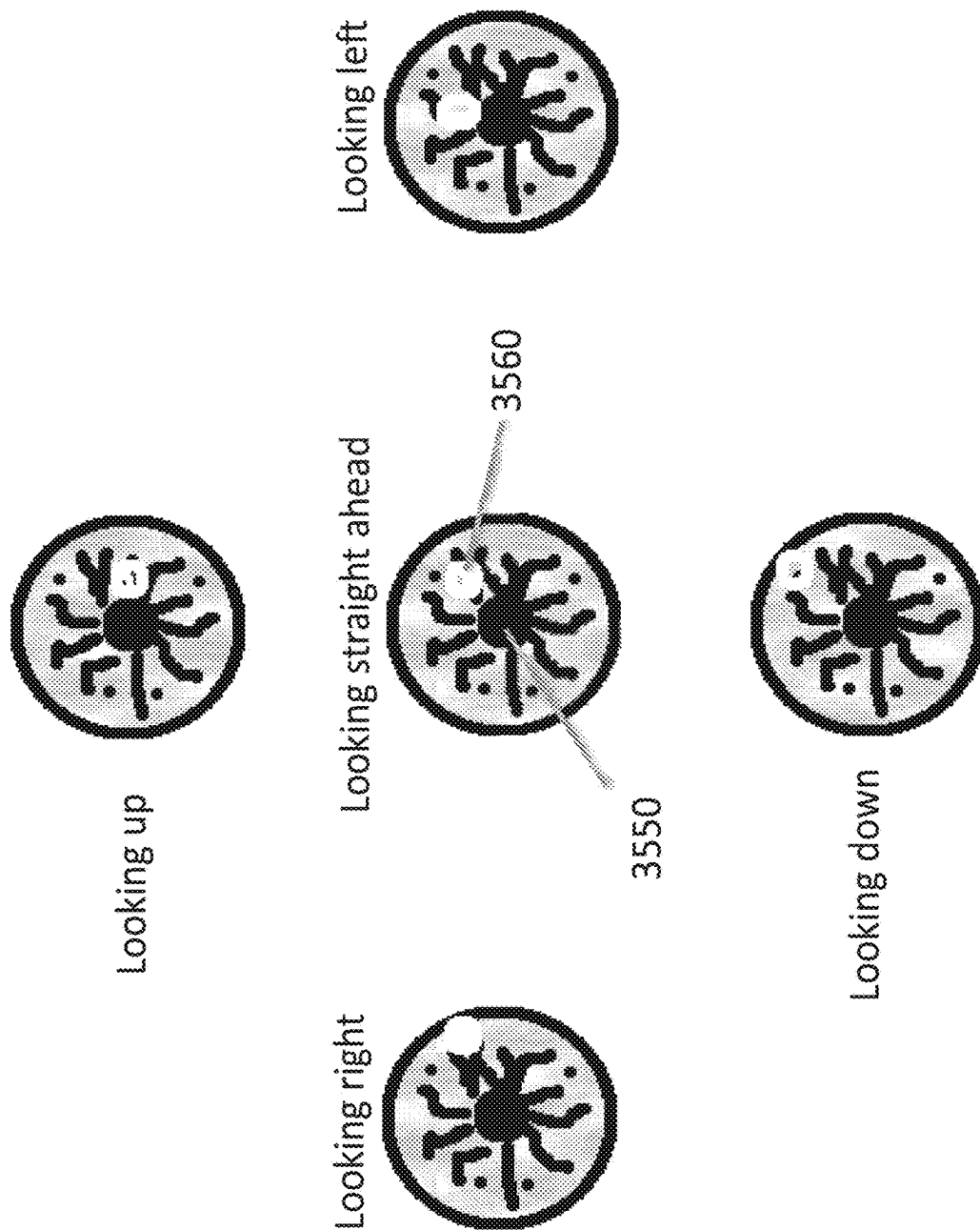


Figure 35

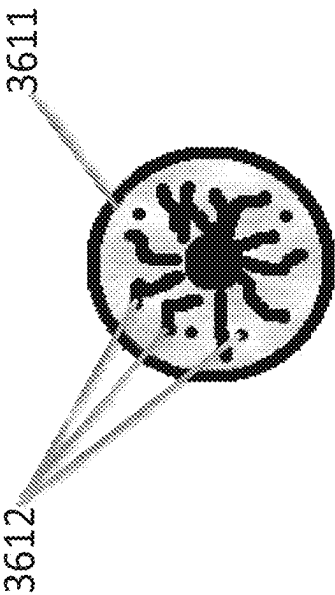


Figure 36a

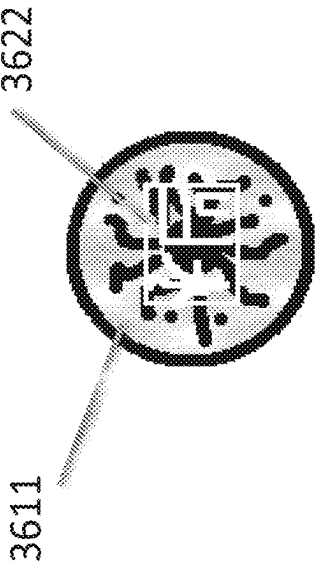


Figure 36b

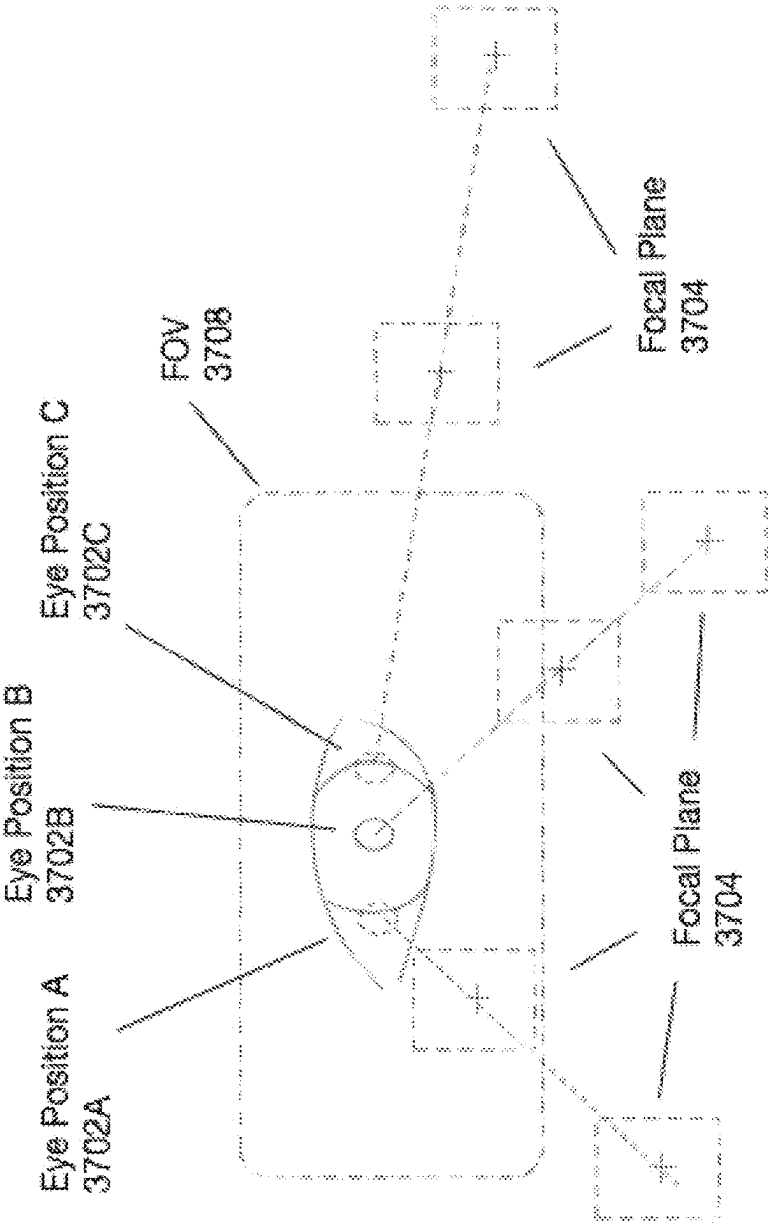


Figure 37

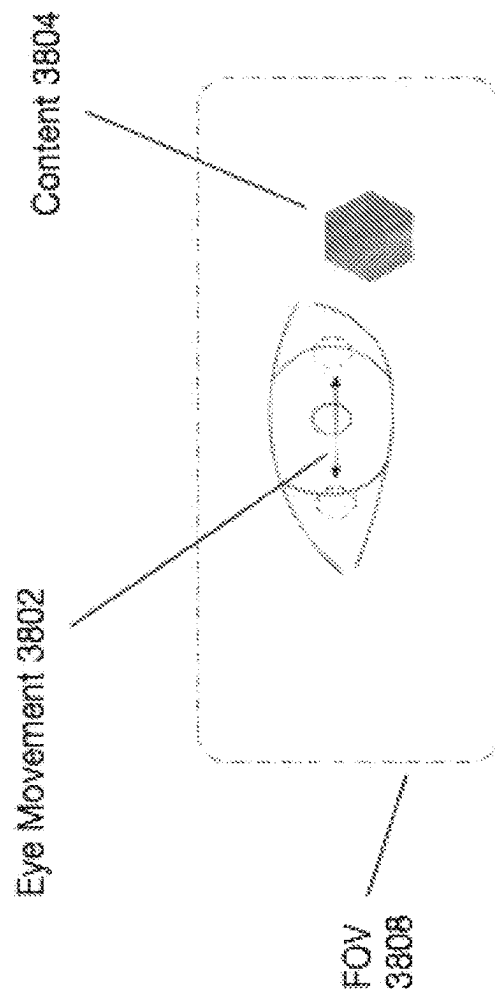


Figure 38

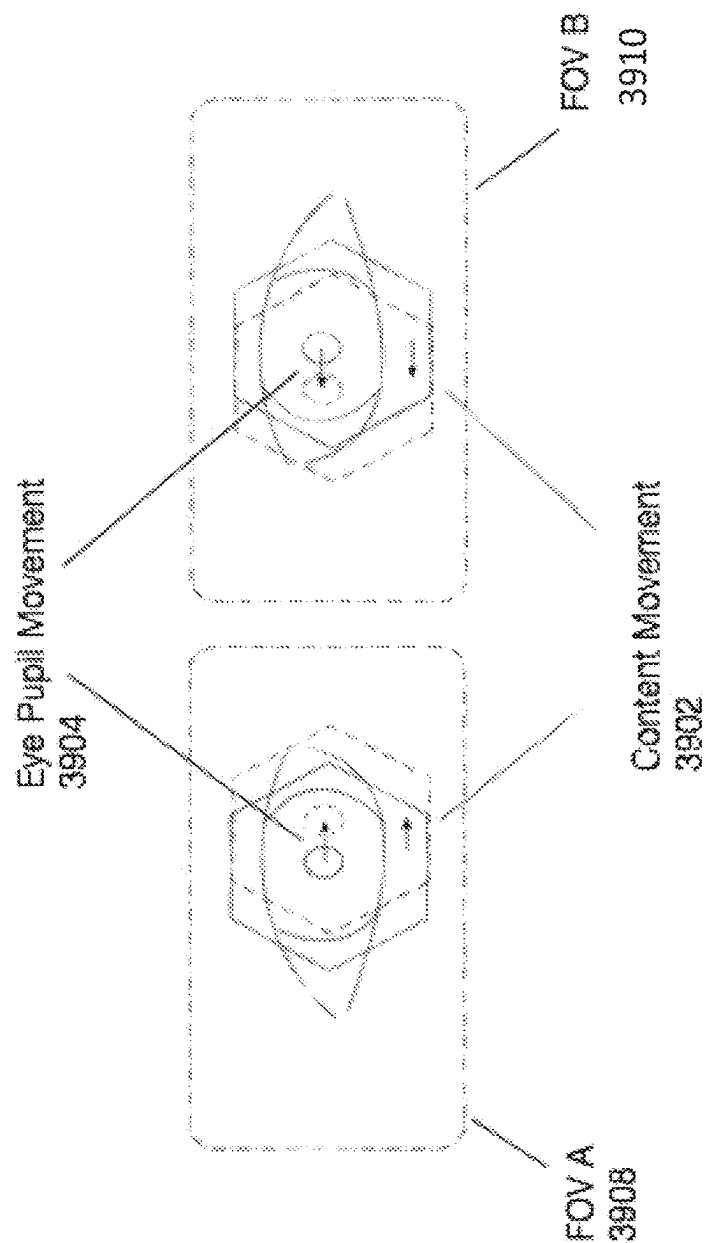


Figure 39

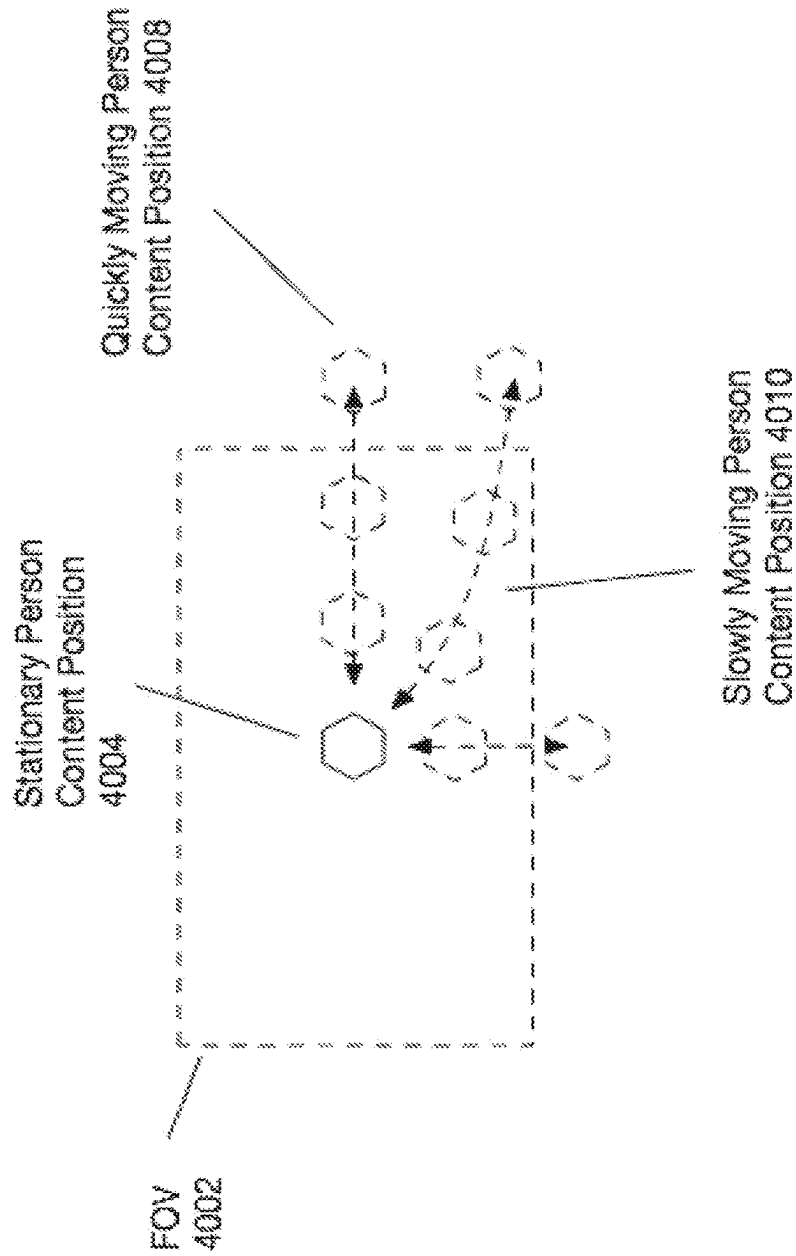


Figure 40

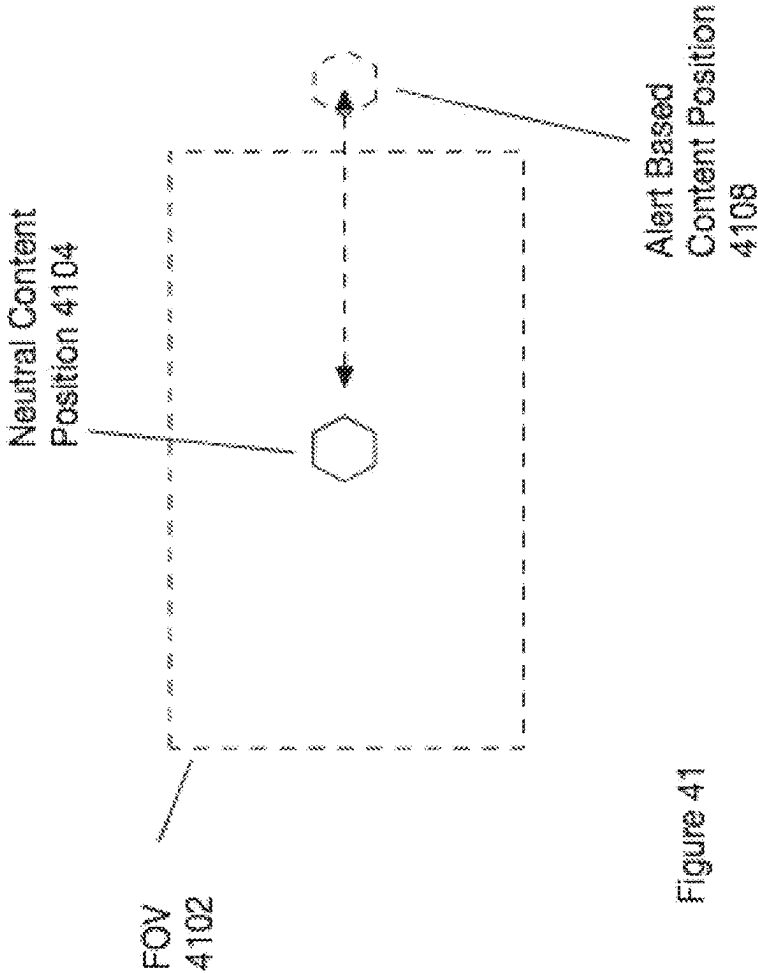


Figure 41

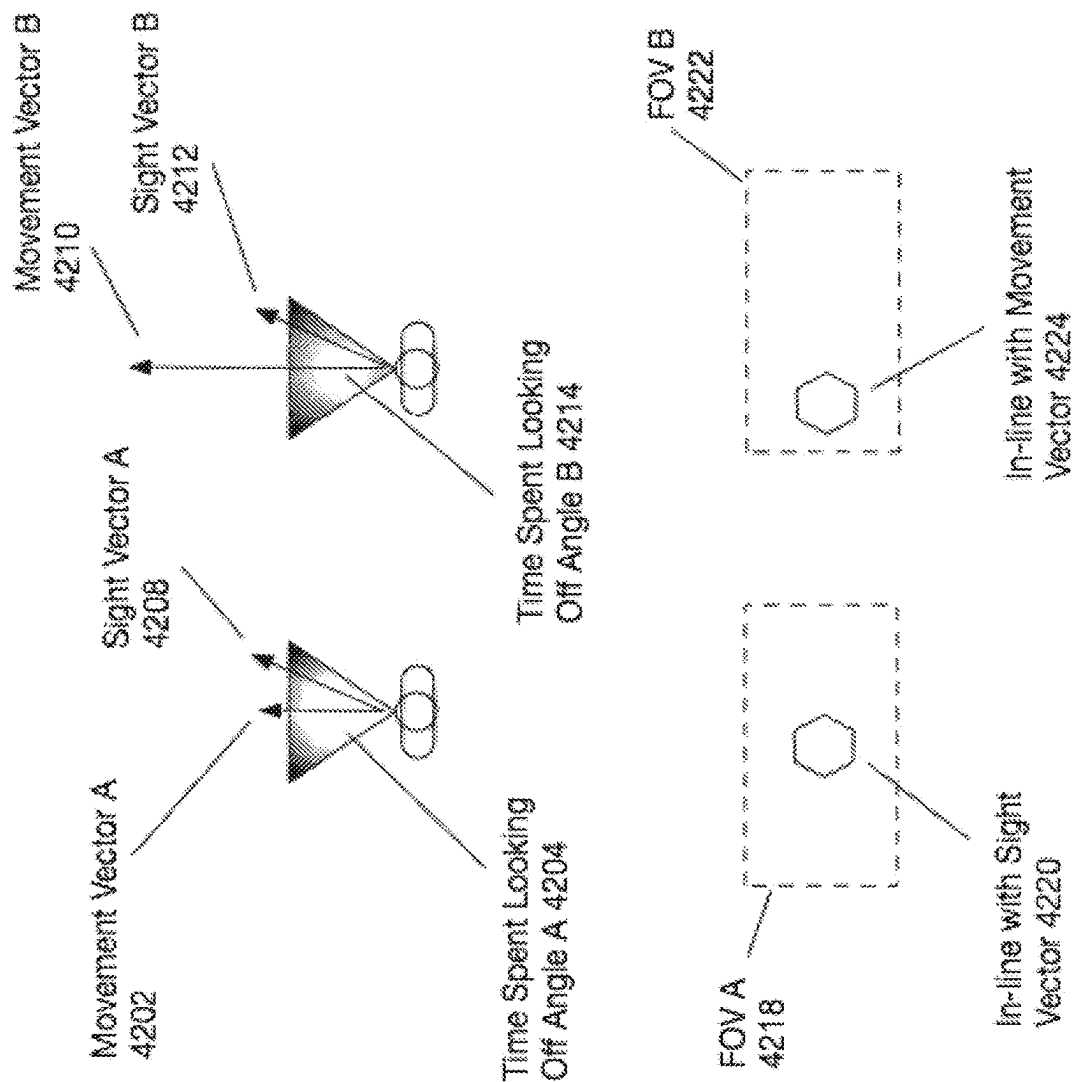


Figure 42

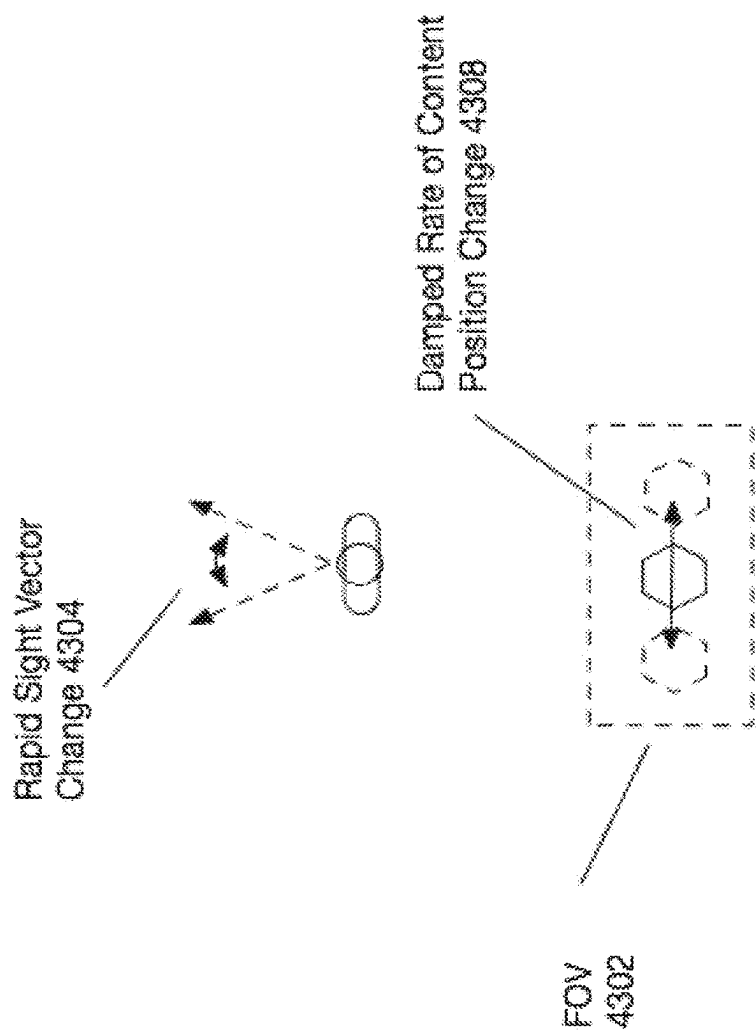


Figure 43

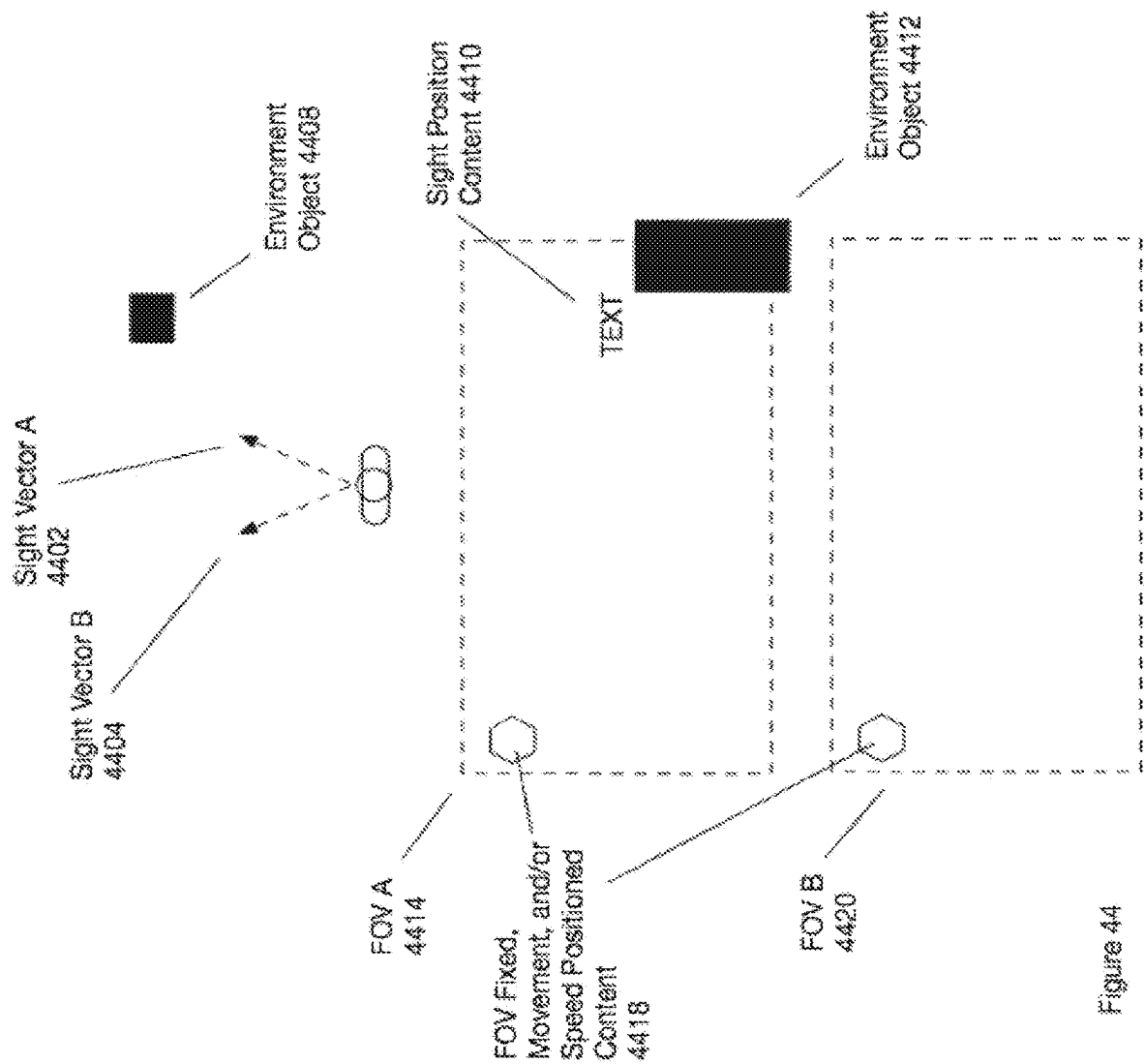


Figure 44

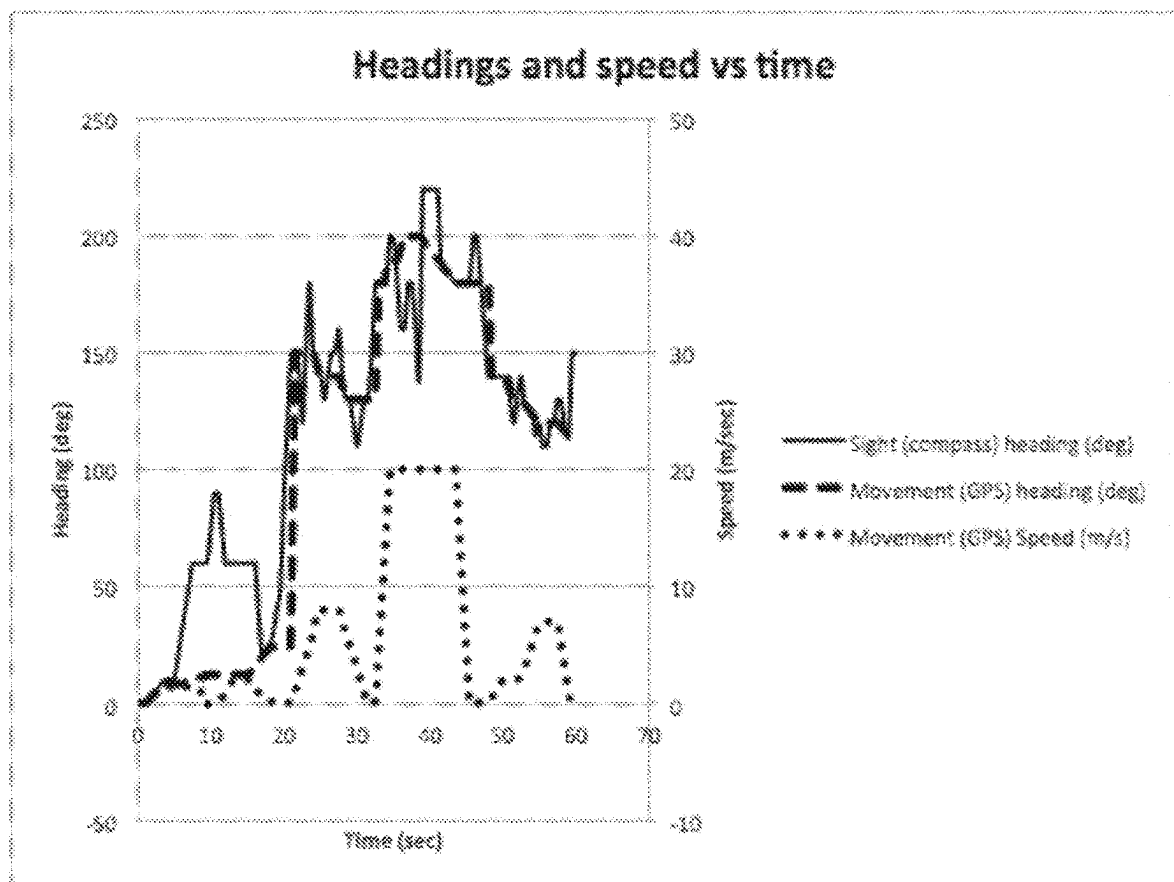


Figure 45

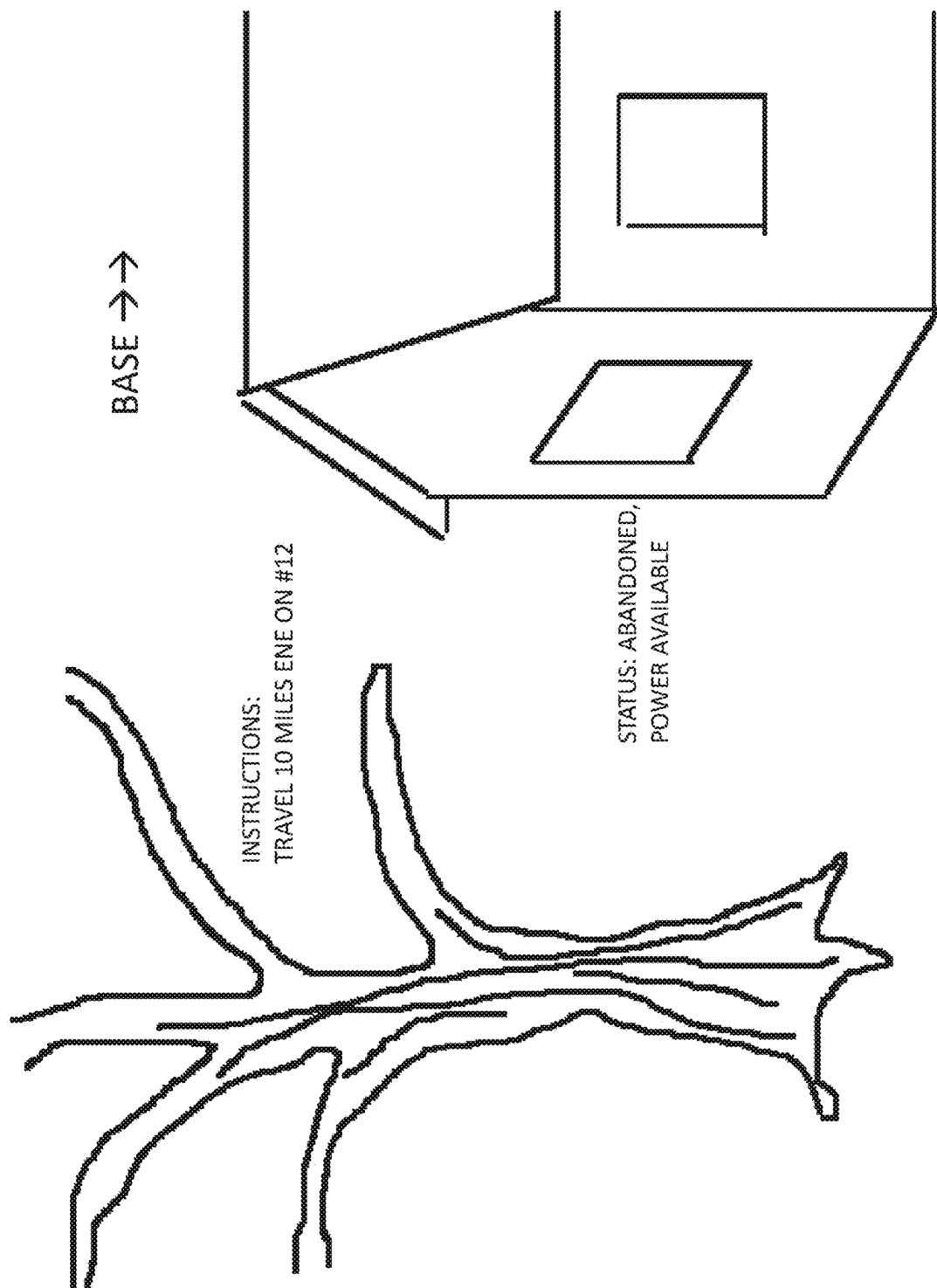


Figure 46

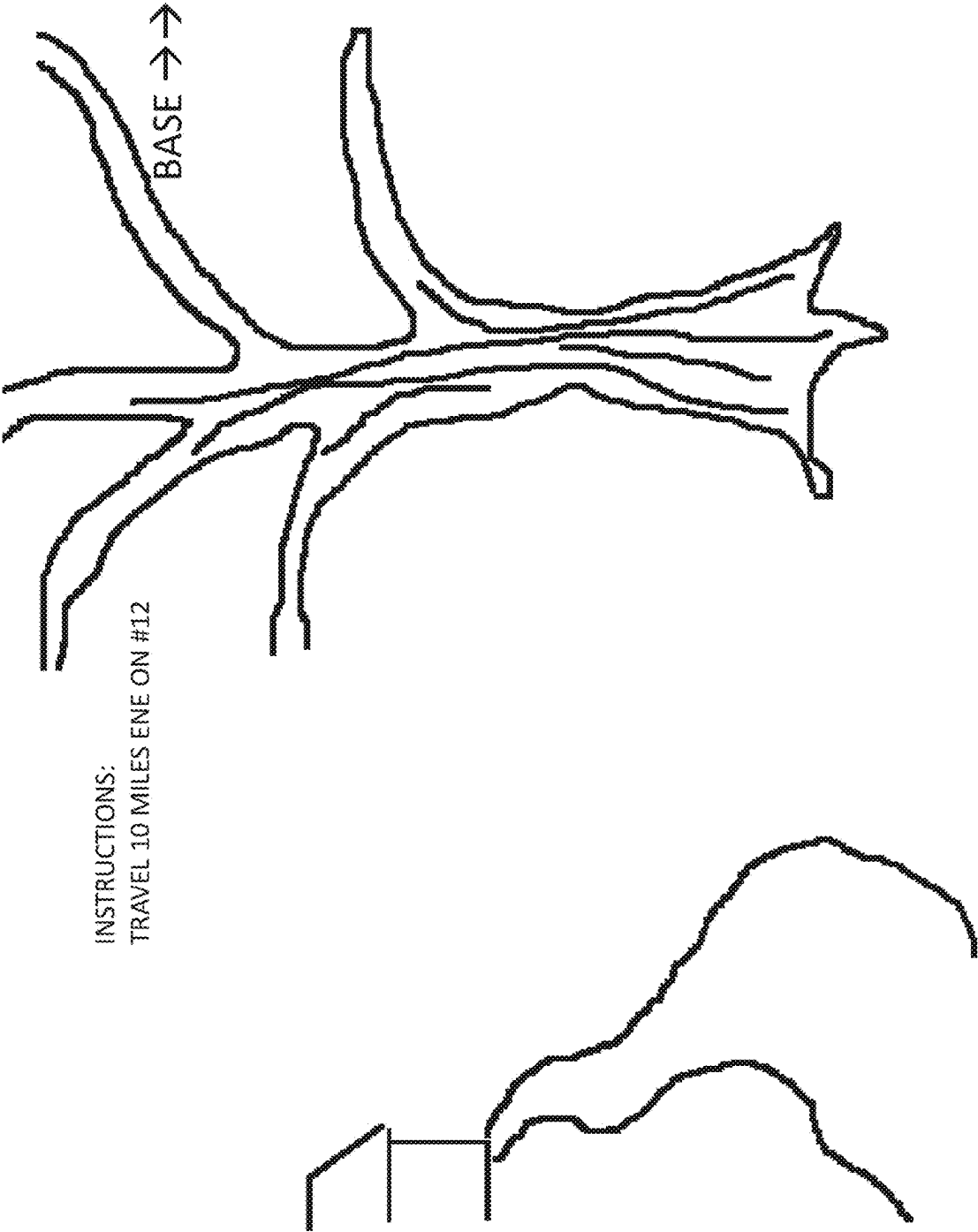


Figure 47

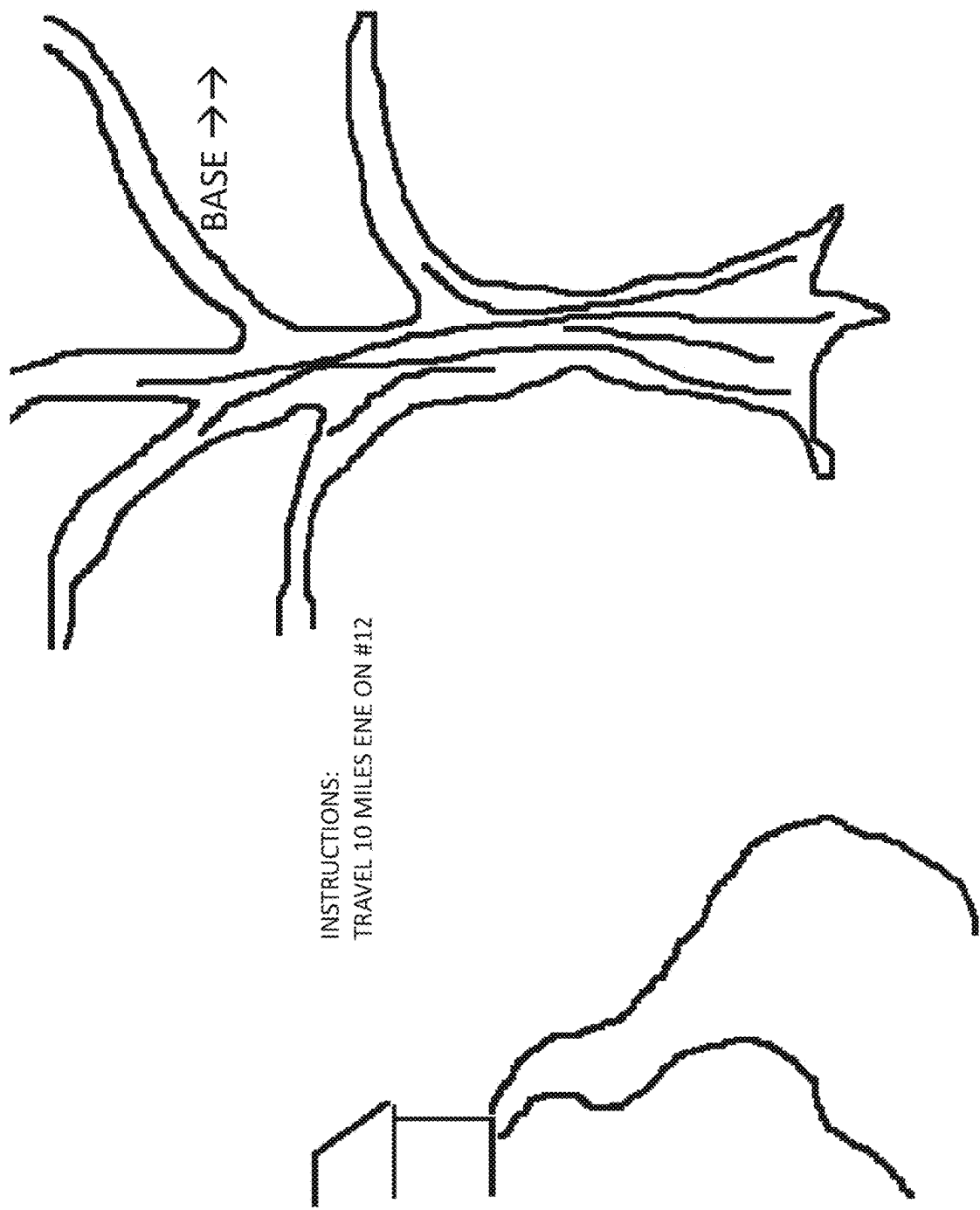


Figure 48

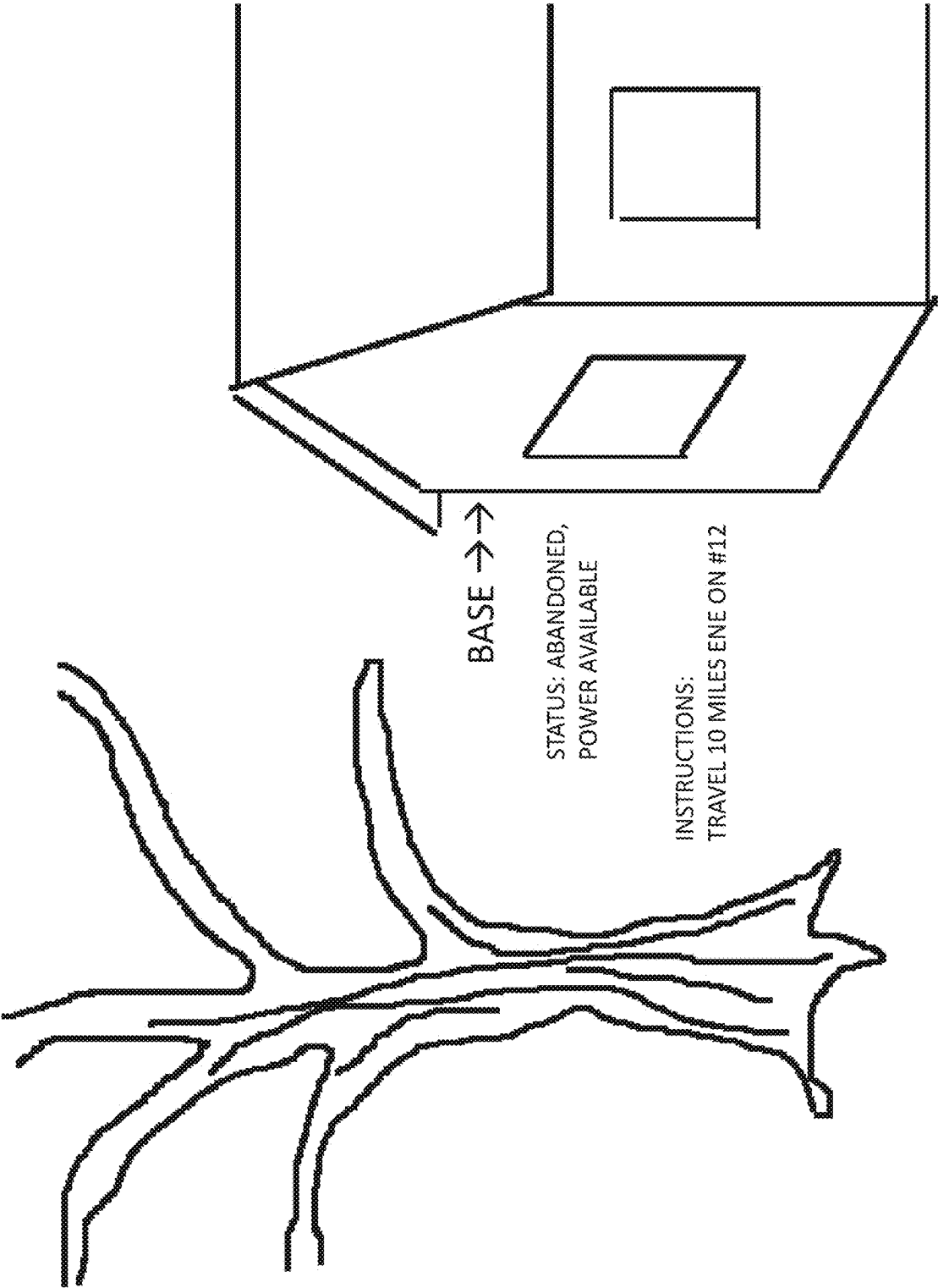


Figure 49

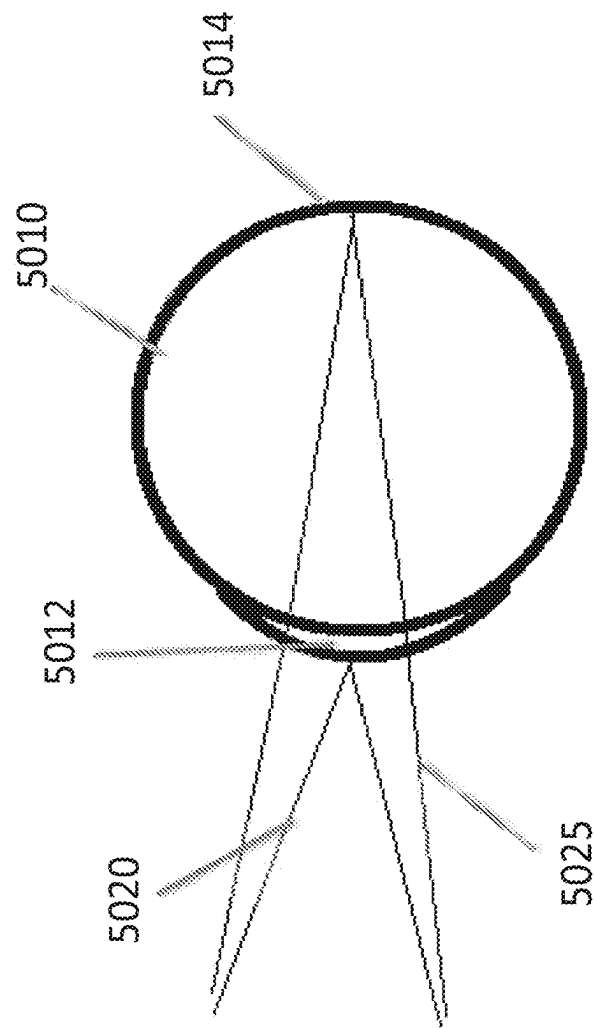


Figure 50



Figure 51

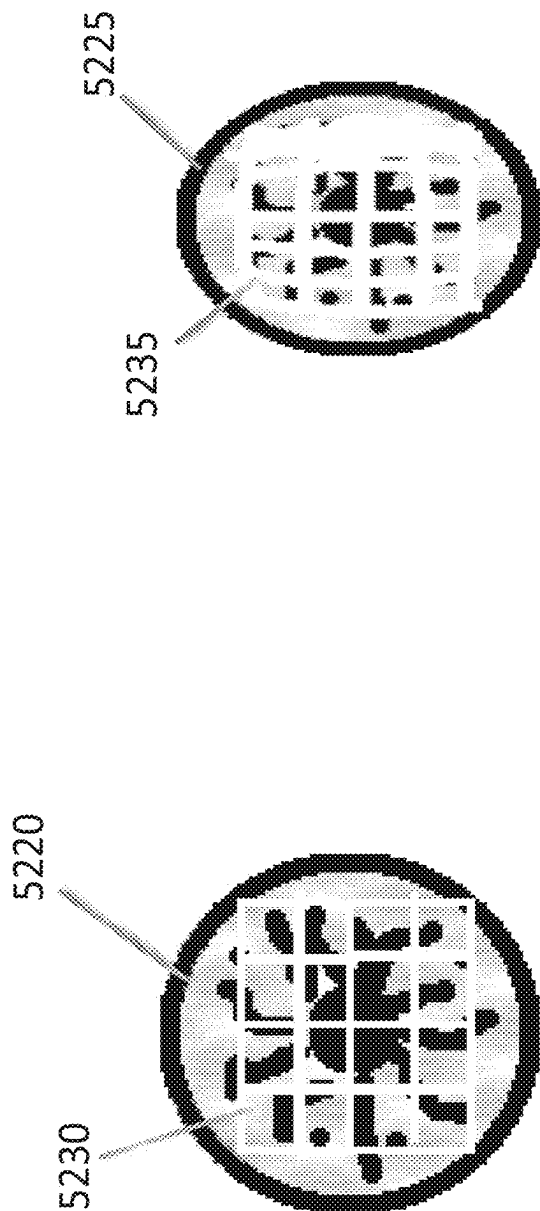


Figure 52a

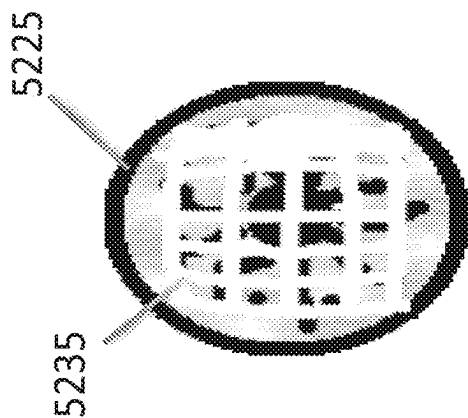


Figure 52b

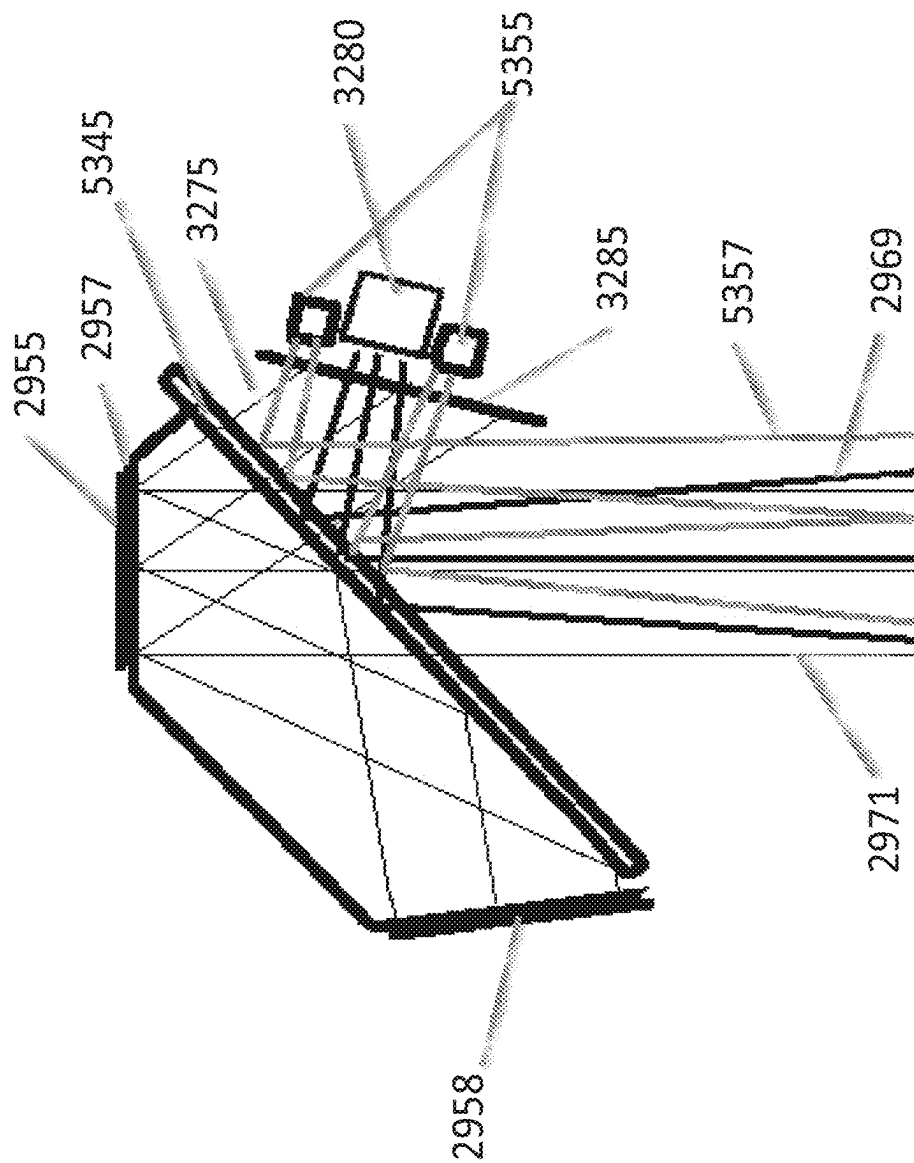


Figure 53

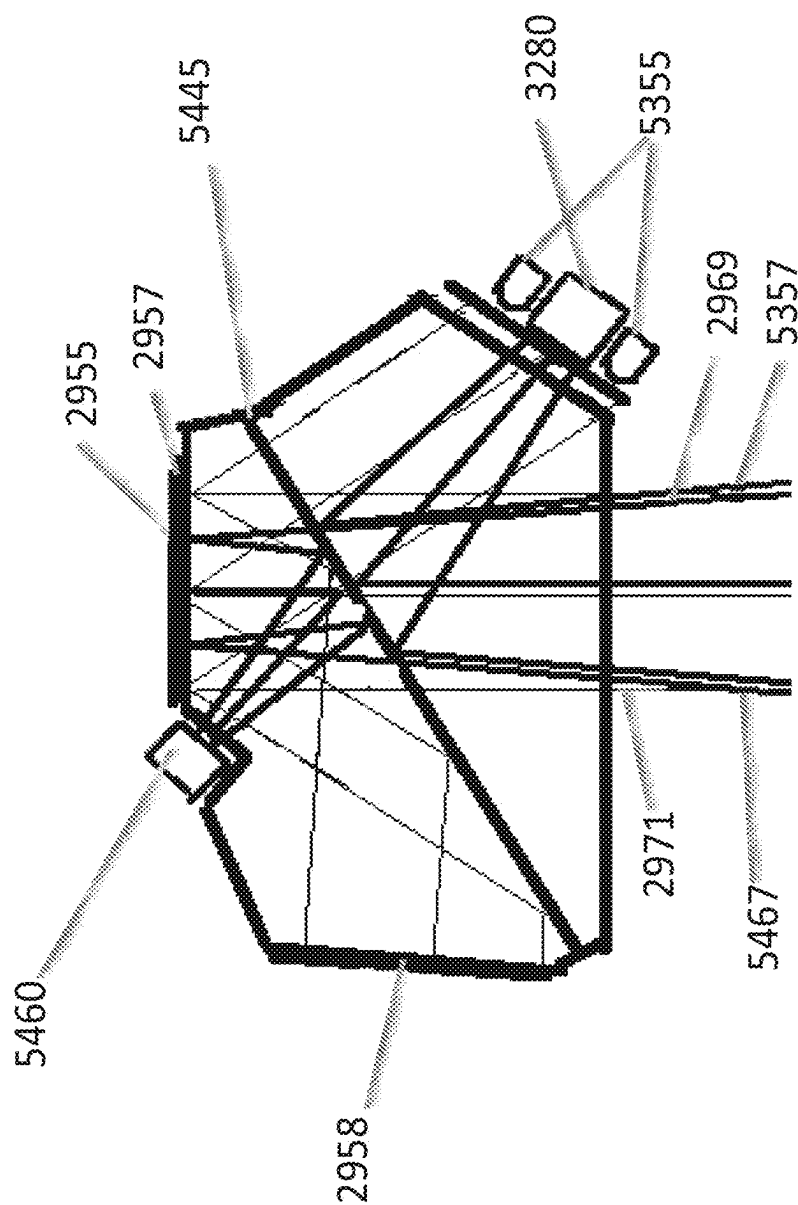


Figure 54

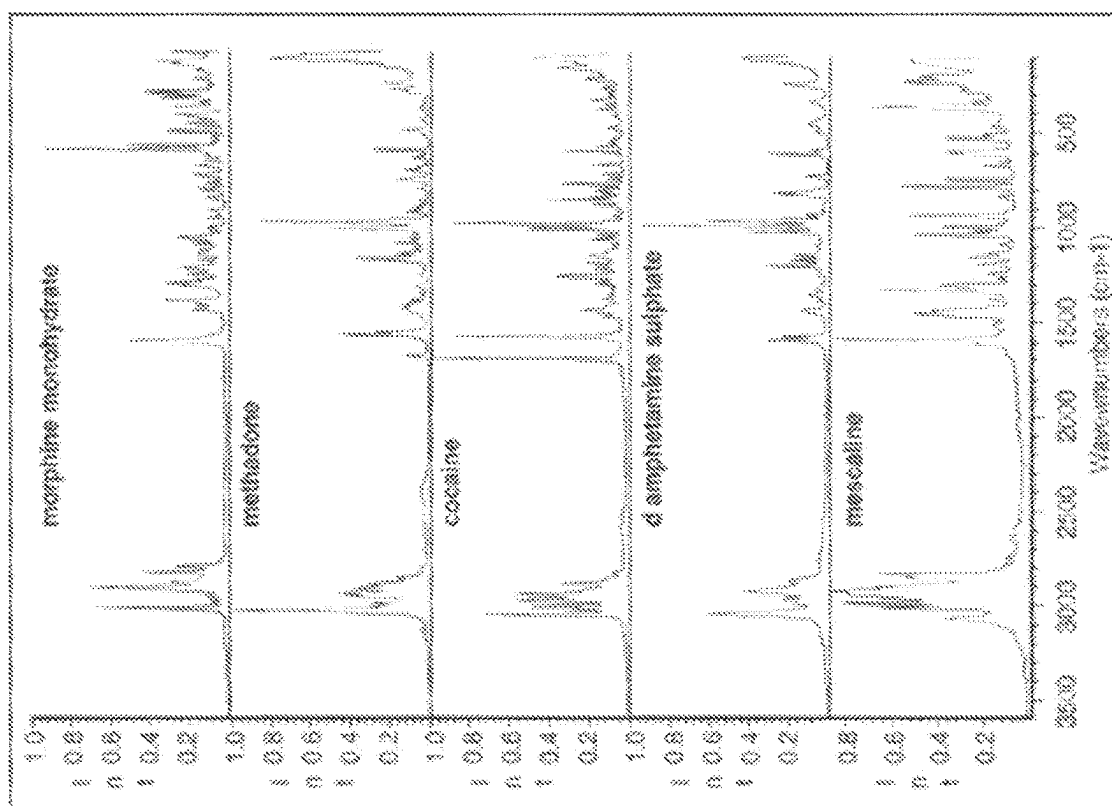


Figure 55

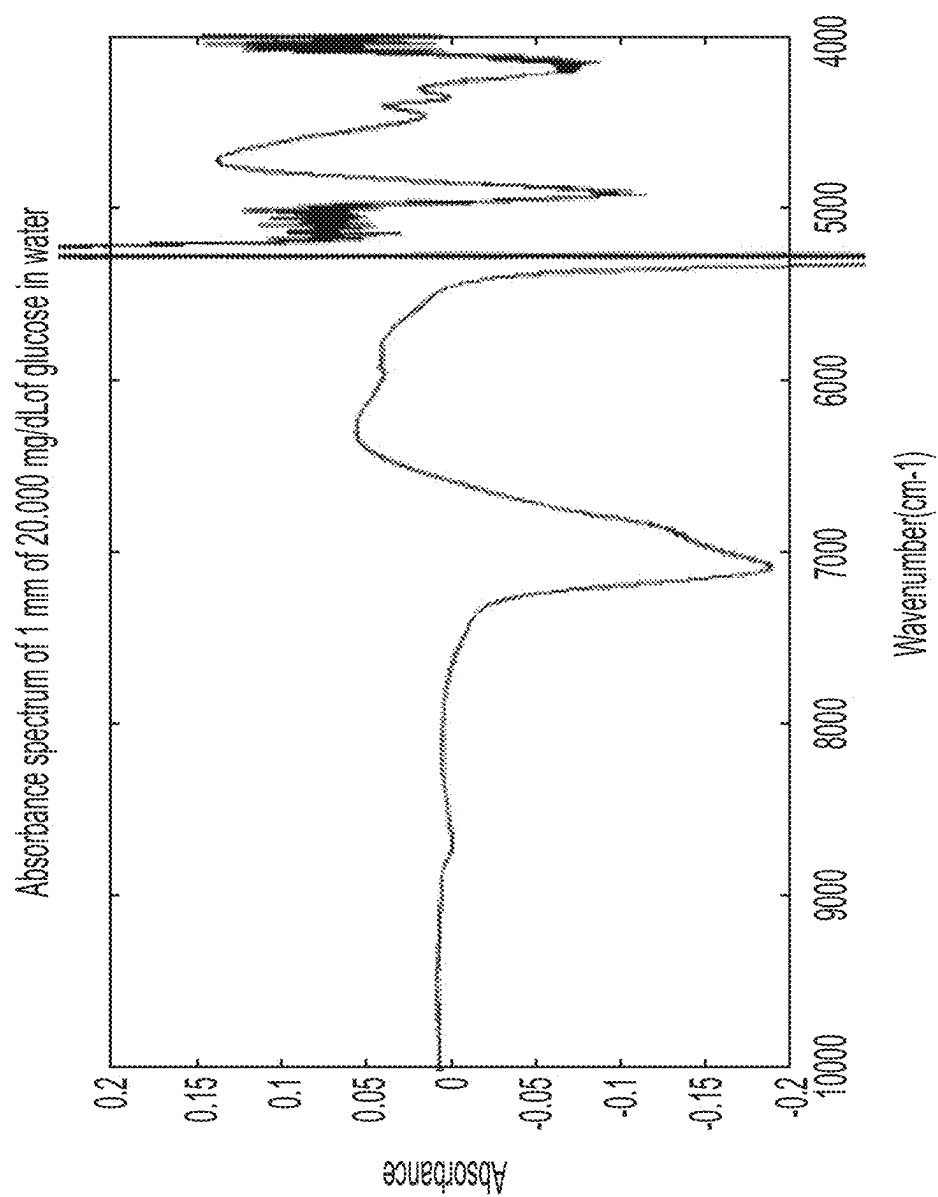
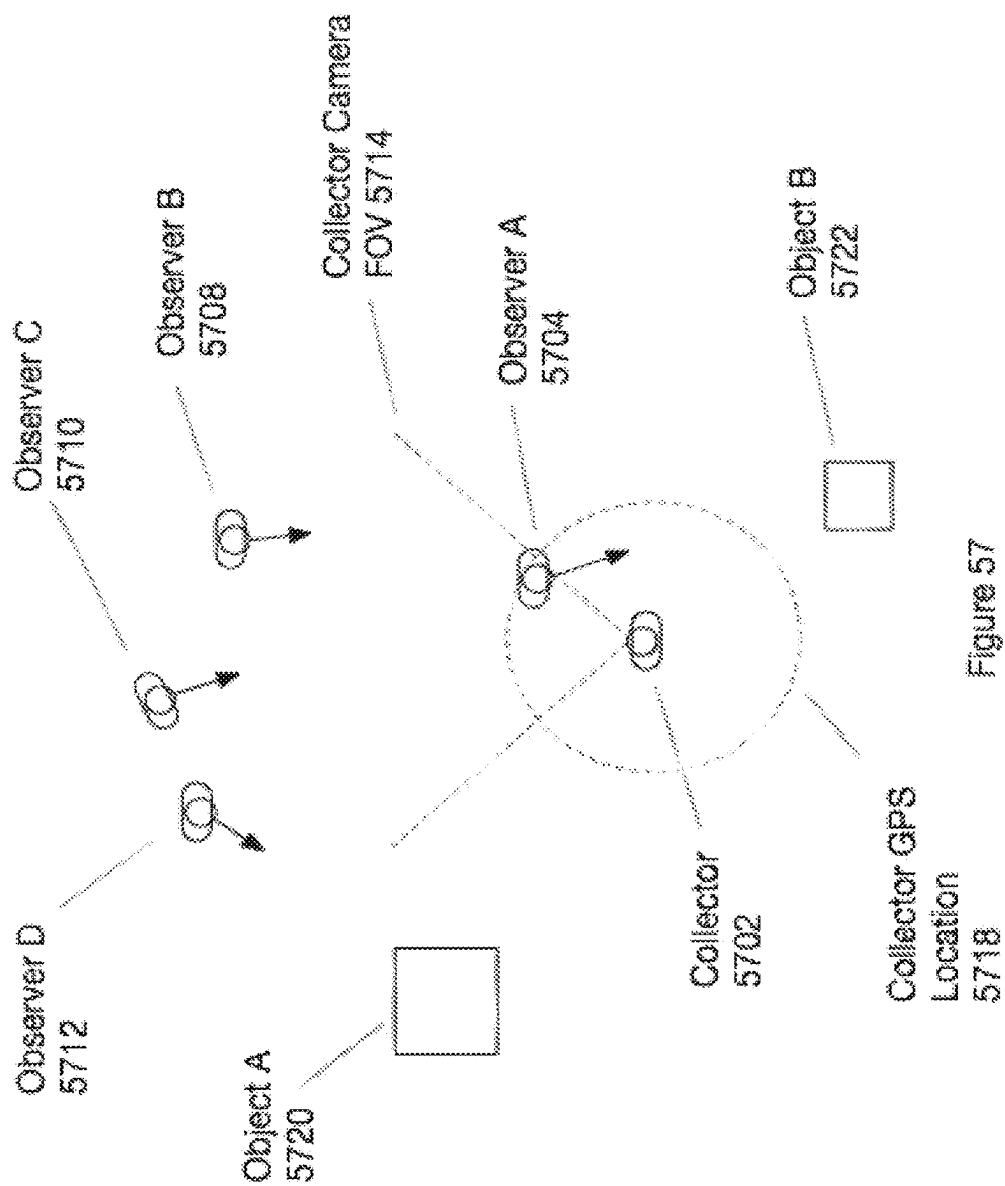
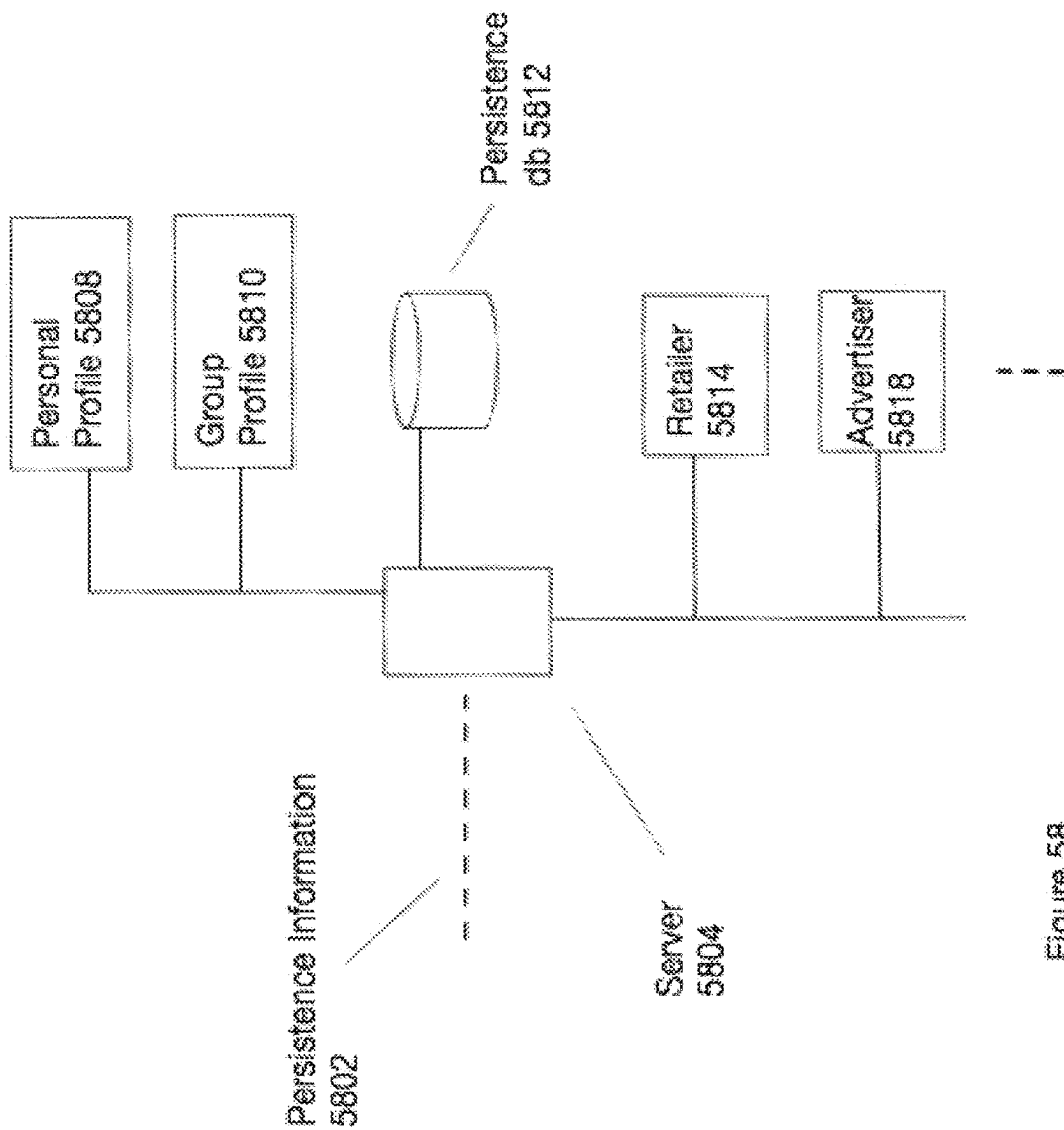
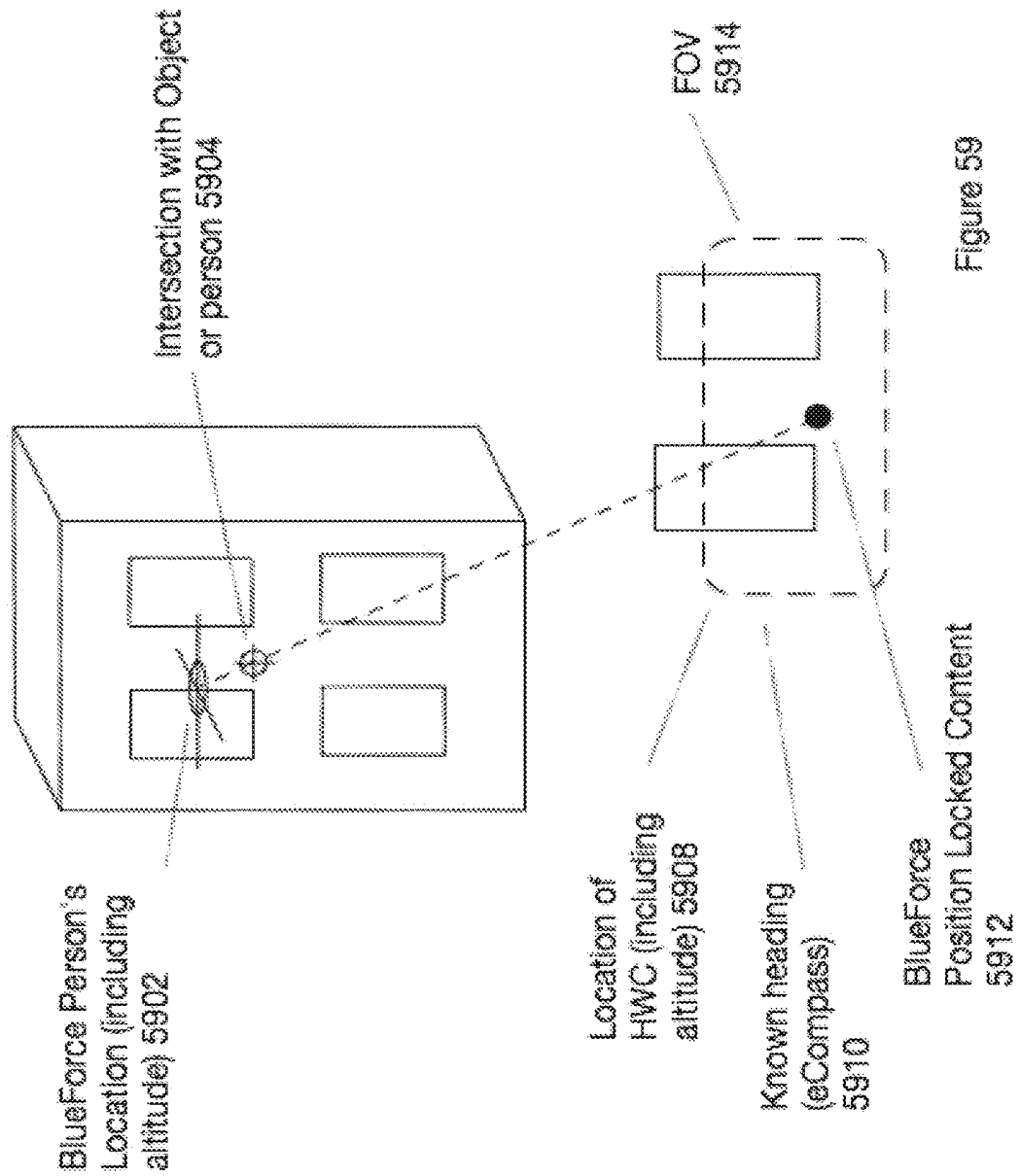


Figure 56







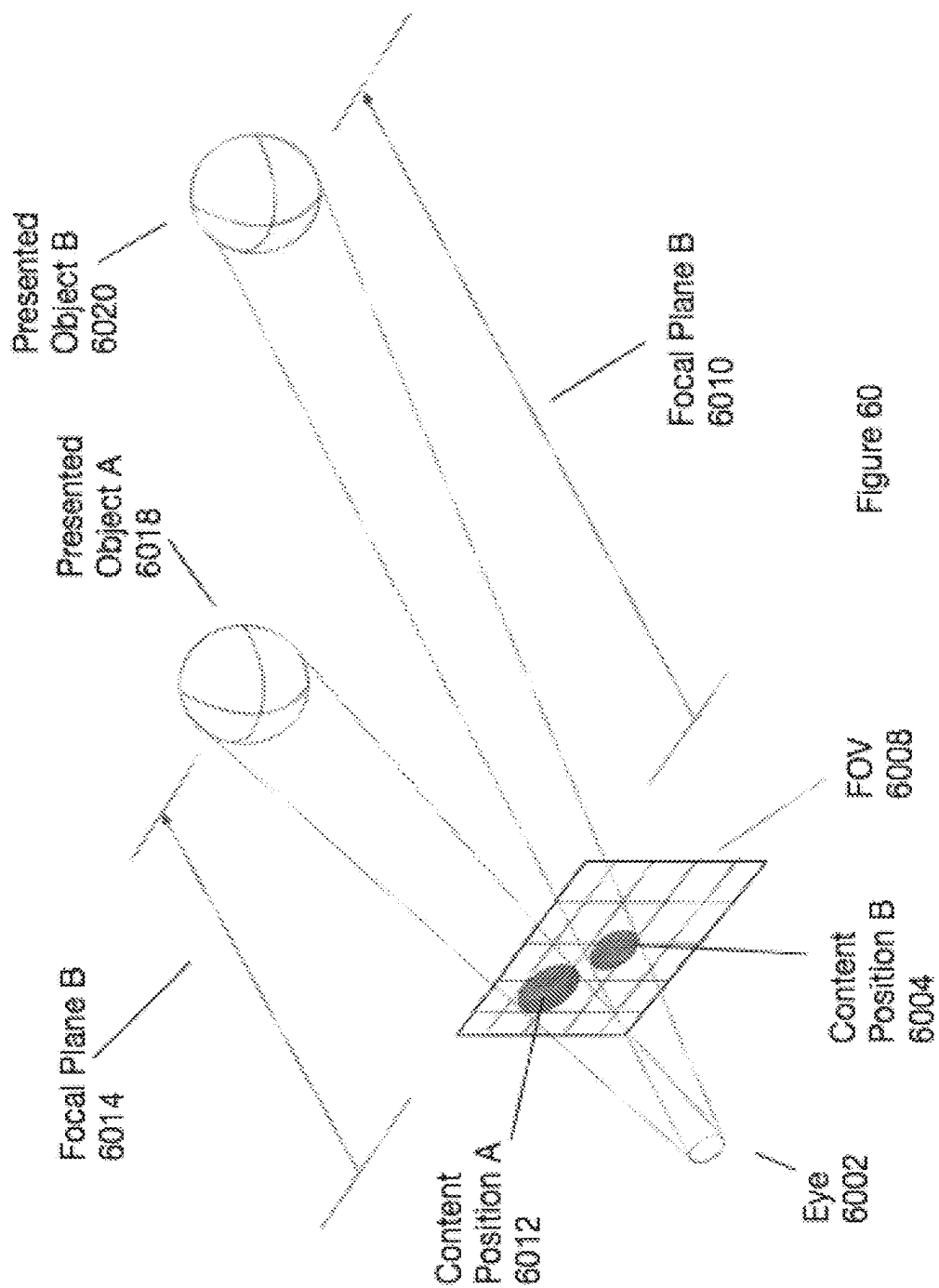


Figure 60

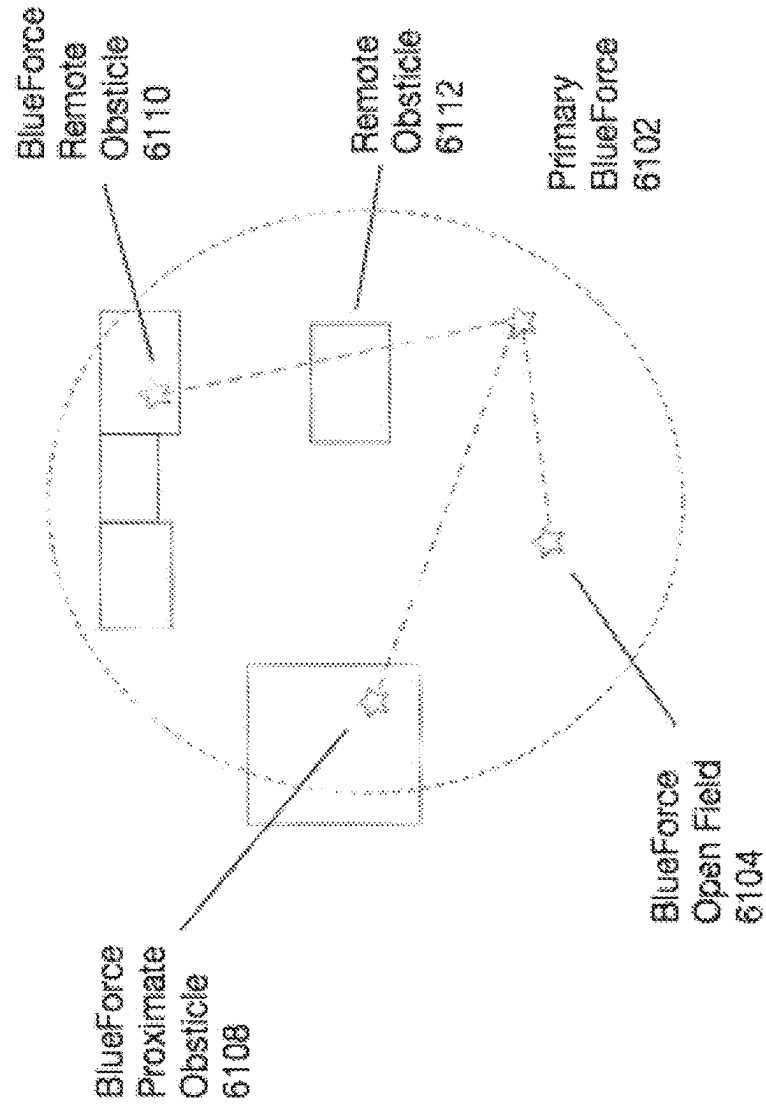
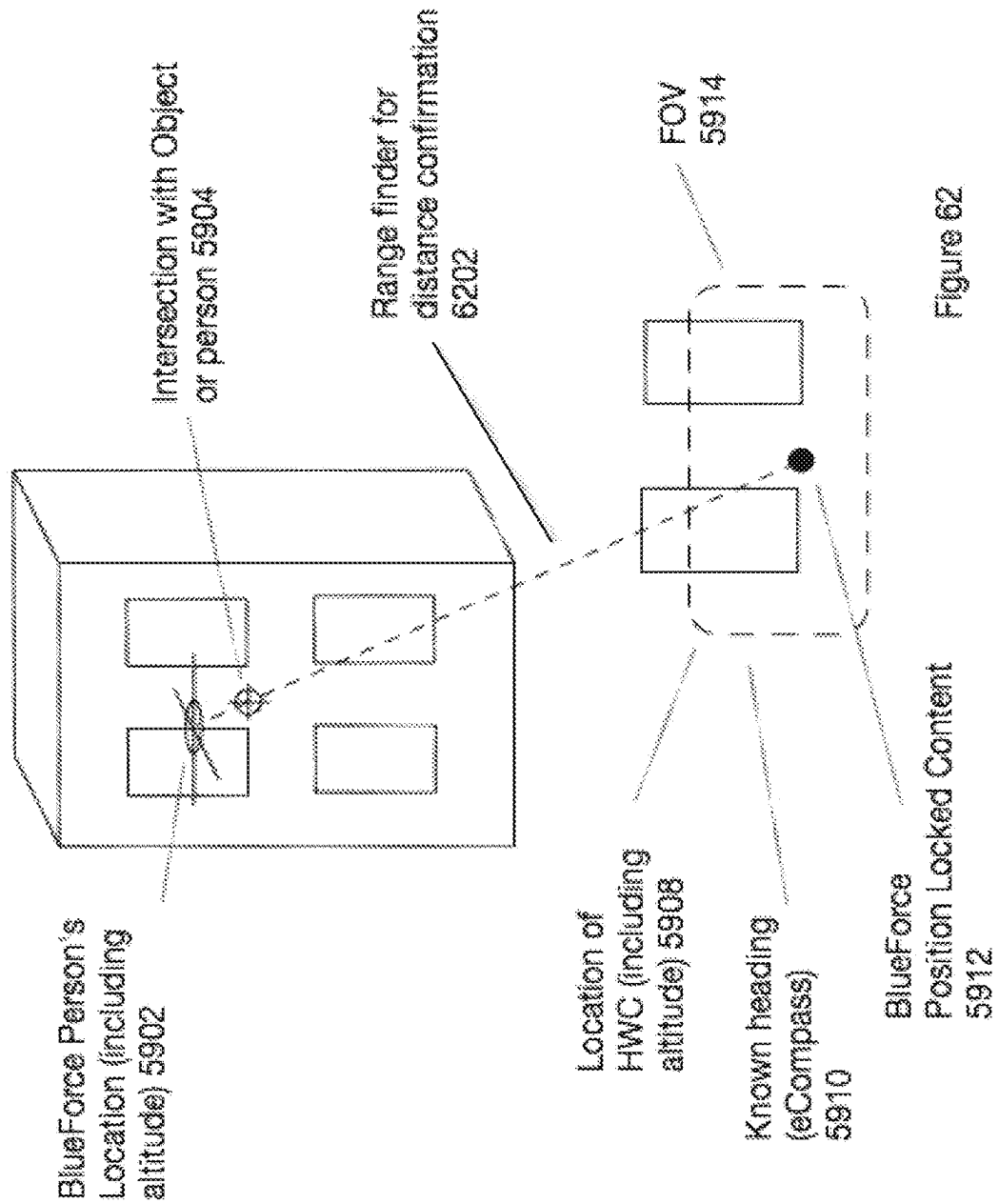


Figure 61



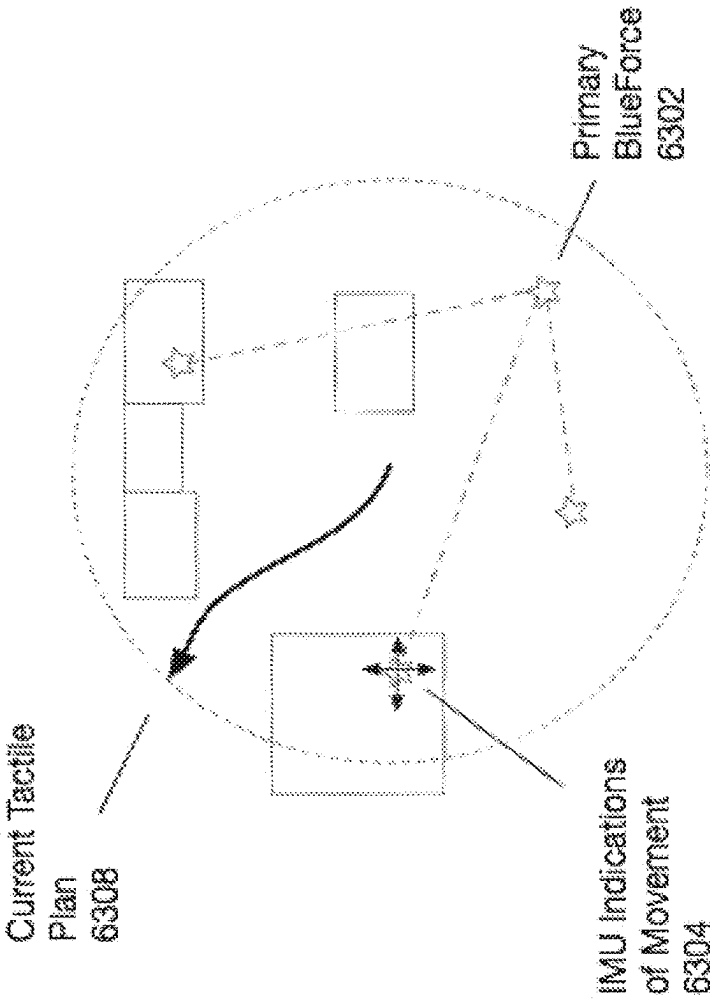


Figure 63

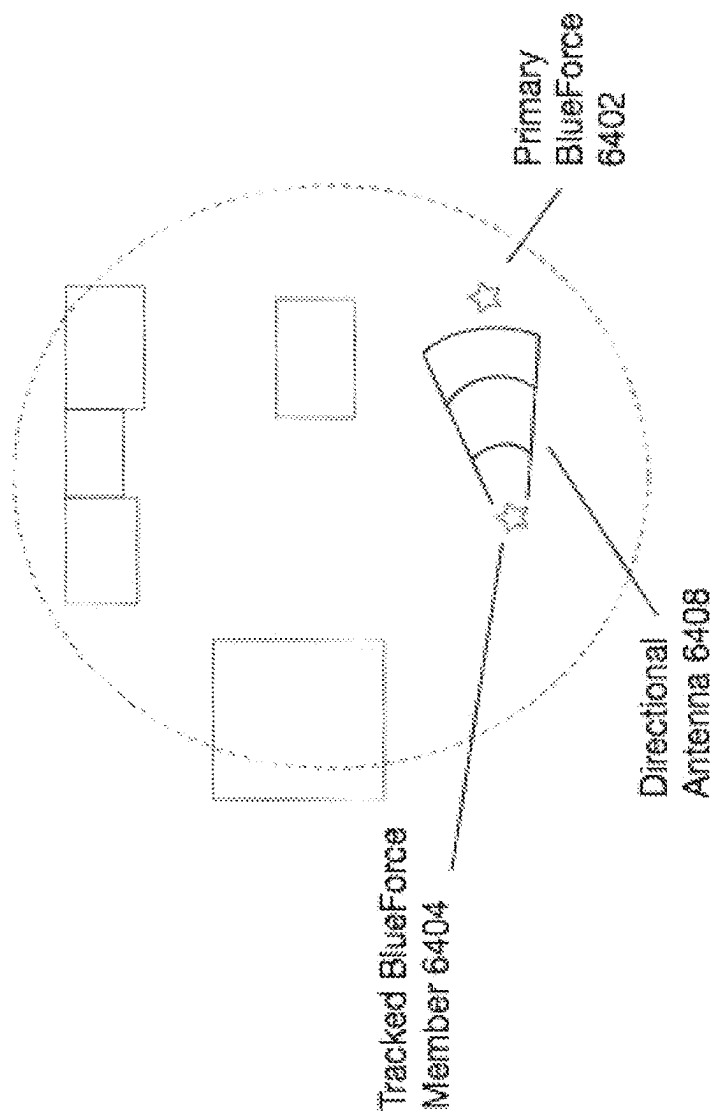


Figure 64

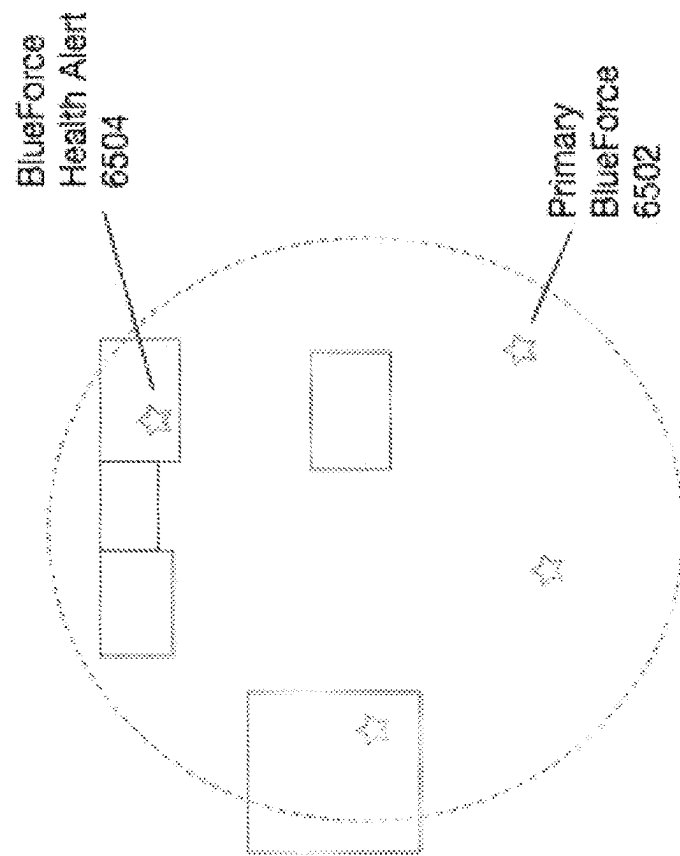


Figure 65

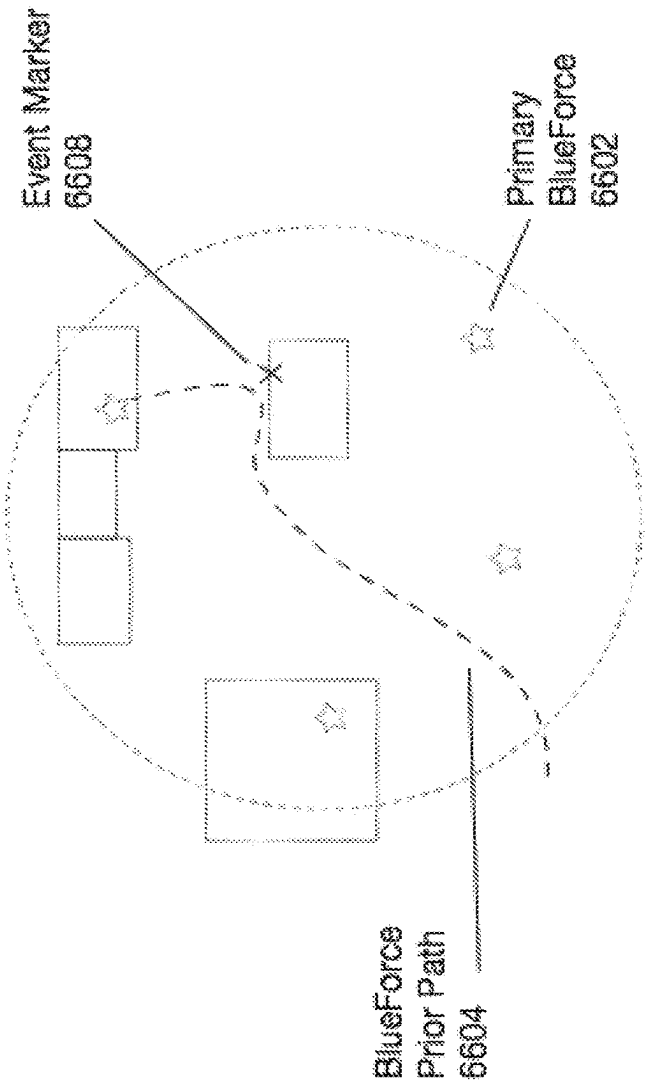


Figure 66

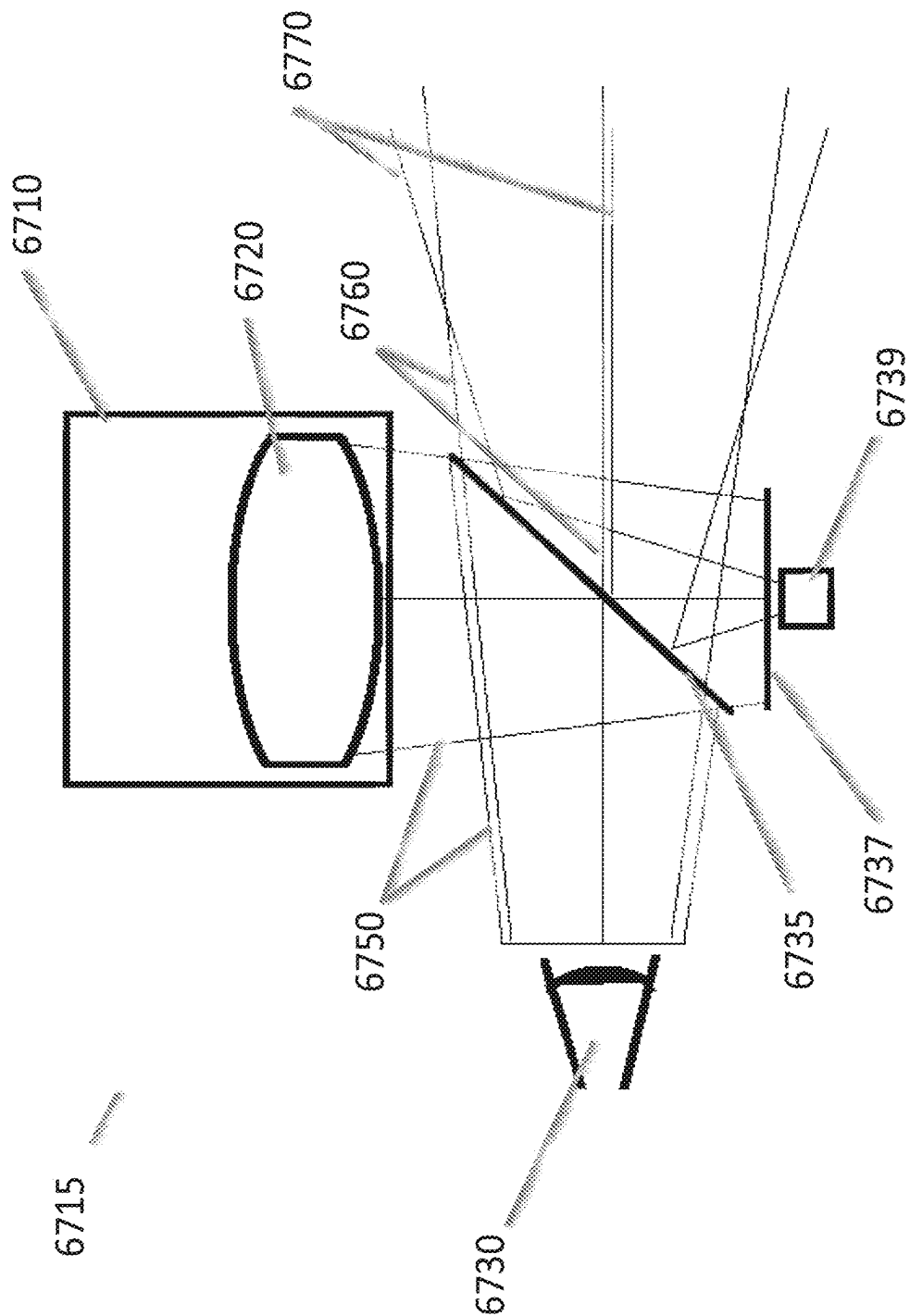


Figure 67

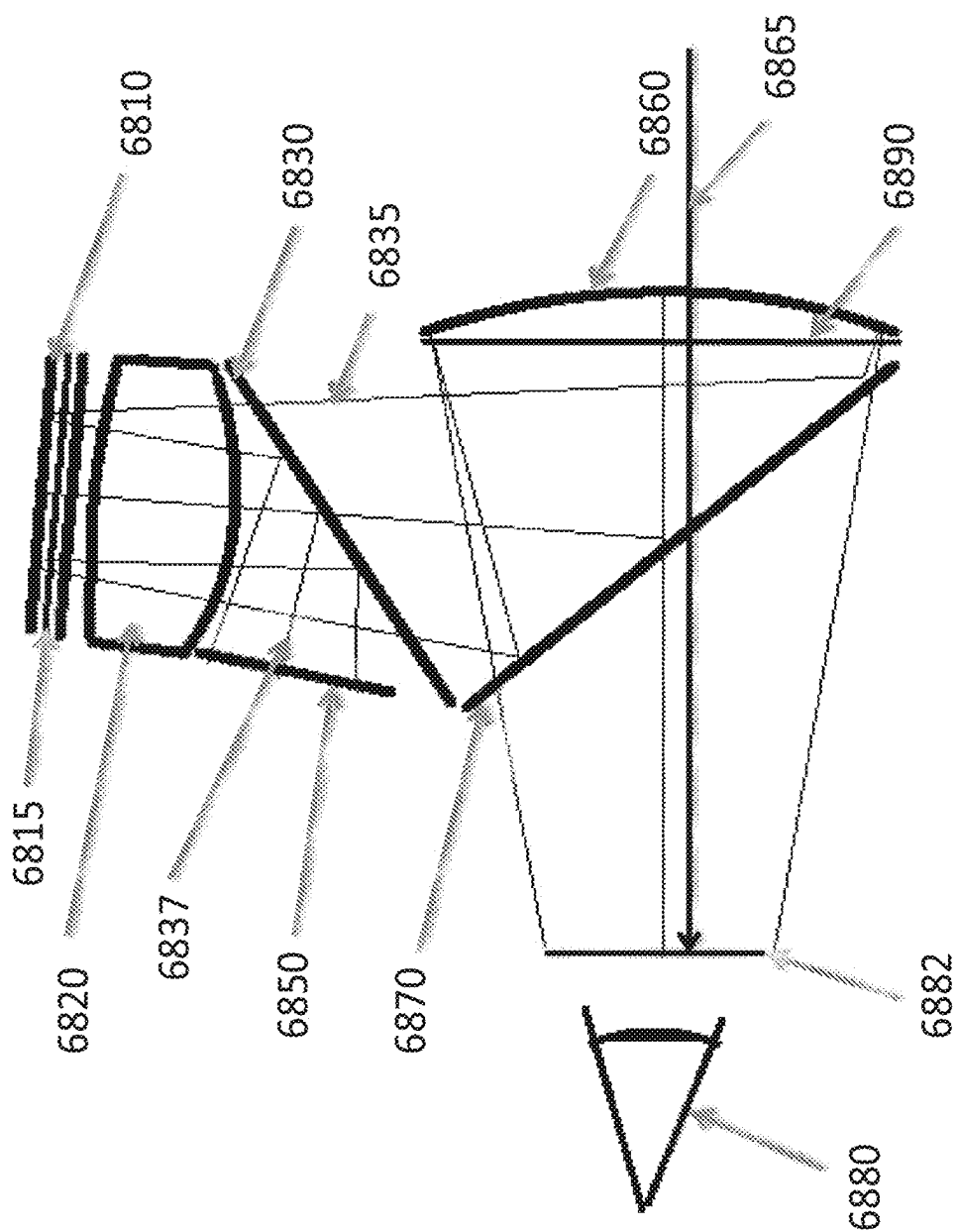


Figure 68

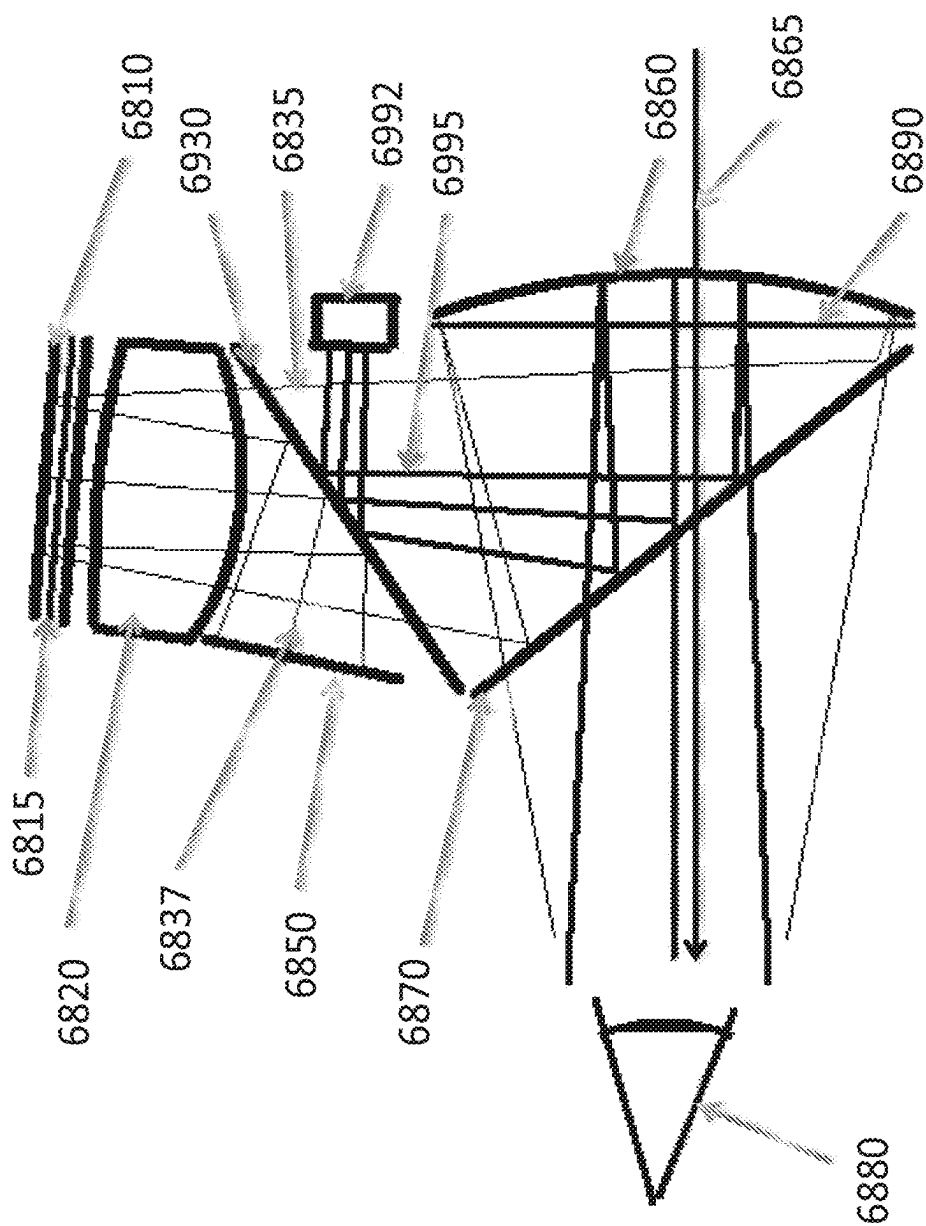


Figure 69

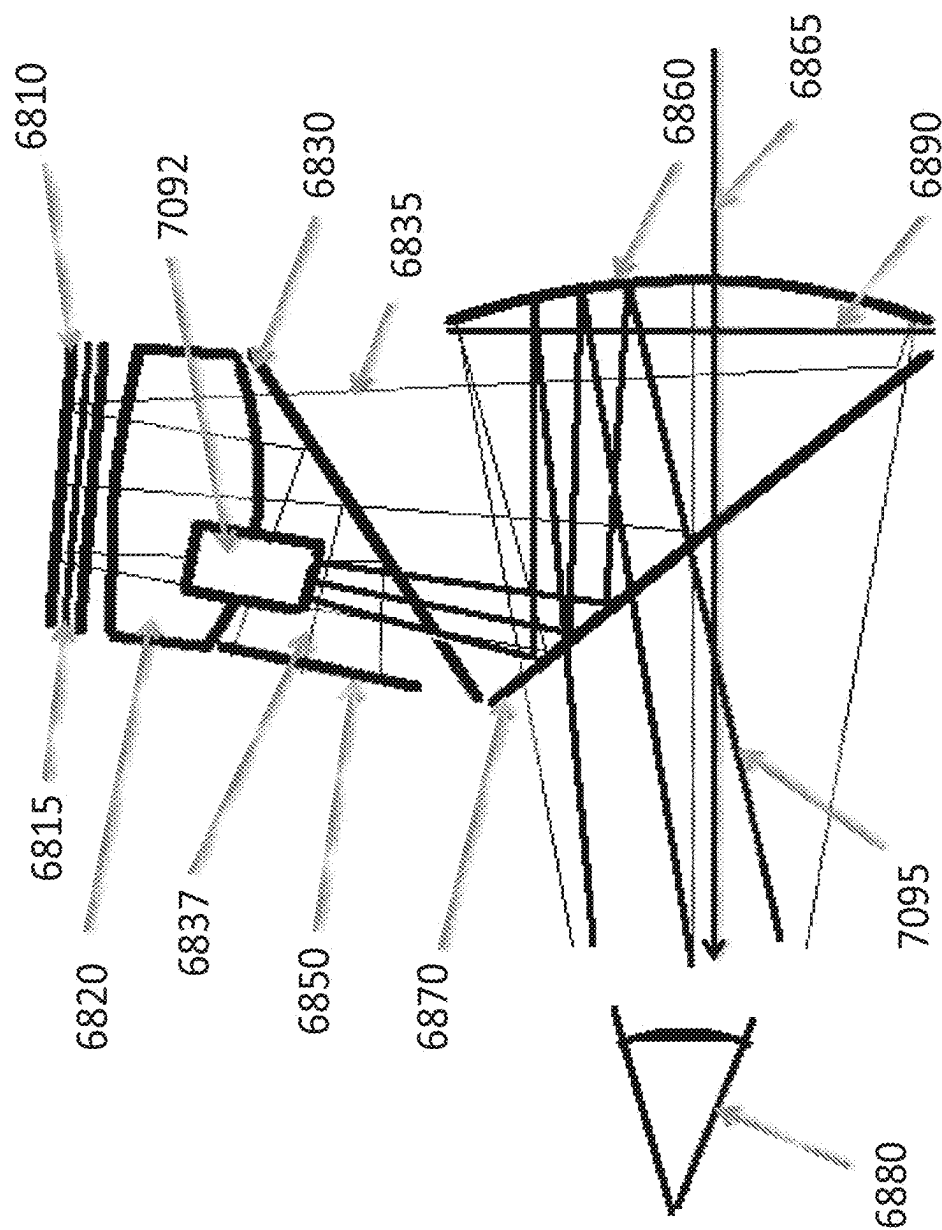


Figure 70

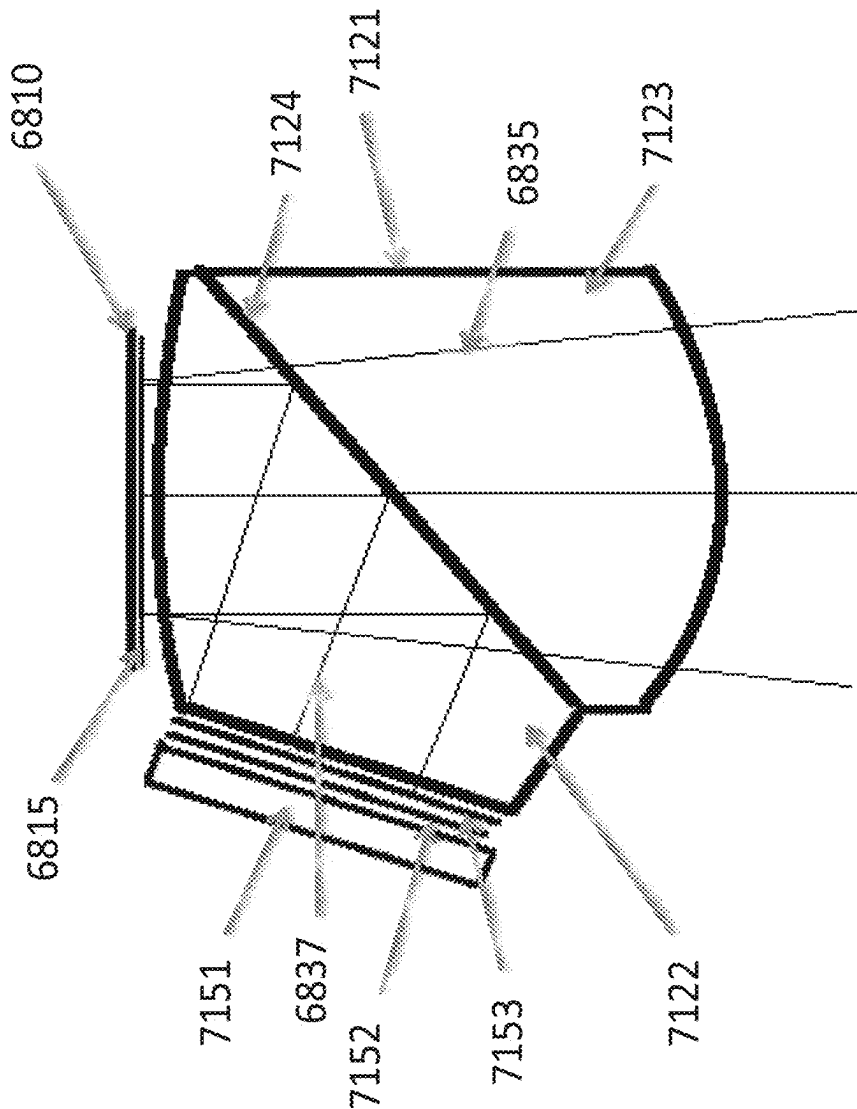


Figure 71

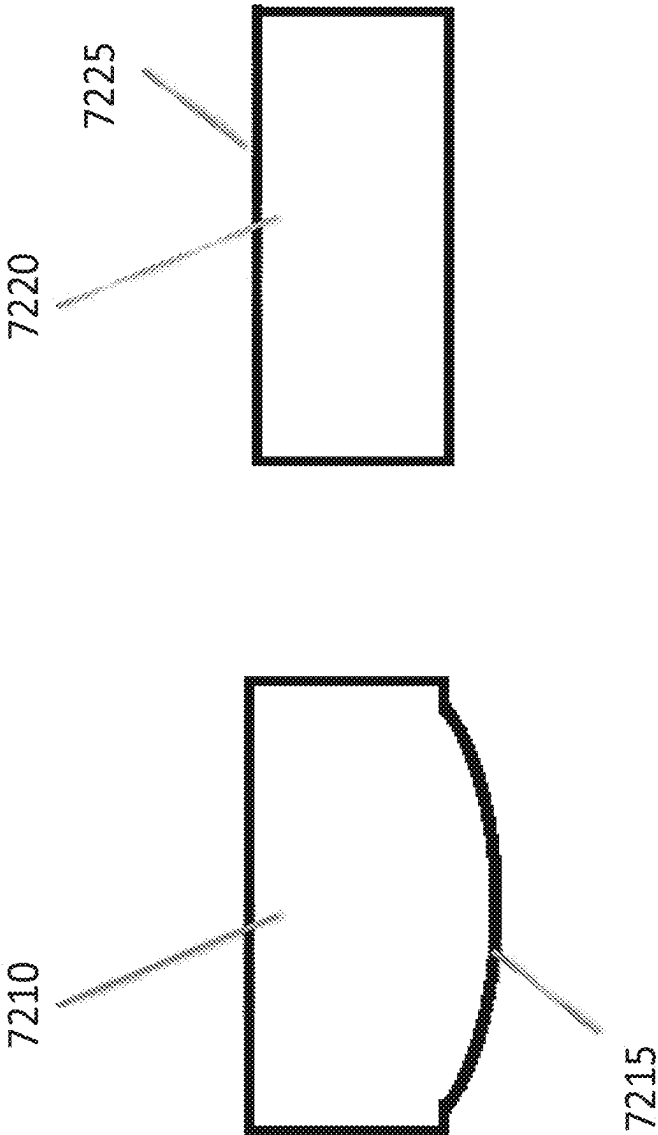


Figure 72

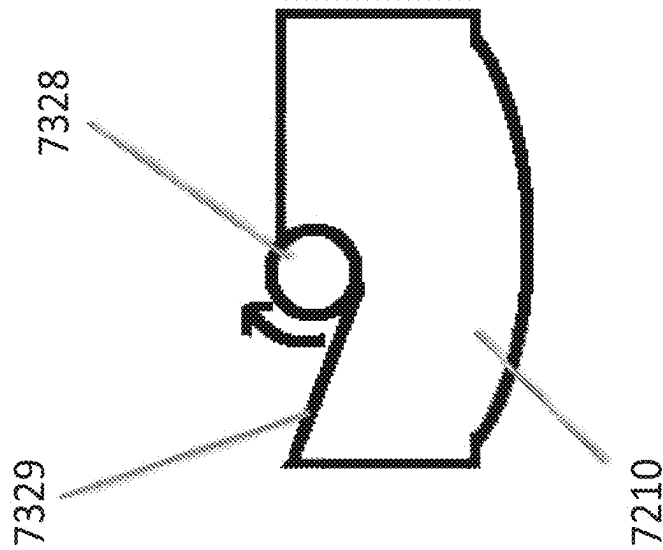


Figure 73

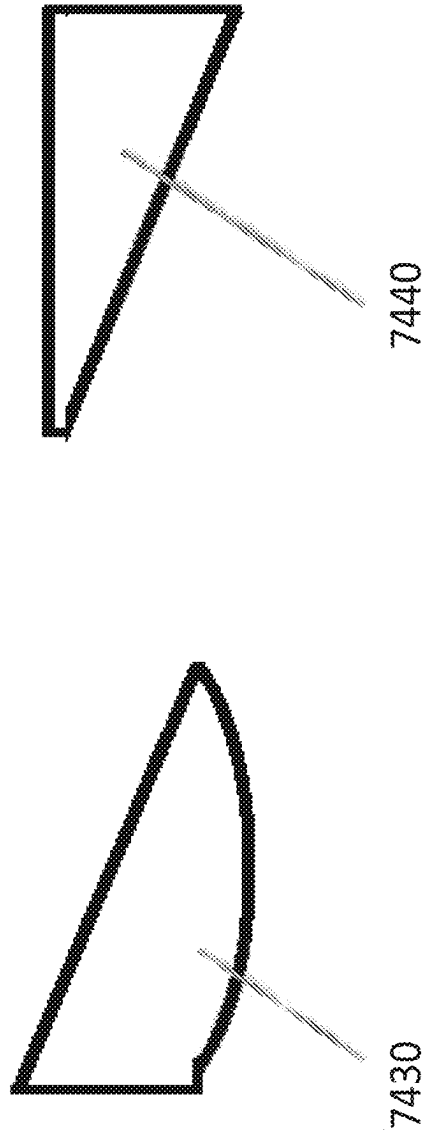


Figure 74

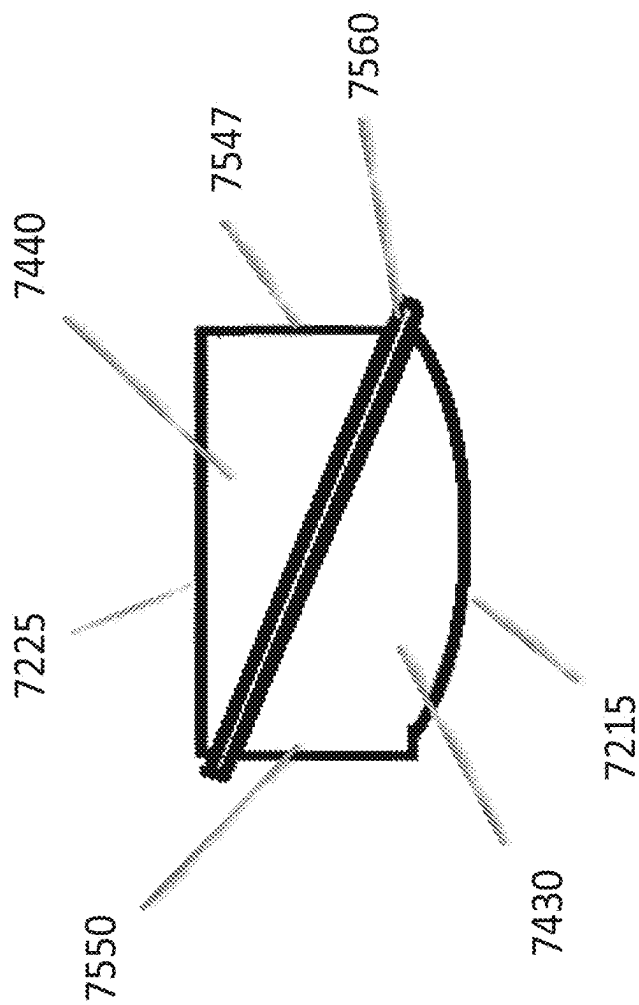


Figure 75

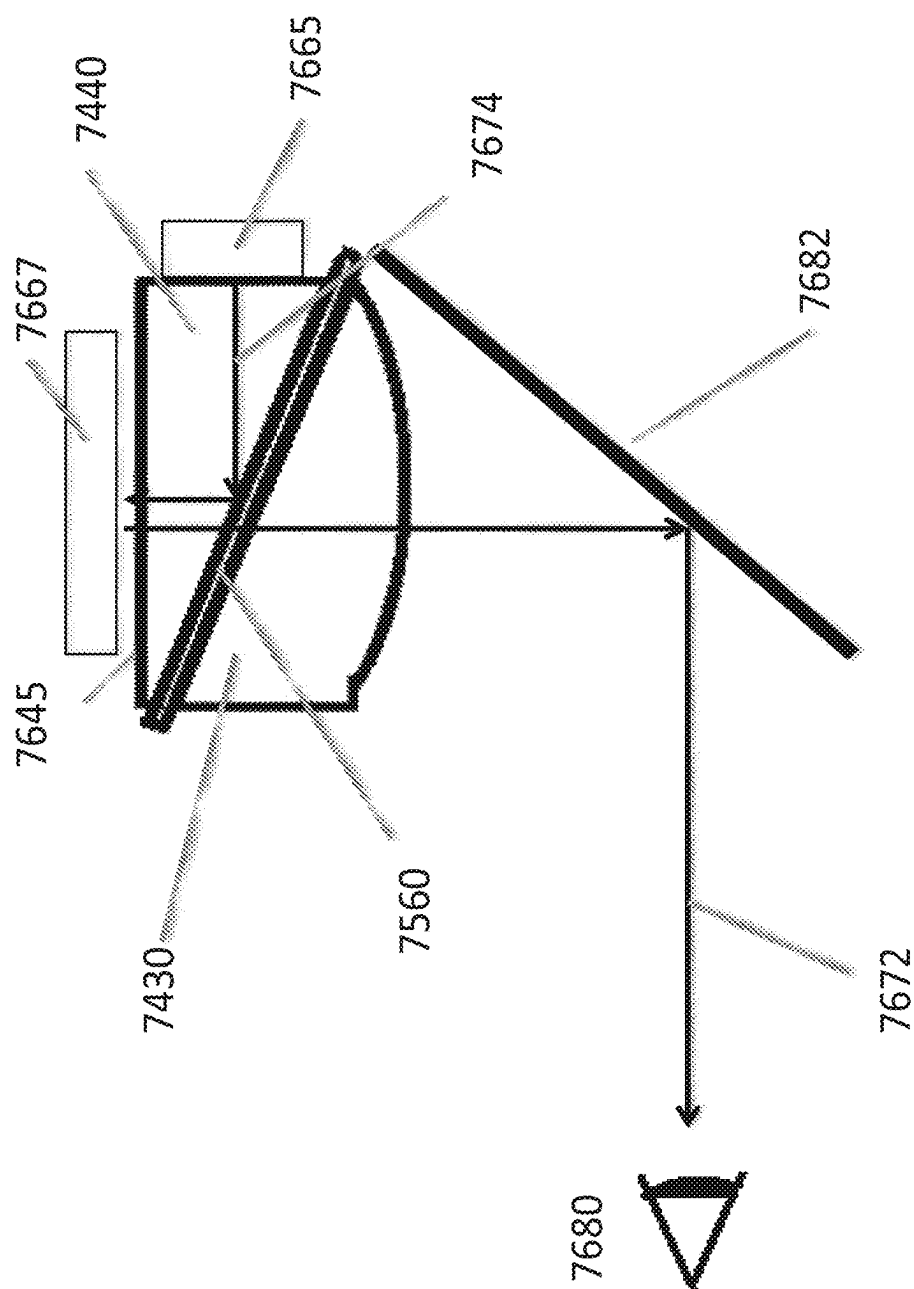


Figure 76

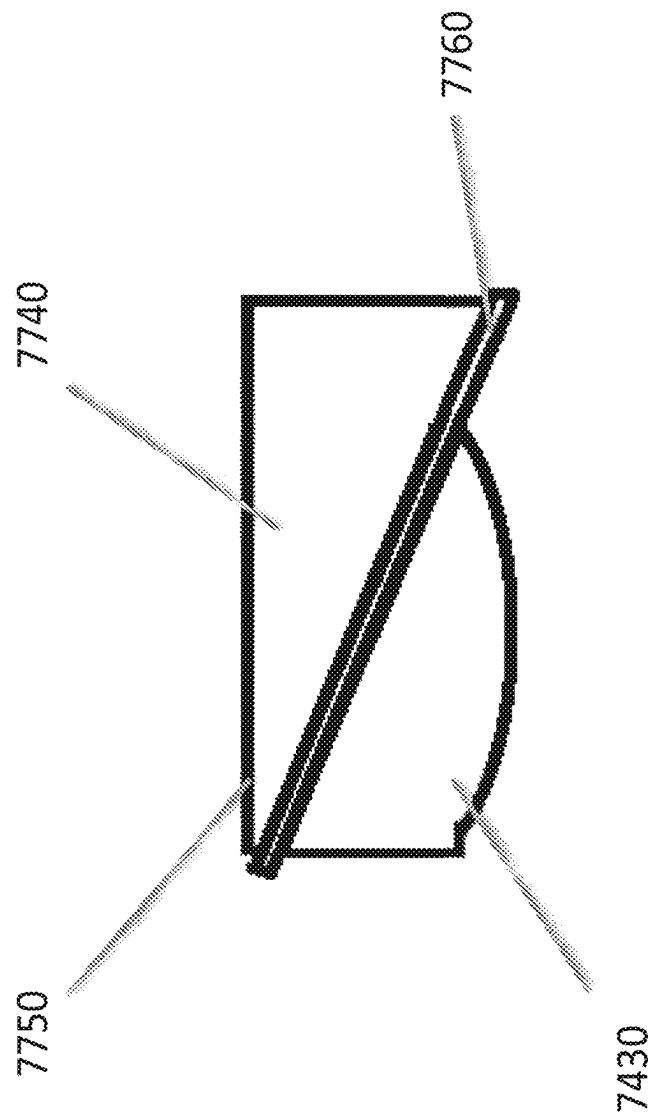


Figure 77

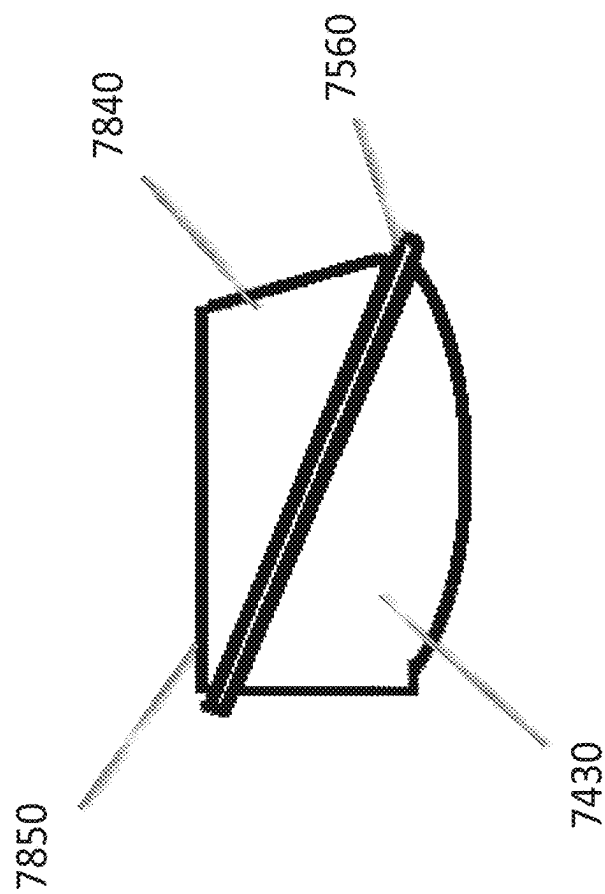


Figure 78

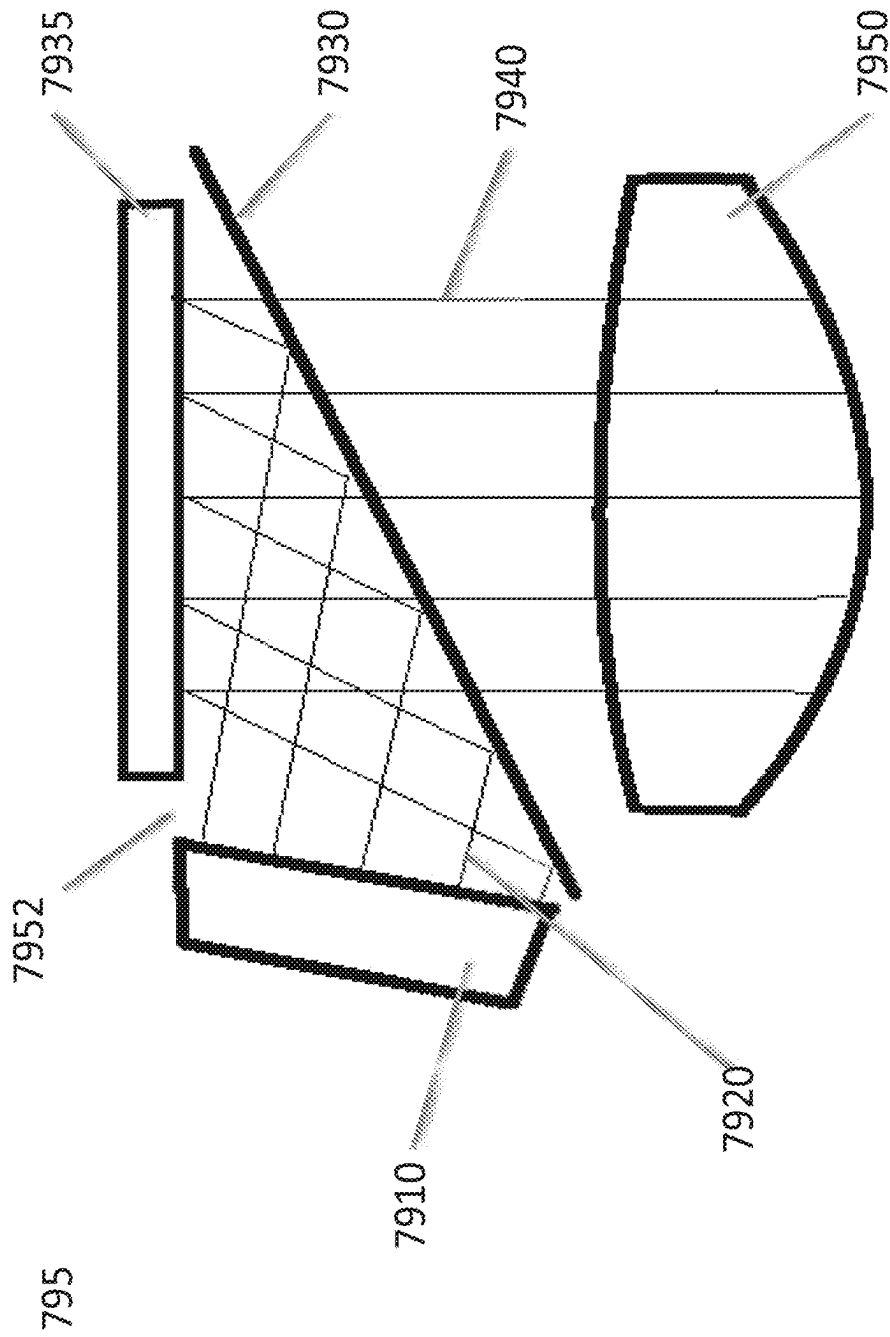


Figure 79

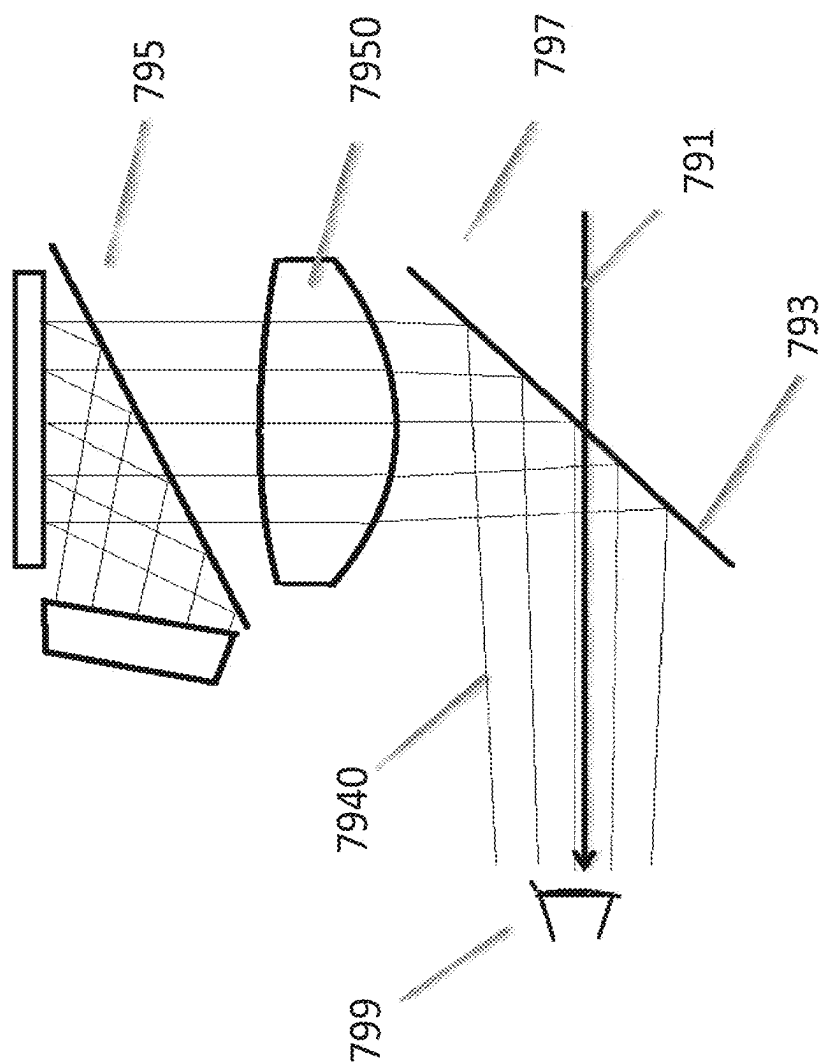


Figure 79a

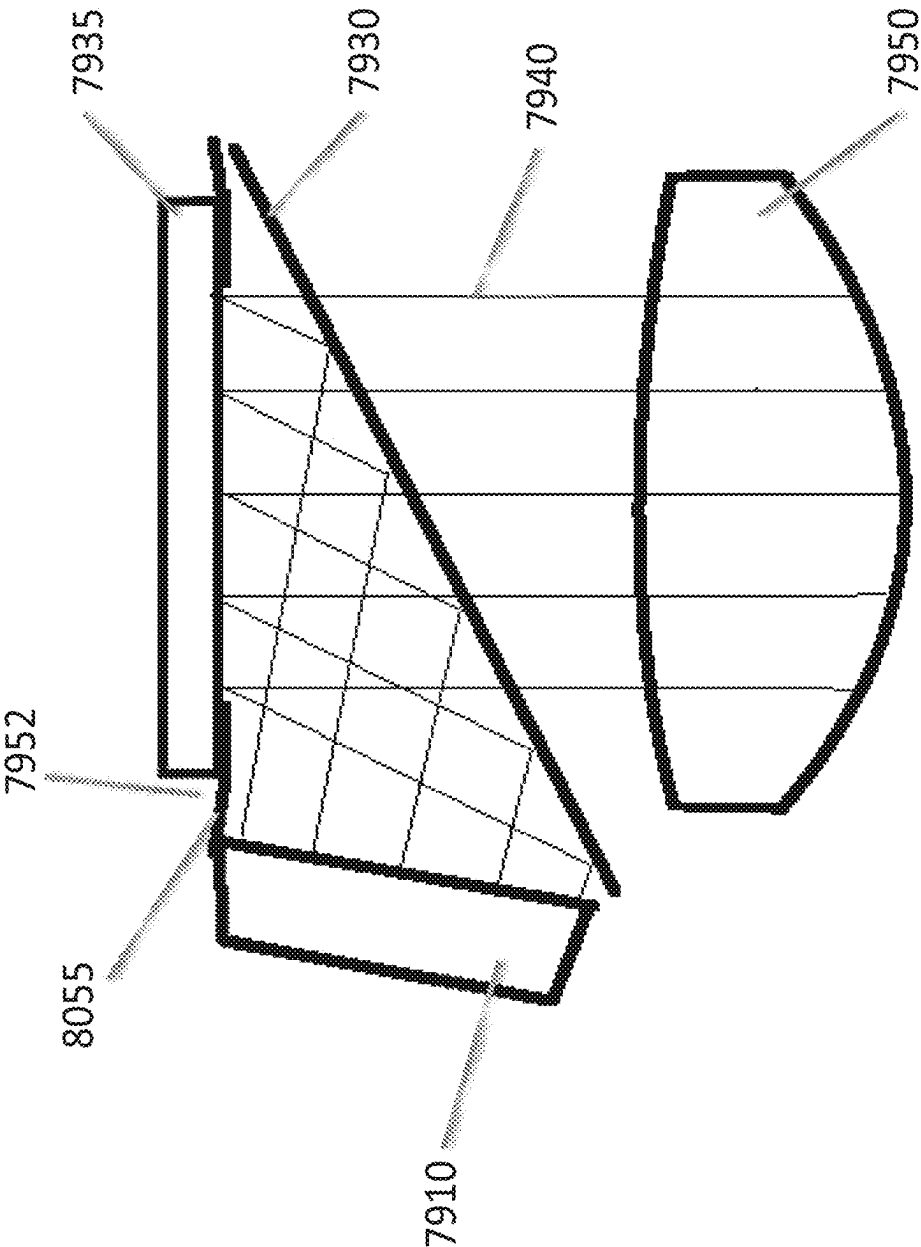
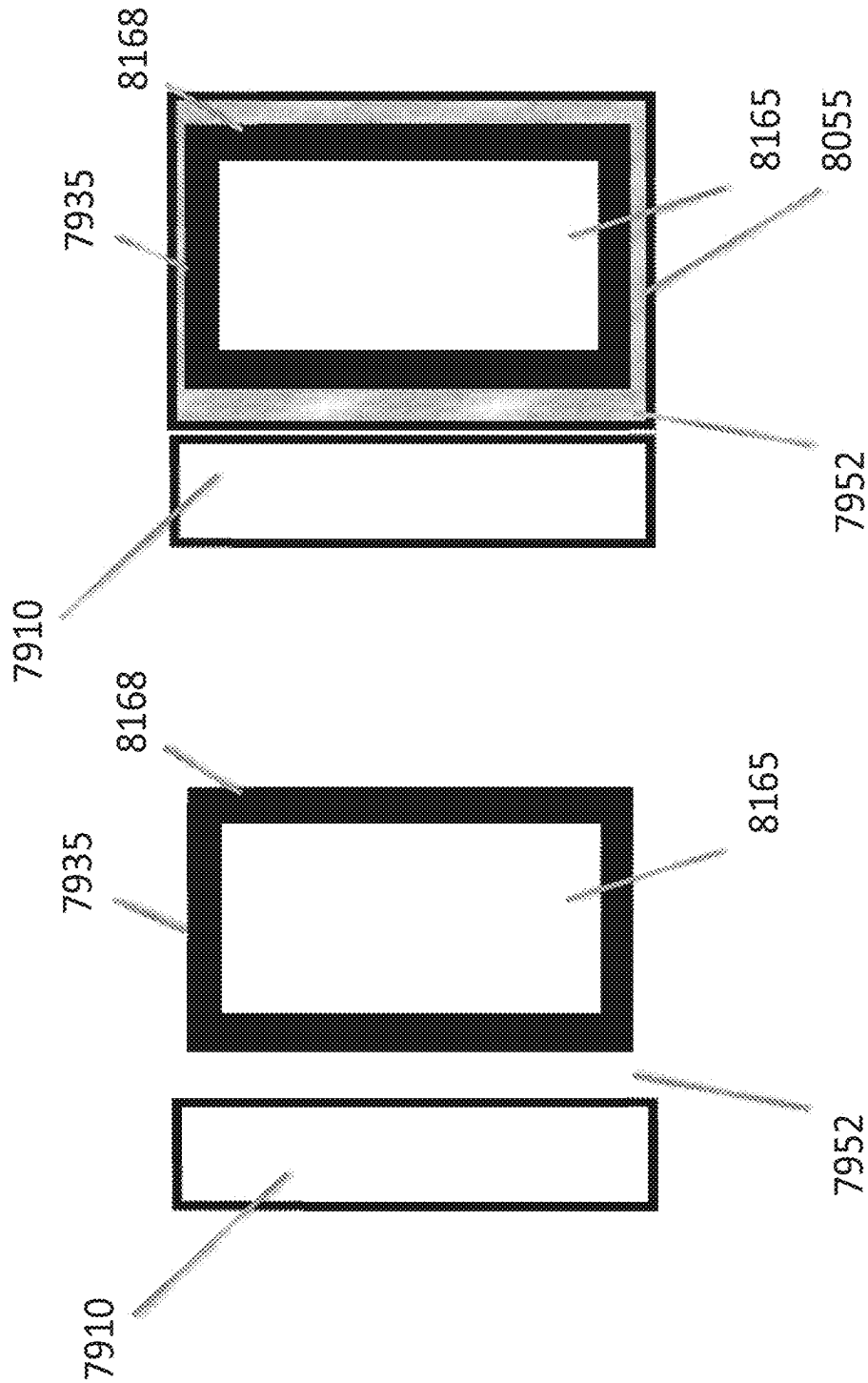


Figure 80



Future

1800-2100

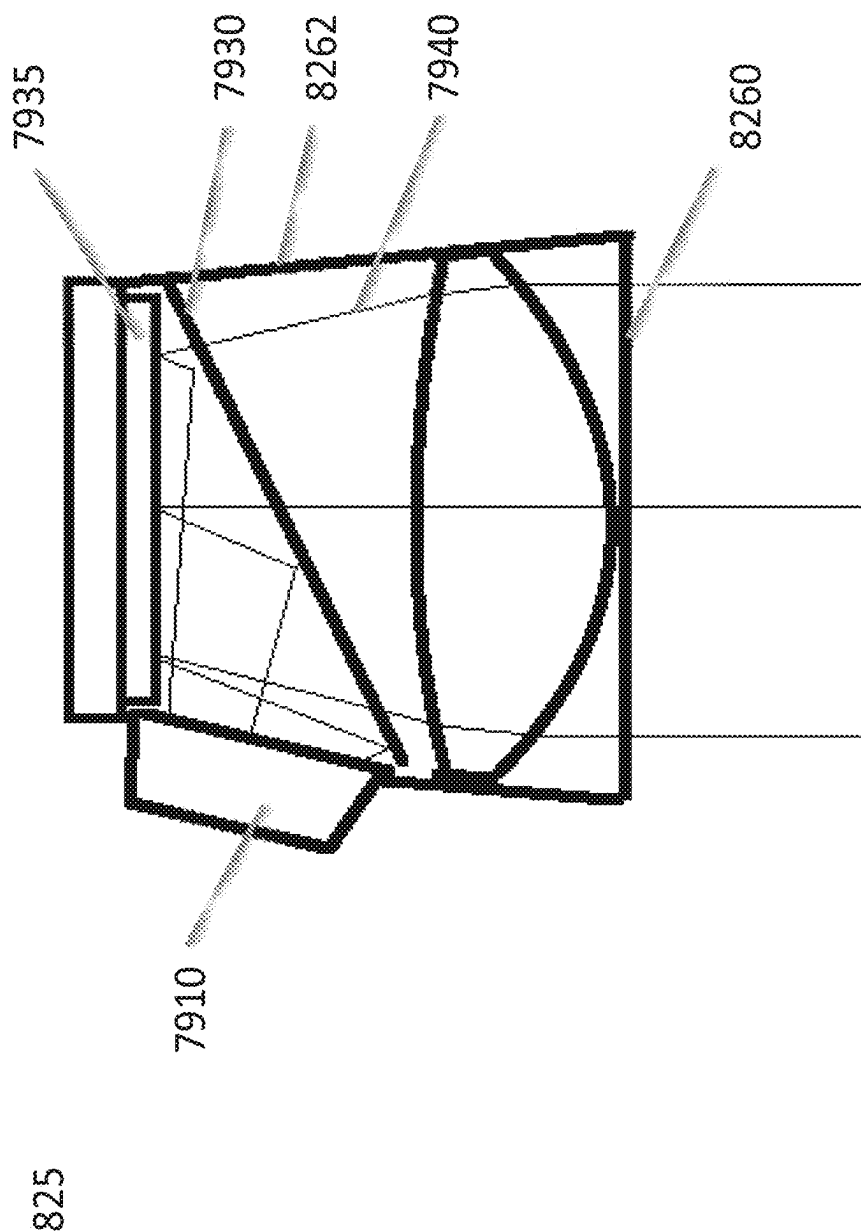


Figure 82

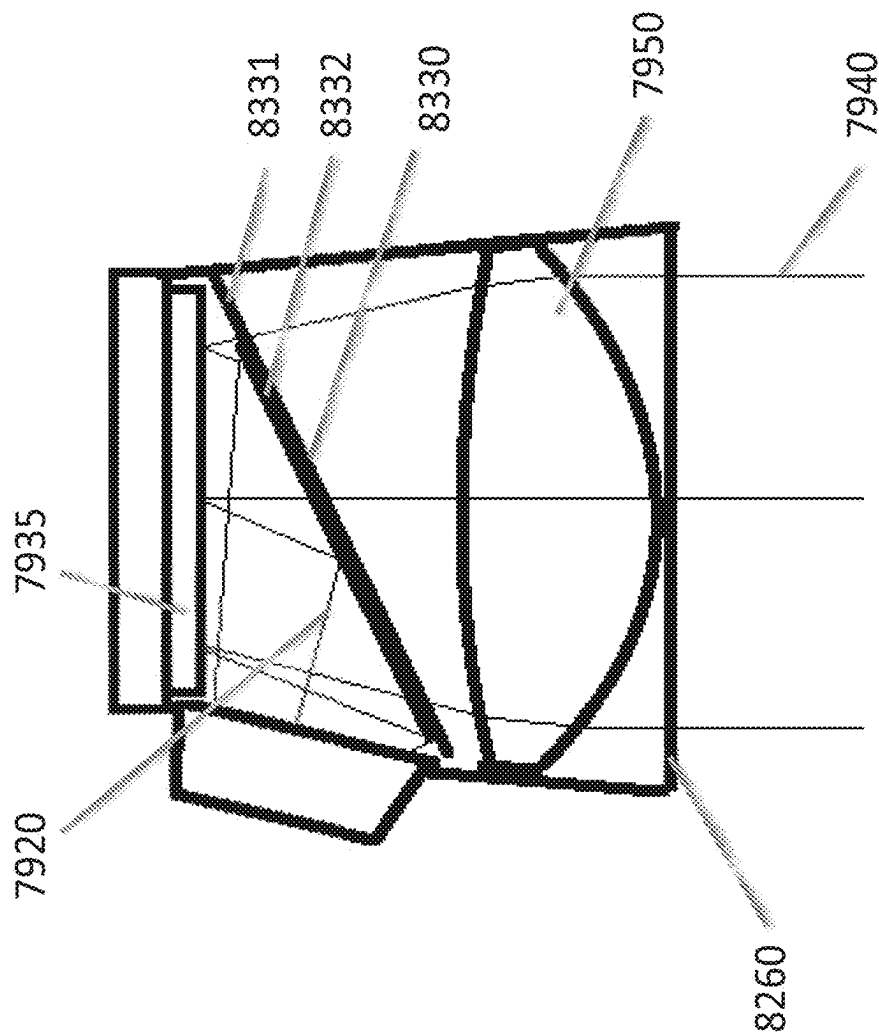


Figure 83

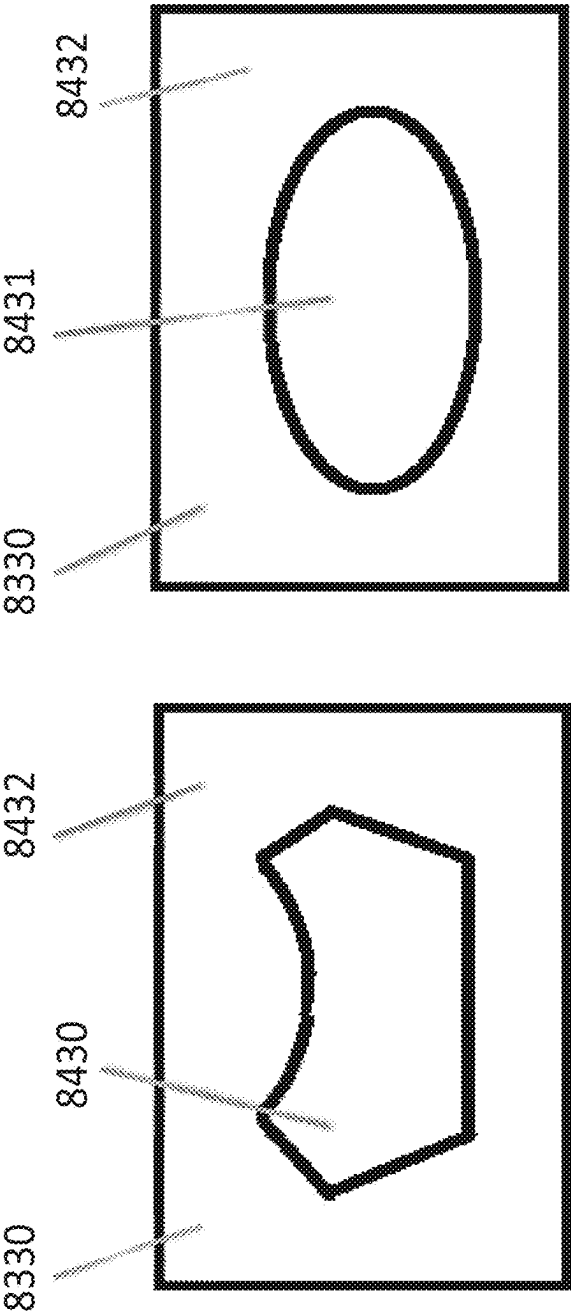


Figure 84a

Figure 84b

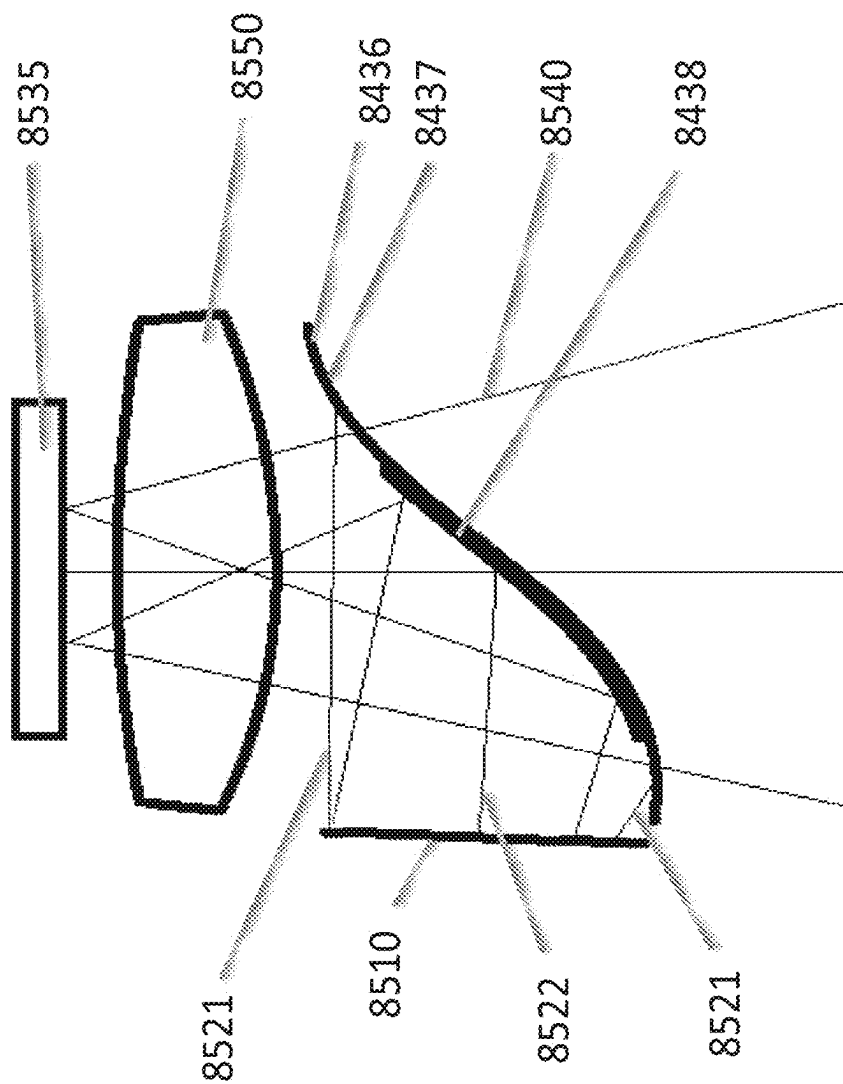


Figure 84c

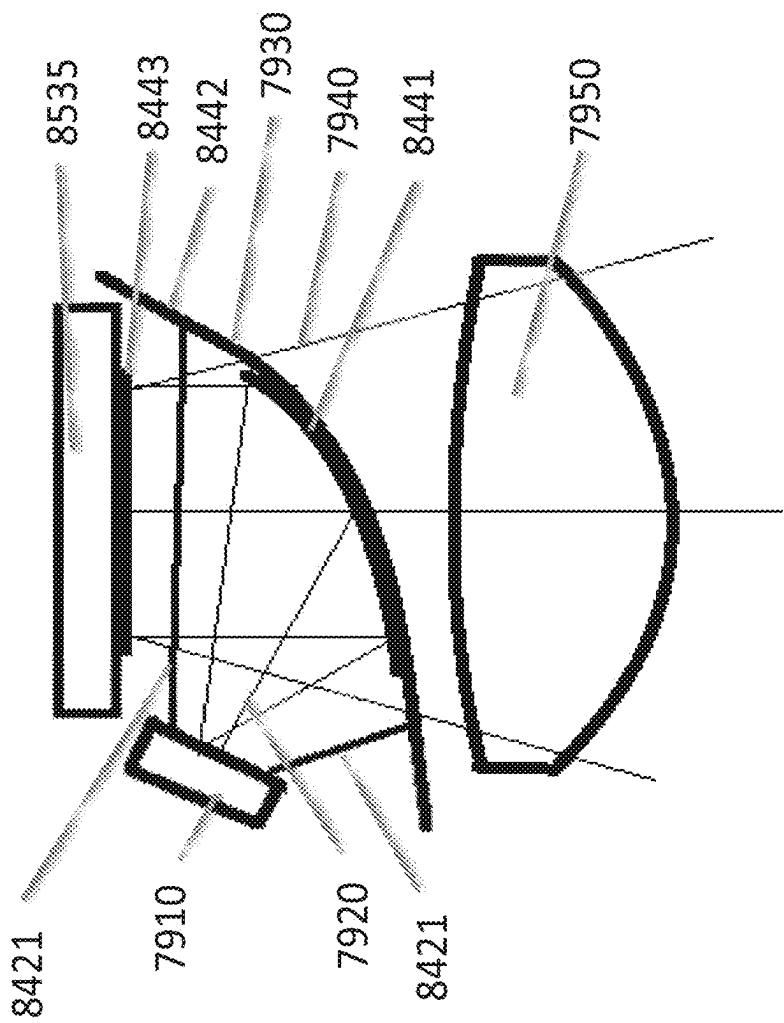


Figure 84d

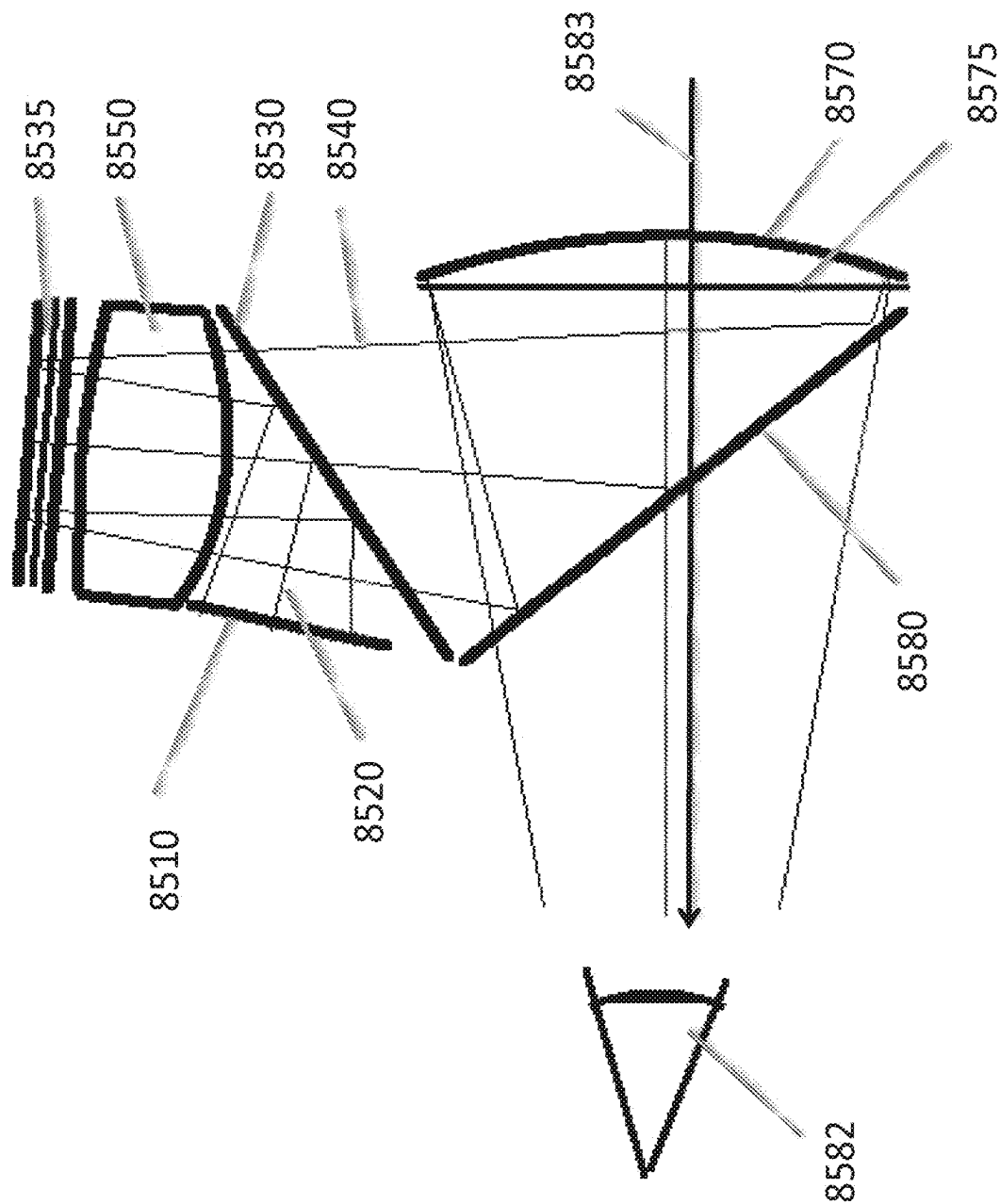


Figure 85

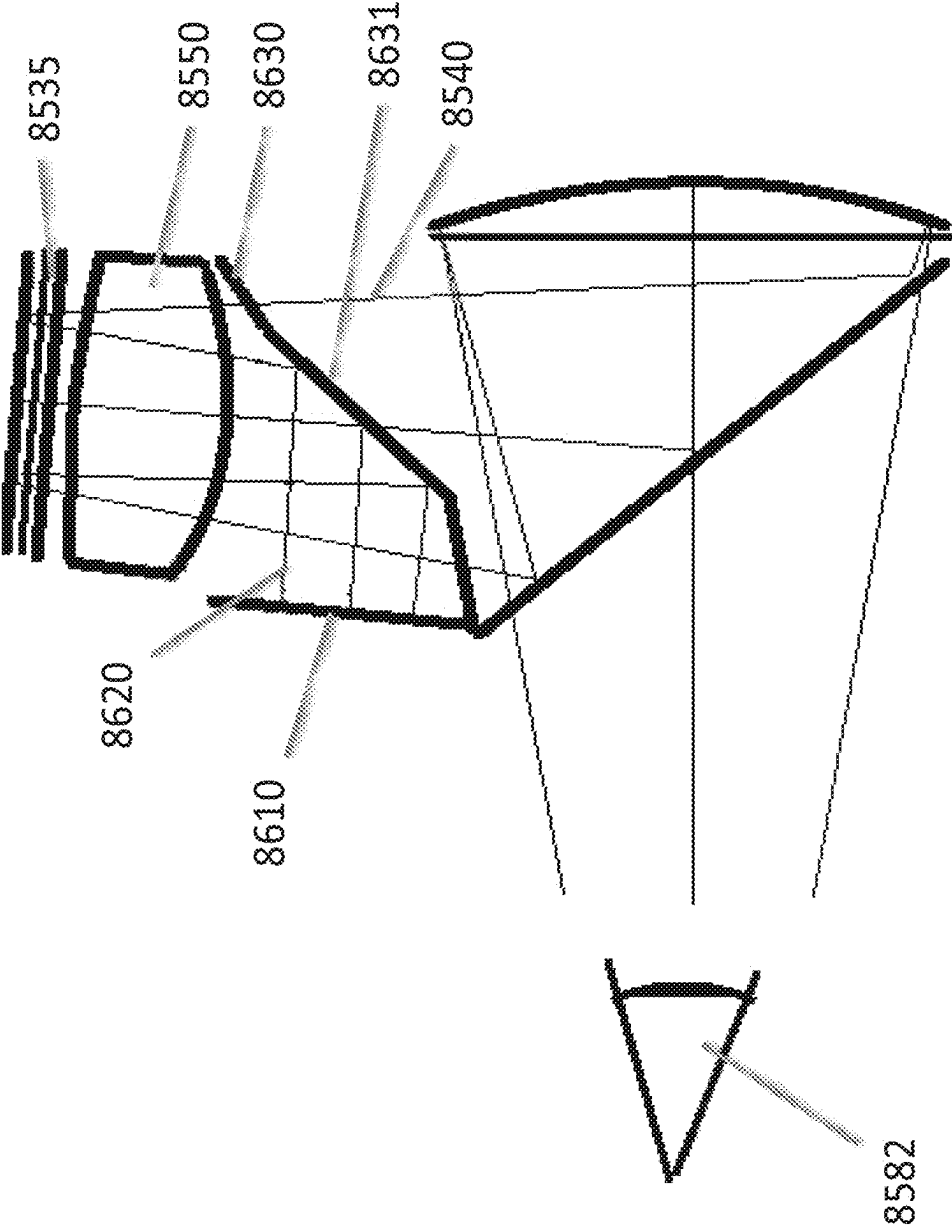


Figure 86

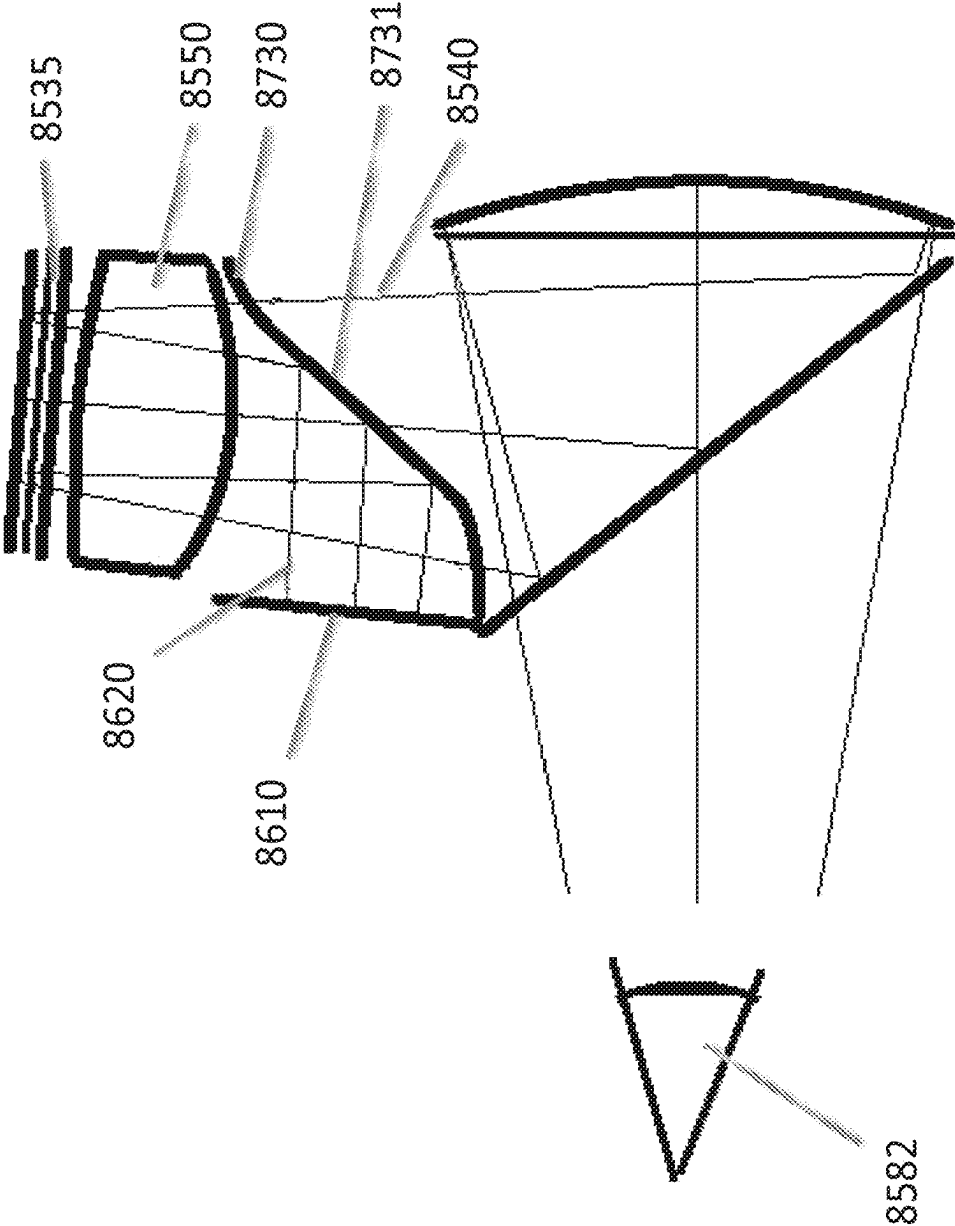


Figure 87

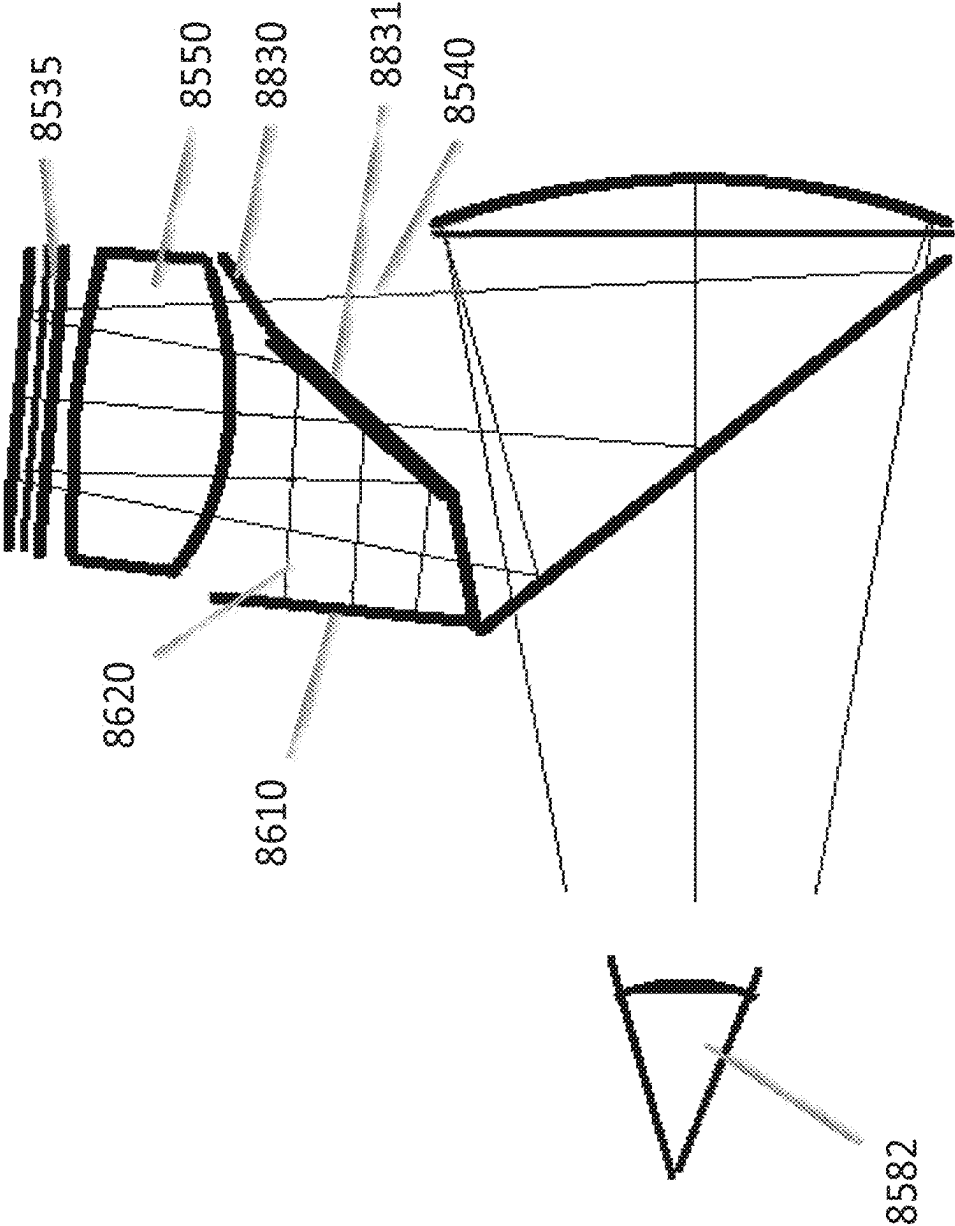


Figure 88

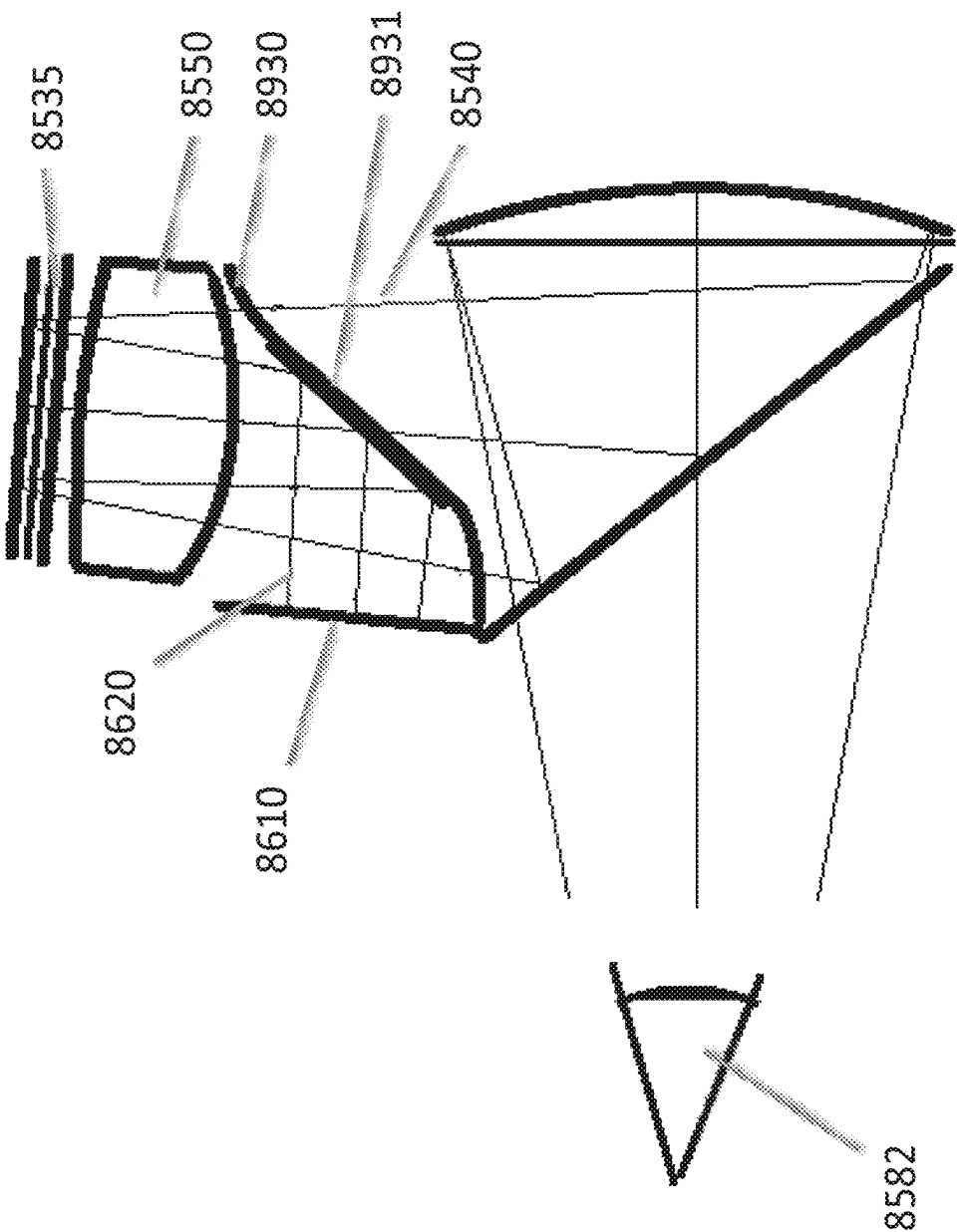


Figure 89

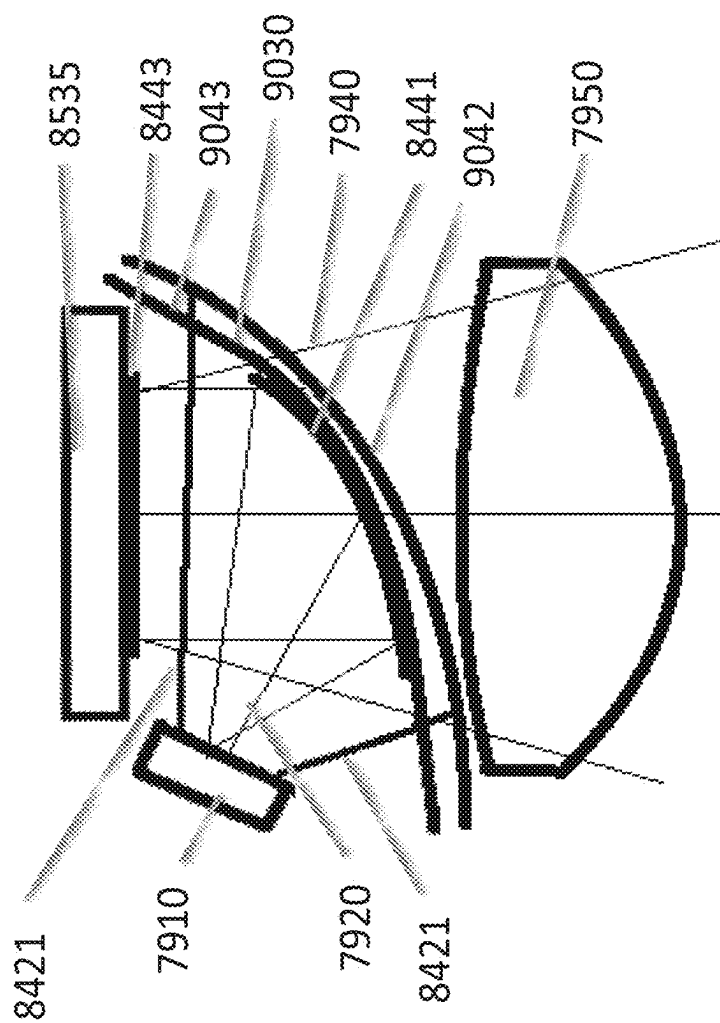


Figure 90

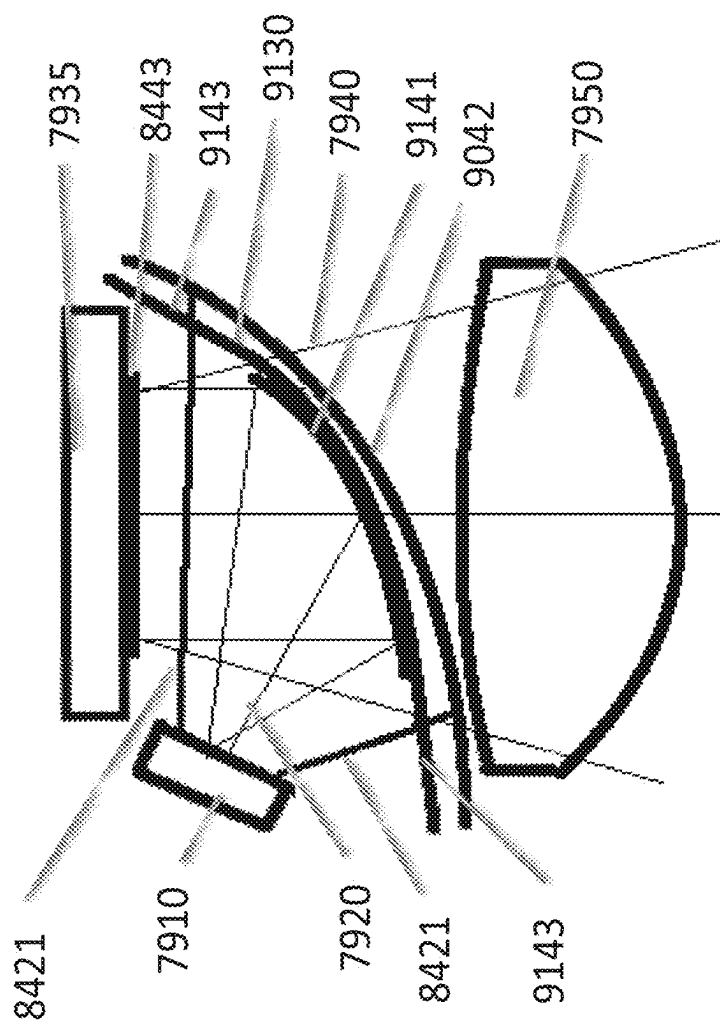


Figure 91

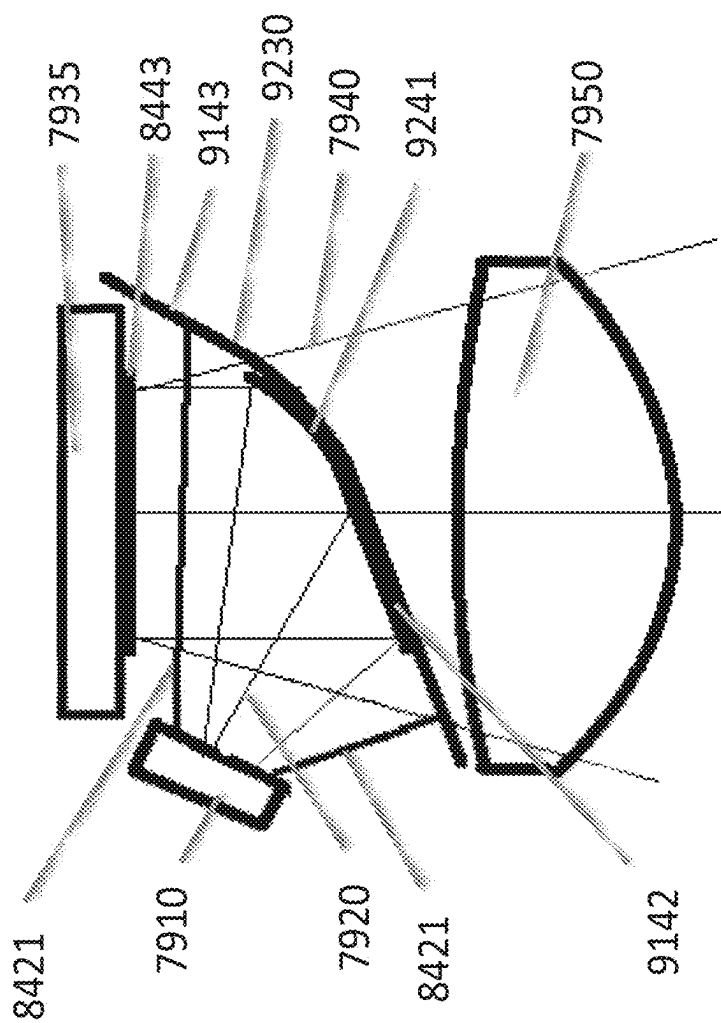


Figure 92

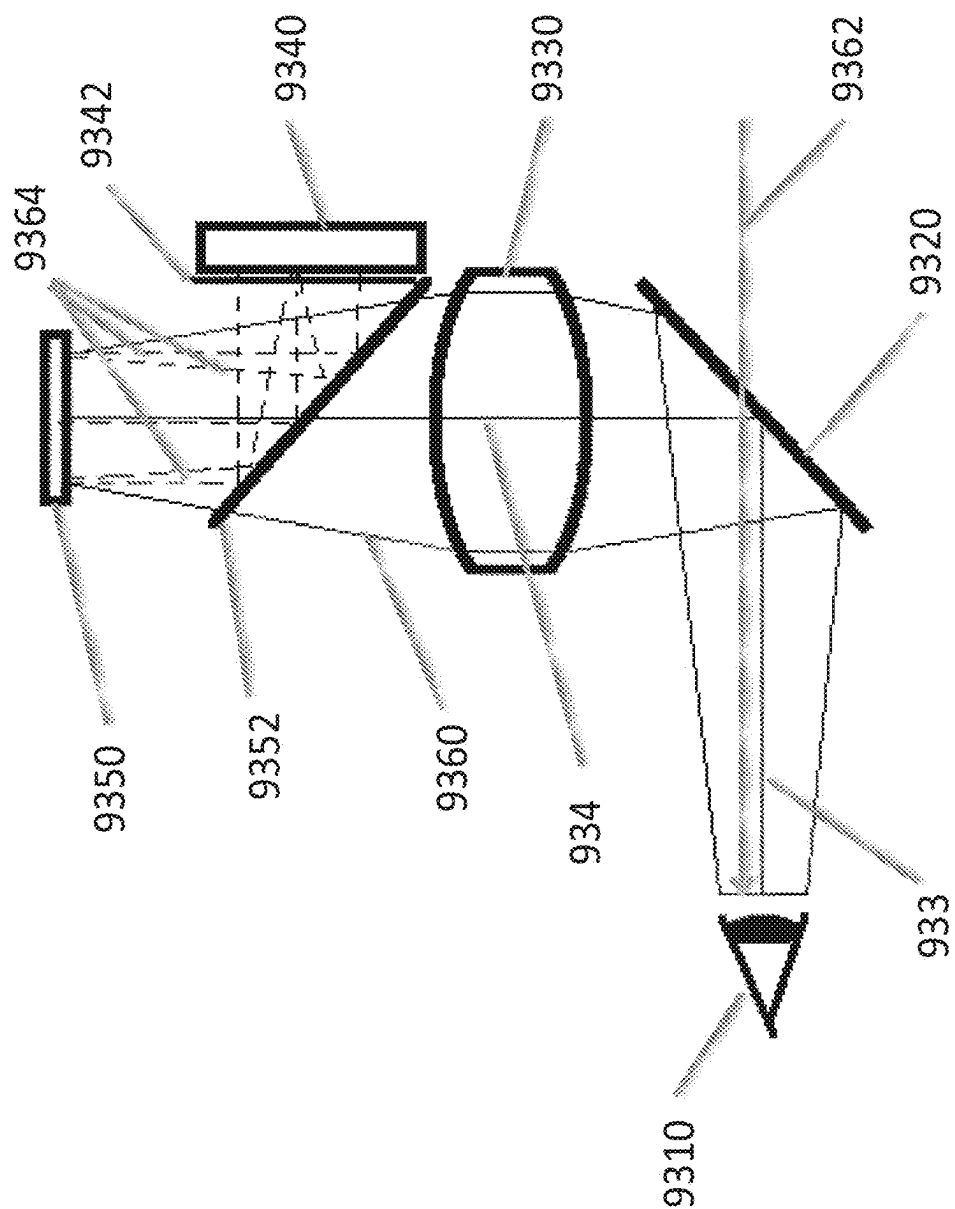


Figure 93

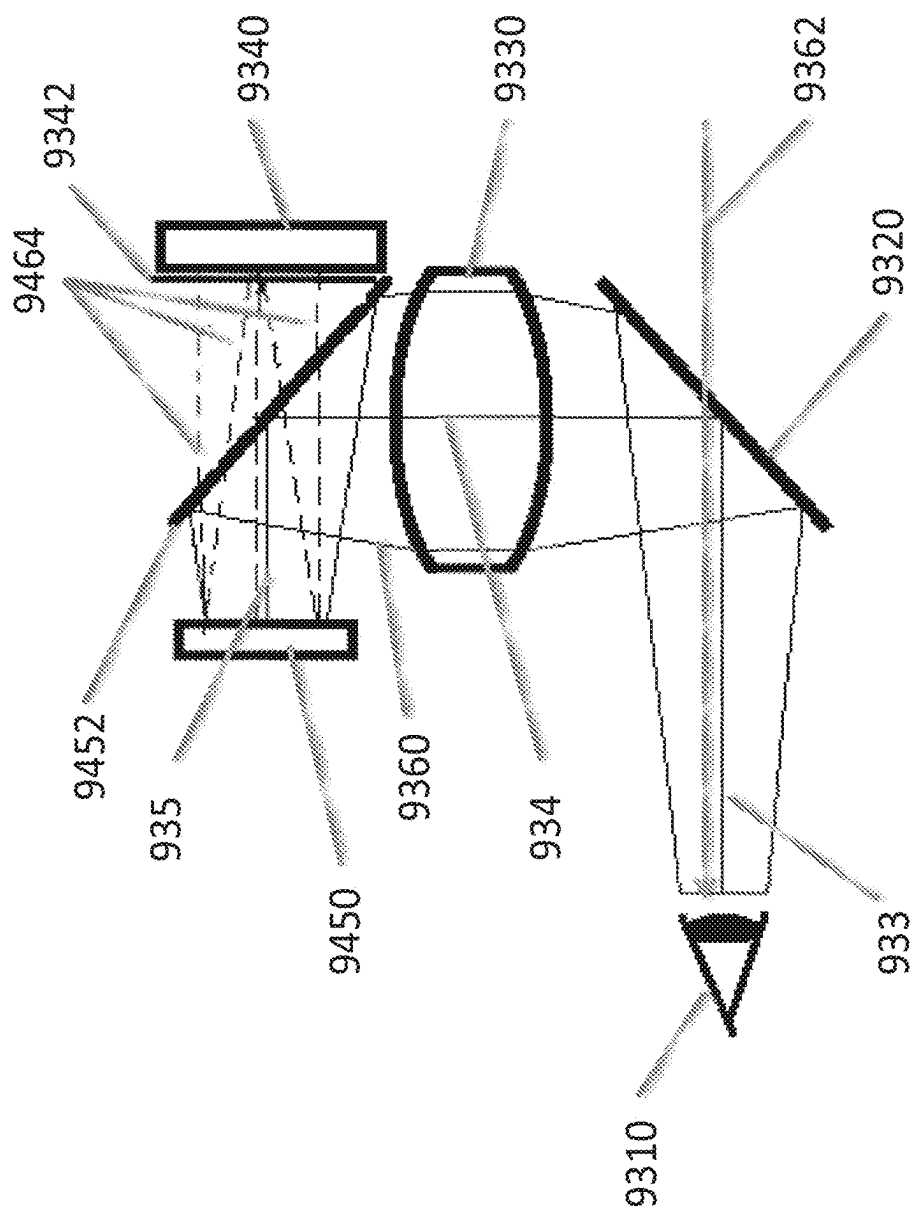


Figure 94

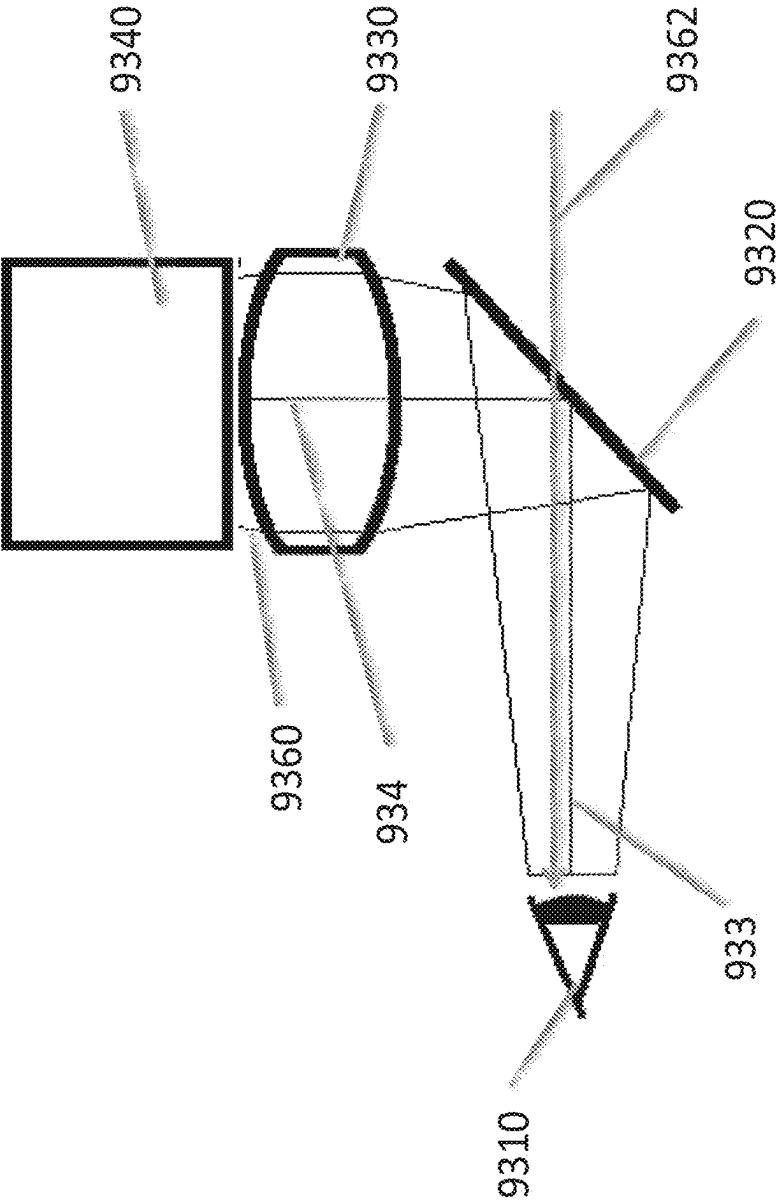


Figure 95

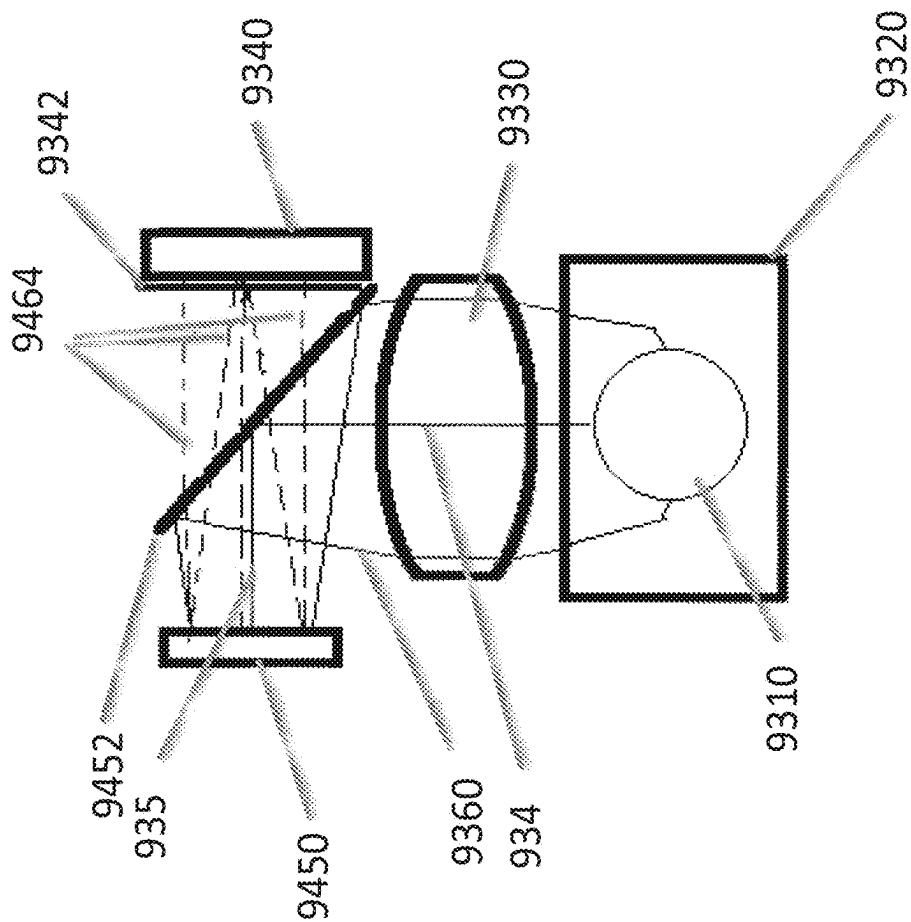
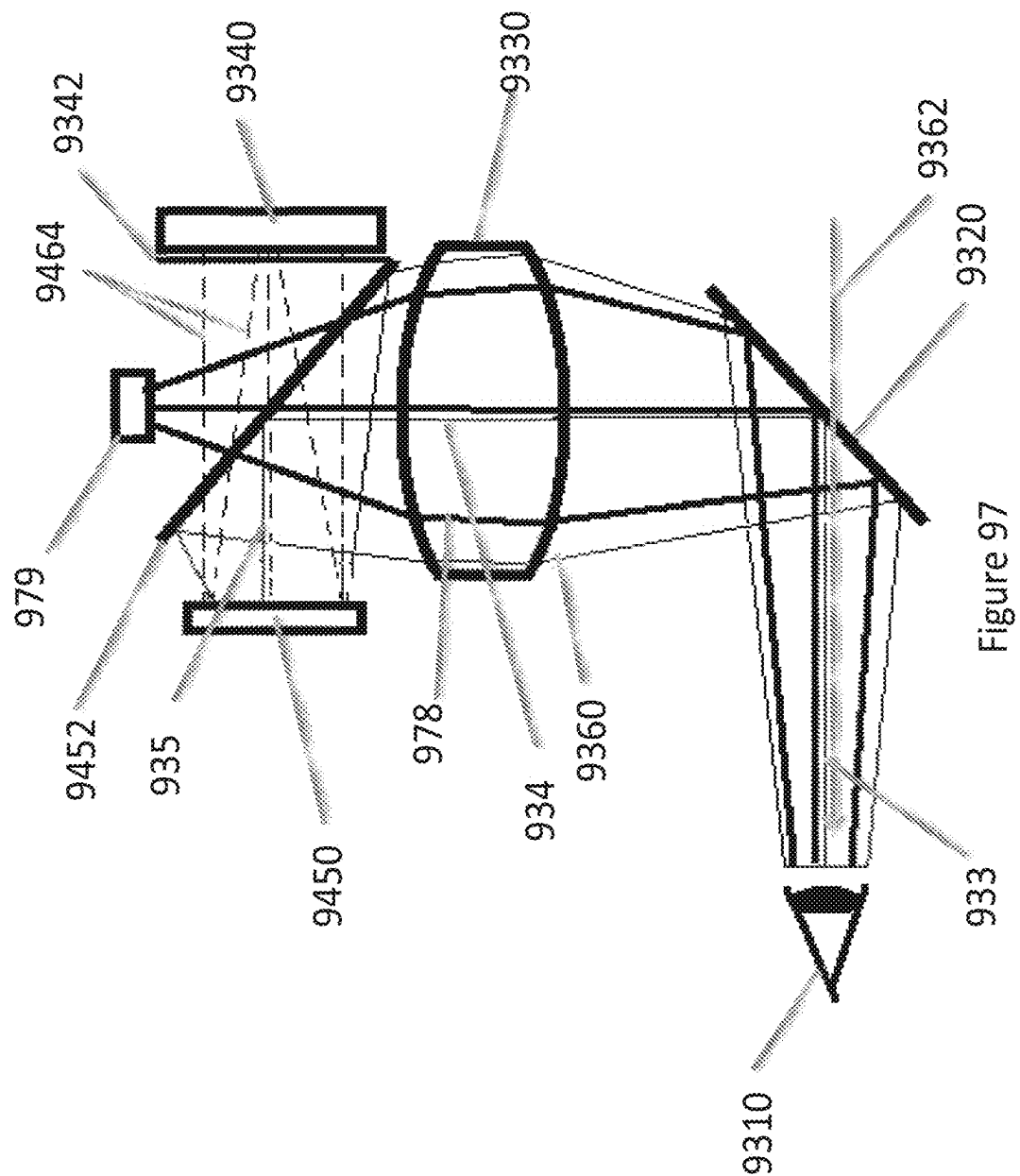


Figure 96



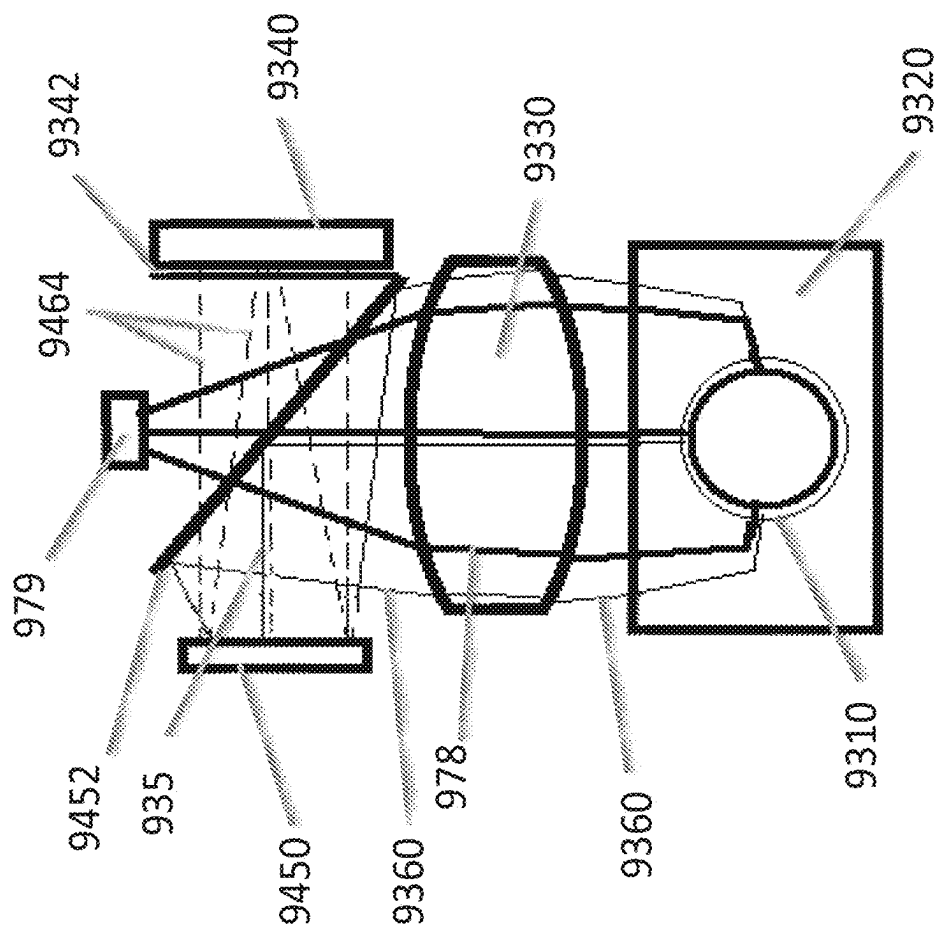


Figure 98

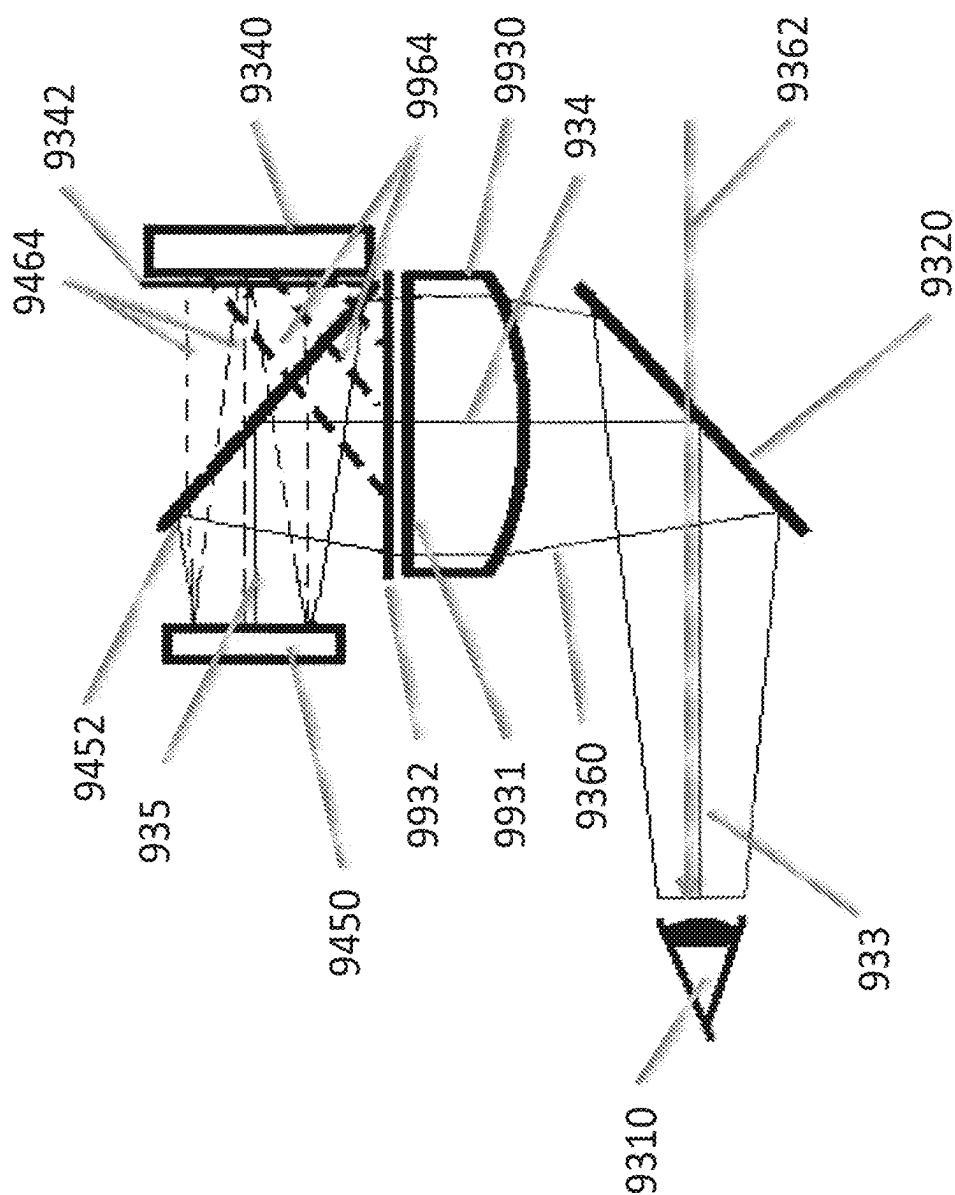


Figure 99

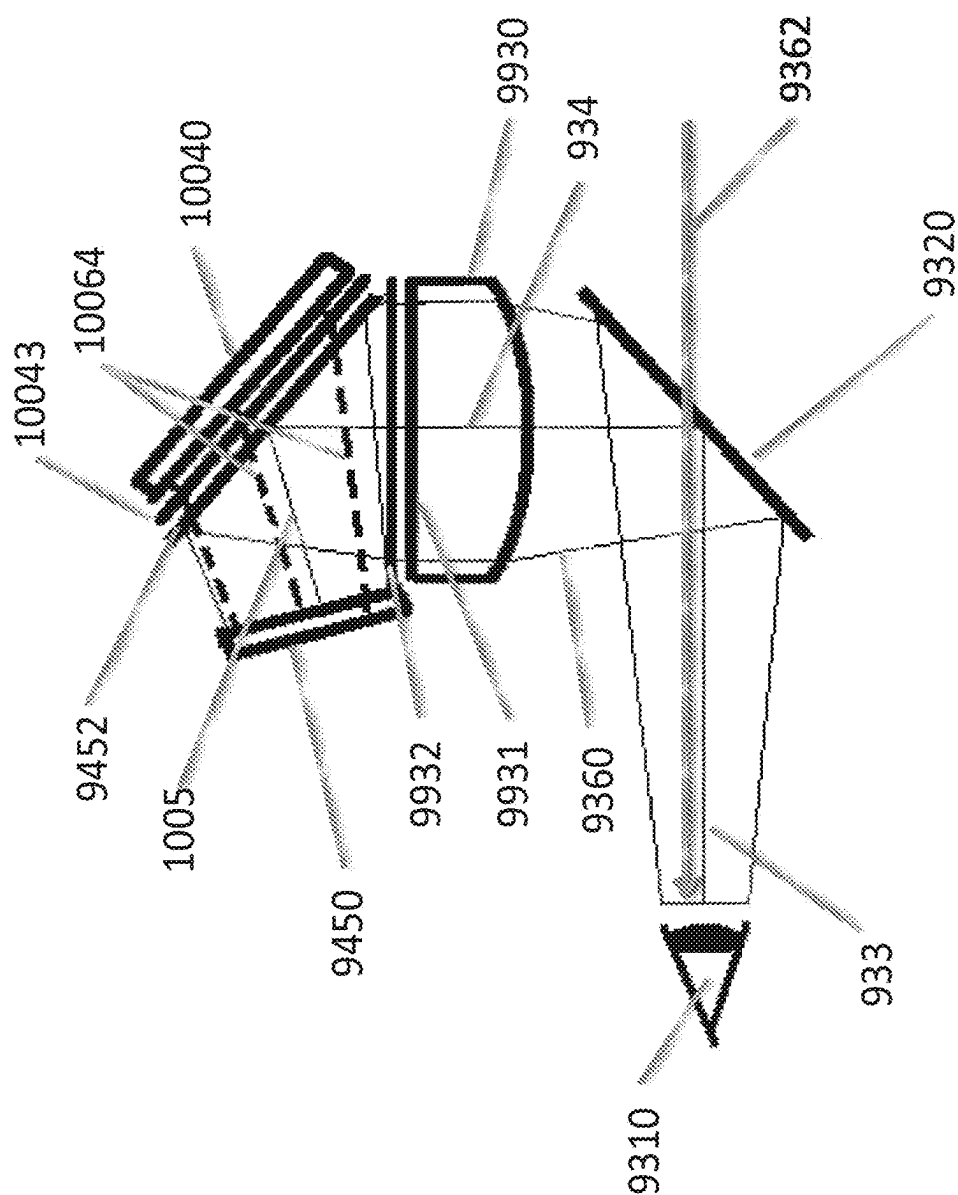


Figure 100

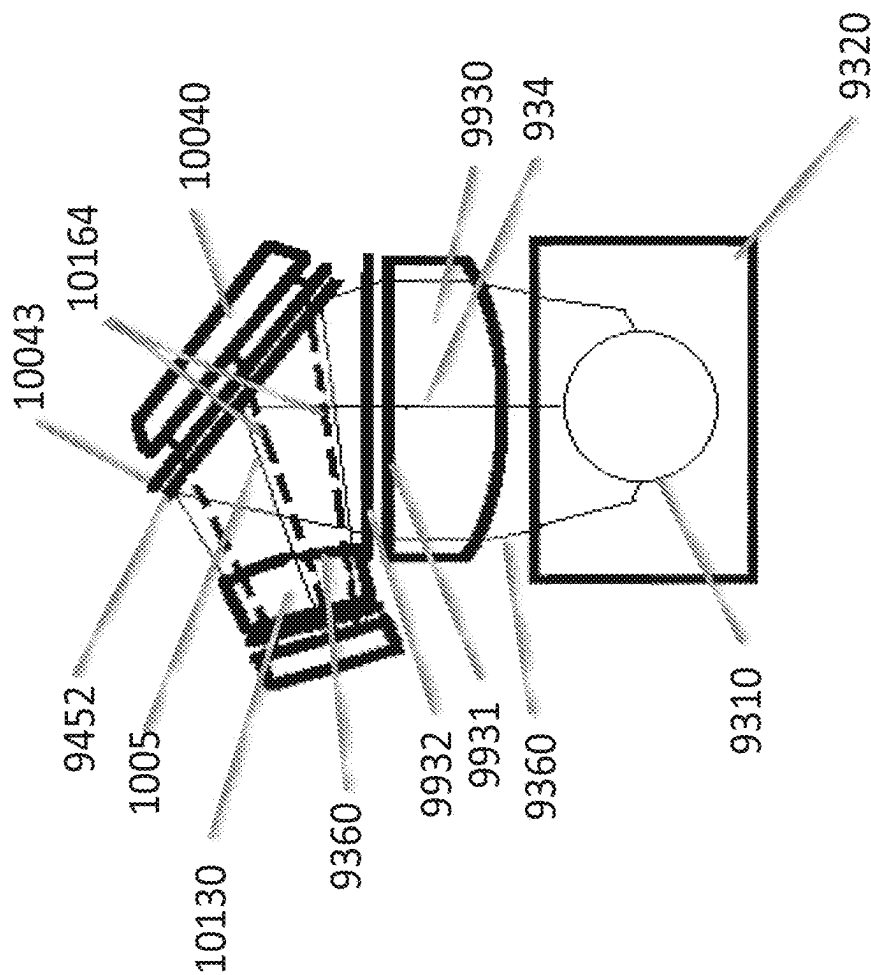


Figure 101

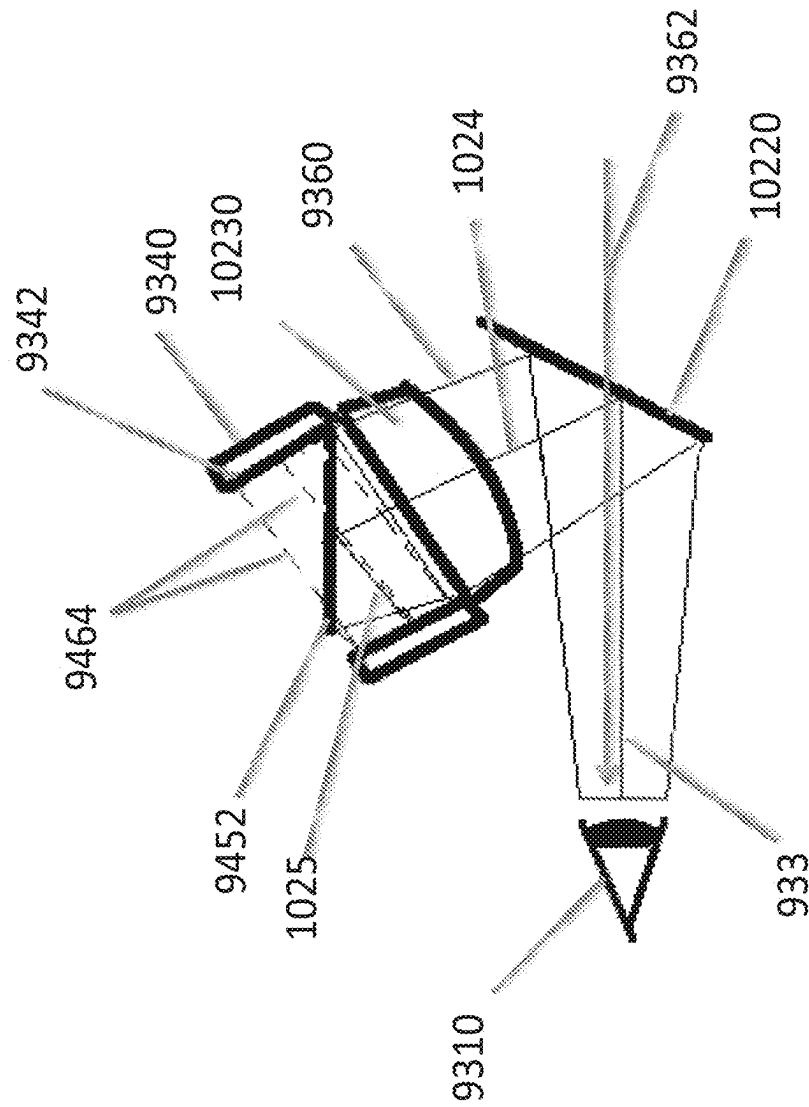


Figure 102

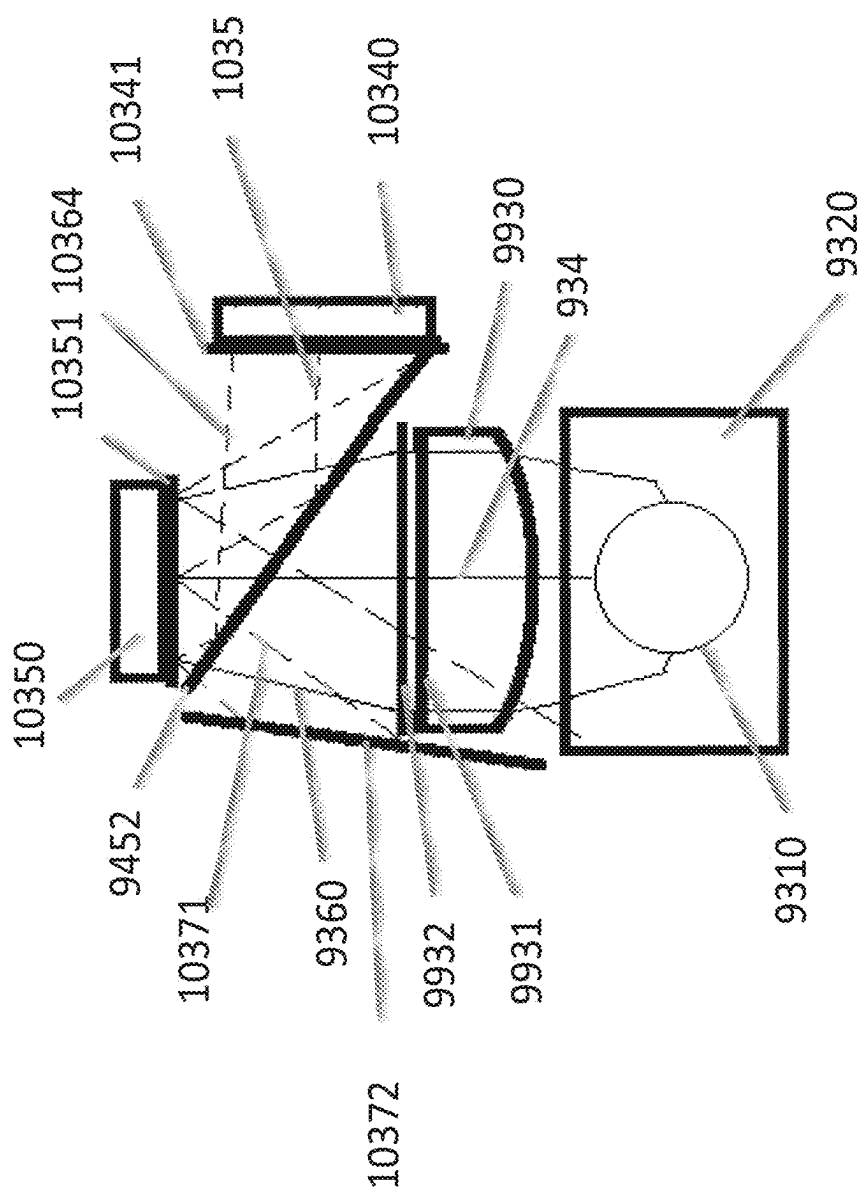


Figure 103

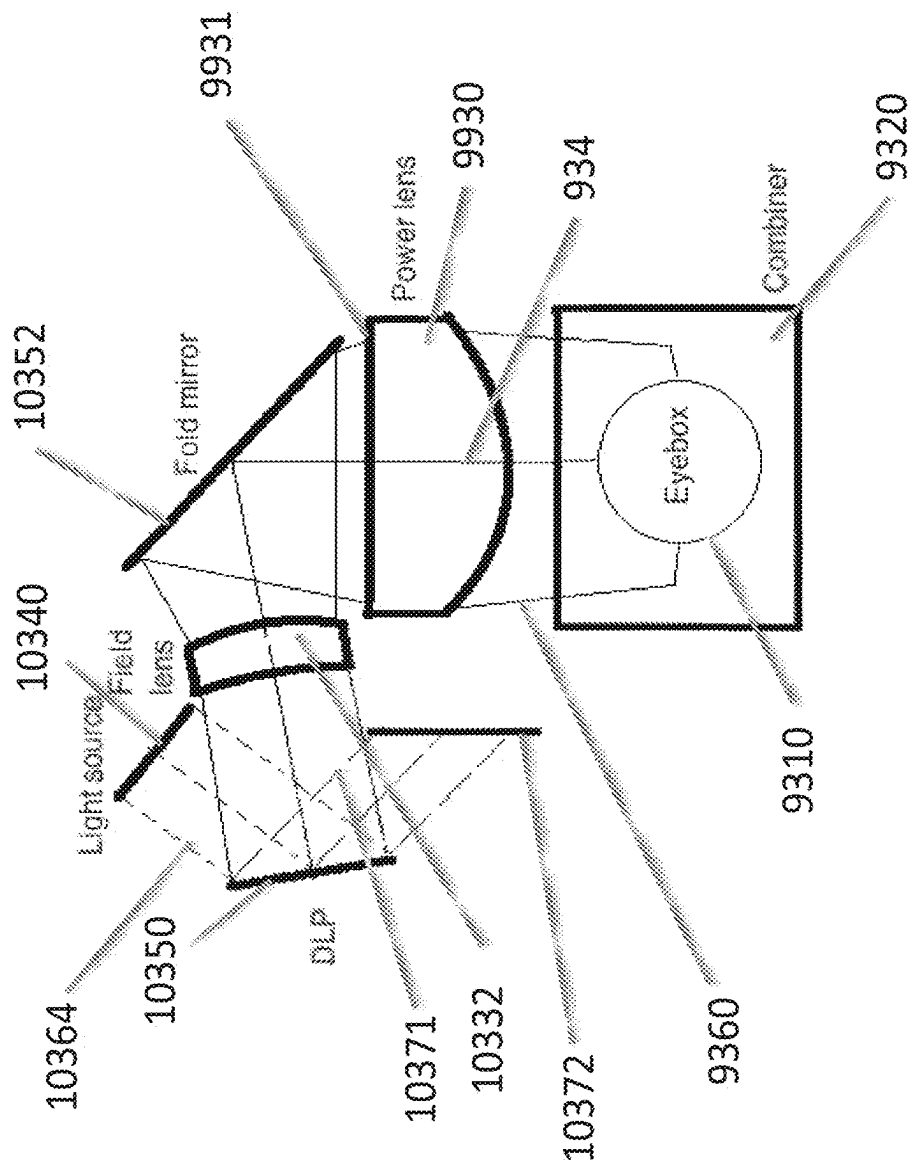


Figure 103a

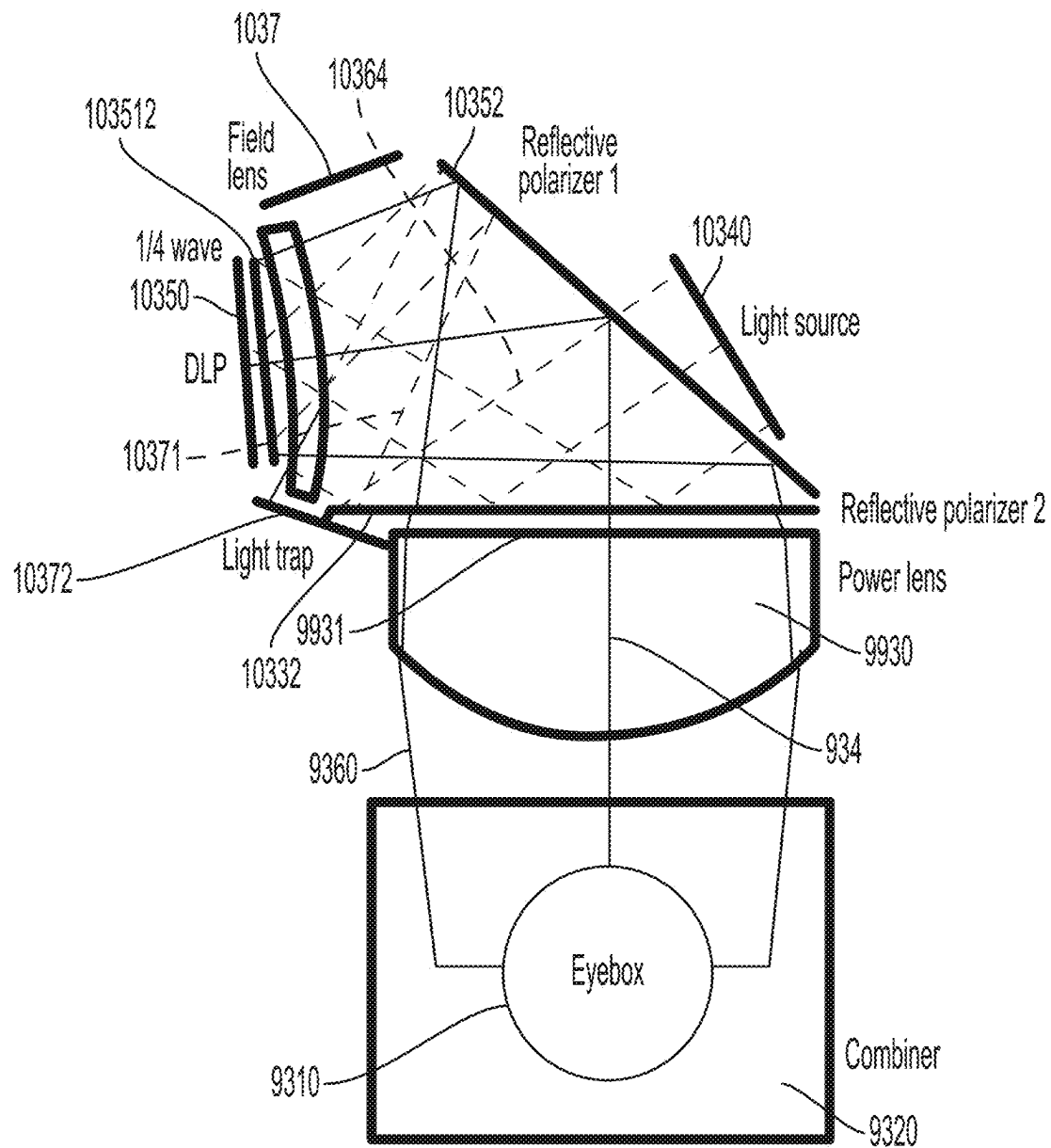


Figure 103b

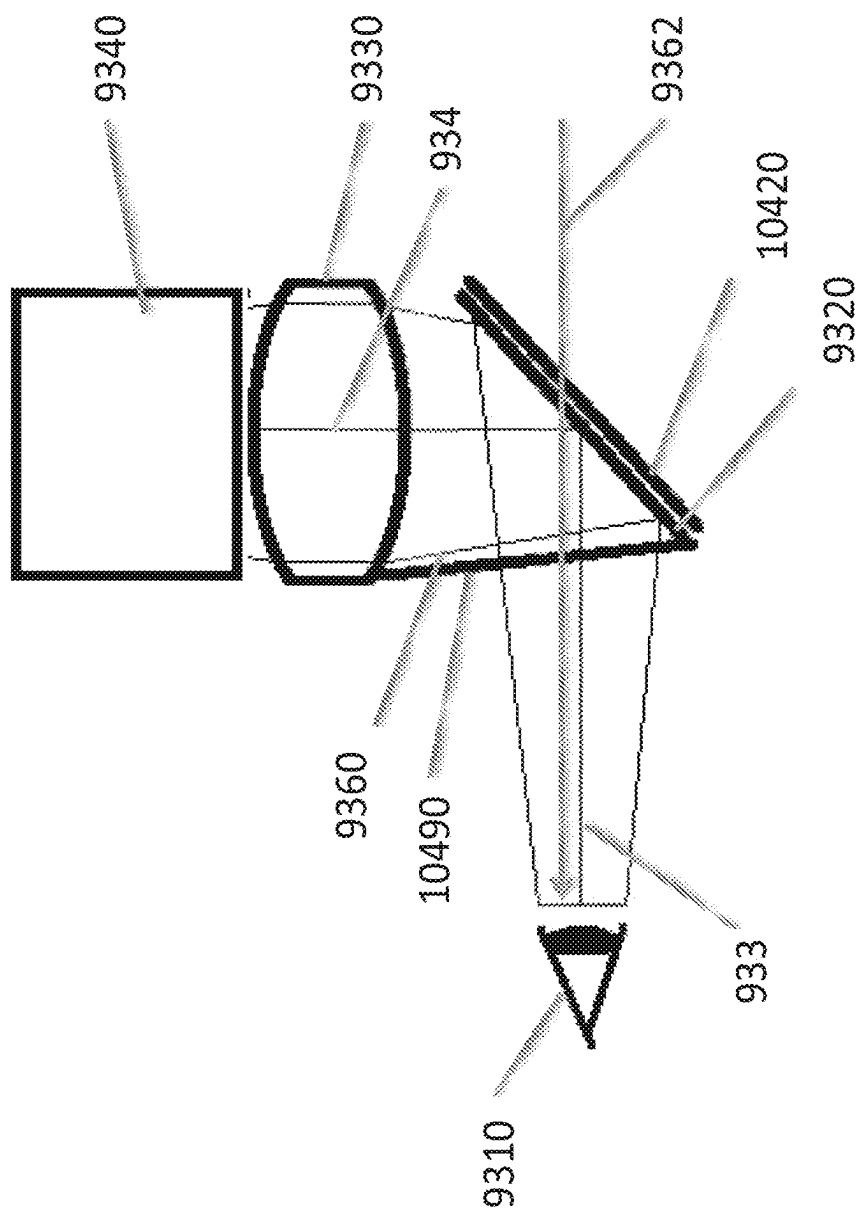


Figure 104

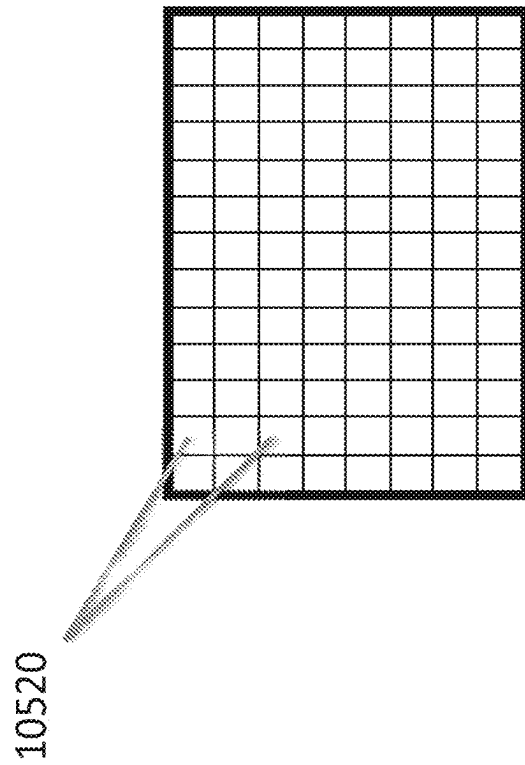


Figure 105

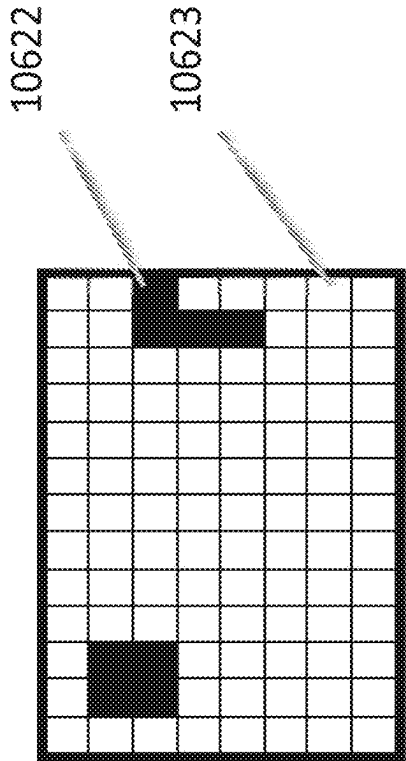


Figure 106a

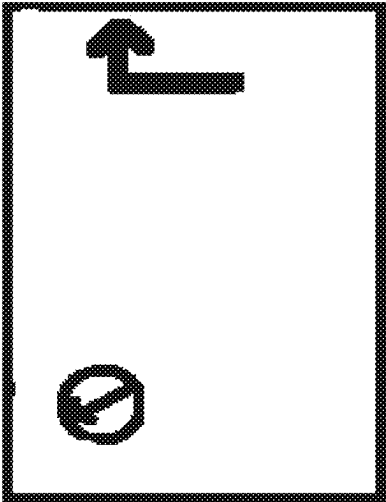


Figure 106b

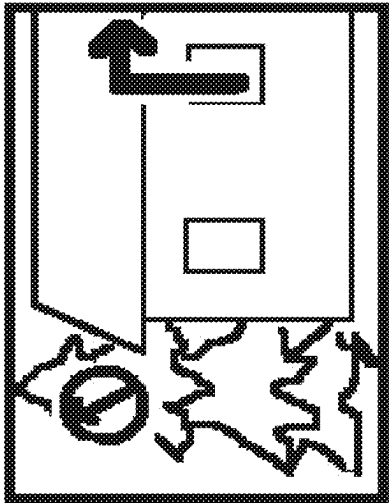
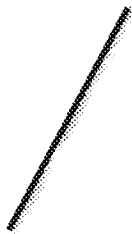


Figure 106c



Figure 107

10700



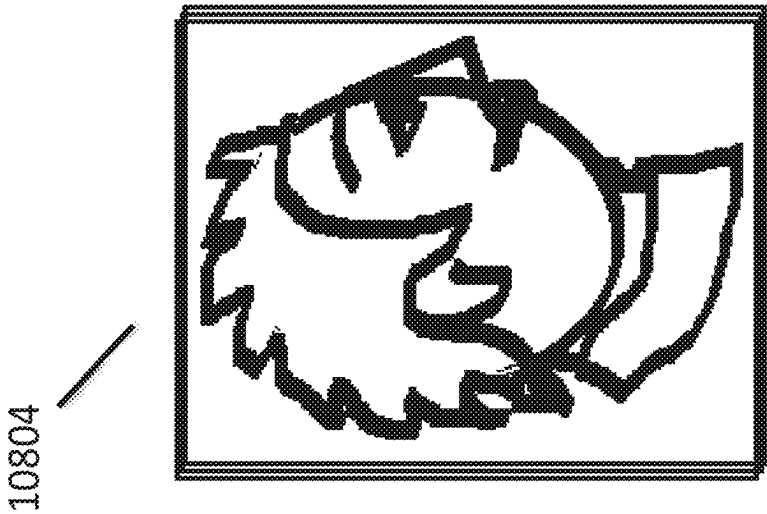


Figure 108B

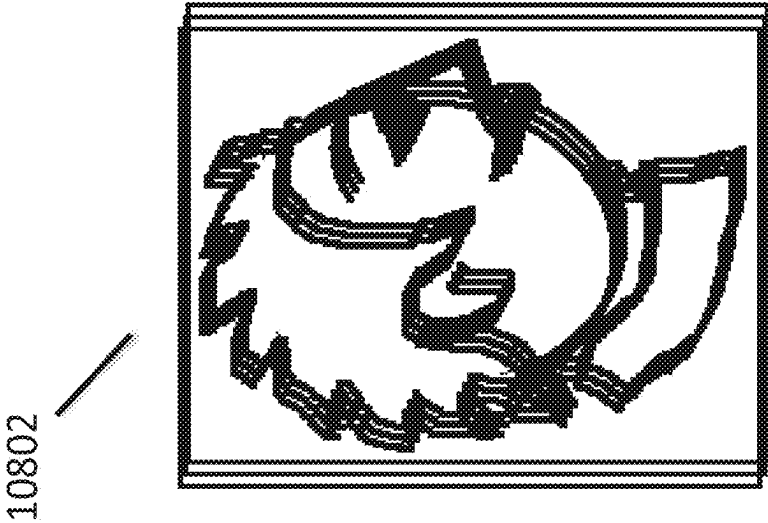


Figure 108A

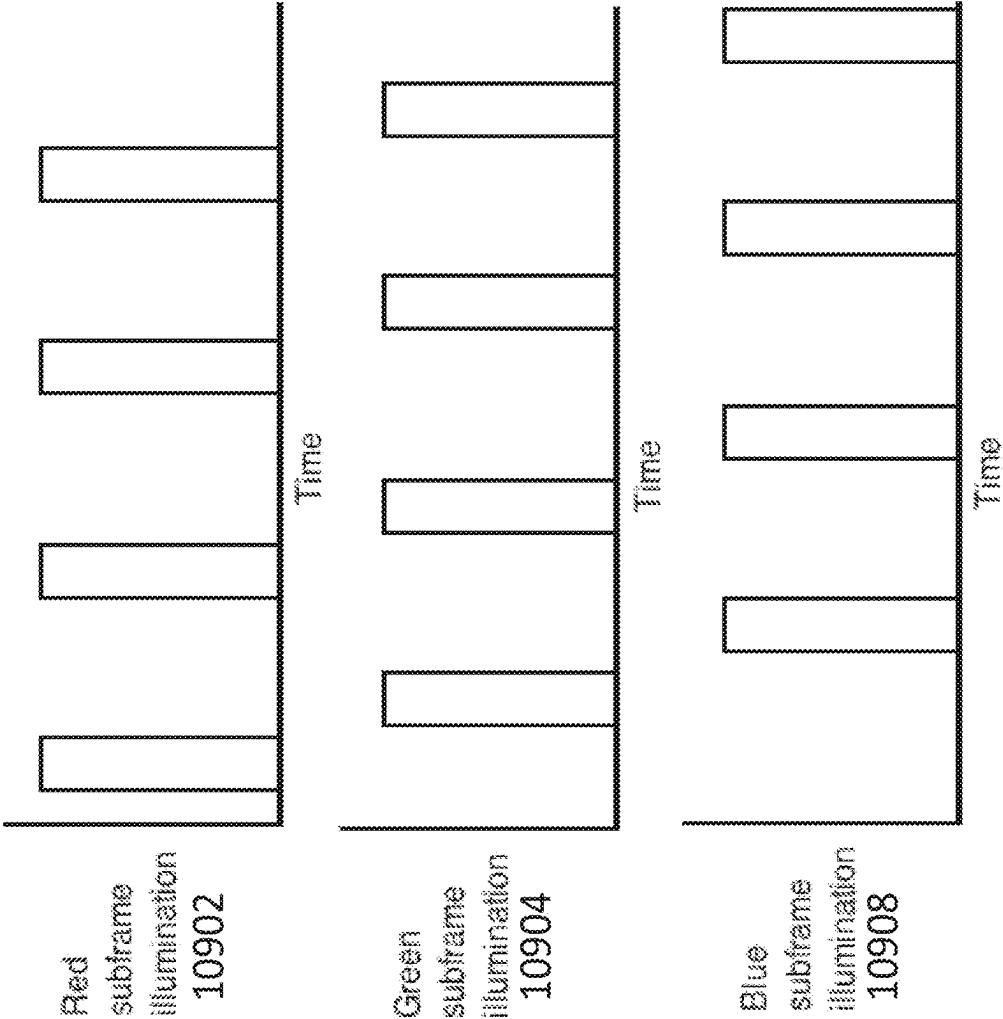


Figure 109

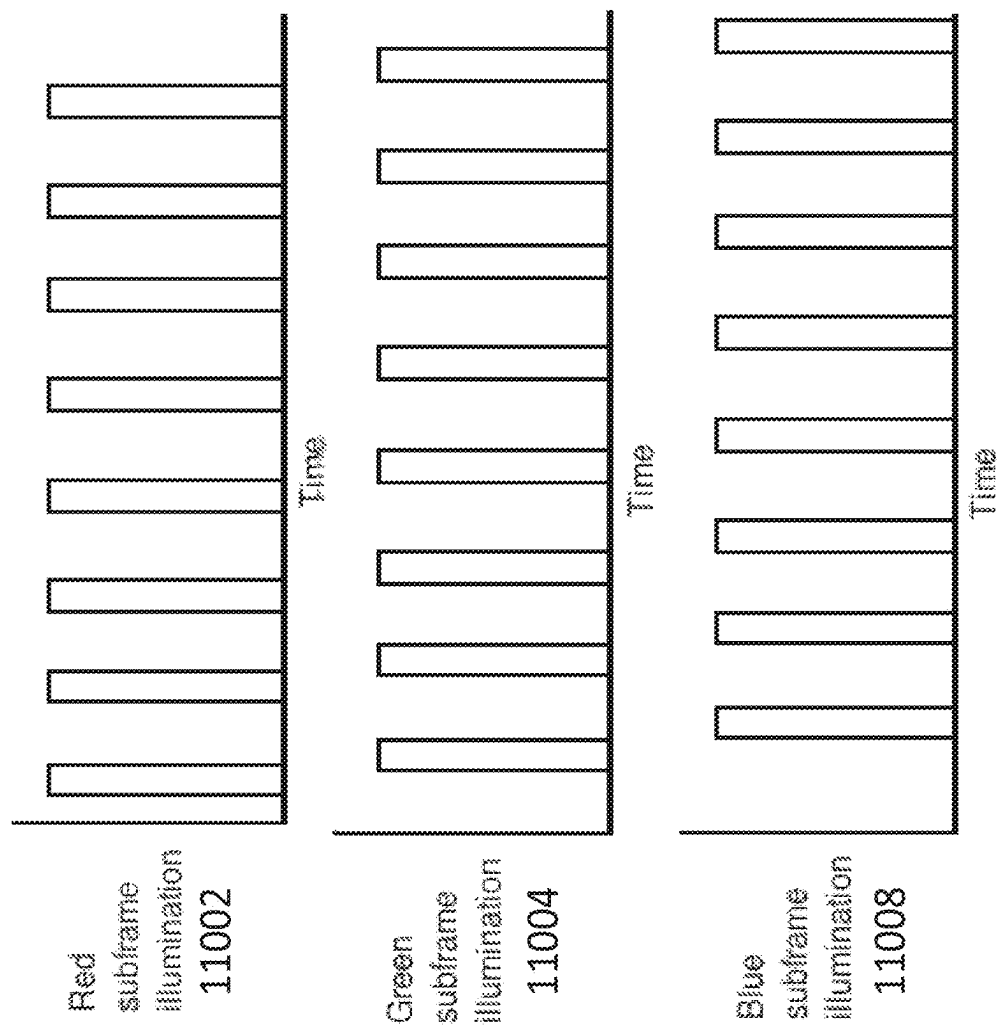


Figure 110

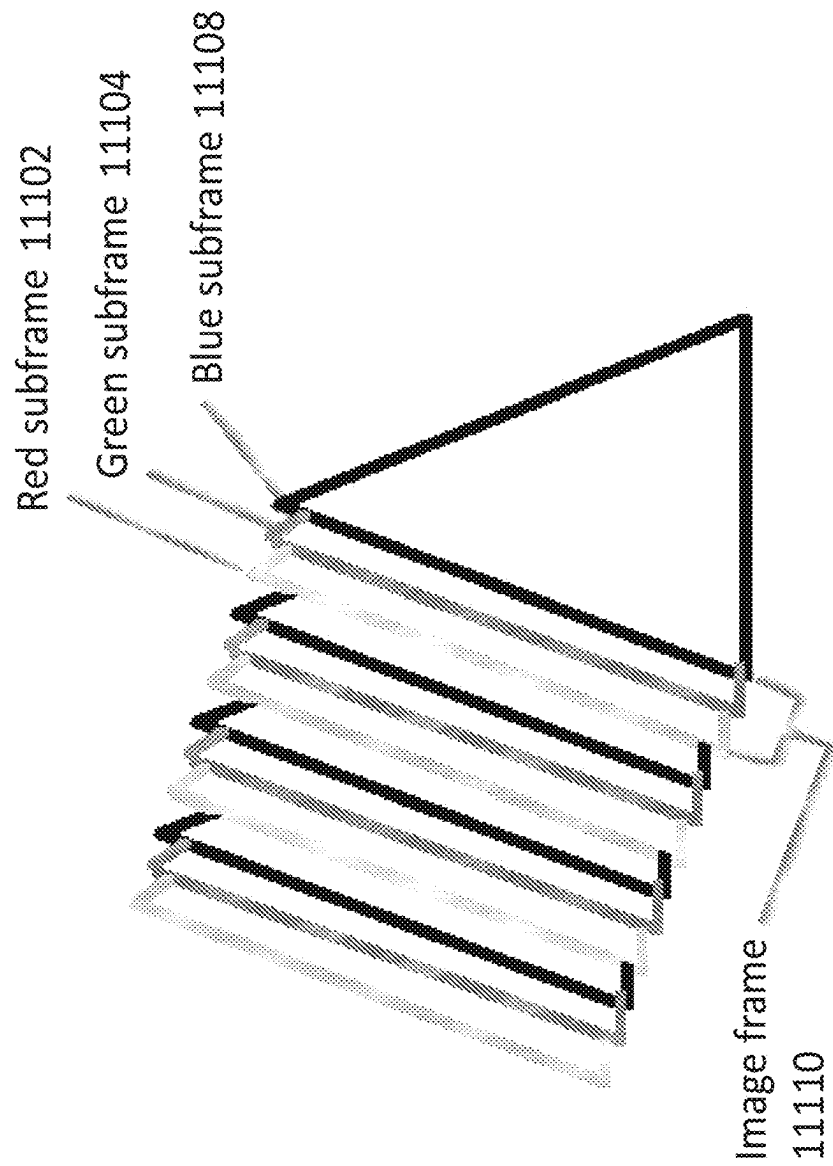
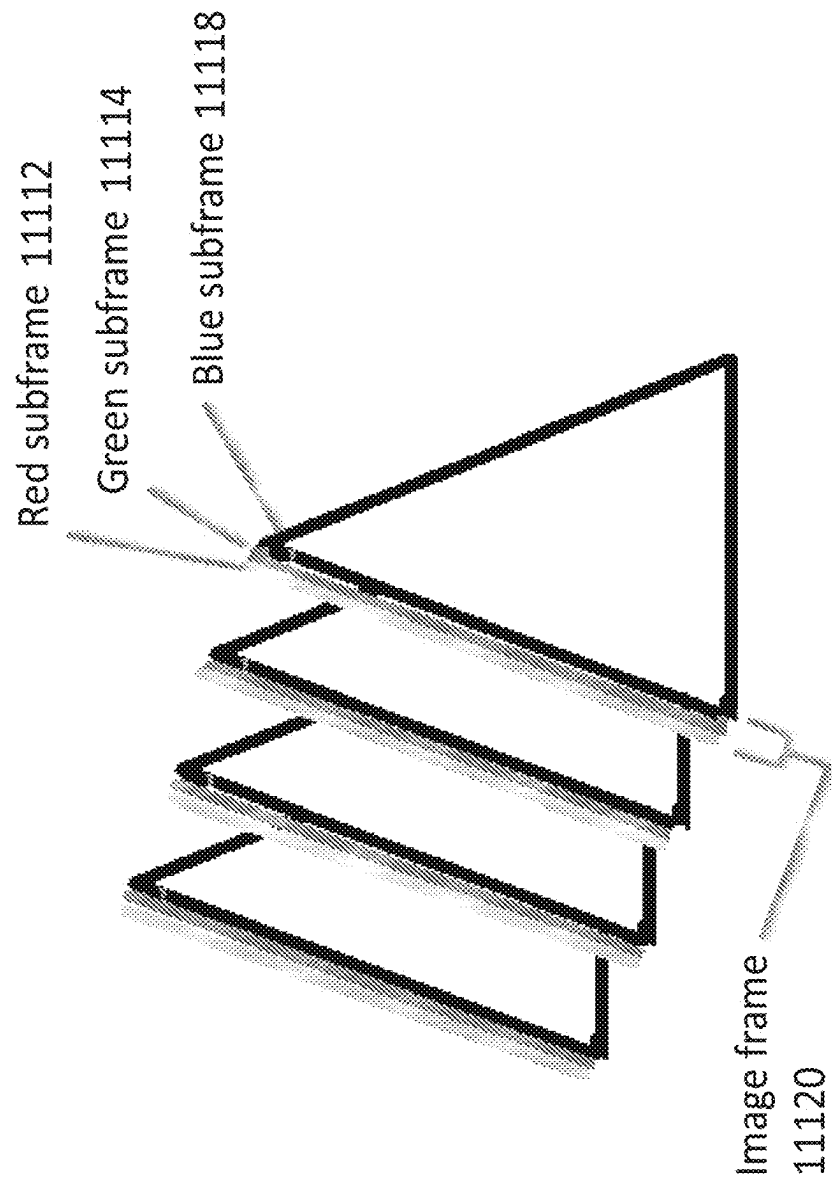


Figure 111a



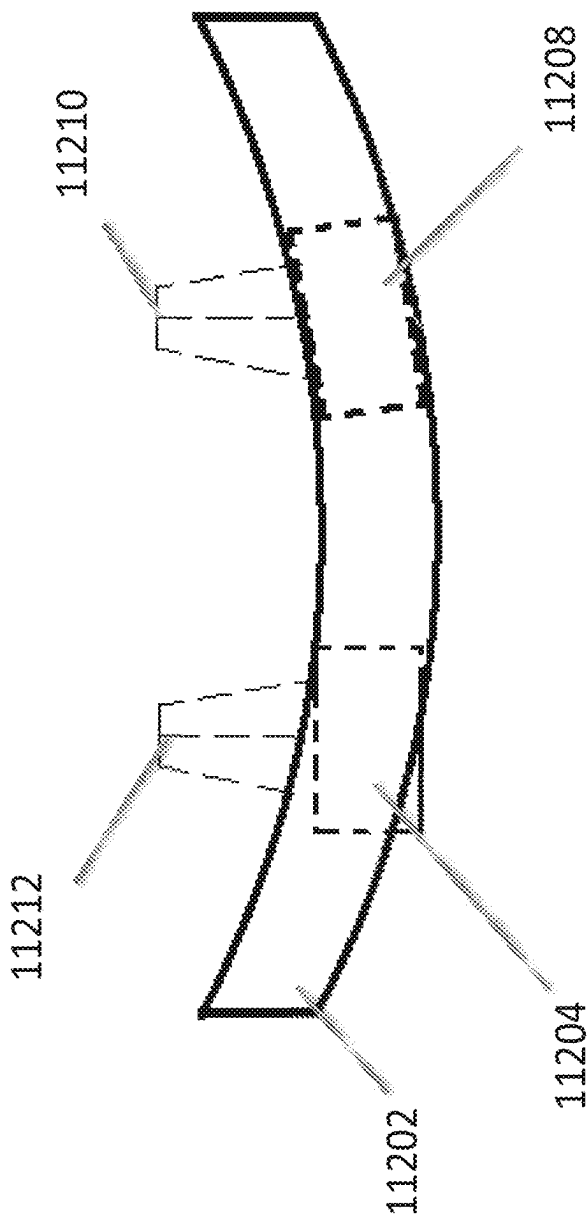


Figure 112

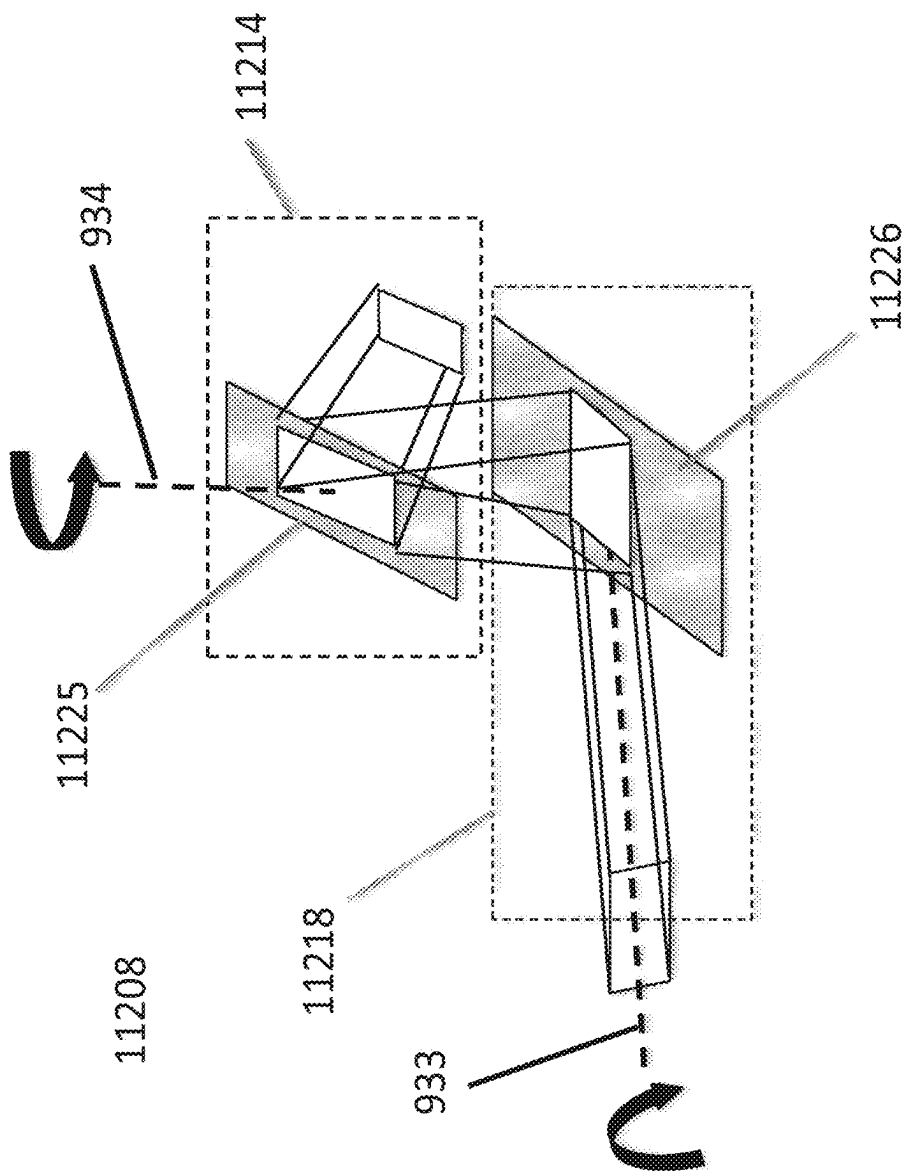


Figure 113

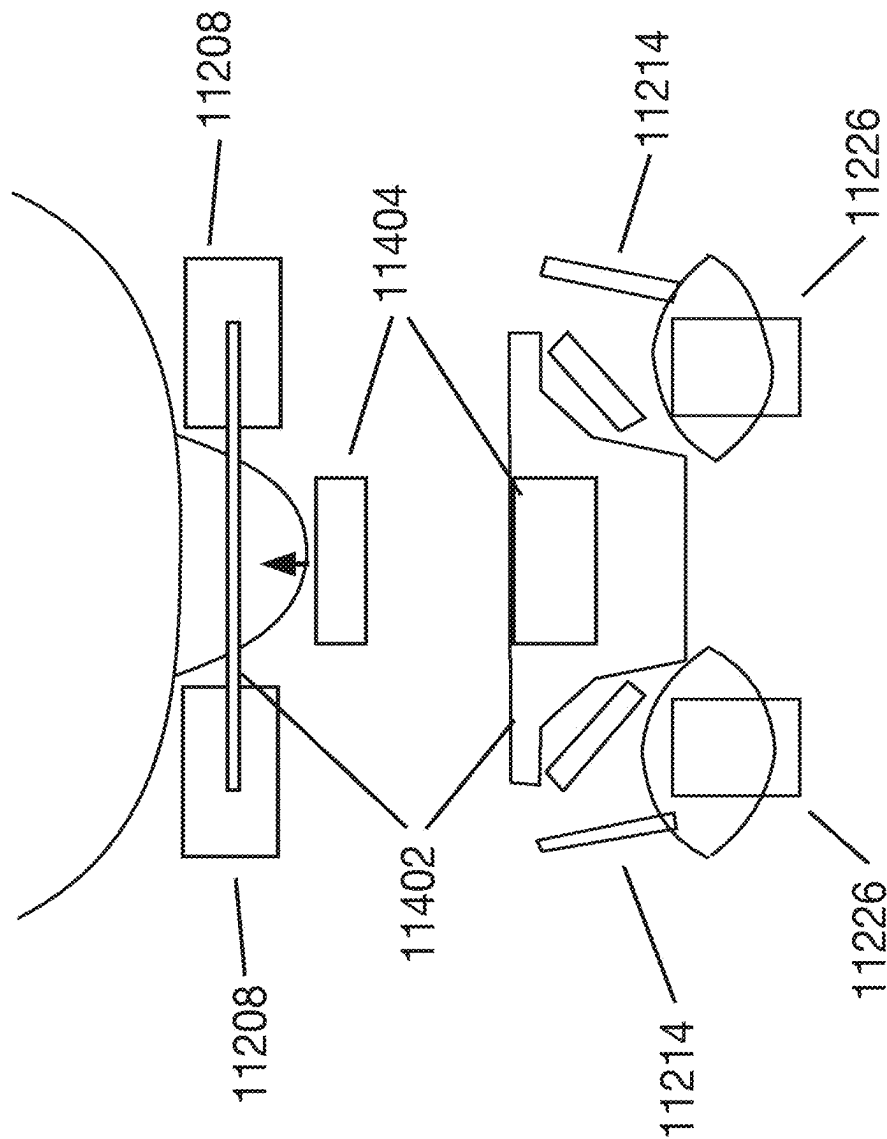


Figure 114

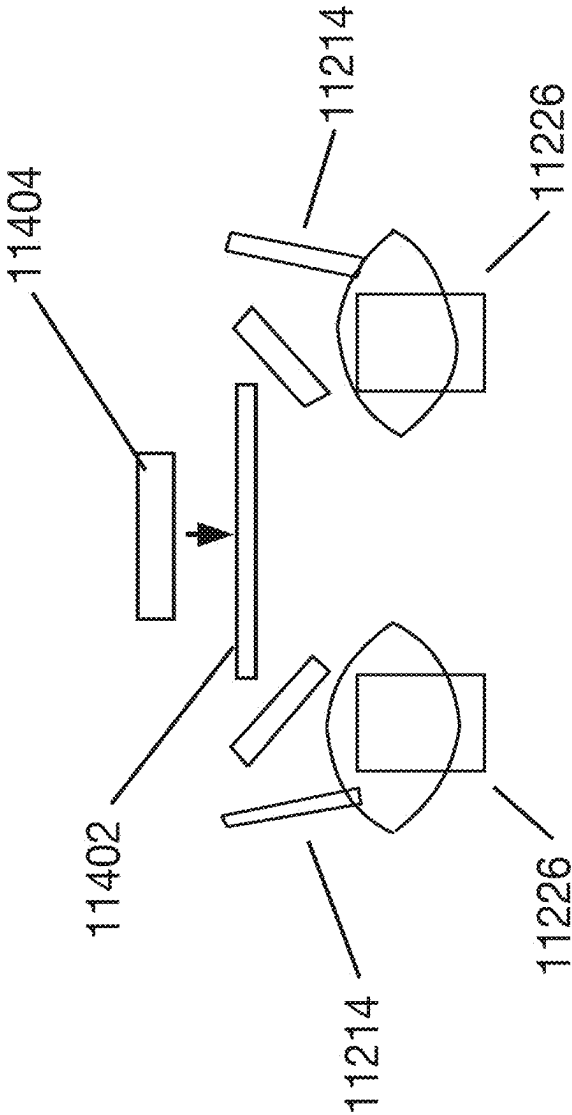


Figure 115

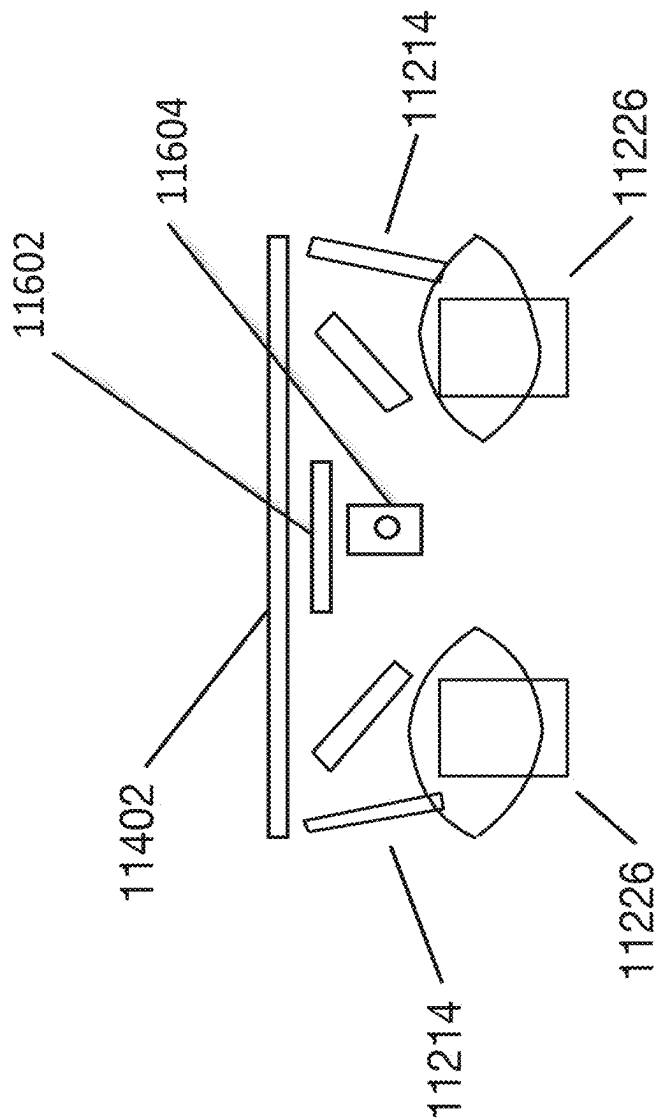


Figure 116

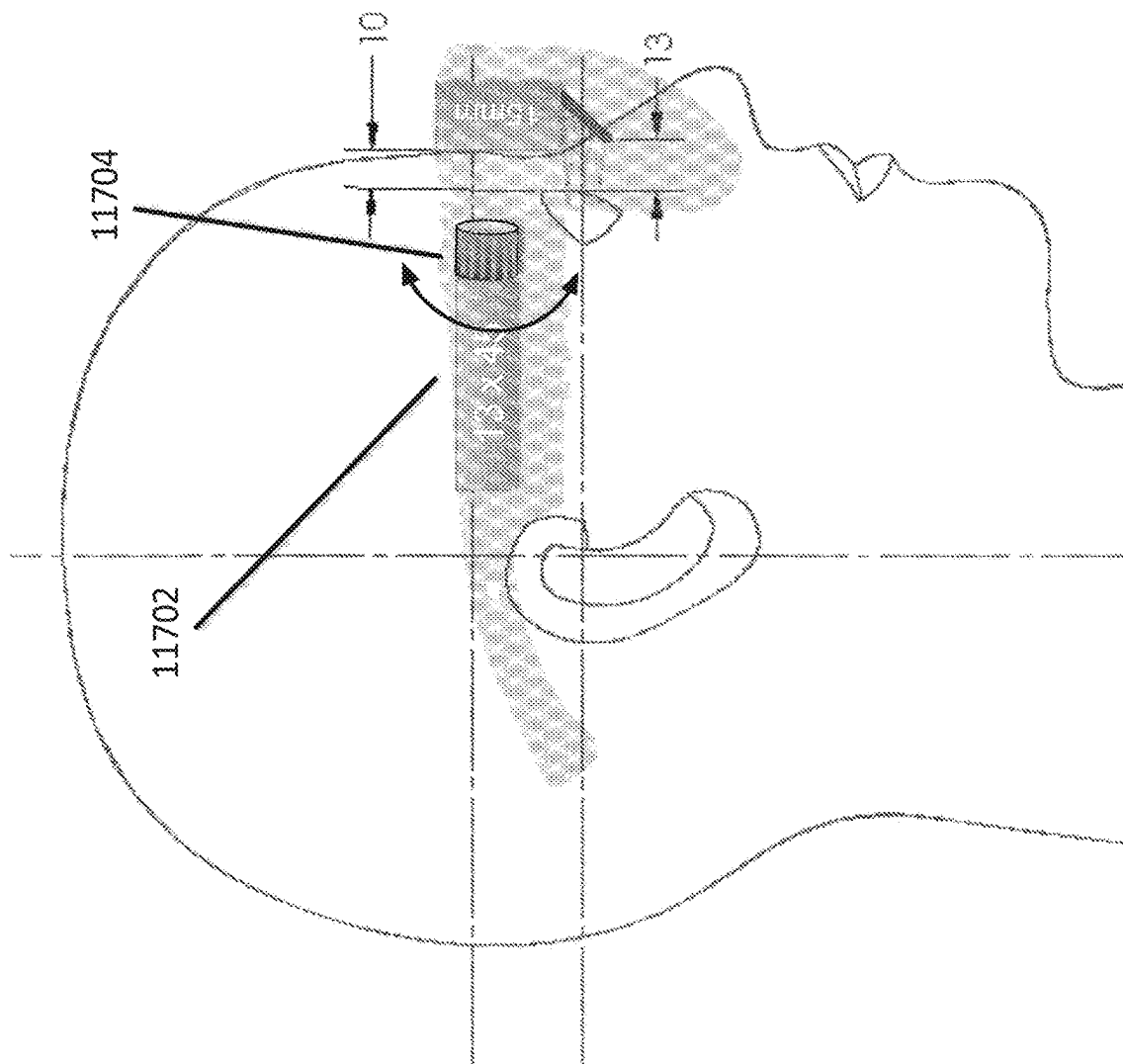


Figure 117

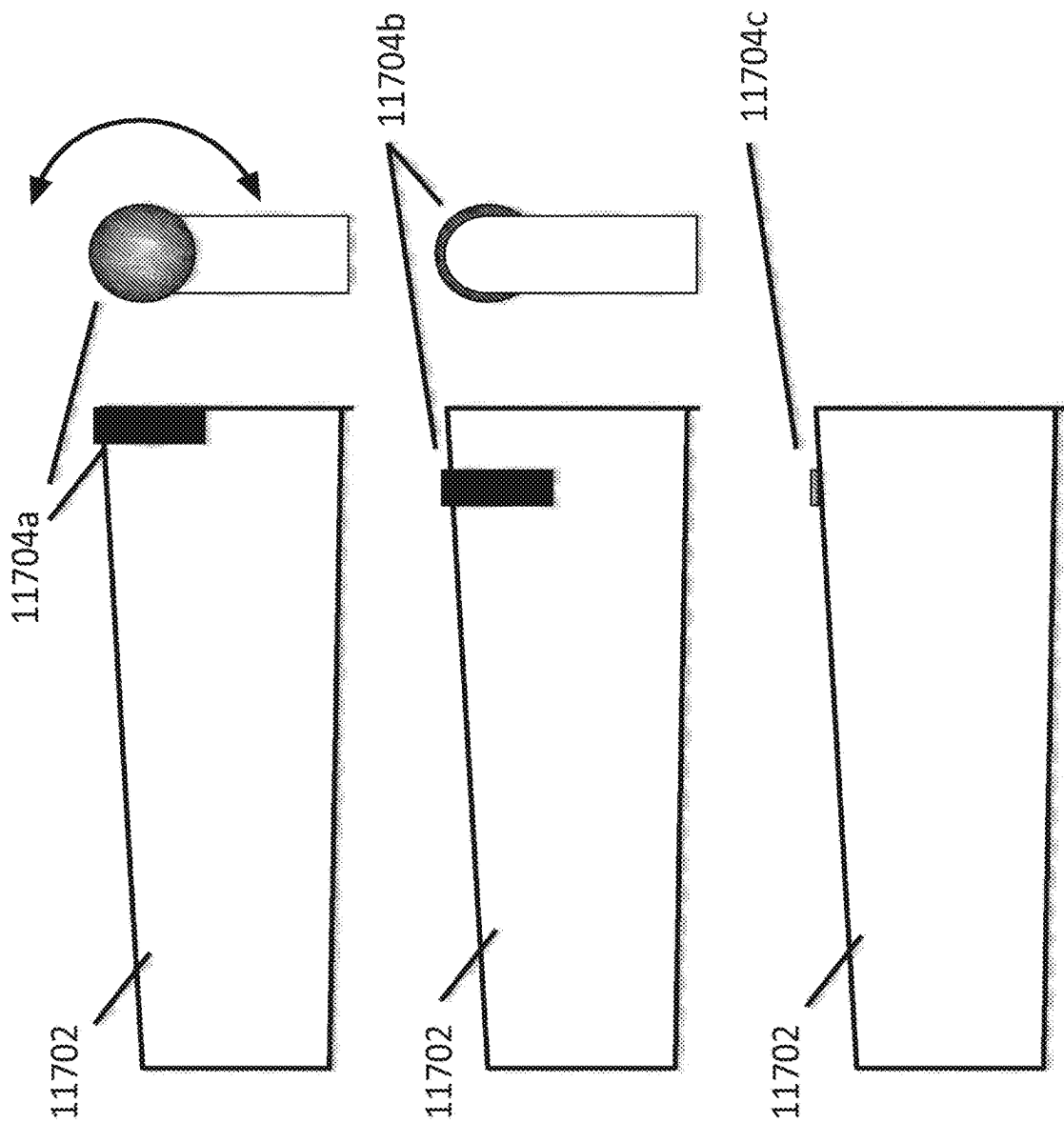


Figure 118

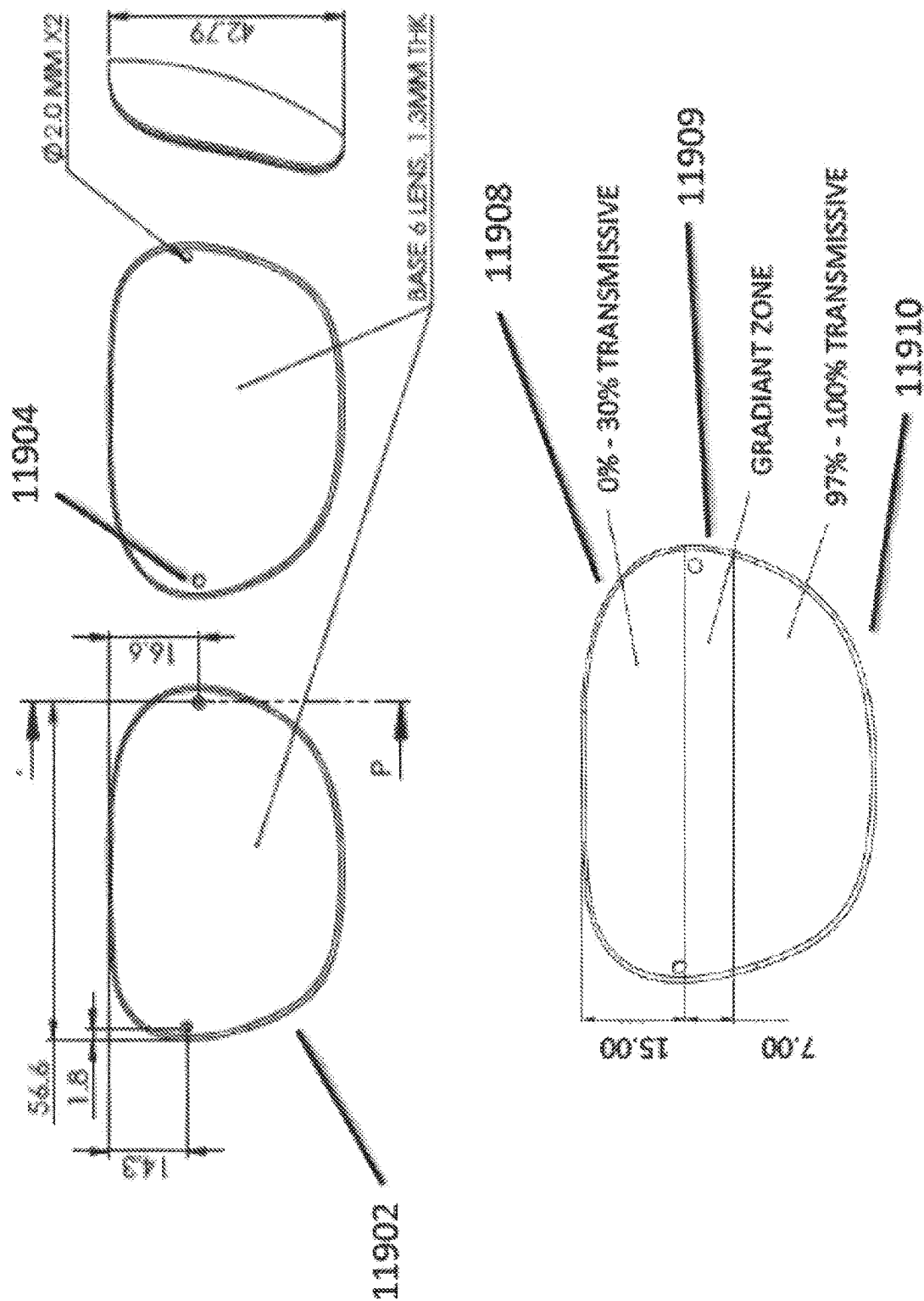


Figure 119

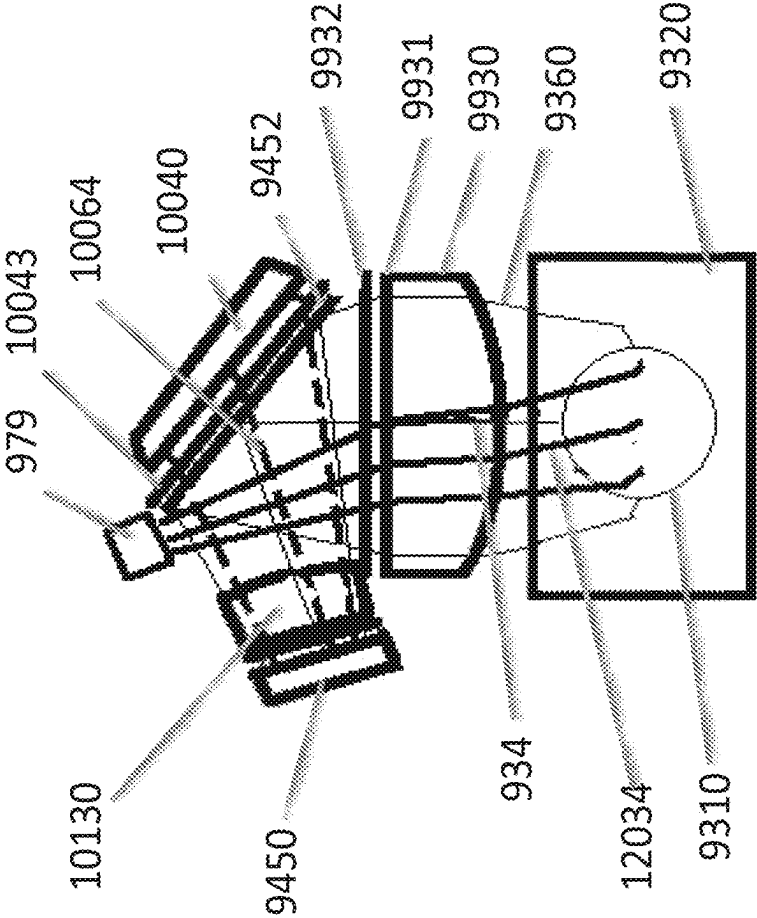


Figure 120

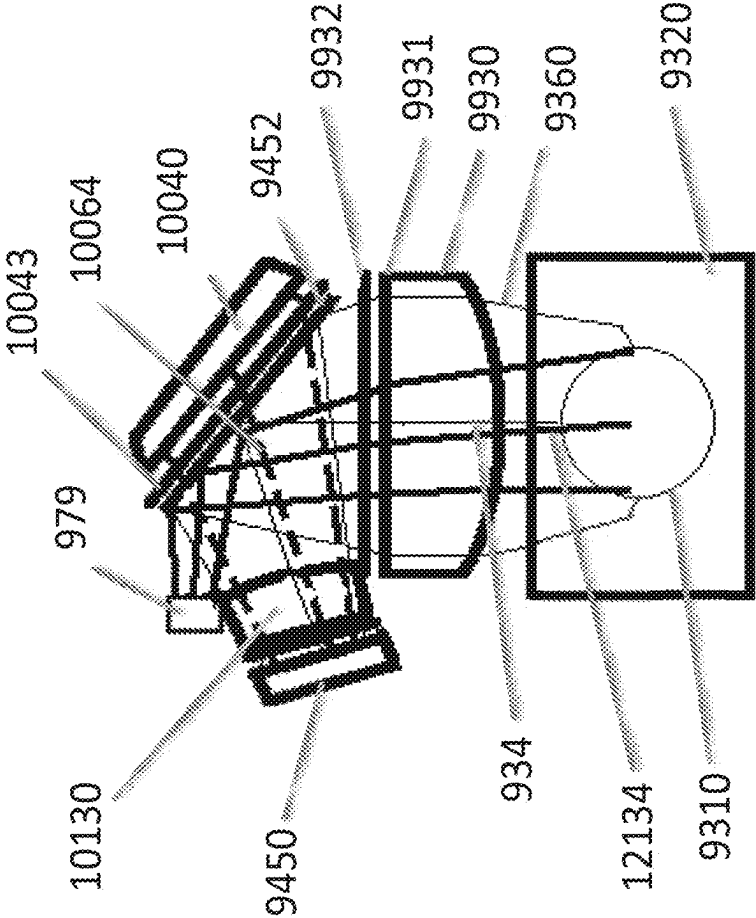


Figure 121

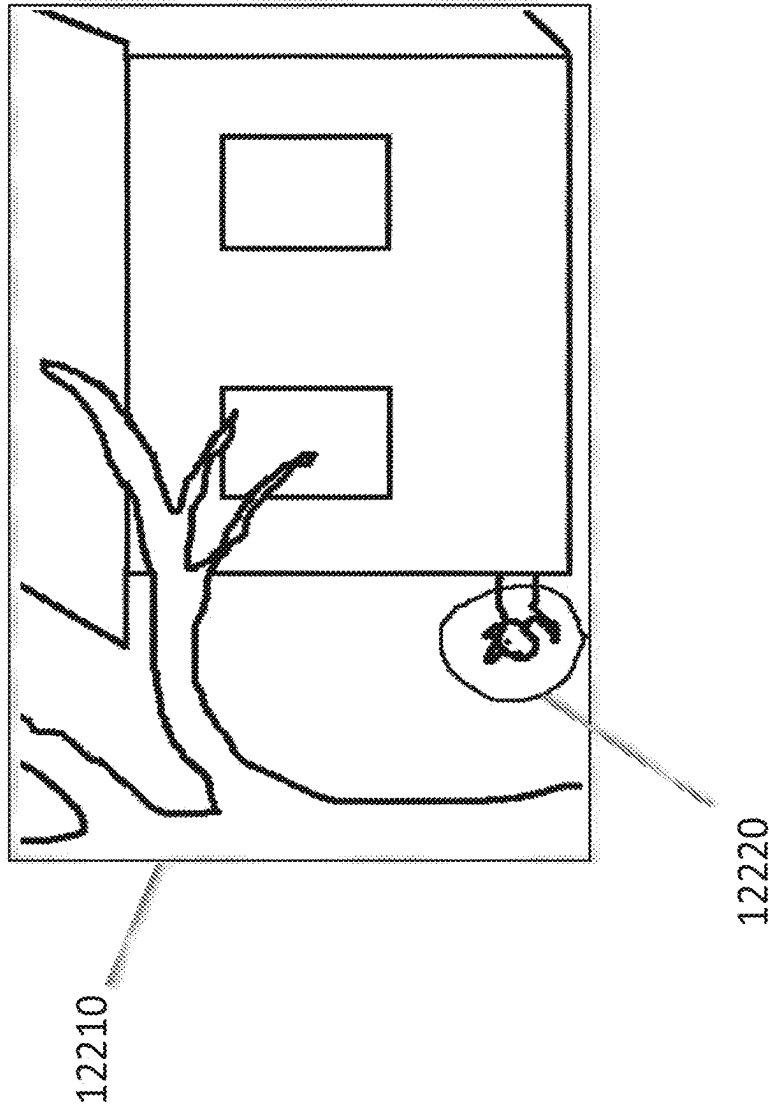


Figure 122

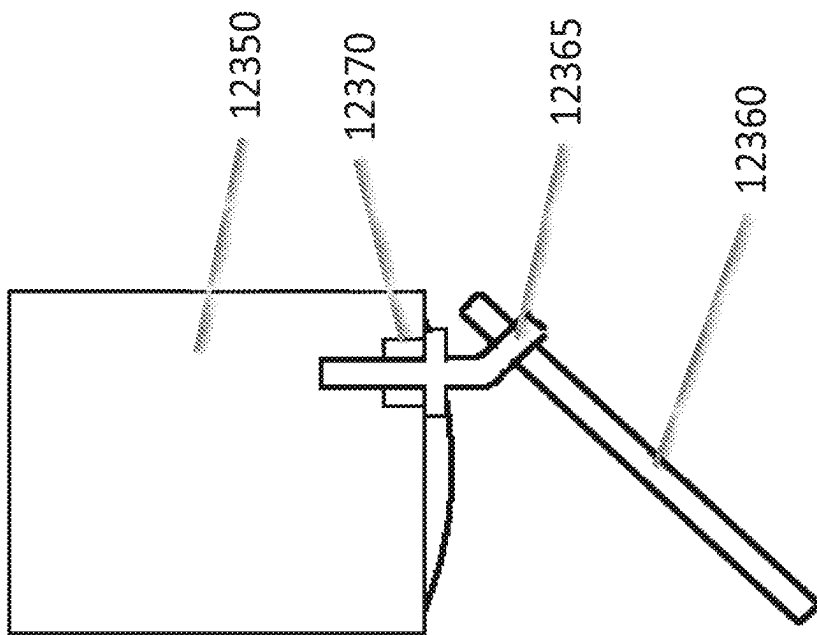


Figure 123

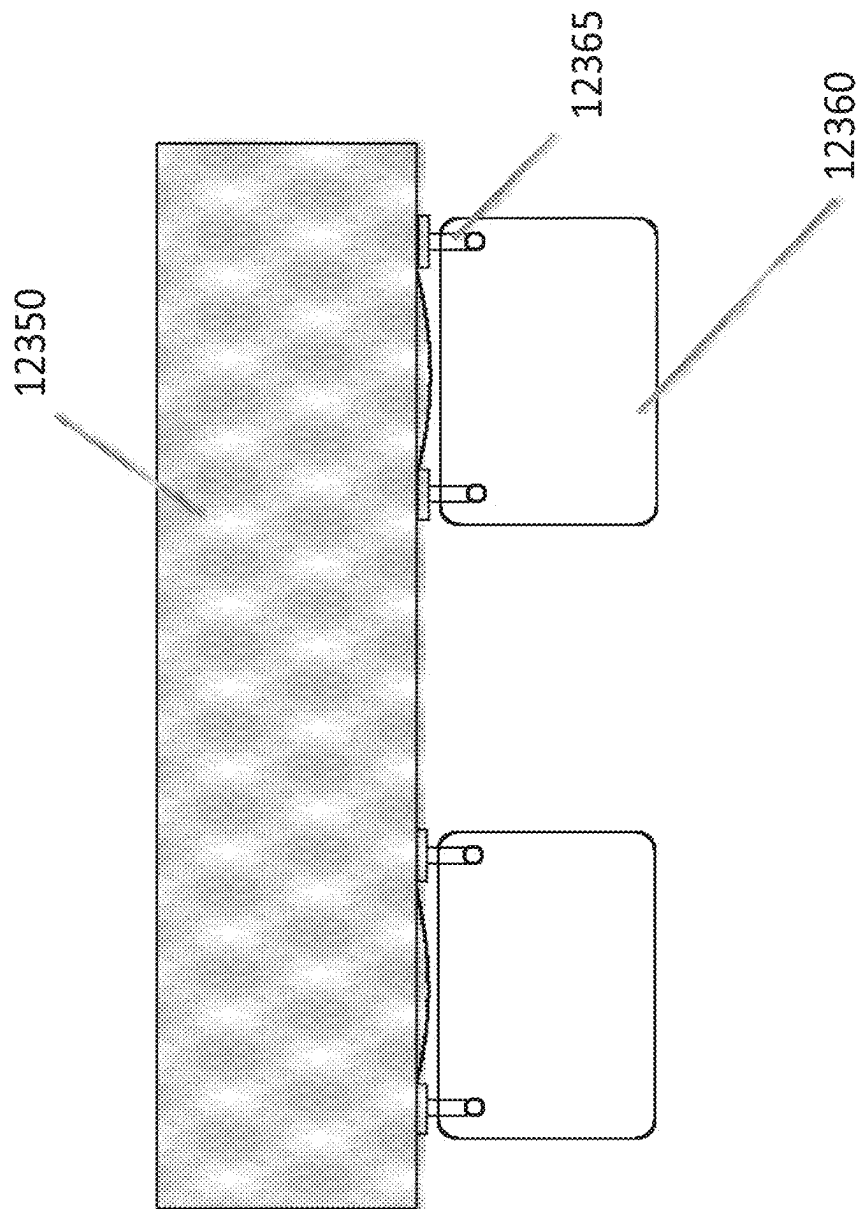


Figure 124

1

**SEE-THROUGH COMPUTER DISPLAY
SYSTEMS****CROSS REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation of U.S. Non-Provisional application Ser. No. 17/389,174, filed on Jul. 29, 2021, which is a continuation of U.S. Non-Provisional application Ser. No. 16/538,720, filed on Aug. 12, 2019, now U.S. Pat. No. 11,126,003, issued on Sep. 21, 2021, which is a continuation of U.S. Non-Provisional application Ser. No. 15/488,916, filed on Apr. 17, 2017, now U.S. Pat. No. 10,379,365, issued on Aug. 13, 2019, which is a continuation of U.S. Non-Provisional application Ser. No. 14/742,249, filed on Jun. 17, 2015, now U.S. Pat. No. 9,746,676, issued on Aug. 29, 2017.

U.S. Non-Provisional application Ser. No. 14/742,249 is a continuation of U.S. Non-Provisional application Ser. No. 14/589,713, filed on Jan. 5, 2015, now U.S. Pat. No. 9,529,195, issued on Dec. 27, 2016.

U.S. Non-Provisional application Ser. No. 14/589,713 is a continuation-in-part of the following:

U.S. Non-Provisional application Ser. No. 14/160,377, filed Jan. 21, 2014, now abandoned.

U.S. Non-Provisional application Ser. No. 14/163,646, filed Jan. 24, 2014, now U.S. Pat. No. 9,400,390, issued on Jul. 26, 2016.

U.S. Non-Provisional application Ser. No. 14/172,901, filed Feb. 4, 2014, now abandoned.

U.S. Non-Provisional application Ser. No. 14/181,459, filed Feb. 14, 2014, now U.S. Pat. No. 9,715,112, issued on Jul. 25, 2017, which is a continuation-in-part of U.S. Non-Provisional application Ser. No. 14/178,047, filed Feb. 11, 2014, now U.S. Pat. No. 9,229,233, issued on Jan. 5, 2016.

U.S. Non-Provisional application Ser. No. 14/589,713 is also a continuation-in-part of the following:

U.S. Non-Provisional application Ser. No. 14/216,175, filed on Mar. 17, 2014, now U.S. Pat. No. 9,298,007, issued on Mar. 29, 2016.

U.S. Non-Provisional application Ser. No. 14/296,699, filed Jun. 5, 2014, now U.S. Pat. No. 9,841,599, issued on Dec. 12, 2017.

U.S. Non-Provisional application Ser. No. 14/325,991, filed Jul. 8, 2014, now U.S. Pat. No. 9,366,867, issued on Jun. 14, 2016.

U.S. Non-Provisional application Ser. No. 14/489,706, filed Sep. 18, 2014, now abandoned.

U.S. Non-Provisional application Ser. No. 14/498,765, filed Sep. 26, 2014, now U.S. Pat. No. 9,366,868, issued on Jun. 14, 2016.

U.S. Non-Provisional application Ser. No. 14/504,723, filed Oct. 2, 2014, now U.S. Pat. No. 9,671,613, issued on Jun. 6, 2017.

U.S. Non-Provisional application Ser. No. 14/559,126, filed Dec. 3, 2014, now U.S. Pat. No. 10,684,687, issued on Jun. 16, 2020.

U.S. Non-Provisional application Ser. No. 14/254,253, filed Apr. 16, 2014, now U.S. Pat. No. 9,952,664, issued on Apr. 24, 2018.

U.S. Non-Provisional application Ser. No. 14/561,146, filed Dec. 4, 2014, now U.S. Pat. No. 9,594,246, issued on Mar. 14, 2017.

U.S. Non-Provisional application Ser. No. 14/554,044, filed Nov. 26, 2014, now U.S. Pat. No. 9,448,409, issued on Sep. 20, 2016.

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All of the above applications are incorporated herein by reference in their entirety.

BACKGROUND**Field of the Invention**

This invention relates to see-through computer display systems.

Description of Related Art

Head mounted displays (HMD) and particularly HMDs that provide a see-through view of the environment are sensitive to the effects of stray light. Where, stray light includes light that is not intended to be included in the displayed image, including light that is scattered or inadvertently reflected from surfaces within the optics of the HMD. This stray light reduces the sharpness and contrast of the displayed image in the HMD. In addition, the stray light causes the black areas of the displayed image to be gray and this effect adversely affects the see-through view of the environment because the see-through view is produced by the combination of the light from the environment and the light from the displayed image, which is in the best case the black portions of displayed image. As a result, it is important to reduce stray light within the optics so that very dark black areas and high contrast can be provided in the images displayed by the HMDs.

SUMMARY

Aspects of the present invention relate to methods and systems for the see-through computer display systems.

These and other systems, methods, objects, features, and advantages of the present invention will be apparent to those skilled in the art from the following detailed description of the preferred embodiment and the drawings. All documents mentioned herein are hereby incorporated in their entirety by reference.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments are described with reference to the following Figures. The same numbers may be used throughout to reference like features and components that are shown in the Figures:

FIG. 1 illustrates a head worn computing system in accordance with the principles of the present invention.

FIG. 2 illustrates a head worn computing system with optical system in accordance with the principles of the present invention.

FIG. 3a illustrates a large prior art optical arrangement.

FIG. 3b illustrates an upper optical module in accordance with the principles of the present invention.

FIG. 4 illustrates an upper optical module in accordance with the principles of the present invention.

FIG. 4a illustrates an upper optical module in accordance with the principles of the present invention.

FIG. 4b illustrates an upper optical module in accordance with the principles of the present invention.

FIG. 5 illustrates an upper optical module in accordance with the principles of the present invention.

FIG. 5a illustrates an upper optical module in accordance with the principles of the present invention.

FIG. 5b illustrates an upper optical module and dark light trap according to the principles of the present invention.

FIG. 5c illustrates an upper optical module and dark light trap according to the principles of the present invention.

FIG. 5d illustrates an upper optical module and dark light trap according to the principles of the present invention.

FIG. 5e illustrates an upper optical module and dark light trap according to the principles of the present invention.

FIG. 6 illustrates upper and lower optical modules in accordance with the principles of the present invention.

FIG. 7 illustrates angles of combiner elements in accordance with the principles of the present invention.

FIG. 8 illustrates upper and lower optical modules in accordance with the principles of the present invention.

FIG. 8a illustrates upper and lower optical modules in accordance with the principles of the present invention.

FIG. 8b illustrates upper and lower optical modules in accordance with the principles of the present invention.

FIG. 8c illustrates upper and lower optical modules in accordance with the principles of the present invention.

FIG. 9 illustrates an eye imaging system in accordance with the principles of the present invention.

FIG. 10 illustrates a light source in accordance with the principles of the present invention.

FIG. 10a illustrates a back lighting system in accordance with the principles of the present invention.

FIG. 10b illustrates a back lighting system in accordance with the principles of the present invention.

FIGS. 11a to 11d illustrate light source and filters in accordance with the principles of the present invention.

FIGS. 12a to 12c illustrate light source and quantum dot systems in accordance with the principles of the present invention.

FIGS. 13a to 13c illustrate peripheral lighting systems in accordance with the principles of the present invention.

FIGS. 14a to 14c illustrate a light suppression systems in accordance with the principles of the present invention.

FIG. 15 illustrates an external user interface in accordance with the principles of the present invention.

FIG. 16a to 16c illustrate distance control systems in accordance with the principles of the present invention.

FIG. 17a to 17c illustrate force interpretation systems in accordance with the principles of the present invention.

FIG. 18a to 18c illustrate user interface mode selection systems in accordance with the principles of the present invention.

FIG. 19 illustrates interaction systems in accordance with the principles of the present invention.

FIG. 20 illustrates external user interfaces in accordance with the principles of the present invention.

FIG. 21 illustrates mD trace representations presented in accordance with the principles of the present invention.

FIG. 22 illustrates mD trace representations presented in accordance with the principles of the present invention.

FIG. 23 illustrates an mD scanned environment in accordance with the principles of the present invention.

FIG. 23a illustrates mD trace representations presented in accordance with the principles of the present invention.

FIG. 24 illustrates a stray light suppression technology in accordance with the principles of the present invention.

FIG. 25 illustrates a stray light suppression technology in accordance with the principles of the present invention.

FIG. 26 illustrates a stray light suppression technology in accordance with the principles of the present invention.

FIG. 27 illustrates a stray light suppression technology in accordance with the principles of the present invention.

FIGS. 28a to 28c illustrate DLP mirror angles.

FIGS. 29 to 33 illustrate eye imaging systems according to the principles of the present invention.

FIGS. 34 and 34a illustrate structured eye lighting systems according to the principles of the present invention.

FIG. 35 illustrates eye glint in the prediction of eye direction analysis in accordance with the principles of the present invention.

FIG. 36a illustrates eye characteristics that may be used in personal identification through analysis of a system according to the principles of the present invention.

FIG. 36b illustrates a digital content presentation reflection off of the wearer's eye that may be analyzed in accordance with the principles of the present invention.

FIG. 37 illustrates eye imaging along various virtual target lines and various focal planes in accordance with the principles of the present invention.

FIG. 38 illustrates content control with respect to eye movement based on eye imaging in accordance with the principles of the present invention.

FIG. 39 illustrates eye imaging and eye convergence in accordance with the principles of the present invention.

FIG. 40 illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

FIG. 41 illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

FIG. 42 illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

FIG. 43 illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

FIG. 44 illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

FIG. 45 illustrates various headings over time in an example.

FIG. 46 illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

FIG. 47 illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

FIG. 48 illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

FIG. 49 illustrates content position dependent on sensor feedback in accordance with the principles of the present invention.

FIG. 50 illustrates light impinging an eye in accordance with the principles of the present invention.

FIG. 51 illustrates a view of an eye in accordance with the principles of the present invention.

FIGS. 52a and 52b illustrate views of an eye with a structured light pattern in accordance with the principles of the present invention.

FIG. 53 illustrates an optics module in accordance with the principles of the present invention.

FIG. 54 illustrates an optics module in accordance with the principles of the present invention.

FIG. 55 shows a series of example spectrum for a variety of controlled substances as measured using a form of infrared spectroscopy.

FIG. 56 shows an infrared absorbance spectrum for glucose.

FIG. 57 illustrates a scene where a person is walking with a HWC mounted on his head.

FIG. 58 illustrates a system for receiving, developing and using movement heading, sight heading, eye heading and/or persistence information from HWC(s).

FIG. 59 illustrates a presentation technology in accordance with the principles of the present invention.

FIG. 60 illustrates a presentation technology in accordance with the principles of the present invention.

FIG. 61 illustrates a presentation technology in accordance with the principles of the present invention.

FIG. 62 illustrates a presentation technology in accordance with the principles of the present invention.

FIG. 63 illustrates a presentation technology in accordance with the principles of the present invention.

FIG. 64 illustrates a presentation technology in accordance with the principles of the present invention.

FIG. 65 illustrates a presentation technology in accordance with the principles of the present invention.

FIG. 66 illustrates a presentation technology in accordance with the principles of the present invention.

FIG. 67 illustrates an optical configuration in accordance with the principles of the present invention.

FIG. 68 illustrates an optical configuration in accordance with the principles of the present invention.

FIG. 69 illustrates an optical configuration in accordance with the principles of the present invention.

FIG. 70 illustrates an optical configuration in accordance with the principles of the present invention.

FIG. 71 illustrates an optical configuration in accordance with the principles of the present invention.

FIG. 72 illustrates an optical element in accordance with the principles of the present invention.

FIG. 73 illustrates an optical element in accordance with the principles of the present invention.

FIG. 74 illustrates an optical element in accordance with the principles of the present invention.

FIG. 75 illustrates an optical element in accordance with the principles of the present invention.

FIG. 76 illustrates an optical element in a see-through computer display in accordance with the principles of the present invention.

FIG. 77 illustrates an optical element in accordance with the principles of the present invention.

FIG. 78 illustrates an optical element in accordance with the principles of the present invention.

FIG. 79a illustrates a schematic of an upper optic in accordance with the principles of the present invention.

FIG. 79 illustrates a schematic of an upper optic in accordance with the principles of the present invention.

FIG. 80 illustrates a stray light control technology in accordance with the principles of the present invention.

FIGS. 81a and 81b illustrate a display with a gap and masked technologies in accordance with the principles of the present invention.

FIG. 82 illustrates an upper module with a trim polarizer in accordance with the principles of the present invention.

FIG. 83 illustrates an optical system with a laminated multiple polarizer film in accordance with the principles of the present invention.

FIGS. 84a and 84b illustrate partially reflective layers in accordance with the principles of the present invention.

FIG. 84c illustrates a laminated multiple polarizer with a complex curve in accordance with the principles of the present invention.

FIG. 84d illustrates a laminated multiple polarizer with a curve in accordance with the principles of the present invention.

FIG. 85 illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

FIG. 86 illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

FIG. 87 illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

FIG. 88 illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

FIG. 89 illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

FIG. 90 illustrates an optical system adapted for a head-mounted display in accordance with the principles of the present invention.

FIG. 91 illustrates an optical system in accordance with the principles of the present invention.

FIG. 92 illustrates an optical system in accordance with the principles of the present invention.

FIG. 93 illustrates an optical system in accordance with the principles of the present invention.

FIG. 94 illustrates an optical system in accordance with the principles of the present invention.

FIG. 95 illustrates an optical system in accordance with the principles of the present invention.

FIG. 96 illustrates an optical system in accordance with the principles of the present invention.

FIG. 97 illustrates an optical system in accordance with the principles of the present invention.

FIG. 98 illustrates an optical system in accordance with the principles of the present invention.

FIG. 99 illustrates an optical system in accordance with the principles of the present invention.

FIG. 100 illustrates an optical system in accordance with the principles of the present invention.

FIG. 101 illustrates an optical system in accordance with the principles of the present invention.

FIG. 102 illustrates an optical system in accordance with the principles of the present invention.

FIGS. 103, 103a and 103b illustrate optical systems in accordance with the principles of the present invention.

FIG. 104 illustrates an optical system in accordance with the principles of the present invention.

FIG. 105 illustrates a blocking optic in accordance with the principles of the present invention.

FIGS. 106a, 106b, and 106c illustrate a blocking optic system in accordance with the principles of the present invention.

FIG. 107 illustrates a full color image in accordance with the principles of the present invention.

FIGS. 108A and 108B illustrate color breakup management in accordance with the principles of the present invention.

FIG. 109 illustrates timing sequences in accordance with the principles of the present invention.

FIG. 110 illustrates timing sequences in accordance with the principles of the present invention.

FIGS. 111a and 111b illustrate sequentially displayed images in accordance with the principles of the present invention.

FIG. 112 illustrates a see-through display with rotated components in accordance with the principles of the present invention.

FIG. 113 illustrates an optics module with twisted reflective surfaces in accordance with the principles of the present invention.

FIG. 114 illustrates PCB and see-through optics module positions within a glasses form factor in accordance with the principles of the present invention.

FIG. 115 illustrates PCB and see-through optics module positions within a glasses form factor in accordance with the principles of the present invention.

FIG. 116 illustrates PCB and see-through optics module positions within a glasses form factor in accordance with the principles of the present invention.

FIG. 117 illustrates a user interface in accordance with the principles of the present invention.

FIG. 118 illustrates a user interface in accordance with the principles of the present invention.

FIG. 119 illustrates a lens arrangement in accordance with the principles of the present invention.

FIGS. 120 and 121 illustrate eye imaging systems in accordance with the principles of the present invention.

FIG. 122 illustrates an identification process in accordance with the principles of the present invention.

FIGS. 123 and 124 illustrate combiner assemblies in accordance with the principles of the present invention.

While the invention has been described in connection with certain preferred embodiments, other embodiments would be understood by one of ordinary skill in the art and are encompassed herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

Aspects of the present invention relate to head-worn computing ("HWC") systems. HWC involves, in some instances, a system that mimics the appearance of head-worn glasses or sunglasses. The glasses may be a fully developed computing platform, such as including computer displays presented in each of the lenses of the glasses to the eyes of the user. In embodiments, the lenses and displays may be configured to allow a person wearing the glasses to see the environment through the lenses while also seeing, simultaneously, digital imagery, which forms an overlaid image that is perceived by the person as a digitally augmented image of the environment, or augmented reality ("AR").

HWC involves more than just placing a computing system on a person's head. The system may need to be designed as a lightweight, compact and fully functional computer display, such as wherein the computer display includes a high resolution digital display that provides a high level of emersion comprised of the displayed digital content and the see-through view of the environmental surroundings. User interfaces and control systems suited to the HWC device may be required that are unlike those used for a more conventional computer such as a laptop. For the HWC and associated systems to be most effective, the glasses may be equipped with sensors to determine environmental conditions, geographic location, relative positioning to other points of interest, objects identified by imaging and movement by the user or other users in a connected group, and the like. The HWC may then change the mode of operation to match the conditions, location, positioning, movements, and the like, in a method generally referred to as a contextually aware HWC. The glasses also may need to be connected, wirelessly or otherwise, to other systems either locally or through a network. Controlling the glasses may be achieved through the use of an external device, automatically through contextually gathered information, through user gestures

captured by the glasses sensors, and the like. Each technique may be further refined depending on the software application being used in the glasses. The glasses may further be used to control or coordinate with external devices that are associated with the glasses.

Referring to FIG. 1, an overview of the HWC system 100 is presented. As shown, the HWC system 100 comprises a HWC 102, which in this instance is configured as glasses to be worn on the head with sensors such that the HWC 102 is aware of the objects and conditions in the environment 114. In this instance, the HWC 102 also receives and interprets control inputs such as gestures and movements 116. The HWC 102 may communicate with external user interfaces 104. The external user interfaces 104 may provide a physical user interface to take control instructions from a user of the HWC 102 and the external user interfaces 104 and the HWC 102 may communicate bi-directionally to affect the user's command and provide feedback to the external device 108. The HWC 102 may also communicate bi-directionally with externally controlled or coordinated local devices 108. For example, an external user interface 104 may be used in connection with the HWC 102 to control an externally controlled or coordinated local device 108. The externally controlled or coordinated local device 108 may provide feedback to the HWC 102 and a customized GUI may be presented in the HWC 102 based on the type of device or specifically identified device 108. The HWC 102 may also interact with remote devices and information sources 112 through a network connection 110. Again, the external user interface 104 may be used in connection with the HWC 102 to control or otherwise interact with any of the remote devices 108 and information sources 112 in a similar way as when the external user interfaces 104 are used to control or otherwise interact with the externally controlled or coordinated local devices 108. Similarly, HWC 102 may interpret gestures 116 (e.g. captured from forward, downward, upward, rearward facing sensors such as camera(s), range finders, IR sensors, etc.) or environmental conditions sensed in the environment 114 to control either local or remote devices 108 or 112.

We will now describe each of the main elements depicted on FIG. 1 in more detail; however, these descriptions are intended to provide general guidance and should not be construed as limiting. Additional description of each element may also be further described herein.

The HWC 102 is a computing platform intended to be worn on a person's head. The HWC 102 may take many different forms to fit many different functional requirements. In some situations, the HWC 102 will be designed in the form of conventional glasses. The glasses may or may not have active computer graphics displays. In situations where the HWC 102 has integrated computer displays the displays may be configured as see-through displays such that the digital imagery can be overlaid with respect to the user's view of the environment 114. There are a number of see-through optical designs that may be used, including ones that have a reflective display (e.g. LCoS, DLP), emissive displays (e.g. OLED, LED), hologram, TIR waveguides, and the like. In embodiments, lighting systems used in connection with the display optics may be solid state lighting systems, such as LED, OLED, quantum dot, quantum dot LED, etc. In addition, the optical configuration may be monocular or binocular. It may also include vision corrective optical components. In embodiments, the optics may be packaged as contact lenses. In other embodiments, the HWC 102 may be in the form of a helmet with a see-through shield, sunglasses, safety glasses, goggles, a mask, fire

helmet with see-through shield, police helmet with see through shield, military helmet with see-through shield, utility form customized to a certain work task (e.g. inventory control, logistics, repair, maintenance, etc.), and the like.

The HWC **102** may also have a number of integrated computing facilities, such as an integrated processor, integrated power management, communication structures (e.g. cell net, WiFi, Bluetooth, local area connections, mesh connections, remote connections (e.g. client server, etc.)), and the like. The HWC **102** may also have a number of positional awareness sensors, such as GPS, electronic compass, altimeter, tilt sensor, IMU, and the like. It may also have other sensors such as a camera, rangefinder, hyperspectral camera, Geiger counter, microphone, spectral illumination detector, temperature sensor, chemical sensor, biological sensor, moisture sensor, ultrasonic sensor, and the like.

The HWC **102** may also have integrated control technologies. The integrated control technologies may be contextual based control, passive control, active control, user control, and the like. For example, the HWC **102** may have an integrated sensor (e.g. camera) that captures user hand or body gestures **116** such that the integrated processing system can interpret the gestures and generate control commands for the HWC **102**. In another example, the HWC **102** may have sensors that detect movement (e.g. a nod, head shake, and the like) including accelerometers, gyros and other inertial measurements, where the integrated processor may interpret the movement and generate a control command in response. The HWC **102** may also automatically control itself based on measured or perceived environmental conditions. For example, if it is bright in the environment the HWC **102** may increase the brightness or contrast of the displayed image. In embodiments, the integrated control technologies may be mounted on the HWC **102** such that a user can interact with it directly. For example, the HWC **102** may have a button(s), touch capacitive interface, and the like.

As described herein, the HWC **102** may be in communication with external user interfaces **104**. The external user interfaces may come in many different forms. For example, a cell phone screen may be adapted to take user input for control of an aspect of the HWC **102**. The external user interface may be a dedicated UI, such as a keyboard, touch surface, button(s), joy stick, and the like. In embodiments, the external controller may be integrated into another device such as a ring, watch, bike, car, and the like. In each case, the external user interface **104** may include sensors (e.g. IMU, accelerometers, compass, altimeter, and the like) to provide additional input for controlling the HWC **102**.

As described herein, the HWC **102** may control or coordinate with other local devices **108**. The external devices **108** may be an audio device, visual device, vehicle, cell phone, computer, and the like. For instance, the local external device **108** may be another HWC **102**, where information may then be exchanged between the separate HWCs **108**.

Similar to the way the HWC **102** may control or coordinate with local devices **106**, the HWC **102** may control or coordinate with remote devices **112**, such as the HWC **102** communicating with the remote devices **112** through a network **110**. Again, the form of the remote device **112** may have many forms. Included in these forms is another HWC **102**. For example, each HWC **102** may communicate its GPS position such that all the HWCs **102** know where all of HWC **102** are located.

FIG. 2 illustrates a HWC **102** with an optical system that includes an upper optical module **202** and a lower optical

module **204**. While the upper and lower optical modules **202** and **204** will generally be described as separate modules, it should be understood that this is illustrative only and the present invention includes other physical configurations, such as that when the two modules are combined into a single module or where the elements making up the two modules are configured into more than two modules. In embodiments, the upper module **202** includes a computer controlled display (e.g. LCoS, DLP, OLED, etc.) and image light delivery optics. In embodiments, the lower module includes eye delivery optics that are configured to receive the upper module's image light and deliver the image light to the eye of a wearer of the HWC. In FIG. 2, it should be noted that while the upper and lower optical modules **202** and **204** are illustrated in one side of the HWC such that image light can be delivered to one eye of the wearer, that it is envisioned by the present invention that embodiments will contain two image light delivery systems, one for each eye.

FIG. 3b illustrates an upper optical module **202** in accordance with the principles of the present invention. In this embodiment, the upper optical module **202** includes a DLP (also known as DMD or digital micromirror device) computer operated display **304** which includes pixels comprised of rotatable mirrors (such as, for example, the DLP3000 available from Texas Instruments), polarized light source **302**, $\frac{1}{4}$ wave retarder film **308**, reflective polarizer **310** and a field lens **312**. The polarized light source **302** provides substantially uniform polarized light that is generally directed towards the reflective polarizer **310**. The reflective polarizer reflects light of one polarization state (e.g. S polarized light) and transmits light of the other polarization state (e.g. P polarized light). The polarized light source **302** and the reflective polarizer **310** are oriented so that the polarized light from the polarized light source **302** is reflected generally towards the DLP **304**. The light then passes through the $\frac{1}{4}$ wave film **308** once before illuminating the pixels of the DLP **304** and then again after being reflected by the pixels of the DLP **304**. In passing through the $\frac{1}{4}$ wave film **308** twice, the light is converted from one polarization state to the other polarization state (e.g. the light is converted from S to P polarized light). The light then passes through the reflective polarizer **310**. In the event that the DLP pixel(s) are in the "on" state (i.e. the mirrors are positioned to reflect light towards the field lens **312**, the "on" pixels reflect the light generally along the optical axis and into the field lens **312**. This light that is reflected by "on" pixels and which is directed generally along the optical axis of the field lens **312** will be referred to as image light **316**. The image light **316** then passes through the field lens to be used by a lower optical module **204**.

The light that is provided by the polarized light source **302**, which is subsequently reflected by the reflective polarizer **310** before it reflects from the DLP **304**, will generally be referred to as illumination light. The light that is reflected by the "off" pixels of the DLP **304** is reflected at a different angle than the light reflected by the "on" pixels, so that the light from the "off" pixels is generally directed away from the optical axis of the field lens **312** and toward the side of the upper optical module **202** as shown in FIG. 3. The light that is reflected by the "off" pixels of the DLP **304** will be referred to as dark state light **314**.

The DLP **304** operates as a computer controlled display and is generally thought of as a MEMs device. The DLP pixels are comprised of small mirrors that can be directed. The mirrors generally flip from one angle to another angle. The two angles are generally referred to as states. When light

is used to illuminate the DLP the mirrors will reflect the light in a direction depending on the state. In embodiments herein, we generally refer to the two states as “on” and “off,” which is intended to depict the condition of a display pixel. “On” pixels will be seen by a viewer of the display as emitting light because the light is directed along the optical axis and into the field lens and the associated remainder of the display system. “Off” pixels will be seen by a viewer of the display as not emitting light because the light from these pixels is directed to the side of the optical housing and into a light trap or light dump where the light is absorbed. The pattern of “on” and “off” pixels produces image light that is perceived by a viewer of the display as a computer generated image. Full color images can be presented to a user by sequentially providing illumination light with complementary colors such as red, green and blue. Where the sequence is presented in a recurring cycle that is faster than the user can perceive as separate images and as a result the user perceives a full color image comprised of the sum of the sequential images. Bright pixels in the image are provided by pixels that remain in the “on” state for the entire time of the cycle, while dimmer pixels in the image are provided by pixels that switch between the “on” state and “off” state within the time of the cycle, or frame time when in a video sequence of images.

FIG. 3a shows an illustration of a system for a DLP 304 in which the unpolarized light source 350 is pointed directly at the DLP 304. In this case, the angle required for the illumination light is such that the field lens 352 must be positioned substantially distant from the DLP 304 to avoid the illumination light from being clipped by the field lens 352. The large distance between the field lens 352 and the DLP 304 along with the straight path of the dark state light 354, means that the light trap for the dark state light 354 is also located at a substantial distance from the DLP. For these reasons, this configuration is larger in size compared to the upper optics module 202 of the preferred embodiments.

The configuration illustrated in FIG. 3b can be lightweight and compact such that it fits into a small portion of a HWC. For example, the upper modules 202 illustrated herein can be physically adapted to mount in an upper frame of a HWC such that the image light can be directed into a lower optical module 204 for presentation of digital content to a wearer's eye. The package of components that combine to generate the image light (i.e. the polarized light source 302, DLP 304, reflective polarizer 310 and ¼ wave film 308) is very light and is compact. The height of the system, excluding the field lens, may be less than 8 mm. The width (i.e. from front to back) may be less than 8 mm. The weight may be less than 2 grams. The compactness of this upper optical module 202 allows for a compact mechanical design of the HWC and the light weight nature of these embodiments help make the HWC lightweight to provide for a HWC that is comfortable for a wearer of the HWC.

The configuration illustrated in FIG. 3b can produce sharp contrast, high brightness and deep blacks, especially when compared to LCD or LCoS displays used in HWC. The “on” and “off” states of the DLP provide for a strong differentiator in the light reflection path representing an “on” pixel and an “off” pixel. As will be discussed in more detail below, the dark state light from the “off” pixel reflections can be managed to reduce stray light in the display system to produce images with high contrast.

FIG. 4 illustrates another embodiment of an upper optical module 202 in accordance with the principles of the present invention. This embodiment includes a light source 404, but in this case, the light source can provide unpolarized illu-

mination light. The illumination light from the light source 404 is directed into a TIR wedge 418 such that the illumination light is incident on an internal surface of the TIR wedge 418 (shown as the angled lower surface of the TIR wedge 418 in FIG. 4) at an angle that is beyond the critical angle as defined by Eqn 1.

$$\text{Critical angle} = \arcsin(1/n)$$

Eqn 1

Where the critical angle is the angle beyond which the illumination light is reflected from the internal surface when the internal surface comprises an interface from a solid with a higher refractive index (n) to air with a refractive index of 1 (e.g. for an interface of acrylic, with a refractive index of n=1.5, to air, the critical angle is 41.8 degrees; for an interface of polycarbonate, with a refractive index of n=1.59, to air the critical angle is 38.9 degrees). Consequently, the TIR wedge 418 is associated with a thin air gap 408 along the internal surface to create an interface between a solid with a higher refractive index and air. By choosing the angle of the light source 404 relative to the DLP 402 in correspondence to the angle of the internal surface of the TIR wedge 418, illumination light is turned toward the DLP 402 at an angle suitable for providing image light 414 as reflected from “on” pixels. Wherein, the illumination light is provided to the DLP 402 at approximately twice the angle of the pixel mirrors in the DLP 402 that are in the “on” state, such that after reflecting from the pixel mirrors, the image light 414 is directed generally along the optical axis of the field lens. Depending on the state of the DLP pixels, the illumination light from “on” pixels may be reflected as image light 414 which is directed towards a field lens and a lower optical module 204, while illumination light reflected from “off” pixels (generally referred to herein as “dark” state light, “off” pixel light or “off” state light) 410 is directed in a separate direction, which may be trapped and not used for the image that is ultimately presented to the wearer's eye.

The light trap for the dark state light 410 may be located along the optical axis defined by the direction of the dark state light 410 and in the side of the housing, with the function of absorbing the dark state light. To this end, the light trap may be comprised of an area outside of the cone of image light 414 from the “on” pixels. The light trap is typically made up of materials that absorb light including coatings of black paints or other light absorbing materials to prevent light scattering from the dark state light degrading the image perceived by the user. In addition, the light trap may be recessed into the wall of the housing or include masks or guards to block scattered light and prevent the light trap from being viewed adjacent to the displayed image.

The embodiment of FIG. 4 also includes a corrective wedge 420 to correct the effect of refraction of the image light 414 as it exits the TIR wedge 418. By including the corrective wedge 420 and providing a thin air gap 408 (e.g. 25 micron), the image light from the “on” pixels can be maintained generally in a direction along the optical axis of the field lens (i.e. the same direction as that defined by the image light 414) so it passes into the field lens and the lower optical module 204. As shown in FIG. 4, the image light 414 from the “on” pixels exits the corrective wedge 420 generally perpendicular to the surface of the corrective wedge 420 while the dark state light exits at an oblique angle. As a result, the direction of the image light 414 from the “on” pixels is largely unaffected by refraction as it exits from the surface of the corrective wedge 420. In contrast, the dark state light 410 is substantially changed in direction by refraction when the dark state light 410 exits the corrective wedge 420.

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The embodiment illustrated in FIG. 4 has the similar advantages of those discussed in connection with the embodiment of FIG. 3b. The dimensions and weight of the upper module 202 depicted in FIG. 4 may be approximately 8x8 mm with a weight of less than 3 grams. A difference in overall performance between the configuration illustrated in FIGS. 3b and the configuration illustrated in FIG. 4 is that the embodiment of FIG. 4 doesn't require the use of polarized light as supplied by the light source 404. This can be an advantage in some situations as will be discussed in more detail below (e.g. increased see-through transparency of the HWC optics from the user's perspective). Polarized light may be used in connection with the embodiment depicted in FIG. 4, in embodiments. An additional advantage of the embodiment of FIG. 4 compared to the embodiment shown in FIG. 3b is that the dark state light (shown as DLP off light 410) is directed at a steeper angle away from the optical axis of the image light 414 due to the added refraction encountered when the dark state light 410 exits the corrective wedge 420. This steeper angle of the dark state light 410 allows for the light trap to be positioned closer to the DLP 402 so that the overall size of the upper module 202 can be reduced. The light trap can also be made larger since the light trap doesn't interfere with the field lens, thereby the efficiency of the light trap can be increased and as a result, stray light can be reduced and the contrast of the image perceived by the user can be increased. FIG. 4a illustrates the embodiment described in connection with FIG. 4 with an example set of corresponding angles at the various surfaces with the reflected angles of a ray of light passing through the upper optical module 202. In this example, the DLP mirrors are provided at 17 degrees to the surface of the DLP device. The angles of the TIR wedge are selected in correspondence to one another to provide TIR reflected illumination light at the correct angle for the DLP mirrors while allowing the image light and dark state light to pass through the thin air gap, various combinations of angles are possible to achieve this.

FIG. 5 illustrates yet another embodiment of an upper optical module 202 in accordance with the principles of the present invention. As with the embodiment shown in FIG. 4, the embodiment shown in FIG. 5 does not require the use of polarized light. Polarized light may be used in connection with this embodiment, but it is not required. The optical module 202 depicted in FIG. 5 is similar to that presented in connection with FIG. 4; however, the embodiment of FIG. 5 includes an off light redirection wedge 502. As can be seen from the illustration, the off light redirection wedge 502 allows the image light 414 to continue generally along the optical axis toward the field lens and into the lower optical module 204 (as illustrated). However, the off light 504 is redirected substantially toward the side of the corrective wedge 420 where it passes into the light trap. This configuration may allow further height compactness in the HWC because the light trap (not illustrated) that is intended to absorb the off light 504 can be positioned laterally adjacent the upper optical module 202 as opposed to below it. In the embodiment depicted in FIG. 5 there is a thin air gap between the TIR wedge 418 and the corrective wedge 420 (similar to the embodiment of FIG. 4). There is also a thin air gap between the corrective wedge 420 and the off light redirection wedge 502. There may be HWC mechanical configurations that warrant the positioning of a light trap for the dark state light elsewhere and the illustration depicted in FIG. 5 should be considered illustrative of the concept that the off light can be redirected to create compactness of the overall HWC. FIG. 5a illustrates an example of the embodi-

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ment described in connection with FIG. 5 with the addition of more details on the relative angles at the various surfaces and a light ray trace for image light and a light ray trace for dark light are shown as it passes through the upper optical module 202. Again, various combinations of angles are possible.

FIG. 4b shows an illustration of a further embodiment in which a solid transparent matched set of wedges 456 is provided with a reflective polarizer 450 at the interface between the wedges. Wherein the interface between the wedges in the wedge set 456 is provided at an angle so that illumination light 452 from the polarized light source 458 is reflected at the proper angle (e.g. 34 degrees for a 17 degree DLP mirror) for the DLP mirror "on" state so that the reflected image light 414 is provided along the optical axis of the field lens. The general geometry of the wedges in the wedge set 456 is similar to that shown in FIGS. 4 and 4a. A quarter wave film 454 is provided on the DLP 402 surface so that the illumination light 452 is one polarization state (e.g. S polarization state) while in passing through the quarter wave film 454, reflecting from the DLP mirror and passing back through the quarter wave film 454, the image light 414 is converted to the other polarization state (e.g. P polarization state). The reflective polarizer is oriented such that the illumination light 452 with its polarization state is reflected and the image light 414 with its other polarization state is transmitted. Since the dark state light from the "off" pixels 410 also passes through the quarter wave film 454 twice, it is also the other polarization state (e.g. P polarization state) so that it is transmitted by the reflective polarizer 450.

The angles of the faces of the wedge set 450 correspond to the needed angles to provide illumination light 452 at the angle needed by the DLP mirrors when in the "on" state so that the reflected image light 414 is reflected from the DLP along the optical axis of the field lens. The wedge set 456 provides an interior interface where a reflective polarizer film can be located to redirect the illumination light 452 toward the mirrors of the DLP 402. The wedge set also provides a matched wedge on the opposite side of the reflective polarizer 450 so that the image light 414 from the "on" pixels exits the wedge set 450 substantially perpendicular to the exit surface, while the dark state light from the "off" pixels 410 exits at an oblique angle to the exit surface. As a result, the image light 414 is substantially unrefracted upon exiting the wedge set 456, while the dark state light from the "off" pixels 410 is substantially refracted upon exiting the wedge set 456 as shown in FIG. 4b.

By providing a solid transparent matched wedge set, the flatness of the interface is reduced, because variations in the flatness have a negligible effect as long as they are within the cone angle of the illuminating light 452. Which can be f #2.2 with a 26 degree cone angle. In a preferred embodiment, the reflective polarizer is bonded between the matched internal surfaces of the wedge set 456 using an optical adhesive so that Fresnel reflections at the interfaces on either side of the reflective polarizer 450 are reduced. The optical adhesive can be matched in refractive index to the material of the wedge set 456 and the pieces of the wedge set 456 can be all made from the same material such as BK7 glass or cast acrylic. Wherein the wedge material can be selected to have low birefringence as well to reduce non-uniformities in brightness. The wedge set 456 and the quarter wave film 454 can also be bonded to the DLP 402 to further reduce Fresnel reflections at the DLP interface losses. In addition, since the image light 414 is substantially normal to the exit surface of the wedge set 456, the flatness of the surface is not critical

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to maintain the wavefront of the image light **414** so that high image quality can be obtained in the displayed image without requiring very tightly toleranced flatness on the exit surface.

A yet further embodiment of the invention that is not illustrated, combines the embodiments illustrated in FIGS. **4b** and FIG. **5**. In this embodiment, the wedge set **456** is comprised of three wedges with the general geometry of the wedges in the wedge set corresponding to that shown in FIGS. **5** and **5a**. A reflective polarizer is bonded between the first and second wedges similar to that shown in FIG. **4b**, however, a third wedge is provided similar to the embodiment of FIG. **5**. Wherein there is an angled thin air gap between the second and third wedges so that the dark state light is reflected by TIR toward the side of the second wedge where it is absorbed in a light trap. This embodiment, like the embodiment shown in FIG. **4b**, uses a polarized light source as has been previously described. The difference in this embodiment is that the image light is transmitted through the reflective polarizer and is transmitted through the angled thin air gap so that it exits normal to the exit surface of the third wedge.

FIG. **5b** illustrates an upper optical module **202** with a dark light trap **514a**. As described in connection with FIGS. **4** and **4a**, image light can be generated from a DLP when using a TIR and corrective lens configuration. The upper module may be mounted in a HWC housing **510** and the housing **510** may include a dark light trap **514a**. The dark light trap **514a** is generally positioned/constructed/formed in a position that is optically aligned with the dark light optical axis **512**. As illustrated, the dark light trap may have depth such that the trap internally reflects dark light in an attempt to further absorb the light and prevent the dark light from combining with the image light that passes through the field lens. The dark light trap may be of a shape and depth such that it absorbs the dark light. In addition, the dark light trap **514b**, in embodiments, may be made of light absorbing materials or coated with light absorbing materials. In embodiments, the recessed light trap **514a** may include baffles to block a view of the dark state light. This may be combined with black surfaces and textured or fibrous surfaces to help absorb the light. The baffles can be part of the light trap, associated with the housing, or field lens, etc.

FIG. **5c** illustrates another embodiment with a light trap **514b**. As can be seen in the illustration, the shape of the trap is configured to enhance internal reflections within the light trap **514b** to increase the absorption of the dark light **512**. FIG. **5d** illustrates another embodiment with a light trap **514c**. As can be seen in the illustration, the shape of the trap **514c** is configured to enhance internal reflections to increase the absorption of the dark light **512**.

FIG. **5e** illustrates another embodiment of an upper optical module **202** with a dark light trap **514d**. This embodiment of upper module **202** includes an off light reflection wedge **502**, as illustrated and described in connection with the embodiment of FIGS. **5** and **5a**. As can be seen in FIG. **5e**, the light trap **514d** is positioned along the optical path of the dark light **512**. The dark light trap **514d** may be configured as described in other embodiments herein. The embodiment of the light trap **514d** illustrated in FIG. **5e** includes a black area on the side wall of the wedge, wherein the side wall is located substantially away from the optical axis of the image light **414**. In addition, baffles **5252** may be added to one or more edges of the field lens **312** to block the view of the light trap **514d** adjacent to the displayed image seen by the user.

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FIG. **6** illustrates a combination of an upper optical module **202** with a lower optical module **204**. In this embodiment, the image light projected from the upper optical module **202** may or may not be polarized. The image light is reflected off a flat combiner element **602** such that it is directed towards the user's eye. Wherein, the combiner element **602** is a partial mirror that reflects image light while transmitting a substantial portion of light from the environment so the user can look through the combiner element and see the environment surrounding the HWC.

The combiner **602** may include a holographic pattern, to form a holographic mirror. If a monochrome image is desired, there may be a single wavelength reflection design for the holographic pattern on the surface of the combiner **602**. If the intention is to have multiple colors reflected from the surface of the combiner **602**, a multiple wavelength holographic mirror may be included on the combiner surface. For example, in a three-color embodiment, where red, green and blue pixels are generated in the image light, the holographic mirror may be reflective to wavelengths substantially matching the wavelengths of the red, green and blue light provided by the light source. This configuration can be used as a wavelength specific mirror where pre-determined wavelengths of light from the image light are reflected to the user's eye. This configuration may also be made such that substantially all other wavelengths in the visible pass through the combiner element **602** so the user has a substantially clear view of the surroundings when looking through the combiner element **602**. The transparency between the user's eye and the surrounding may be approximately 80% when using a combiner that is a holographic mirror. Wherein holographic mirrors can be made using lasers to produce interference patterns in the holographic material of the combiner where the wavelengths of the lasers correspond to the wavelengths of light that are subsequently reflected by the holographic mirror.

In another embodiment, the combiner element **602** may include a notch mirror comprised of a multilayer coated substrate wherein the coating is designed to substantially reflect the wavelengths of light provided by the light source and substantially transmit the remaining wavelengths in the visible spectrum. For example, in the case where red, green and blue light is provided by the light source to enable full color images to be provided to the user, the notch mirror is a tristimulus notch mirror wherein the multilayer coating is designed to reflect narrow bands of red, green and blue light that are matched to the what is provided by the light source and the remaining visible wavelengths are transmitted through the coating to enable a view of the environment through the combiner. In another example where monochrome images are provided to the user, the notch mirror is designed to reflect a single narrow band of light that is matched to the wavelength range of the light provided by the light source while transmitting the remaining visible wavelengths to enable a see-thru view of the environment. The combiner **602** with the notch mirror would operate, from the user's perspective, in a manner similar to the combiner that includes a holographic pattern on the combiner element **602**. The combiner, with the tristimulus notch mirror, would reflect the "on" pixels to the eye because of the match between the reflective wavelengths of the notch mirror and the color of the image light, and the wearer would be able to see with high clarity the surroundings. The transparency between the user's eye and the surrounding may be approximately 80% when using the tristimulus notch mirror. In addition, the image provided by the upper optical module **202** with the notch mirror combiner can provide higher

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contrast images than the holographic mirror combiner due to less scattering of the imaging light by the combiner.

Light can escape through the combiner **602** and may produce face glow as the light is generally directed downward onto the cheek of the user. When using a holographic mirror combiner or a tristimulus notch mirror combiner, the escaping light can be trapped to avoid face glow. In embodiments, if the image light is polarized before the combiner, a linear polarizer can be laminated, or otherwise associated, to the combiner, with the transmission axis of the polarizer oriented relative to the polarized image light so that any escaping image light is absorbed by the polarizer. In embodiments, the image light would be polarized to provide S polarized light to the combiner for better reflection. As a result, the linear polarizer on the combiner would be oriented to absorb S polarized light and pass P polarized light. This provides the preferred orientation of polarized sunglasses as well.

If the image light is unpolarized, a microlouvered film such as a privacy filter can be used to absorb the escaping image light while providing the user with a see-thru view of the environment. In this case, the absorbance or transmittance of the microlouvered film is dependent on the angle of the light. Where steep angle light is absorbed and light at less of an angle is transmitted. For this reason, in an embodiment, the combiner with the microlouver film is angled at greater than 45 degrees to the optical axis of the image light (e.g. the combiner can be oriented at 50 degrees so the image light from the file lens is incident on the combiner at an oblique angle).

FIG. 7 illustrates an embodiment of a combiner element **602** at various angles when the combiner element **602** includes a holographic mirror. Normally, a mirrored surface reflects light at an angle equal to the angle that the light is incident to the mirrored surface. Typically, this necessitates that the combiner element be at 45 degrees, **602a**, if the light is presented vertically to the combiner so the light can be reflected horizontally towards the wearer's eye. In embodiments, the incident light can be presented at angles other than vertical to enable the mirror surface to be oriented at other than 45 degrees, but in all cases wherein a mirrored surface is employed (including the tristimulus notch mirror described previously), the incident angle equals the reflected angle. As a result, increasing the angle of the combiner **602a** requires that the incident image light be presented to the combiner **602a** at a different angle which positions the upper optical module **202** to the left of the combiner as shown in FIG. 7. In contrast, a holographic mirror combiner, included in embodiments, can be made such that light is reflected at a different angle from the angle that the light is incident onto the holographic mirrored surface. This allows freedom to select the angle of the combiner element **602b** independent of the angle of the incident image light and the angle of the light reflected into the wearer's eye. In embodiments, the angle of the combiner element **602b** is greater than 45 degrees (shown in FIG. 7) as this allows a more laterally compact HWC design. The increased angle of the combiner element **602b** decreases the front to back width of the lower optical module **204** and may allow for a thinner HWC display (i.e. the furthest element from the wearer's eye can be closer to the wearer's face).

FIG. 8 illustrates another embodiment of a lower optical module **204**. In this embodiment, polarized image light provided by the upper optical module **202**, is directed into the lower optical module **204**. The image light reflects off a polarized mirror **804** and is directed to a focusing partially reflective mirror **802**, which is adapted to reflect the polar-

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ized light. An optical element such as a $\frac{1}{4}$ wave film located between the polarized mirror **804** and the partially reflective mirror **802**, is used to change the polarization state of the image light such that the light reflected by the partially reflective mirror **802** is transmitted by the polarized mirror **804** to present image light to the eye of the wearer. The user can also see through the polarized mirror **804** and the partially reflective mirror **802** to see the surrounding environment. As a result, the user perceives a combined image comprised of the displayed image light overlaid onto the see-thru view of the environment.

While many of the embodiments of the present invention have been referred to as upper and lower modules containing certain optical components, it should be understood that the image light and dark light production and management functions described in connection with the upper module may be arranged to direct light in other directions (e.g. upward, sideward, etc.). In embodiments, it may be preferred to mount the upper module **202** above the wearer's eye, in which case the image light would be directed downward. In other embodiments it may be preferred to produce light from the side of the wearer's eye, or from below the wearer's eye. In addition, the lower optical module is generally configured to deliver the image light to the wearer's eye and allow the wearer to see through the lower optical module, which may be accomplished through a variety of optical components.

FIG. 8a illustrates an embodiment of the present invention where the upper optical module **202** is arranged to direct image light into a TIR waveguide **810**. In this embodiment, the upper optical module **202** is positioned above the wearer's eye **812** and the light is directed horizontally into the TIR waveguide **810**. The TIR waveguide is designed to internally reflect the image light in a series of downward TIR reflections until it reaches the portion in front of the wearer's eye, where the light passes out of the TIR waveguide **812** into the wearer's eye. In this embodiment, an outer shield **814** is positioned in front of the TIR waveguide **810**.

FIG. 8b illustrates an embodiment of the present invention where the upper optical module **202** is arranged to direct image light into a TIR waveguide **818**. In this embodiment, the upper optical module **202** is arranged on the side of the TIR waveguide **818**. For example, the upper optical module may be positioned in the arm or near the arm of the HWC when configured as a pair of head worn glasses. The TIR waveguide **818** is designed to internally reflect the image light in a series of TIR reflections until it reaches the portion in front of the wearer's eye, where the light passes out of the TIR waveguide **812** into the wearer's eye.

FIG. 8c illustrates yet further embodiments of the present invention where an upper optical module **202** is directing polarized image light into an optical guide **828** where the image light passes through a polarized reflector **824**, changes polarization state upon reflection of the optical element **822** which includes a $\frac{1}{4}$ wave film for example and then is reflected by the polarized reflector **824** towards the wearer's eye, due to the change in polarization of the image light. The upper optical module **202** may be positioned to direct light to a mirror **820**, to position the upper optical module **202** laterally, in other embodiments, the upper optical module **202** may direct the image light directly towards the polarized reflector **824**. It should be understood that the present invention comprises other optical arrangements intended to direct image light into the wearer's eye.

Another aspect of the present invention relates to eye imaging. In embodiments, a camera is used in connection with an upper optical module **202** such that the wearer's eye

can be imaged using pixels in the “off” state on the DLP. FIG. 9 illustrates a system where the eye imaging camera 802 is mounted and angled such that the field of view of the eye imaging camera 802 is redirected toward the wearer’s eye by the mirror pixels of the DLP 402 that are in the “off” state. In this way, the eye imaging camera 802 can be used to image the wearer’s eye along the same optical axis as the displayed image that is presented to the wearer. Wherein, image light that is presented to the wearer’s eye illuminates the wearer’s eye so that the eye can be imaged by the eye imaging camera 802. In the process, the light reflected by the eye passes back through the optical train of the lower optical module 204 and a portion of the upper optical module to where the light is reflected by the “off” pixels of the DLP 402 toward the eye imaging camera 802.

In embodiments, the eye imaging camera may image the wearer’s eye at a moment in time where there are enough “off” pixels to achieve the required eye image resolution. In another embodiment, the eye imaging camera collects eye image information from “off” pixels over time and forms a time lapsed image. In another embodiment, a modified image is presented to the user wherein enough “off” state pixels are included that the camera can obtain the desired resolution and brightness for imaging the wearer’s eye and the eye image capture is synchronized with the presentation of the modified image.

The eye imaging system may be used for security systems. The HWC may not allow access to the HWC or other system if the eye is not recognized (e.g. through eye characteristics including retina or iris characteristics, etc.). The HWC may be used to provide constant security access in some embodiments. For example, the eye security confirmation may be a continuous, near-continuous, real-time, quasi real-time, periodic, etc. process so the wearer is effectively constantly being verified as known. In embodiments, the HWC may be worn and eye security tracked for access to other computer systems.

The eye imaging system may be used for control of the HWC. For example, a blink, wink, or particular eye movement may be used as a control mechanism for a software application operating on the HWC or associated device.

The eye imaging system may be used in a process that determines how or when the HWC 102 delivers digitally displayed content to the wearer. For example, the eye imaging system may determine that the user is looking in a direction and then HWC may change the resolution in an area of the display or provide some content that is associated with something in the environment that the user may be looking at. Alternatively, the eye imaging system may identify different user’s and change the displayed content or enabled features provided to the user. User’s may be identified from a database of users eye characteristics either located on the HWC 102 or remotely located on the network 110 or on a server 112. In addition, the HWC may identify a primary user or a group of primary users from eye characteristics wherein the primary user(s) are provided with an enhanced set of features and all other user’s are provided with a different set of features. Thus in this use case, the HWC 102 uses identified eye characteristics to either enable features or not and eye characteristics need only be analyzed in comparison to a relatively small database of individual eye characteristics.

FIG. 10 illustrates a light source that may be used in association with the upper optics module 202 (e.g. polarized light source if the light from the solid state light source is polarized such as polarized light source 302 and 458), and light source 404. In embodiments, to provide a uniform

surface of light 1008 to be directed into the upper optical module 202 and towards the DLP of the upper optical module, either directly or indirectly, the solid state light source 1002 may be projected into a backlighting optical system 1004. The solid state light source 1002 may be one or more LEDs, laser diodes, OLEDs. In embodiments, the backlighting optical system 1004 includes an extended section with a length/distance ratio of greater than 3, wherein the light undergoes multiple reflections from the sidewalls to mix of homogenize the light as supplied by the solid state light source 1002. The backlighting optical system 1004 can also include structures on the surface opposite (on the left side as shown in FIG. 10) to where the uniform light 1008 exits the backlight 1004 to change the direction of the light toward the DLP 302 and the reflective polarizer 310 or the DLP 402 and the TIR wedge 418. The backlighting optical system 1004 may also include structures to collimate the uniform light 1008 to provide light to the DLP with a smaller angular distribution or narrower cone angle. Diffusers or polarizers can be used on the entrance or exit surface of the backlighting optical system. Diffusers can be used to spread or uniformize the exiting light from the backlight to improve the uniformity or increase the angular spread of the uniform light 1008. Elliptical diffusers that diffuse the light more in some directions and less in others can be used to improve the uniformity or spread of the uniform light 1008 in directions orthogonal to the optical axis of the uniform light 1008. Linear polarizers can be used to convert unpolarized light as supplied by the solid state light source 1002 to polarized light so the uniform light 1008 is polarized with a desired polarization state. A reflective polarizer can be used on the exit surface of the backlight 1004 to polarize the uniform light 1008 to the desired polarization state, while reflecting the other polarization state back into the backlight where it is recycled by multiple reflections within the backlight 1004 and at the solid state light source 1002. Therefore by including a reflective polarizer at the exit surface of the backlight 1004, the efficiency of the polarized light source is improved.

FIGS. 10a and 10b show illustrations of structures in backlight optical systems 1004 that can be used to change the direction of the light provided to the entrance face 1045 by the light source and then collimates the light in a direction lateral to the optical axis of the exiting uniform light 1008. Structure 1060 includes an angled sawtooth pattern in a transparent waveguide wherein the left edge of each sawtooth clips the steep angle rays of light thereby limiting the angle of the light being redirected. The steep surface at the right (as shown) of each sawtooth then redirects the light so that it reflects off the left angled surface of each sawtooth and is directed toward the exit surface 1040. The sawtooth surfaces shown on the lower surface in FIGS. 10a and 10b, can be smooth and coated (e.g. with an aluminum coating or a dielectric mirror coating) to provide a high level of reflectivity without scattering. Structure 1050 includes a curved face on the left side (as shown) to focus the rays after they pass through the exit surface 1040, thereby providing a mechanism for collimating the uniform light 1008. In a further embodiment, a diffuser can be provided between the solid state light source 1002 and the entrance face 1045 to homogenize the light provided by the solid state light source 1002. In yet a further embodiment, a polarizer can be used between the diffuser and the entrance face 1045 of the backlight 1004 to provide a polarized light source. Because the sawtooth pattern provides smooth reflective surfaces, the polarization state of the light can be preserved from the entrance face 1045 to the exit face 1040. In this embodiment,

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the light entering the backlight from the solid state light source **1002** passes through the polarizer so that it is polarized with the desired polarization state. If the polarizer is an absorptive linear polarizer, the light of the desired polarization state is transmitted while the light of the other polarization state is absorbed. If the polarizer is a reflective polarizer, the light of the desired polarization state is transmitted into the backlight **1004** while the light of the other polarization state is reflected back into the solid state light source **1002** where it can be recycled as previously described, to increase the efficiency of the polarized light source.

FIG. **11a** illustrates a light source **1100** that may be used in association with the upper optics module **202**. In embodiments, the light source **1100** may provide light to a backlighting optical system **1004** as described above in connection with FIG. **10**. In embodiments, the light source **1100** includes a tristimulus notch filter **1102**. The tristimulus notch filter **1102** has narrow band pass filters for three wavelengths, as indicated in FIG. **11c** in a transmission graph **1108**. The graph shown in FIG. **11b**, as **1104** illustrates an output of three different colored LEDs. One can see that the bandwidths of emission are narrow, but they have long tails. The tristimulus notch filter **1102** can be used in connection with such LEDs to provide a light source **1100** that emits narrow filtered wavelengths of light as shown in FIG. **11d** as the transmission graph **1110**. Wherein the clipping effects of the tristimulus notch filter **1102** can be seen to have cut the tails from the LED emission graph **1104** to provide narrower wavelength bands of light to the upper optical module **202**. The light source **1100** can be used in connection with a combiner **602** with a holographic mirror or tristimulus notch mirror to provide narrow bands of light that are reflected toward the wearer's eye with less waste light that does not get reflected by the combiner, thereby improving efficiency and reducing escaping light that can cause faceglow.

FIG. **12a** illustrates another light source **1200** that may be used in association with the upper optics module **202**. In embodiments, the light source **1200** may provide light to a backlighting optical system **1004** as described above in connection with FIG. **10**. In embodiments, the light source **1200** includes a quantum dot cover glass **1202**. Where the quantum dots absorb light of a shorter wavelength and emit light of a longer wavelength (FIG. **12b** shows an example wherein a UV spectrum **1202** applied to a quantum dot results in the quantum dot emitting a narrow band shown as a PL spectrum **1204**) that is dependent on the material makeup and size of the quantum dot. As a result, quantum dots in the quantum dot cover glass **1202** can be tailored to provide one or more bands of narrow bandwidth light (e.g. red, green and blue emissions dependent on the different quantum dots included as illustrated in the graph shown in FIG. **12c** where three different quantum dots are used. In embodiments, the LED driver light emits UV light, deep blue or blue light. For sequential illumination of different colors, multiple light sources **1200** would be used where each light source **1200** would include a quantum dot cover glass **1202** with a quantum dot selected to emit at one of the desired colors. The light source **1100** can be used in connection with a combiner **602** with a holographic mirror or tristimulus notch mirror to provide narrow transmission bands of light that are reflected toward the wearer's eye with less waste light that does not get reflected.

Another aspect of the present invention relates to the generation of peripheral image lighting effects for a person wearing a HWC. In embodiments, a solid state lighting system (e.g. LED, OLED, etc), or other lighting system, may

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be included inside the optical elements of an lower optical module **204**. The solid state lighting system may be arranged such that lighting effects outside of a field of view (FOV) of the presented digital content is presented to create an emersive effect for the person wearing the HWC. To this end, the lighting effects may be presented to any portion of the HWC that is visible to the wearer. The solid state lighting system may be digitally controlled by an integrated processor on the HWC. In embodiments, the integrated processor will control the lighting effects in coordination with digital content that is presented within the FOV of the HWC. For example, a movie, picture, game, or other content, may be displayed or playing within the FOV of the HWC. The content may show a bomb blast on the right side of the FOV and at the same moment, the solid state lighting system inside of the upper module optics may flash quickly in concert with the FOV image effect. The effect may not be fast, it may be more persistent to indicate, for example, a general glow or color on one side of the user. The solid state lighting system may be color controlled, with red, green and blue LEDs, for example, such that color control can be coordinated with the digitally presented content within the field of view.

FIG. **13a** illustrates optical components of a lower optical module **204** together with an outer lens **1302**. FIG. **13a** also shows an embodiment including effects LED's **1308a** and **1308b**. FIG. **13a** illustrates image light **1312**, as described herein elsewhere, directed into the upper optical module where it will reflect off of the combiner element **1304**, as described herein elsewhere. The combiner element **1304** in this embodiment is angled towards the wearer's eye at the top of the module and away from the wearer's eye at the bottom of the module, as also illustrated and described in connection with FIG. **8** (e.g. at a 45 degree angle). The image light **1312** provided by an upper optical module **202** (not shown in FIG. **13a**) reflects off of the combiner element **1304** towards the collimating mirror **1310**, away from the wearer's eye, as described herein elsewhere. The image light **1312** then reflects and focuses off of the collimating mirror **1304**, passes back through the combiner element **1304**, and is directed into the wearer's eye. The wearer can also view the surrounding environment through the transparency of the combiner element **1304**, collimating mirror **1310**, and outer lens **1302** (if it is included). As described herein elsewhere, various surfaces are polarized to create the optical path for the image light and to provide transparency of the elements such that the wearer can view the surrounding environment. The wearer will generally perceive that the image light forms an image in the FOV **1305**. In embodiments, the outer lens **1302** may be included. The outer lens **1302** is an outer lens that may or may not be corrective and it may be designed to conceal the lower optical module components in an effort to make the HWC appear to be in a form similar to standard glasses or sunglasses.

In the embodiment illustrated in FIG. **13a**, the effects LEDs **1308a** and **1308b** are positioned at the sides of the combiner element **1304** and the outer lens **1302** and/or the collimating mirror **1310**. In embodiments, the effects LEDs **1308a** are positioned within the confines defined by the combiner element **1304** and the outer lens **1302** and/or the collimating mirror. The effects LEDs **1308a** and **1308b** are also positioned outside of the FOV **1305**. In this arrangement, the effects LEDs **1308a** and **1308b** can provide lighting effects within the lower optical module outside of the FOV **1305**. In embodiments the light emitted from the effects LEDs **1308a** and **1308b** may be polarized such that the light passes through the combiner element **1304** toward the wearer's eye and does not pass through the outer lens

1302 and/or the collimating mirror **1310**. This arrangement provides peripheral lighting effects to the wearer in a more private setting by not transmitting the lighting effects through the front of the HWC into the surrounding environment. However, in other embodiments, the effects LEDs **1308a** and **1308b** may be unpolarized so the lighting effects provided are made to be purposefully viewable by others in the environment for entertainment such as giving the effect of the wearer's eye glowing in correspondence to the image content being viewed by the wearer.

FIG. **13b** illustrates a cross section of the embodiment described in connection with FIG. **13a**. As illustrated, the effects LED **1308a** is located in the upper-front area inside of the optical components of the lower optical module. It should be understood that the effects LED **1308a** position in the described embodiments is only illustrative and alternate placements are encompassed by the present invention. Additionally, in embodiments, there may be one or more effects LEDs **1308a** in each of the two sides of HWC to provide peripheral lighting effects near one or both eyes of the wearer.

FIG. **13c** illustrates an embodiment where the combiner element **1304** is angled away from the eye at the top and towards the eye at the bottom (e.g. in accordance with the holographic or notch filter embodiments described herein). In this embodiment, the effects LED **1308a** is located on the outer lens **1302** side of the combiner element **1304** to provide a concealed appearance of the lighting effects. As with other embodiments, the effects LED **1308a** of FIG. **13c** may include a polarizer such that the emitted light can pass through a polarized element associated with the combiner element **1304** and be blocked by a polarized element associated with the outer lens **1302**.

Another aspect of the present invention relates to the mitigation of light escaping from the space between the wearer's face and the HWC itself. Another aspect of the present invention relates to maintaining a controlled lighting environment in proximity to the wearer's eyes. In embodiments, both the maintenance of the lighting environment and the mitigation of light escape are accomplished by including a removable and replaceable flexible shield for the HWC. Wherein the removable and replaceable shield can be provided for one eye or both eyes in correspondence to the use of the displays for each eye. For example, in a night vision application, the display to only one eye could be used for night vision while the display to the other eye is turned off to provide good see-thru when moving between areas where visible light is available and dark areas where night vision enhancement is needed.

FIG. **14a** illustrates a removable and replaceable flexible eye cover **1402** with an opening **1408** that can be attached and removed quickly from the HWC **102** through the use of magnets. Other attachment methods may be used, but for illustration of the present invention we will focus on a magnet implementation. In embodiments, magnets may be included in the eye cover **1402** and magnets of an opposite polarity may be included (e.g. embedded) in the frame of the HWC **102**. The magnets of the two elements would attract quite strongly with the opposite polarity configuration. In another embodiment, one of the elements may have a magnet and the other side may have metal for the attraction. In embodiments, the eye cover **1402** is a flexible elastomeric shield. In embodiments, the eye cover **1402** may be an elastomeric bellows design to accommodate flexibility and more closely align with the wearer's face. FIG. **14b** illustrates a removable and replaceable flexible eye cover **1404** that is adapted as a single eye cover. In embodiments, a

single eye cover may be used for each side of the HWC to cover both eyes of the wearer. In embodiments, the single eye cover may be used in connection with a HWC that includes only one computer display for one eye. These configurations prevent light that is generated and directed generally towards the wearer's face by covering the space between the wearer's face and the HWC. The opening **1408** allows the wearer to look through the opening **1408** to view the displayed content and the surrounding environment through the front of the HWC. The image light in the lower optical module **204** can be prevented from emitting from the front of the HWC through internal optics polarization schemes, as described herein, for example.

FIG. **14c** illustrates another embodiment of a light suppression system. In this embodiment, the eye cover **1410** may be similar to the eye cover **1402**, but eye cover **1410** includes a front light shield **1412**. The front light shield **1412** may be opaque to prevent light from escaping the front lens of the HWC. In other embodiments, the front light shield **1412** is polarized to prevent light from escaping the front lens. In a polarized arrangement, in embodiments, the internal optical elements of the HWC (e.g. of the lower optical module **204**) may polarize light transmitted towards the front of the HWC and the front light shield **1412** may be polarized to prevent the light from transmitting through the front light shield **1412**.

In embodiments, an opaque front light shield **1412** may be included and the digital content may include images of the surrounding environment such that the wearer can visualize the surrounding environment. One eye may be presented with night vision environmental imagery and this eye's surrounding environment optical path may be covered using an opaque front light shield **1412**. In other embodiments, this arrangement may be associated with both eyes.

Another aspect of the present invention relates to automatically configuring the lighting system(s) used in the HWC **102**. In embodiments, the display lighting and/or effects lighting, as described herein, may be controlled in a manner suitable for when an eye cover **1408** is attached or removed from the HWC **102**. For example, at night, when the light in the environment is low, the lighting system(s) in the HWC may go into a low light mode to further control any amounts of stray light escaping from the HWC and the areas around the HWC. Covert operations at night, while using night vision or standard vision, may require a solution which prevents as much escaping light as possible so a user may clip on the eye cover(s) **1408** and then the HWC may go into a low light mode. The low light mode may, in some embodiments, only go into a low light mode when the eye cover **1408** is attached if the HWC identifies that the environment is in low light conditions (e.g. through environment light level sensor detection). In embodiments, the low light level may be determined to be at an intermediate point between full and low light dependent on environmental conditions.

Another aspect of the present invention relates to automatically controlling the type of content displayed in the HWC when eye covers **1408** are attached or removed from the HWC. In embodiments, when the eye cover(s) **1408** is attached to the HWC, the displayed content may be restricted in amount or in color amounts. For example, the display(s) may go into a simple content delivery mode to restrict the amount of information displayed. This may be done to reduce the amount of light produced by the display(s). In an embodiment, the display(s) may change from color displays to monochrome displays to reduce the amount of light produced. In an embodiment, the mono-

chrome lighting may be red to limit the impact on the wearer's eyes to maintain an ability to see better in the dark.

Referring to FIG. 15, we now turn to describe a particular external user interface 104, referred to generally as a pen 1500. The pen 1500 is a specially designed external user interface 104 and can operate as a user interface, such as to many different styles of HWC 102. The pen 1500 generally follows the form of a conventional pen, which is a familiar user handled device and creates an intuitive physical interface for many of the operations to be carried out in the HWC system 100. The pen 1500 may be one of several user interfaces 104 used in connection with controlling operations within the HWC system 100. For example, the HWC 102 may watch for and interpret hand gestures 116 as control signals, where the pen 1500 may also be used as a user interface with the same HWC 102. Similarly, a remote keyboard may be used as an external user interface 104 in concert with the pen 1500. The combination of user interfaces or the use of just one control system generally depends on the operation(s) being executed in the HWC's system 100.

While the pen 1500 may follow the general form of a conventional pen, it contains numerous technologies that enable it to function as an external user interface 104. FIG. 15 illustrates technologies comprised in the pen 1500. As can be seen, the pen 1500 may include a camera 1508, which is arranged to view through lens 1502. The camera may then be focused, such as through lens 1502, to image a surface upon which a user is writing or making other movements to interact with the HWC 102. There are situations where the pen 1500 will also have an ink, graphite, or other system such that what is being written can be seen on the writing surface. There are other situations where the pen 1500 does not have such a physical writing system so there is no deposit on the writing surface, where the pen would only be communicating data or commands to the HWC 102. The lens configuration is described in greater detail herein. The function of the camera is to capture information from an unstructured writing surface such that pen strokes can be interpreted as intended by the user. To assist in the prediction of the intended stroke path, the pen 1500 may include a sensor, such as an IMU 1512. Of course, the IMU could be included in the pen 1500 in its separate parts (e.g. gyro, accelerometer, etc.) or an IMU could be included as a single unit. In this instance, the IMU 1512 is used to measure and predict the motion of the pen 1500. In turn, the integrated microprocessor 1510 would take the IMU information and camera information as inputs and process the information to form a prediction of the pen tip movement.

The pen 1500 may also include a pressure monitoring system 1504, such as to measure the pressure exerted on the lens 1502. As will be described in greater detail herein, the pressure measurement can be used to predict the user's intention for changing the weight of a line, type of a line, type of brush, click, double click, and the like. In embodiments, the pressure sensor may be constructed using any force or pressure measurement sensor located behind the lens 1502, including for example, a resistive sensor, a current sensor, a capacitive sensor, a voltage sensor such as a piezoelectric sensor, and the like.

The pen 1500 may also include a communications module 1518, such as for bi-directional communication with the HWC 102. In embodiments, the communications module 1518 may be a short distance communication module (e.g. Bluetooth). The communications module 1518 may be security matched to the HWC 102. The communications module 1518 may be arranged to communicate data and commands

to and from the microprocessor 1510 of the pen 1500. The microprocessor 1510 may be programmed to interpret data generated from the camera 1508, IMU 1512, and pressure sensor 1504, and the like, and then pass a command onto the HWC 102 through the communications module 1518, for example. In another embodiment, the data collected from any of the input sources (e.g. camera 1508, IMU 1512, pressure sensor 1504) by the microprocessor may be communicated by the communication module 1518 to the HWC 102, and the HWC 102 may perform data processing and prediction of the user's intention when using the pen 1500. In yet another embodiment, the data may be further passed on through a network 110 to a remote device 112, such as a server, for the data processing and prediction. The commands may then be communicated back to the HWC 102 for execution (e.g. display writing in the glasses display, make a selection within the UI of the glasses display, control a remote external device 112, control a local external device 108), and the like. The pen may also include memory 1514 for long or short term uses.

The pen 1500 may also include a number of physical user interfaces, such as quick launch buttons 1522, a touch sensor 1520, and the like. The quick launch buttons 1522 may be adapted to provide the user with a fast way of jumping to a software application in the HWC system 100. For example, the user may be a frequent user of communication software packages (e.g. email, text, Twitter, Instagram, Facebook, Google+, and the like), and the user may program a quick launch button 1522 to command the HWC 102 to launch an application. The pen 1500 may be provided with several quick launch buttons 1522, which may be user programmable or factory programmable. The quick launch button 1522 may be programmed to perform an operation. For example, one of the buttons may be programmed to clear the digital display of the HWC 102. This would create a fast way for the user to clear the screens on the HWC 102 for any reason, such as for example to better view the environment. The quick launch button functionality will be discussed in further detail below. The touch sensor 1520 may be used to take gesture style input from the user. For example, the user may be able to take a single finger and run it across the touch sensor 1520 to affect a page scroll.

The pen 1500 may also include a laser pointer 1524. The laser pointer 1524 may be coordinated with the IMU 1512 to coordinate gestures and laser pointing. For example, a user may use the laser 1524 in a presentation to help with guiding the audience with the interpretation of graphics and the IMU 1512 may, either simultaneously or when the laser 1524 is off, interpret the user's gestures as commands or data input.

FIGS. 16A-C illustrate several embodiments of lens and camera arrangements 1600 for the pen 1500. One aspect relates to maintaining a constant distance between the camera and the writing surface to enable the writing surface to be kept in focus for better tracking of movements of the pen 1500 over the writing surface. Another aspect relates to maintaining an angled surface following the circumference of the writing tip of the pen 1500 such that the pen 1500 can be rolled or partially rolled in the user's hand to create the feel and freedom of a conventional writing instrument.

FIG. 16A illustrates an embodiment of the writing lens end of the pen 1500. The configuration includes a ball lens 1604, a camera or image capture surface 1602, and a domed cover lens 1608. In this arrangement, the camera views the writing surface through the ball lens 1604 and dome cover lens 1608. The ball lens 1604 causes the camera to focus such that the camera views the writing surface when the pen

1500 is held in the hand in a natural writing position, such as with the pen **1500** in contact with a writing surface. In embodiments, the ball lens **1604** should be separated from the writing surface to obtain the highest resolution of the writing surface at the camera **1602**. In embodiments, the ball lens **1604** is separated by approximately 1 to 3 mm. In this configuration, the domed cover lens **1608** provides a surface that can keep the ball lens **1604** separated from the writing surface at a constant distance, such as substantially independent of the angle used to write on the writing surface. For instance, in embodiments the field of view of the camera in this arrangement would be approximately 60 degrees.

The domed cover lens, or other lens **1608** used to physically interact with the writing surface, will be transparent or transmissive within the active bandwidth of the camera **1602**. In embodiments, the domed cover lens **1608** may be spherical or other shape and comprised of glass, plastic, sapphire, diamond, and the like. In other embodiments where low resolution imaging of the surface is acceptable. The pen **1500** can omit the domed cover lens **1608** and the ball lens **1604** can be in direct contact with the surface.

FIG. **16B** illustrates another structure where the construction is somewhat similar to that described in connection with FIG. **16A**; however this embodiment does not use a dome cover lens **1608**, but instead uses a spacer **1610** to maintain a predictable distance between the ball lens **1604** and the writing surface, wherein the spacer may be spherical, cylindrical, tubular or other shape that provides spacing while allowing for an image to be obtained by the camera **1602** through the lens **1604**. In a preferred embodiment, the spacer **1610** is transparent. In addition, while the spacer **1610** is shown as spherical, other shapes such as an oval, doughnut shape, half sphere, cone, cylinder or other form may be used.

FIG. **16C** illustrates yet another embodiment, where the structure includes a post **1614**, such as running through the center of the lensed end of the pen **1500**. The post **1614** may be an ink deposition system (e.g. ink cartridge), graphite deposition system (e.g. graphite holder), or a dummy post whose purpose is mainly only that of alignment. The selection of the post type is dependent on the pen's use. For instance, in the event the user wants to use the pen **1500** as a conventional ink depositing pen as well as a fully functional external user interface **104**, the ink system post would be the best selection. If there is no need for the 'writing' to be visible on the writing surface, the selection would be the dummy post. The embodiment of FIG. **16C** includes camera(s) **1602** and an associated lens **1612**, where the camera **1602** and lens **1612** are positioned to capture the writing surface without substantial interference from the post **1614**. In embodiments, the pen **1500** may include multiple cameras **1602** and lenses **1612** such that more or all of the circumference of the tip **1614** can be used as an input system. In an embodiment, the pen **1500** includes a contoured grip that keeps the pen aligned in the user's hand so that the camera **1602** and lens **1612** remains pointed at the surface.

Another aspect of the pen **1500** relates to sensing the force applied by the user to the writing surface with the pen **1500**. The force measurement may be used in a number of ways. For example, the force measurement may be used as a discrete value, or discontinuous event tracking, and compared against a threshold in a process to determine a user's intent. The user may want the force interpreted as a 'click' in the selection of an object, for instance. The user may intend multiple force exertions interpreted as multiple clicks. There may be times when the user holds the pen **1500** in a certain position or holds a certain portion of the pen **1500** (e.g. a button or touch pad) while clicking to affect a

certain operation (e.g. a 'right click'). In embodiments, the force measurement may be used to track force and force trends. The force trends may be tracked and compared to threshold limits, for example. There may be one such threshold limit, multiple limits, groups of related limits, and the like. For example, when the force measurement indicates a fairly constant force that generally falls within a range of related threshold values, the microprocessor **1510** may interpret the force trend as an indication that the user desires to maintain the current writing style, writing tip type, line weight, brush type, and the like. In the event that the force trend appears to have gone outside of a set of threshold values intentionally, the microprocessor may interpret the action as an indication that the user wants to change the current writing style, writing tip type, line weight, brush type, and the like. Once the microprocessor has made a determination of the user's intent, a change in the current writing style, writing tip type, line weight, brush type, and the like may be executed. In embodiments, the change may be noted to the user (e.g. in a display of the HWC **102**), and the user may be presented with an opportunity to accept the change.

FIG. **17A** illustrates an embodiment of a force sensing surface tip **1700** of a pen **1500**. The force sensing surface tip **1700** comprises a surface connection tip **1702** (e.g. a lens as described herein elsewhere) in connection with a force or pressure monitoring system **1504**. As a user uses the pen **1500** to write on a surface or simulate writing on a surface the force monitoring system **1504** measures the force or pressure the user applies to the writing surface and the force monitoring system communicates data to the microprocessor **1510** for processing. In this configuration, the microprocessor **1510** receives force data from the force monitoring system **1504** and processes the data to make predictions of the user's intent in applying the particular force that is currently being applied. In embodiments, the processing may be provided at a location other than on the pen (e.g. at a server in the HWC system **100**, on the HWC **102**). For clarity, when reference is made herein to processing information on the microprocessor **1510**, the processing of information contemplates processing the information at a location other than on the pen. The microprocessor **1510** may be programmed with force threshold(s), force signature(s), force signature library and/or other characteristics intended to guide an inference program in determining the user's intentions based on the measured force or pressure. The microprocessor **1510** may be further programmed to make inferences from the force measurements as to whether the user has attempted to initiate a discrete action (e.g. a user interface selection 'click') or is performing a constant action (e.g. writing within a particular writing style). The inferencing process is important as it causes the pen **1500** to act as an intuitive external user interface **104**.

FIG. **17B** illustrates a force **1708** versus time **1710** trend chart with a single threshold **1718**. The threshold **1718** may be set at a level that indicates a discrete force exertion indicative of a user's desire to cause an action (e.g. select an object in a GUI). Event **1712**, for example, may be interpreted as a click or selection command because the force quickly increased from below the threshold **1718** to above the threshold **1718**. The event **1714** may be interpreted as a double click because the force quickly increased above the threshold **1718**, decreased below the threshold **1718** and then essentially repeated quickly. The user may also cause the force to go above the threshold **1718** and hold for a period indicating that the user is intending to select an object

in the GUI (e.g. a GUI presented in the display of the HWC 102) and 'hold' for a further operation (e.g. moving the object).

While a threshold value may be used to assist in the interpretation of the user's intention, a signature force event trend may also be used. The threshold and signature may be used in combination or either method may be used alone. For example, a single-click signature may be represented by a certain force trend signature or set of signatures. The single-click signature(s) may require that the trend meet a criteria of a rise time between λ any y values, a hold time of between a and b values and a fall time of between c and d values, for example. Signatures may be stored for a variety of functions such as click, double click, right click, hold, move, etc. The microprocessor 1510 may compare the real-time force or pressure tracking against the signatures from a signature library to make a decision and issue a command to the software application executing in the GUI.

FIG. 17C illustrates a force 1708 versus time 1710 trend chart with multiple thresholds 1718. By way of example, the force trend is plotted on the chart with several pen force or pressure events. As noted, there are both presumably intentional events 1720 and presumably non-intentional events 1722. The two thresholds 1718 of FIG. 4C create three zones of force: a lower, middle and higher range. The beginning of the trend indicates that the user is placing a lower zone amount of force. This may mean that the user is writing with a given line force weight and does not intend to change the weight, the user is writing. Then the trend shows a significant increase 1720 in force into the middle force range. This force change appears, from the trend to have been sudden and thereafter it is sustained. The microprocessor 1510 may interpret this as an intentional change and as a result change the operation in accordance with preset rules (e.g. change line width, increase line weight, etc.). The trend then continues with a second apparently intentional event 1720 into the higher-force range. During the performance in the higher-force range, the force dips below the upper threshold 1718. This may indicate an unintentional force change and the microprocessor may detect the change in range however not affect a change in the operations being coordinated by the pen 1500. As indicated above, the trend analysis may be done with thresholds and/or signatures.

Generally, in the present disclosure, instrument stroke parameter changes may be referred to as a change in line type, line weight, tip type, brush type, brush width, brush pressure, color, and other forms of writing, coloring, painting, and the like.

Another aspect of the pen 1500 relates to selecting an operating mode for the pen 1500 dependent on contextual information and/or selection interface(s). The pen 1500 may have several operating modes. For instance, the pen 1500 may have a writing mode where the user interface(s) of the pen 1500 (e.g. the writing surface end, quick launch buttons 1522, touch sensor 1520, motion based gesture, and the like) is optimized or selected for tasks associated with writing. As another example, the pen 1500 may have a wand mode where the user interface(s) of the pen is optimized or selected for tasks associated with software or device control (e.g. the HWC 102, external local device, remote device 112, and the like). The pen 1500, by way of another example, may have a presentation mode where the user interface(s) is optimized or selected to assist a user with giving a presentation (e.g. pointing with the laser pointer 1524 while using the button(s) 1522 and/or gestures to control the presentation or applications relating to the presentation). The pen may, for example, have a mode that is optimized or selected for

a particular device that a user is attempting to control. The pen 1500 may have a number of other modes and an aspect of the present invention relates to selecting such modes.

FIG. 18A illustrates an automatic user interface(s) mode selection based on contextual information. The microprocessor 1510 may be programmed with IMU thresholds 1814 and 1812. The thresholds 1814 and 1812 may be used as indications of upper and lower bounds of an angle 1804 and 1802 of the pen 1500 for certain expected positions during certain predicted modes. When the microprocessor 1510 determines that the pen 1500 is being held or otherwise positioned within angles 1802 corresponding to writing thresholds 1814, for example, the microprocessor 1510 may then institute a writing mode for the pen's user interfaces. Similarly, if the microprocessor 1510 determines (e.g. through the IMU 1512) that the pen is being held at an angle 1804 that falls between the predetermined wand thresholds 1812, the microprocessor may institute a wand mode for the pen's user interface. Both of these examples may be referred to as context based user interface mode selection as the mode selection is based on contextual information (e.g. position) collected automatically and then used through an automatic evaluation process to automatically select the pen's user interface(s) mode.

As with other examples presented herein, the microprocessor 1510 may monitor the contextual trend (e.g. the angle of the pen over time) in an effort to decide whether to stay in a mode or change modes. For example, through signatures, thresholds, trend analysis, and the like, the microprocessor may determine that a change is an unintentional change and therefore no user interface mode change is desired.

FIG. 18B illustrates an automatic user interface(s) mode selection based on contextual information. In this example, the pen 1500 is monitoring (e.g. through its microprocessor) whether or not the camera at the writing surface end 1508 is imaging a writing surface in close proximity to the writing surface end of the pen 1500. If the pen 1500 determines that a writing surface is within a predetermined relatively short distance, the pen 1500 may decide that a writing surface is present 1820 and the pen may go into a writing mode user interface(s) mode. In the event that the pen 1500 does not detect a relatively close writing surface 1822, the pen may predict that the pen is not currently being used to as a writing instrument and the pen may go into a non-writing user interface(s) mode.

FIG. 18C illustrates a manual user interface(s) mode selection. The user interface(s) mode may be selected based on a twist of a section 1824 of the pen 1500 housing, clicking an end button 1828, pressing a quick launch button 1522, interacting with touch sensor 1520, detecting a predetermined action at the pressure monitoring system (e.g. a click), detecting a gesture (e.g. detected by the IMU), etc. The manual mode selection may involve selecting an item in a GUI associated with the pen 1500 (e.g. an image presented in the display of HWC 102).

In embodiments, a confirmation selection may be presented to the user in the event a mode is going to change. The presentation may be physical (e.g. a vibration in the pen 1500), through a GUI, through a light indicator, etc.

FIG. 19 illustrates a couple pen use-scenarios 1900 and 1901. There are many use scenarios and we have presented a couple in connection with FIG. 19 as a way of illustrating use scenarios to further the understanding of the reader. As such, the use-scenarios should be considered illustrative and non-limiting.

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Use scenario **1900** is a writing scenario where the pen **1500** is used as a writing instrument. In this example, quick launch button **122A** is pressed to launch a note application **1910** in the GUI **1908** of the HWC **102** display **1904**. Once the quick launch button **122A** is pressed, the HWC **102** launches the note program **1910** and puts the pen into a writing mode. The user uses the pen **1500** to scribe symbols **1902** on a writing surface, the pen records the scribing and transmits the scribing to the HWC **102** where symbols representing the scribing are displayed **1912** within the note application **1910**.

Use scenario **1901** is a gesture scenario where the pen **1500** is used as a gesture capture and command device. In this example, the quick launch button **122B** is activated and the pen **1500** activates a wand mode such that an application launched on the HWC **102** can be controlled. Here, the user sees an application chooser **1918** in the display(s) of the HWC **102** where different software applications can be chosen by the user. The user gestures (e.g. swipes, spins, turns, etc.) with the pen to cause the application chooser **1918** to move from application to application. Once the correct application is identified (e.g. highlighted) in the chooser **1918**, the user may gesture or click or otherwise interact with the pen **1500** such that the identified application is selected and launched. Once an application is launched, the wand mode may be used to scroll, rotate, change applications, select items, initiate processes, and the like, for example.

In an embodiment, the quick launch button **122A** may be activated and the HWC **102** may launch an application chooser presenting to the user a set of applications. For example, the quick launch button may launch a chooser to show all communication programs (e.g. SMS, Twitter, Instagram, Facebook, email, etc.) available for selection such that the user can select the program the user wants and then go into a writing mode. By way of further example, the launcher may bring up selections for various other groups that are related or categorized as generally being selected at a given time (e.g. Microsoft Office products, communication products, productivity products, note products, organizational products, and the like).

FIG. **20** illustrates yet another embodiment of the present invention. FIG. **2000** illustrates a watchband clip on controller **2000**. The watchband clip on controller may be a controller used to control the HWC **102** or devices in the HWC system **100**. The watchband clip on controller **2000** has a fastener **2018** (e.g. rotatable clip) that is mechanically adapted to attach to a watchband, as illustrated at **2004**.

The watchband controller **2000** may have quick launch interfaces **2008** (e.g. to launch applications and choosers as described herein), a touch pad **2014** (e.g. to be used as a touch style mouse for GUI control in a HWC **102** display) and a display **2012**. The clip **2018** may be adapted to fit a wide range of watchbands so it can be used in connection with a watch that is independently selected for its function. The clip, in embodiments, is rotatable such that a user can position it in a desirable manner. In embodiments the clip may be a flexible strap. In embodiments, the flexible strap may be adapted to be stretched to attach to a hand, wrist, finger, device, weapon, and the like.

In embodiments, the watchband controller may be configured as a removable and replacable watchband. For example, the controller may be incorporated into a band with a certain width, segment spacing's, etc. such that the watchband, with its incorporated controller, can be attached to a watch body. The attachment, in embodiments, may be mechanically adapted to attach with a pin upon which the

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watchband rotates. In embodiments, the watchband controller may be electrically connected to the watch and/or watch body such that the watch, watch body and/or the watchband controller can communicate data between them.

The watchband controller may have 3-axis motion monitoring (e.g. through an IMU, accelerometers, magnetometers, gyroscopes, etc.) to capture user motion. The user motion may then be interpreted for gesture control.

In embodiments, the watchband controller may comprise fitness sensors and a fitness computer. The sensors may track heart rate, calories burned, strides, distance covered, and the like. The data may then be compared against performance goals and/or standards for user feedback.

Another aspect of the present invention relates to visual display techniques relating to micro Doppler ("mD") target tracking signatures ("mD signatures"). mD is a radar technique that uses a series of angle dependent electromagnetic pulses that are broadcast into an environment and return pulses are captured. Changes between the broadcast pulse and return pulse are indicative of changes in the shape, distance and angular location of objects or targets in the environment. These changes provide signals that can be used to track a target and identify the target through the mD signature. Each target or target type has a unique mD signature. Shifts in the radar pattern can be analyzed in the time domain and frequency domain based on mD techniques to derive information about the types of targets present (e.g. whether people are present), the motion of the targets and the relative angular location of the targets and the distance to the targets. By selecting a frequency used for the mD pulse relative to known objects in the environment, the pulse can penetrate the known objects to enable information about targets to be gathered even when the targets are visually blocked by the known objects. For example, pulse frequencies can be used that will penetrate concrete buildings to enable people to be identified inside the building. Multiple pulse frequencies can be used as well in the mD radar to enable different types of information to be gathered about the objects in the environment. In addition, the mD radar information can be combined with other information such as distance measurements or images captured of the environment that are analyzed jointly to provide improved object identification and improved target identification and tracking. In embodiments, the analysis can be performed on the HWC or the information can be transmitted to a remote network for analysis and results transmitted back to the HWC. Distance measurements can be provided by laser range finding, structured lighting, stereoscopic depth maps or sonar measurements. Images of the environment can be captured using one or more cameras capable of capturing images from visible, ultraviolet or infrared light. The mD radar can be attached to the HWC, located adjacently (e.g. in a vehicle) and associated wirelessly with the HWC or located remotely. Maps or other previously determined information about the environment can also be used in the analysis of the mD radar information. Embodiments of the present invention relate to visualizing the mD signatures in useful ways.

FIG. **21** illustrates a FOV **2102** of a HWC **102** from a wearer's perspective. The wearer, as described herein elsewhere, has a see-through FOV **2102** wherein the wearer views adjacent surroundings, such as the buildings illustrated in FIG. **21**. The wearer, as described herein elsewhere, can also see displayed digital content presented within a portion of the FOV **2102**. The embodiment illustrated in FIG. **21** is indicating that the wearer can see the buildings and other surrounding elements in the environment and

digital content representing traces, or travel paths, of bullets being fired by different people in the area. The surroundings are viewed through the transparency of the FOV **2102**. The traces are presented via the digital computer display, as described herein elsewhere. In embodiments, the trace presented is based on a mD signature that is collected and communicated to the HWC in real time. The mD radar itself may be on or near the wearer of the HWC **102** or it may be located remote from the wearer. In embodiments, the mD radar scans the area, tracks and identifies targets, such as bullets, and communicates traces, based on locations, to the HWC **102**.

There are several traces **2108** and **2104** presented to the wearer in the embodiment illustrated in FIG. **21**. The traces communicated from the mD radar may be associated with GPS locations and the GPS locations may be associated with objects in the environment, such as people, buildings, vehicles, etc, both in latitude and longitude perspective and an elevation perspective. The locations may be used as markers for the HWC such that the traces, as presented in the FOV, can be associated, or fixed in space relative to the markers. For example, if the friendly fire trace **2108** is determined, by the mD radar, to have originated from the upper right window of the building on the left, as illustrated in FIG. **21**, then a virtual marker may be set on or near the window. When the HWC views, through its camera or other sensor, for example, the building's window, the trace may then virtually anchor with the virtual marker on the window. Similarly, a marker may be set near the termination position or other flight position of the friendly fire trace **2108**, such as the upper left window of the center building on the right, as illustrated in FIG. **21**. This technique fixes in space the trace such that the trace appears fixed to the environmental positions independent of where the wearer is looking. So, for example, as the wearer's head turns, the trace appears fixed to the marked locations.

In embodiments, certain user positions may be known and thus identified in the FOV. For example, the shooter of the friendly fire trace **2108** may be from a known friendly combatant and as such his location may be known. The position may be known based on his GPS location based on a mobile communication system on him, such as another HWC **102**. In other embodiments, the friendly combatant may be marked by another friendly. For example, if the friendly position in the environment is known through visual contact or communicated information, a wearer of the HWC **102** may use a gesture or external user interface **104** to mark the location. If a friendly combatant location is known the originating position of the friendly fire trace **2108** may be color coded or otherwise distinguished from unidentified traces on the displayed digital content. Similarly, enemy fire traces **2104** may be color coded or otherwise distinguished on the displayed digital content. In embodiments, there may be an additional distinguished appearance on the displayed digital content for unknown traces.

In addition to situationally associated trace appearance, the trace colors or appearance may be different from the originating position to the terminating position. This path appearance change may be based on the mD signature. The mD signature may indicate that the bullet, for example, is slowing as it propagates and this slowing pattern may be reflected in the FOV **2102** as a color or pattern change. This can create an intuitive understanding of wear the shooter is located. For example, the originating color may be red, indicative of high speed, and it may change over the course of the trace to yellow, indicative of a slowing trace. This pattern changing may also be different for a friendly, enemy

and unknown combatant. The enemy may go blue to green for a friendly trace, for example.

FIG. **21** illustrates an embodiment where the user sees the environment through the FOV and may also see color coded traces, which are dependent on bullet speed and combatant type, where the traces are fixed in environmental positions independent on the wearer's perspective. Other information, such as distance, range, range rings, time of day, date, engagement type (e.g. hold, stop firing, back away, etc.) may also be displayed in the FOV.

Another aspect of the present invention relates to mD radar techniques that trace and identify targets through other objects, such as walls (referred to generally as through wall mD), and visualization techniques related therewith. FIG. **22** illustrates a through wall mD visualization technique according to the principles of the present invention. As described herein elsewhere, the mD radar scanning the environment may be local or remote from the wearer of a HWC **102**. The mD radar may identify a target (e.g. a person) that is visible **2204** and then track the target as he goes behind a wall **2208**. The tracking may then be presented to the wearer of a HWC **102** such that digital content reflective of the target and the target's movement, even behind the wall, is presented in the FOV **2202** of the HWC **102**. In embodiments, the target, when out of visible sight, may be represented by an avatar in the FOV to provide the wearer with imagery representing the target.

mD target recognition methods can identify the identity of a target based on the vibrations and other small movements of the target. This can provide a personal signature for the target. In the case of humans, this may result in a personal identification of a target that has been previously characterized. The cardio, heart beat, lung expansion and other small movements within the body may be unique to a person and if those attributes are pre-identified they may be matched in real time to provide a personal identification of a person in the FOV **2202**. The person's mD signatures may be determined based on the position of the person. For example, the database of personal mD signature attributes may include mD signatures for a person standing, sitting, laying down, running, walking, jumping, etc. This may improve the accuracy of the personal data match when a target is tracked through mD signature techniques in the field. In the event a person is personally identified, a specific indication of the person's identity may be presented in the FOV **2202**. The indication may be a color, shape, shade, name, indication of the type of person (e.g. enemy, friendly, etc.), etc. to provide the wearer with intuitive real time information about the person being tracked. This may be very useful in a situation where there is more than one person in an area of the person being tracked. If just one person in the area is personally identified, that person or the avatar of that person can be presented differently than other people in the area.

FIG. **23** illustrates an mD scanned environment **2300**. An mD radar may scan an environment in an attempt to identify objects in the environment. In this embodiment, the mD scanned environment reveals two vehicles **2302a** and **2302b**, an enemy combatant **2309**, two friendly combatants **2308a** and **2308b** and a shot trace **2318**. Each of these objects may be personally identified or type identified. For example, the vehicles **2302a** and **2302b** may be identified through the mD signatures as a tank and heavy truck. The enemy combatant **2309** may be identified as a type (e.g. enemy combatant) or more personally (e.g. by name). The friendly combatants may be identified as a type (e.g. friendly combatant) or more

personally (e.g. by name). The shot trace **2318** may be characterized by type of projectile or weapon type for the projectile, for example.

FIG. **23a** illustrates two separate HWC **102** FOV display techniques according to the principles of the present invention. FOV **2312** illustrates a map view **2310** where the mD scanned environment is presented. Here, the wearer has a perspective on the mapped area so he can understand all tracked targets in the area. This allows the wearer to traverse the area with knowledge of the targets. FOV **2312** illustrates a heads-up view to provide the wearer with an augmented reality style view of the environment that is in proximity of the wearer.

An aspect of the present invention relates to suppression of extraneous or stray light. As discussed herein elsewhere, eyeglow and faceglow are two such artifacts that develop from such light. Eyeglow and faceglow can be caused by image light escaping from the optics module. The escaping light is then visible, particularly in dark environments when the user is viewing bright displayed images with the HWC. Light that escapes through the front of the HWC is visible as eyeglow as it that light that is visible in the region of the user's eyes. Eyeglow can appear in the form of a small version of the displayed image that the user is viewing. Light that escapes from the bottom of the HWC shines onto the user's face, cheek or chest so that these portions of the user appear to glow. Eyeglow and faceglow can both increase the visibility of the user and highlight the use of the HWC, which may be viewed negatively by the user. As such, reducing eyeglow and faceglow is advantageous. In combat situations (e.g. the mD trace presentation scenarios described herein) and certain gaming situations, the suppression of extraneous or stray light is very important.

The disclosure relating to FIG. **6** shows an example where a portion of the image light passes through the combiner **602** such that the light shines onto the user's face, thereby illuminating a portion of the user's face in what is generally referred to herein as faceglow. Faceglow be caused by any portion of light from the HWC that illuminates the user's face.

An example of the source for the faceglow light can come from wide cone angle light associated with the image light incident onto the combiner **602**. Where the combiner can include a holographic mirror or a notch mirror in which the narrow bands of high reflectivity are matched to wavelengths of light by the light source. The wide cone angle associated with the image light corresponds with the field of view provided by the HWC. Typically the reflectivity of holographic mirrors and notch mirrors is reduced as the cone angle of the incident light is increased above 8 degrees. As a result, for a field of view of 30 degrees, substantial image light can pass through the combiner and cause faceglow.

FIG. **24** shows an illustration of a light trap **2410** for the faceglow light. In this embodiment, an extension of the outer shield lens of the HWC is coated with a light absorbing material in the region where the converging light responsible for faceglow is absorbed in a light trap **2410**. The light absorbing material can be black or it can be a filter designed to absorb only the specific wavelengths of light provided by the light source(s) in the HWC. In addition, the surface of the light trap **2410** may be textured or fibrous to further improve the absorption.

FIG. **25** illustrates an optical system for a HWC that includes an outer absorptive polarizer **2520** to block the faceglow light. In this embodiment, the image light is polarized and as a result the light responsible for faceglow is similarly polarized. The absorptive polarizer is oriented

with a transmission axis such that the faceglow light is absorbed and not transmitted. In this case, the rest of the imaging system in the HWC may not require polarized image light and the image light may be polarized at any point before the combiner. In embodiments, the transmission axis of the absorptive polarizer **2520** is oriented vertically so that external glare from water (S polarized light) is absorbed and correspondingly, the polarization of the image light is selected to be horizontal (S polarization). Consequently, image light that passes through the combiner **602** and is then incident onto the absorptive polarizer **2520**, is absorbed. In FIG. **25** the absorptive polarizer **2520** is shown outside the shield lens, alternatively the absorptive polarizer **2520** can be located inside the shield lens.

FIG. **26** illustrates an optical system for a HWC that includes a film with an absorptive notch filter **2620**. In this case, the absorptive notch filter absorbs narrow bands of light that are selected to match the light provided by the optical system's light source. As a result, the absorptive notch filter is opaque with respect to the faceglow light and is transparent to the remainder of the wavelengths included in the visible spectrum so that the user has a clear view of the surrounding environment. A triple notch filter suitable for this approach is available from Iridian Spectral Technologies, Ottawa, ON: <http://www.ilphotonics.com/cdv2/Iridian-Interference%20Filters/New%20filters/Triple%20Notch%20Filter.pdf>.

In embodiments, the combiner **602** may include a notch mirror coating to reflect the wavelengths of light in the image light and a notch filter **2620** can be selected in correspondence to the wavelengths of light provided by the light source and the narrow bands of high reflectivity provided by the notch mirror. In this way, image light that is not reflected by the notch mirror is absorbed by the notch filter **2620**. In embodiments of the invention the light source can provide one narrow band of light for a monochrome imaging or three narrow bands of light for full color imaging. The notch mirror and associated notch filter would then each provide one narrow band or three narrow bands of high reflectivity and absorption respectively.

FIG. **27** includes a microlouver film **2750** to block the faceglow light. Microlouver film is sold by 3M as ALCF-P, for example and is typically used as a privacy filter for computer. See [http://multimedia.3m.com/mws/mediaeewebserver?mwsid=SSSSSuH8gc7nZxtUoY_x1Y_eevUqel7zHvTSevTSevTSeSSSSS--&fn=ALCF-P_ABR2_Control_Film_DS.pdf](http://multimedia.3m.com/mws/mediaeewebserver?mwsid=SSSSSuH8gc7nZxtUoY_x1Y_eevUqel7zHvTSevTSeSSSSS--&fn=ALCF-P_ABR2_Control_Film_DS.pdf) The microlouver film transmits light within a somewhat narrow angle (e.g. 30 degrees of normal and absorbs light beyond 30 degrees of normal). In FIG. **27**, the microlouver film **2750** is positioned such that the faceglow light **2758** is incident beyond 30 degrees from normal while the see-through light **2755** is incident within 30 degrees of normal to the microlouver film **2750**. As such, the faceglow light **2758** is absorbed by the microlouver film and the see-through light **2755** is transmitted so that the user has a bright see-thru view of the surrounding environment.

We now turn back to a description of eye imaging technologies. Aspects of the present invention relate to various methods of imaging the eye of a person wearing the HWC **102**. In embodiments, technologies for imaging the eye using an optical path involving the "off" state and "no power" state, which is described in detail below, are described. In embodiments, technologies for imaging the eye with optical configurations that do not involve reflecting the eye image off of DLP mirrors is described. In embodiments, unstructured light, structured light, or controlled

lighting conditions, are used to predict the eye's position based on the light reflected off of the front of the wearer's eye. In embodiments, a reflection of a presented digital content image is captured as it reflects off of the wearer's eye and the reflected image may be processed to determine the quality (e.g. sharpness) of the image presented. In embodiments, the image may then be adjusted (e.g. focused differently) to increase the quality of the image presented based on the image reflection.

FIGS. 28a, 28b and 28c show illustrations of the various positions of the DLP mirrors. FIG. 28a shows the DLP mirrors in the "on" state 2815. With the mirror in the "on" state 2815, illumination light 2810 is reflected along an optical axis 2820 that extends into the lower optical module 204. FIG. 28b shows the DLP mirrors in the "off" state 2825. With the mirror in the "off" state 2825, illumination light 2810 is reflected along an optical axis 2830 that is substantially to the side of optical axis 2820 so that the "off" state light is directed toward a dark light trap as has been described herein elsewhere. FIG. 28c shows the DLP mirrors in a third position, which occurs when no power is applied to the DLP. This "no power" state differs from the "on" and "off" states in that the mirror edges are not in contact with the substrate and as such are less accurately positioned. FIG. 28c shows all of the DLP mirrors in the "no power" state 2835. The "no power" state is achieved by simultaneously setting the voltage to zero for the "on" contact and "off" contact for a DLP mirror, as a result, the mirror returns to a no stress position where the DLP mirror is in the plane of the DLP platform as shown in FIG. 28c. Although not normally done, it is also possible to apply the "no power" state to individual DLP mirrors. When the DLP mirrors are in the "no power" state they do not contribute image content. Instead, as shown in FIG. 28c, when the DLP mirrors are in the "no power" state, the illumination light 2810 is reflected along an optical axis 2840 that is between the optical axes 2820 and 2830 that are respectively associated with the "on" and "off" states and as such this light doesn't contribute to the displayed image as a bright or dark pixel. This light can however contribute scattered light into the lower optical module 204 and as a result the displayed image contrast can be reduced or artifacts can be created in the image that detract from the image content. Consequently, it is generally desirable, in embodiments, to limit the time associated with the "no power" state to times when images are not displayed or to reduce the time associated with having DLP mirrors in the "no power" state so that the affect of the scattered light is reduced.

FIG. 29 shows an embodiment of the invention that can be used for displaying digital content images to a wearer of the HWC 102 and capturing images of the wearer's eye. In this embodiment, light from the eye 2971 passes back through the optics in the lower module 204, the solid corrective wedge 2966, at least a portion of the light passes through the partially reflective layer 2960, the solid illumination wedge 2964 and is reflected by a plurality of DLP mirrors on the DLP 2955 that are in the "no power" state. The reflected light then passes back through the illumination wedge 2964 and at least a portion of the light is reflected by the partially reflective layer 2960 and the light is captured by the camera 2980.

For comparison, illuminating light rays 2973 from the light source 2958 are also shown being reflected by the partially reflective layer 2960. Where the angle of the illuminating light 2973 is such that the DLP mirrors, when in the "on" state, reflect the illuminating light 2973 to form image light 2969 that substantially shares the same optical

axis as the light from the wearer's eye 2971. In this way, images of the wearer's eye are captured in a field of view that overlaps the field of view for the displayed image content. In contrast, light reflected by DLP mirrors in the "off" state form dark light 2975 which is directed substantially to the side of the image light 2969 and the light from eye 2971. Dark light 2975 is directed toward a light trap 2962 that absorbs the dark light to improve the contrast of the displayed image as has been described above in this specification.

[In an embodiment, partially reflective layer 2960 is a reflective polarizer. The light that is reflected from the eye 2971 can then be polarized prior to entering the corrective wedge 2966 (e.g. with an absorptive polarizer between the upper module 202 and the lower module 204), with a polarization orientation relative to the reflective polarizer that enables the light reflected from the eye 2971 to substantially be transmitted by the reflective polarizer. A quarter wave retarder layer 2957 is then included adjacent to the DLP 2955 (as previously disclosed in FIG. 3b) so that the light reflected from the eye 2971 passes through the quarter wave retarder layer 2957 once before being reflected by the plurality of DLP mirrors in the "no power" state and then passes through a second time after being reflected. By passing through the quarter wave retarder layer 2957 twice, the polarization state of the light from the eye 2971 is reversed, such that when it is incident upon the reflective polarizer, the light from the eye 2971 is then substantially reflected toward the camera 2980. By using a partially reflective layer 2960 that is a reflective polarizer and polarizing the light from the eye 2971 prior to entering the corrective wedge 2964, losses attributed to the partially reflective layer 2960 are reduced.

FIG. 28c shows the case wherein the DLP mirrors are simultaneously in the "no power" state, this mode of operation can be particularly useful when the HWC 102 is first put onto the head of the wearer. When the HWC 102 is first put onto the head of the wearer, it is not necessary to display an image yet. As a result, the DLP can be in a "no power" state for all the DLP mirrors and an image of the wearer's eyes can be captured. The captured image of the wearer's eye can then be compared to a database, using iris identification techniques, or other eye pattern identification techniques to determine, for example, the identity of the wearer.

In a further embodiment illustrated by FIG. 29 all of the DLP mirrors are put into the "no power" state for a portion of a frame time (e.g. 50% of a frame time for the displayed digital content image) and the capture of the eye image is synchronized to occur at the same time and for the same duration. By reducing the time that the DLP mirrors are in the "no power" state, the time where light is scattered by the DLP mirrors being in the "no power" state is reduced such that the wearer doesn't perceive a change in the displayed image quality. This is possible because the DLP mirrors have a response time on the order of microseconds while typical frame times for a displayed image are on the order of 0.016 seconds. This method of capturing images of the wearer's eye can be used periodically to capture repetitive images of the wearer's eye. For example, eye images could be captured for 50% of the frame time of every 10th frame displayed to the wearer. In another example, eye images could be captured for 10% of the frame time of every frame displayed to the wearer.

Alternately, the "no power" state can be applied to a subset of the DLP mirrors (e.g. 10% of the DLP mirrors) within while another subset is in busy generating image light for content to be displayed. This enables the capture of an

eye image(s) during the display of digital content to the wearer. The DLP mirrors used for eye imaging can, for example, be distributed randomly across the area of the DLP to minimize the impact on the quality of the digital content being displayed to the wearer. To improve the displayed image perceived by the wearer, the individual DLP mirrors put into the “no power” state for capturing each eye image, can be varied over time such as in a random pattern, for example. In yet a further embodiment, the DLP mirrors put into the “no power” state for eye imaging may be coordinated with the digital content in such a way that the “no power” mirrors are taken from a portion of the image that requires less resolution.

In the embodiments of the invention as illustrated in FIGS. 9 and 29, in both cases the reflective surfaces provided by the DLP mirrors do not preserve the wavefront of the light from the wearer’s eye so that the image quality of captured image of the eye is somewhat limited. It may still be useful in certain embodiments, but it is somewhat limited. This is due to the DLP mirrors not being constrained to be on the same plane. In the embodiment illustrated in FIG. 9, the DLP mirrors are tilted so that they form rows of DLP mirrors that share common planes. In the embodiment illustrated in FIG. 29, the individual DLP mirrors are not accurately positioned to be in the same plane since they are not in contact with the substrate. Examples of advantages of the embodiments associated with FIG. 29 are: first, the camera 2980 can be located between the DLP 2955 and the illumination light source 2958 to provide a more compact upper module 202. Second, the polarization state of the light reflected from the eye 2971 can be the same as that of the image light 2969 so that the optical path of the light reflected from the eye and the image light can be the same in the lower module 204.

FIG. 30 shows an illustration of an embodiment for displaying images to the wearer and simultaneously capturing images of the wearer’s eye, wherein light from the eye 2971 is reflected towards a camera 3080 by the partially reflective layer 2960. The partially reflective layer 2960 can be an optically flat layer such that the wavefront of the light from the eye 2971 is preserved and as a result, higher quality images of the wearer’s eye can be captured. In addition, since the DLP 2955 is not included in the optical path for the light from the eye 2971, and the eye imaging process shown in FIG. 30 does not interfere with the displayed image, images of the wearer’s eye can be captured independently (e.g. with independent of timing, impact on resolution, or pixel count used in the image light) from the displayed images.

In the embodiment illustrated in FIG. 30, the partially reflective layer 2960 is a reflective polarizer, the illuminating light 2973 is polarized, the light from the eye 2971 is polarized and the camera 3080 is located behind a polarizer 3085. The polarization axis of the illuminating light 2973 and the polarization axis of the light from the eye are oriented perpendicular to the transmission axis of the reflective polarizer so that they are both substantially reflected by the reflective polarizer. The illumination light 2973 passes through a quarter wave layer 2957 before being reflected by the DLP mirrors in the DLP 2955. The reflected light passes back through the quarter wave layer 2957 so that the polarization states of the image light 2969 and dark light 2975 are reversed in comparison to the illumination light 2973. As such, the image light 2969 and dark light 2975 are substantially transmitted by the reflective polarizer. Where the DLP mirrors in the “on” state provide the image light 2969 along an optical axis that extends into the lower optical

module 204 to display an image to the wearer. At the same time, DLP mirrors in the “off” state provide the dark light 2975 along an optical axis that extends to the side of the upper optics module 202. In the region of the corrective wedge 2966 where the dark light 2975 is incident on the side of the upper optics module 202, an absorptive polarizer 3085 is positioned with its transmission axis perpendicular to the polarization axis of the dark light and parallel to the polarization axis of the light from the eye so that the dark light 2975 is absorbed and the light from the eye 2971 is transmitted to the camera 3080.

FIG. 31 shows an illustration of another embodiment of a system for displaying images and simultaneously capturing image of the wearer’s eye that is similar to the one shown in FIG. 30. The difference in the system shown in FIG. 31 is that the light from the eye 2971 is subjected to multiple reflections before being captured by the camera 3180. To enable the multiple reflections, a mirror 3187 is provided behind the absorptive polarizer 3185. Therefore, the light from the eye 2971 is polarized prior to entering the corrective wedge 2966 with a polarization axis that is perpendicular to the transmission axis of the reflective polarizer that comprises the partially reflective layer 2960. In this way, the light from the eye 2971 is reflected first by the reflective polarizer, reflected second by the mirror 3187 and reflected third by the reflective polarizer before being captured by the camera 3180. While the light from the eye 2971 passes through the absorptive polarizer 3185 twice, since the polarization axis of the light from the eye 2971 is oriented parallel to the polarization axis of the light from the eye 2971, it is substantially transmitted by the absorptive polarizer 3185. As with the system described in connection with FIG. 30, the system shown in FIG. 31 includes an optically flat partially reflective layer 2960 that preserves the wavefront of the light from the eye 2971 so that higher quality images of the wearer’s eye can be captured. Also, since the DLP 2955 is not included in the optical path for the light reflected from the eye 2971 and the eye imaging process shown in FIG. 31 does not interfere with the displayed image, images of the wearer’s eye can be captured independently from the displayed images.

FIG. 32 shows an illustration of a system for displaying images and simultaneously capturing images of the wearer’s eye that includes a beam splitter plate 3212 comprised of a reflective polarizer, which is held in air between the light source 2958, the DLP 2955 and the camera 3280. The illumination light 2973 and the light from the eye 2971 are both polarized with polarization axes that are perpendicular to the transmission axis of the reflective polarizer. As a result, both the illumination light 2973 and the light from the eye 2971 are substantially reflected by the reflective polarizer. The illumination light 2973 is reflected toward the DLP 2955 by the reflective polarizer and split into image light 2969 and dark light 3275 depending on whether the individual DLP mirrors are respectively in the “on” state or the “off” state. By passing through the quarter wave layer 2957 twice, the polarization state of the illumination light 2973 is reversed in comparison to the polarization state of the image light 2969 and the dark light 3275. As a result, the image light 2969 and the dark light 3275 are then substantially transmitted by the reflective polarizer. The absorptive polarizer 3285 at the side of the beam splitter plate 3212 has a transmission axis that is perpendicular to the polarization axis of the dark light 3275 and parallel to the polarization axis of the light from the eye 2971 so that the dark light 3275 is absorbed and the light from the eye 2971 is transmitted to the camera 3280. As in the system shown in FIG. 30, the

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system shown in FIG. 31 includes an optically flat beam splitter plate 3212 that preserves the wavefront of the light from the eye 2971 so that higher quality images of the wearer's eye can be captured. Also, since the DLP 2955 is not included in the optical path for the light from the eye 2971 and the eye imaging process shown in FIG. 31 does not interfere with the displayed image, images of the wearer's eye can be captured independently from the displayed images.

Eye imaging systems where the polarization state of the light from the eye 2971 needs to be opposite to that of the image light 2969 (as shown in FIGS. 30, 31 and 32), need to be used with lower modules 204 that include combiners that will reflect both polarization states. As such, these upper modules 202 are best suited for use with the lower modules 204 that include combiners that are reflective regardless of polarization state, examples of these lower modules are shown in FIGS. 6, 8a, 8b, 8c and 24-27.

In a further embodiment shown in FIG. 33, the partially reflective layer 3360 is comprised of a reflective polarizer on the side facing the illumination light 2973 and a short pass dichroic mirror on the side facing the light from the eye 3371 and the camera 3080. Where the short pass dichroic mirror is a dielectric mirror coating that transmits visible light and reflects infrared light. The partially reflective layer 3360 can be comprised of a reflective polarizer bonded to the inner surface of the illumination wedge 2964 and a short pass dichroic mirror coating on the opposing inner surface of the corrective wedge 2966, wherein the illumination wedge 2964 and the corrective wedge 2966 are then optically bonded together. Alternatively, the partially reflective layer 3360 can be comprised of a thin substrate that has a reflective polarizer bonded to one side and a short pass dichroic mirror coating on the other side, where the partially reflective layer 3360 is then bonded between the illumination wedge 2964 and the corrective wedge 2966. In this embodiment, an infrared light is included to illuminate the eye so that the light from the eye and the images captured of the eye are substantially comprised of infrared light. The wavelength of the infrared light is then matched to the reflecting wavelength of the shortpass dichroic mirror and the wavelength that the camera can capture images, for example an 800 nm wavelength can be used. In this way, the short pass dichroic mirror transmits the image light and reflects the light from the eye. The camera 3080 is then positioned at the side of the corrective wedge 2966 in the area of the absorbing light trap 3382, which is provided to absorb the dark light 2975. By positioning the camera 3080 in a depression in the absorbing light trap 3382, scattering of the dark light 2975 by the camera 3080 can be reduced so that higher contrast images can be displayed to the wearer. An advantage of this embodiment is that the light from the eye need not be polarized, which can simplify the optical system and increase efficiency for the eye imaging system.

In yet another embodiment shown in FIG. 32a a beam splitter plate 3222 is comprised of a reflective polarizer on the side facing the illumination light 2973 and a short pass dichroic mirror on the side facing the light from the eye 3271 and the camera 3280. An absorbing surface 3295 is provided to trap the dark light 3275 and the camera 3280 is positioned in an opening in the absorbing surface 3295. In this way the system of FIG. 32 can be made to function with unpolarized light from the eye 3271.

In embodiments directed to capturing images of the wearer's eye, light to illuminate the wearer's eye can be provided by several different sources including: light from the displayed image (i.e. image light); light from the envi-

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ronment that passes through the combiner or other optics; light provided by a dedicated eye light, etc. FIGS. 34 and 34a show illustrations of dedicated eye illumination lights 3420. FIG. 34 shows an illustration from a side view in which the dedicated illumination eye light 3420 is positioned at a corner of the combiner 3410 so that it doesn't interfere with the image light 3415. The dedicated eye illumination light 3420 is pointed so that the eye illumination light 3425 illuminates the eyebox 3427 where the eye 3430 is located when the wearer is viewing displayed images provided by the image light 3415. FIG. 34a shows an illustration from the perspective of the eye of the wearer to show how the dedicated eye illumination light 3420 is positioned at the corner of the combiner 3410. While the dedicated eye illumination light 3420 is shown at the upper left corner of the combiner 3410, other positions along one of the edges of the combiner 3410, or other optical or mechanical components, are possible as well. In other embodiments, more than one dedicated eye light 3420 with different positions can be used. In an embodiment, the dedicated eye light 3420 is an infrared light that is not visible by the wearer (e.g. 800 nm) so that the eye illumination light 3425 doesn't interfere with the displayed image perceived by the wearer.

FIG. 35 shows a series of illustrations of captured eye images that show the eye glint (i.e. light that reflects off the front of the eye) produced by a dedicated eye light. In this embodiment of the invention, captured images of the wearer's eye are analyzed to determine the relative positions of the iris 3550, pupil, or other portion of the eye, and the eye glint 3560. The eye glint is a reflected image of the dedicated eye light 3420 when the dedicated light is used. FIG. 35 illustrates the relative positions of the iris 3550 and the eye glint 3560 for a variety of eye positions. By providing a dedicated eye light 3420 in a fixed position, combined with the fact that the human eye is essentially spherical, or at least a reliably repeatable shape, the eye glint provides a fixed reference point against which the determined position of the iris can be compared to determine where the wearer is looking, either within the displayed image or within the see-through view of the surrounding environment. By positioning the dedicated eye light 3420 at a corner of the combiner 3410, the eye glint 3560 is formed away from the iris 3550 in the captured images. As a result, the positions of the iris and the eye glint can be determined more easily and more accurately during the analysis of the captured images, since they do not interfere with one another. In a further embodiment, the combiner includes an associated cut filter that prevents infrared light from the environment from entering the HWC and the camera is an infrared camera, so that the eye glint is only provided by light from the dedicated eye light. For example, the combiner can include a low pass filter that passes visible light while absorbing infrared light and the camera can include a high pass filter that absorbs visible light while passing infrared light.

In an embodiment of the eye imaging system, the lens for the camera is designed to take into account the optics associated with the upper module 202 and the lower module 204. This is accomplished by designing the camera to include the optics in the upper module 202 and optics in the lower module 204, so that a high MTF image is produced, at the image sensor in the camera, of the wearer's eye. In yet a further embodiment, the camera lens is provided with a large depth of field to eliminate the need for focusing the camera to enable sharp image of the eye to be captured. Where a large depth of field is typically provided by a high f/# lens (e.g. $f/\# > 5$). In this case, the reduced light gathering

associated with high $f/\#$ lenses is compensated by the inclusion of a dedicated eye light to enable a bright image of the eye to be captured. Further, the brightness of the dedicated eye light can be modulated and synchronized with the capture of eye images so that the dedicated eye light has a reduced duty cycle and the brightness of infrared light on the wearer's eye is reduced.

In a further embodiment, FIG. 36a shows an illustration of an eye image that is used to identify the wearer of the HWC. In this case, an image of the wearer's eye 3611 is captured and analyzed for patterns of identifiable features 3612. The patterns are then compared to a database of eye images to determine the identity of the wearer. After the identity of the wearer has been verified, the operating mode of the HWC and the types of images, applications, and information to be displayed can be adjusted and controlled in correspondence to the determined identity of the wearer. Examples of adjustments to the operating mode depending on who the wearer is determined to be or not be include: making different operating modes or feature sets available, shutting down or sending a message to an external network, allowing guest features and applications to run, etc.

FIG. 36b is an illustration of another embodiment using eye imaging, in which the sharpness of the displayed image is determined based on the eye glint produced by the reflection of the displayed image from the wearer's eye surface. By capturing images of the wearer's eye 3611, an eye glint 3622, which is a small version of the displayed image can be captured and analyzed for sharpness. If the displayed image is determined to not be sharp, then an automated adjustment to the focus of the HWC optics can be performed to improve the sharpness. This ability to perform a measurement of the sharpness of a displayed image at the surface of the wearer's eye can provide a very accurate measurement of image quality. Having the ability to measure and automatically adjust the focus of displayed images can be very useful in augmented reality imaging where the focus distance of the displayed image can be varied in response to changes in the environment or changes in the method of use by the wearer.

An aspect of the present invention relates to controlling the HWC 102 through interpretations of eye imagery. In embodiments, eye-imaging technologies, such as those described herein, are used to capture an eye image or series of eye images for processing. The image(s) may be processed to determine a user intended action, an HWC predetermined reaction, or other action. For example, the imagery may be interpreted as an affirmative user control action for an application on the HWC 102. Or, the imagery may cause, for example, the HWC 102 to react in a pre-determined way such that the HWC 102 is operating safely, intuitively, etc.

FIG. 37 illustrates a eye imagery process that involves imaging the HWC 102 wearer's eye(s) and processing the images (e.g. through eye imaging technologies described herein) to determine in what position 3702 the eye is relative to its neutral or forward looking position and/or the FOV 3708. The process may involve a calibration step where the user is instructed, through guidance provided in the FOV of the HWC 102, to look in certain directions such that a more accurate prediction of the eye position relative to areas of the FOV can be made. In the event the wearer's eye is determined to be looking towards the right side of the FOV 3708 (as illustrated in FIG. 37, the eye is looking out of the page) a virtual target line may be established to project what in the environment the wearer may be looking towards or at. The virtual target line may be used in connection with an image captured by camera on the HWC 102 that images the

surrounding environment in front of the wearer. In embodiments, the field of view of the camera capturing the surrounding environment matches, or can be matched (e.g. digitally), to the FOV 3708 such that making the comparison is made more clear. For example, with the camera capturing the image of the surroundings in an angle that matches the FOV 3708 the virtual line can be processed (e.g. in 2d or 3d, depending on the camera images capabilities and/or the processing of the images) by projecting what surrounding environment objects align with the virtual target line. In the event there are multiple objects along the virtual target line, focal planes may be established corresponding to each of the objects such that digital content may be placed in an area in the FOV 3708 that aligns with the virtual target line and falls at a focal plane of an intersecting object. The user then may see the digital content when he focuses on the object in the environment, which is at the same focal plane. In embodiments, objects in line with the virtual target line may be established by comparison to mapped information of the surroundings.

In embodiments, the digital content that is in line with the virtual target line may not be displayed in the FOV until the eye position is in the right position. This may be a predetermined process. For example, the system may be set up such that a particular piece of digital content (e.g. an advertisement, guidance information, object information, etc.) will appear in the event that the wearer looks at a certain object(s) in the environment. A virtual target line(s) may be developed that virtually connects the wearer's eye with an object(s) in the environment (e.g. a building, portion of a building, mark on a building, gps location, etc.) and the virtual target line may be continually updated depending on the position and viewing direction of the wearer (e.g. as determined through GPS, e-compass, IMU, etc.) and the position of the object. When the virtual target line suggests that the wearer's pupil is substantially aligned with the virtual target line or about to be aligned with the virtual target line, the digital content may be displayed in the FOV 3704.

In embodiments, the time spent looking along the virtual target line and/or a particular portion of the FOV 3708 may indicate that the wearer is interested in an object in the environment and/or digital content being displayed. In the event there is no digital content being displayed at the time a predetermined period of time is spent looking at a direction, digital content may be presented in the area of the FOV 3708. The time spent looking at an object may be interpreted as a command to display information about the object, for example. In other embodiments, the content may not relate to the object and may be presented because of the indication that the person is relatively inactive. In embodiments, the digital content may be positioned in proximity to the virtual target line, but not in-line with it such that the wearer's view of the surroundings are not obstructed but information can augment the wearer's view of the surroundings. In embodiments, the time spent looking along a target line in the direction of displayed digital content may be an indication of interest in the digital content. This may be used as a conversion event in advertising. For example, an advertiser may pay more for an add placement if the wearer of the HWC 102 looks at a displayed advertisement for a certain period of time. As such, in embodiments, the time spent looking at the advertisement, as assessed by comparing eye position with the content placement, target line or other appropriate position may be used to determine a rate of conversion or other compensation amount due for the presentation.

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An aspect of the invention relates to removing content from the FOV of the HWC **102** when the wearer of the HWC **102** apparently wants to view the surrounding environments clearly. FIG. **38** illustrates a situation where eye imagery suggests that the eye has or is moving quickly so the digital content **3804** in the FOV **3808** is removed from the FOV **3808**. In this example, the wearer may be looking quickly to the side indicating that there is something on the side in the environment that has grabbed the wearer's attention. This eye movement **3802** may be captured through eye imaging techniques (e.g. as described herein) and if the movement matches a predetermined movement (e.g. speed, rate, pattern, etc.) the content may be removed from view. In embodiments, the eye movement is used as one input and HWC movements indicated by other sensors (e.g. IMU in the HWC) may be used as another indication. These various sensor movements may be used together to project an event that should cause a change in the content being displayed in the FOV.

Another aspect of the present invention relates to determining a focal plane based on the wearer's eye convergence. Eyes are generally converged slightly and converge more when the person focuses on something very close. This is generally referred to as convergence. In embodiments, convergence is calibrated for the wearer. That is, the wearer may be guided through certain focal plane exercises to determine how much the wearer's eyes converge at various focal planes and at various viewing angles. The convergence information may then be stored in a database for later reference. In embodiments, a general table may be used in the event there is no calibration step or the person skips the calibration step. The two eyes may then be imaged periodically to determine the convergence in an attempt to understand what focal plane the wearer is focused on. In embodiments, the eyes may be imaged to determine a virtual target line and then the eye's convergence may be determined to establish the wearer's focus, and the digital content may be displayed or altered based thereon.

FIG. **39** illustrates a situation where digital content is moved **3902** within one or both of the FOVs **3908** and **3910** to align with the convergence of the eyes as determined by the pupil movement **3904**. By moving the digital content to maintain alignment, in embodiments, the overlapping nature of the content is maintained so the object appears properly to the wearer. This can be important in situations where 3D content is displayed.

An aspect of the present invention relates to controlling the HWC **102** based on events detected through eye imaging. A wearer winking, blinking, moving his eyes in a certain pattern, etc. may, for example, control an application of the HWC **102**. Eye imaging (e.g. as described herein) may be used to monitor the eye(s) of the wearer and once a predetermined pattern is detected an application control command may be initiated.

An aspect of the invention relates to monitoring the health of a person wearing a HWC **102** by monitoring the wearer's eye(s). Calibrations may be made such that the normal performance, under various conditions (e.g. lighting conditions, image light conditions, etc.) of a wearer's eyes may be documented. The wearer's eyes may then be monitored through eye imaging (e.g. as described herein) for changes in their performance. Changes in performance may be indicative of a health concern (e.g. concussion, brain injury, stroke, loss of blood, etc.). If detected the data indicative of the change or event may be communicated from the HWC **102**.

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Aspects of the present invention relate to security and access of computer assets (e.g. the HWC itself and related computer systems) as determined through eye image verification. As discussed herein elsewhere, eye imagery may be compared to known person eye imagery to confirm a person's identity. Eye imagery may also be used to confirm the identity of people wearing the HWCs **102** before allowing them to link together or share files, streams, information, etc.

A variety of use cases for eye imaging are possible based on technologies described herein. An aspect of the present invention relates to the timing of eye image capture. The timing of the capture of the eye image and the frequency of the capture of multiple images of the eye can vary dependent on the use case for the information gathered from the eye image. For example, capturing an eye image to identify the user of the HWC may be required only when the HWC has been turned ON or when the HWC determines that the HWC has been put onto a wearer's head, to control the security of the HWC and the associated information that is displayed to the user. Wherein, the orientation, movement pattern, stress or position of the earhorns (or other portions of the HWC) of the HWC can be used to determine that a person has put the HWC onto their head with the intention to use the HWC. Those same parameters may be monitored in an effort to understand when the HWC is dismounted from the user's head. This may enable a situation where the capture of an eye image for identifying the wearer may be completed only when a change in the wearing status is identified. In a contrasting example, capturing eye images to monitor the health of the wearer may require images to be captured periodically (e.g. every few seconds, minutes, hours, days, etc.). For example, the eye images may be taken in minute intervals when the images are being used to monitor the health of the wearer when detected movements indicate that the wearer is exercising. In a further contrasting example, capturing eye images to monitor the health of the wearer for long-term effects may only require that eye images be captured monthly. Embodiments of the invention relate to selection of the timing and rate of capture of eye images to be in correspondence with the selected use scenario associated with the eye images. These selections may be done automatically, as with the exercise example above where movements indicate exercise, or these selections may be set manually. In a further embodiment, the selection of the timing and rate of eye image capture is adjusted automatically depending on the mode of operation of the HWC. The selection of the timing and rate of eye image capture can further be selected in correspondence with input characteristics associated with the wearer including age and health status, or sensed physical conditions of the wearer including heart rate, chemical makeup of the blood and eye blink rate.

FIG. **40** illustrates an embodiment in which digital content presented in a see-through FOV is positioned based on the speed in which the wearer is moving. When the person is not moving, as measured by sensor(s) in the HWC **102** (e.g. IMU, GPS based tracking, etc.), digital content may be presented at the stationary person content position **4004**. The content position **4004** is indicated as being in the middle of the see-through FOV **4002**; however, this is meant to illustrate that the digital content is positioned within the see-through FOV at a place that is generally desirable knowing that the wearer is not moving and as such the wearer's surrounding see through view can be somewhat obstructed. So, the stationary person content position, or neutral position, may not be centered in the see-through FOV; it may be positioned somewhere in the see-through FOV deemed desirable and the sensor feedback may shift the digital

content from the neutral position. The movement of the digital content for a quickly moving person is also shown in FIG. 40 wherein as the person turns their head to the side, the digital content moves out of the see-through FOV to content position 4008 and then moves back as the person turns their head back. For a slowly moving person, the head movement can be more complex and as such the movement of the digital content in an out of the see-through FOV can follow a path such as that shown by content position 4010.

In embodiments, the sensor that assesses the wearer's movements may be a GPS sensor, IMU, accelerometer, etc. The content position may be shifted from a neutral position to a position towards a side edge of the field of view as the forward motion increases. The content position may be shifted from a neutral position to a position towards a top or bottom edge of the field of view as the forward motion increases. The content position may shift based on a threshold speed of the assessed motion. The content position may shift linearly based on the speed of the forward motion. The content position may shift non-linearly based on the speed of the forward motion. The content position may shift outside of the field of view. In embodiments, the content is no longer displayed if the speed of movement exceeds a predetermined threshold and will be displayed again once the forward motion slows.

In embodiments, the content position may generally be referred to as shifting; it should be understood that the term shifting encompasses a process where the movement from one position to another within the see-through FOV or out of the FOV is visible to the wearer (e.g. the content appears to slowly or quickly move and the user perceives the movement itself) or the movement from one position to another may not be visible to the wearer (e.g. the content appears to jump in a discontinuous fashion or the content disappears and then reappears in the new position).

Another aspect of the present invention relates to removing the content from the field of view or shifting it to a position within the field of view that increases the wearer's view of the surrounding environment when a sensor causes an alert command to be issued. In embodiments, the alert may be due to a sensor or combination of sensors that sense a condition above a threshold value. For example, if an audio sensor detects a loud sound of a certain pitch, content in the field of view may be removed or shifted to provide a clear view of the surrounding environment for the wearer. In addition to the shifting of the content, in embodiments, an indication of why the content was shifted may be presented in the field of view or provided through audio feedback to the wearer. For instance, if a carbon monoxide sensor detects a high concentration in the area, content in the field of view may be shifted to the side of the field of view or removed from the field of view and an indication may be provided to the wearer that there is a high concentration of carbon monoxide in the area. This new information, when presented in the field of view, may similarly be shifted within or outside of the field of view depending on the movement speed of the wearer.

FIG. 41 illustrates how content may be shifted from a neutral position 4104 to an alert position 4108. In this embodiment, the content is shifted outside of the see-through FOV 4102. In other embodiments, the content may be shifted as described herein.

Another aspect of the present invention relates to identification of various vectors or headings related to the HWC 102, along with sensor inputs, to determine how to position content in the field of view. In embodiments, the speed of movement of the wearer is detected and used as an input for

position of the content and, depending on the speed, the content may be positioned with respect to a movement vector or heading (i.e. the direction of the movement), or a sight vector or heading (i.e. the direction of the wearer's sight direction). For example, if the wearer is moving very fast the content may be positioned within the field of view with respect to the movement vector because the wearer is only going to be looking towards the sides of himself periodically and for short periods of time. As another example, if the wearer is moving slowly, the content may be positioned with respect to the sight heading because the user may more freely be shifting his view from side to side.

FIG. 42 illustrates two examples where the movement vector may effect content positioning. Movement vector A 4202 is shorter than movement vector B 4210 indicating that the forward speed and/or acceleration of movement of the person associated with movement vector A 4202 is lower than the person associated with movement vector B 4210. Each person is also indicated as having a sight vector or heading 4208 and 4212. The sight vectors A 4208 and B 4210 are the same from a relative perspective. The white area inside of the black triangle in front of each person is indicative of how much time each person likely spends looking at a direction that is not in line with the movement vector. The time spent looking off angle A 4204 is indicated as being more than that of the time spent looking off angle B 4214. This may be because the movement vector speed A is lower than movement vector speed B. The faster the person moves forward the more the person tends to look in the forward direction, typically. The FOVs A 4218 and B 4222 illustrate how content may be aligned depending on the movement vectors 4202 and 4210 and sight vectors 4208 and 4212. FOV A 4218 is illustrated as presenting content in-line with the sight vector 4220. This may be due to the lower speed of the movement vector A 4202. This may also be due to the prediction of a larger amount of time spent looking off angle A 4204. FOV B 4222 is illustrated as presenting content in line with the movement vector 4224. This may be due to the higher speed of movement vector B 4210. This may also be due to the prediction of a shorter amount of time spent looking off angle B 4214.

Another aspect of the present invention relates to damping a rate of content position change within the field of view. As illustrated in FIG. 43, the sight vector may undergo a rapid change 4304. This rapid change may be an isolated event or it may be made at or near a time when other sight vector changes are occurring. The wearer's head may be turning back and forth for some reason. In embodiments, the rapid successive changes in sight vector may cause a damped rate of content position change 4308 within the FOV 4302. For example, the content may be positioned with respect to the sight vector, as described herein, and the rapid change in sight vector may normally cause a rapid content position change; however, since the sight vector is successively changing, the rate of position change with respect to the sight vector may be damped, slowed, or stopped. The position rate change may be altered based on the rate of change of the sight vector, average of the sight vector changes, or otherwise altered.

Another aspect of the present invention relates to simultaneously presenting more than one content in the field of view of a see-through optical system of a HWC 102 and positioning one content with the sight heading and one content with the movement heading. FIG. 44 illustrates two FOV's A 4414 and B 4420, which correspond respectively to the two identified sight vectors A 4402 and B 4404. FIG. 44 also illustrates an object in the environment 4408 at a

position relative to the sight vectors A 4402 and B 4404. When the person is looking along sight vector A 4402, the environment object 4408 can be seen through the field of view A 4414 at position 4412. As illustrated, sight heading aligned content is presented as TEXT in proximity with the environment object 4412. At the same time, other content 4418 is presented in the field of view A 4414 at a position aligned in correspondence with the movement vector. As the movement speed increases, the content 4418 may shift as described herein. When the sight vector of the person is sight vector B 4404 the environmental object 4408 is not seen in the field of view B 4420. As a result, the sight aligned content 4410 is not presented in field of view B 4420; however, the movement aligned content 4418 is presented and is still dependent on the speed of the motion.

FIG. 45 shows an example set of data for a person moving through an environment over a path that starts with a movement heading of 0 degrees and ends with a movement heading of 114 degrees during which time the speed of movement varies from 0m/sec to 20 m/sec. The sight heading can be seen to vary on either side of the movement heading while moving as the person looks from side to side. Large changes in sight heading occur when the movement speed is 0m/sec when the person is standing still, followed by step changes in movement heading.

Embodiments provide a process for determining the display heading that takes into account the way a user moves through an environment and provides a display heading that makes it easy for the user to find the displayed information while also providing unencumbered see-through views of the environment in response to different movements, speed of movement or different types of information being displayed.

FIG. 46 illustrates a see-through view as may be seen when using a HWC wherein information is overlaid onto a see-through view of the environment. The tree and the building are actually in the environment and the text is displayed in the see-through display such that it appears overlaid on the environment. In addition to text information such as, for example, instructions and weather information, some augmented reality information is shown that relates to nearby objects in the environment.

In an embodiment, the display heading is determined based on speed of movement. At low speeds, the display heading may be substantially the same as the sight heading while at high speed the display heading may be substantially the same as the movement heading. In embodiments, as long as the user remains stationary, the displayed information is presented directly in front of the user and HWC. However, as the movement speed increases (e.g. above a threshold or continually, etc.) the display heading becomes substantially the same as the movement heading regardless of the direction the user is looking, so that when the user looks in the direction of movement, the displayed information is directly in front of the user and HMD and when the user looks to the side the displayed information is not visible.

Rapid changes in sight heading can be followed by a slower change in the display heading to provide a damped response to head rotation. Alternatively, the display heading can be substantially the time averaged sight heading so that the displayed information is presented at a heading that is in the middle of a series of sight headings over a period of time. In this embodiment, if the user stops moving their head, the display heading gradually becomes the same as the sight heading and the displayed information moves into the display field of view in front of the user and HMD. In embodiments, when there is a high rate of sight heading

change, the process delays the effect of the time averaged sight heading on the display heading. In this way, the effect of rapid head movements on display heading is reduced and the positioning of the displayed information within the display field of view is stabilized laterally.

In another embodiment, display heading is determined based on speed of movement where at high-speed, the display heading is substantially the same as the movement heading. At mid-speed the display heading is substantially the same as a time averaged sight heading so that rapid head rotations are damped out and the display heading is in the middle of back and forth head movements.

In yet another embodiment, the type of information being displayed is included in determining how the information should be displayed. Augmented reality information that is connected to objects in the environment is given a display heading that substantially matches the sight heading. In this way, as the user rotates their head, augmented reality information comes into view that is related to objects that are in the see-through view of the environment. At the same time, information that is not connected to objects in the environment is given a display heading that is determined based on the type of movements and speed of movements as previously described in this specification.

In yet a further embodiment, when the speed of movement is determined to be above a threshold, the information displayed is moved downward in the display field of view so that the upper portion of the display field of view has less information or no information displayed to provide the user with an unencumbered see-through view of the environment.

FIGS. 47 and 48 show illustrations of a see-through view including overlaid displayed information. FIG. 47 shows the see-through view immediately after a rapid change in sight heading from the sight heading associated with the see-through view shown in FIG. 46 wherein the change in sight heading comes from a head rotation. In this case, the display heading is delayed. FIG. 48 shows how at a later time, the display heading catches up to the sight heading. The augmented reality information remains in positions within the display field of view where the association with objects in the environment can be readily made by the user.

FIG. 49 shows an illustration of a see-through view example including overlaid displayed information that has been shifted downward in the display field of view to provide an unencumbered see-through view in the upper portion of the see-through view. At the same time, augmented reality labels have been maintained in locations within the display field of view so they can be readily associated with objects in the environment.

In a further embodiment, in an operating mode such as when the user is moving in an environment, digital content is presented at the side of the user's see-through FOV so that the user can only view the digital content by turning their head. In this case, when the user is looking straight ahead, such as when the movement heading matches the sight heading, the see-through view FOV does not include digital content. The user then accesses the digital content by turning their head to the side whereupon the digital content moves laterally into the user's see-through FOV. In another embodiment, the digital content is ready for presentation and will be presented if an indication for its presentation is received. For example, the information may be ready for presentation and if the sight heading or predetermined position of the HWC 102 is achieved the content may then be presented. The wearer may look to the side and the content may be presented. In another embodiment, the user

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may cause the content to move into an area in the field of view by looking in a direction for a predetermined period of time, blinking, winking, or displaying some other pattern that can be captured through eye imaging technologies (e.g. as described herein elsewhere).

In yet another embodiment, an operating mode is provided wherein the user can define sight headings wherein the associated see-through FOV includes digital content or does not include digital content. In an example, this operating mode can be used in an office environment where when the user is looking at a wall digital content is provided within the FOV, whereas when the user is looking toward a hallway, the FOV is unencumbered by digital content. In another example, when the user is looking horizontally digital content is provided within the FOV, but when the user looks down (e.g. to look at a desktop or a cellphone) the digital content is removed from the FOV.

Another aspect of the present invention relates to collecting and using eye position and sight heading information. Head worn computing with motion heading, sight heading, and/or eye position prediction (sometimes referred to as "eye heading" herein) may be used to identify what a wearer of the HWC **102** is apparently interested in and the information may be captured and used. In embodiments, the information may be characterized as viewing information because the information apparently relates to what the wearer is looking at. The viewing information may be used to develop a personal profile for the wearer, which may indicate what the wearer tends to look at. The viewing information from several or many HWC's **102** may be captured such that group or crowd viewing trends may be established. For example, if the movement heading and sight heading are known, a prediction of what the wearer is looking at may be made and used to generate a personal profile or portion of a crowd profile. In another embodiment, if the eye heading and location, sight heading and/or movement heading are known, a prediction of what is being looked at may be predicted. The prediction may involve understanding what is in proximity of the wearer and this may be understood by establishing the position of the wearer (e.g. through GPS or other location technology) and establishing what mapped objects are known in the area. The prediction may involve interpreting images captured by the camera or other sensors associated with the HWC **102**. For example, if the camera captures an image of a sign and the camera is in-line with the sight heading, the prediction may involve assessing the likelihood that the wearer is viewing the sign. The prediction may involve capturing an image or other sensory information and then performing object recognition analysis to determine what is being viewed. For example, the wearer may be walking down a street and the camera that is in the HWC **102** may capture an image and a processor, either on-board or remote from the HWC **102**, may recognize a face, object, marker, image, etc. and it may be determined that the wearer may have been looking at it or towards it.

FIG. **50** illustrates a cross section of an eyeball of a wearer of an HWC with focus points that can be associated with the eye imaging system of the invention. The eyeball **5010** includes an iris **5012** and a retina **5014**. Because the eye imaging system of the invention provides coaxial eye imaging with a display system, images of the eye can be captured from a perspective directly in front of the eye and inline with where the wearer is looking. In embodiments of the invention, the eye imaging system can be focused at the iris **5012** and/or the retina **5014** of the wearer, to capture images of the external surface of the iris **5012** or the internal portions of

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the eye, which includes the retina **5014**. FIG. **50** shows light rays **5020** and **5025** that are respectively associated with capturing images of the iris **5012** or the retina **5014** wherein the optics associated with the eye imaging system are respectively focused at the iris **5012** or the retina **5014**. Illuminating light can also be provided in the eye imaging system to illuminate the iris **5012** or the retina **5014**. FIG. **51** shows an illustration of an eye including an iris **5130** and a sclera **5125**. In embodiments, the eye imaging system can be used to capture images that include the iris **5130** and portions the sclera **5125**. The images can then be analyzed to determine color, shapes and patterns that are associated with the user. In further embodiments, the focus of the eye imaging system is adjusted to enable images to be captured of the iris **5012** or the retina **5014**. Illuminating light can also be adjusted to illuminate the iris **5012** or to pass through the pupil of the eye to illuminate the retina **5014**. The illuminating light can be visible light to enable capture of colors of the iris **5012** or the retina **5014**, or the illuminating light can be ultraviolet (e.g. 340 nm), near infrared (e.g. 850 nm) or mid-wave infrared (e.g. 5000 nm) light to enable capture of hyperspectral characteristics of the eye.

FIG. **53** illustrates a display system that includes an eye imaging system. The display system includes a polarized light source **2958**, a DLP **2955**, a quarter wave film **2957** and a beam splitter plate **5345**. The eye imaging system includes a camera **3280**, illuminating lights **5355** and beam splitter plate **5345**. Where the beam splitter plate **5345** can be a reflective polarizer on the side facing the polarized light source **2958** and a hot mirror on the side facing the camera **3280**. Wherein the hot mirror reflects infrared light (e.g. wavelengths 700 to 2000 nm) and transmits visible light (e.g. wavelengths 400 to 670 nm). The beam splitter plate **5345** can be comprised of multiple laminated films, a substrate film with coatings or a rigid transparent substrate with films on either side. By providing a reflective polarizer on the one side, the light from the polarized light source **2958** is reflected toward the DLP **2955** where it passes through the quarter wave film **2957** once, is reflected by the DLP mirrors in correspondence with the image content being displayed by the DLP **2955** and then passes back through the quarter wave film **2957**. In so doing, the polarization state of the light from the polarized light source is changed, so that it is transmitted by the reflective polarizer on the beam splitter plate **5345** and the image light **2971** passes into the lower optics module **204** where the image is displayed to the user. At the same time, infrared light **5357** from the illuminating lights **5355** is reflected by the hot mirror so that it passes into the lower optics module **204** where it illuminates the user's eye. Portions of the infrared light **2969** are reflected by the user's eye and this light passes back through the lower optics module **204**, is reflected by the hot mirror on the beam splitter plate **5345** and is captured by the camera **3280**. In this embodiment, the image light **2971** is polarized while the infrared light **5357** and **2969** can be unpolarized. In an embodiment, the illuminating lights **5355** provide two different infrared wavelengths and eye images are captured in pairs, wherein the pairs of eye images are analyzed together to improve the accuracy of identification of the user based on iris analysis.

FIG. **54** shows an illustration of a further embodiment of a display system with an eye imaging system. In addition to the features of FIG. **53**, this system includes a second camera **5460**. Wherein the second camera **5460** is provided to capture eye images in the visible wavelengths. Illumination of the eye can be provided by the displayed image or by see-through light from the environment. Portions of the

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displayed image can be modified to provide improved illumination of the user's eye when images of the eye are to be captured such as by increasing the brightness of the displayed image or increasing the white areas within the displayed image. Further, modified displayed images can be presented briefly for the purpose of capturing eye images and the display of the modified images can be synchronized with the capture of the eye images. As shown in FIG. 54, visible light 5467 is polarized when it is captured by the second camera 5460 since it passes through the beam splitter 5445 and the beam splitter 5445 is a reflective polarizer on the side facing the second camera 5460. In this eye imaging system, visible eye images can be captured by the second camera 5460 at the same time that infrared eye images are captured by the camera 3280. Wherein, the characteristics of the camera 3280 and the second camera 5460 and the associated respective images captured can be different in terms of resolution and capture rate.

FIGS. 52a and 52b illustrate captured images of eyes where the eyes are illuminated with structured light patterns. In FIG. 52a, an eye 5220 is shown with a projected structured light pattern 5230, where the light pattern is a grid of lines. A light pattern of such as 5230 can be provided by the light source 5355 shown in FIG. 53 by including a diffractive or a refractive device to modify the light 5357 as are known by those skilled in the art. A visible light source can also be included for the second camera 5460 shown in FIG. 54 which can include a diffractive or refractive to modify the light 5467 to provide a light pattern. FIG. 52b illustrates how the structured light pattern of 5230 becomes distorted to 5235 when the user's eye 5225 looks to the side. This distortion comes from the fact that the human eye is not spherical in shape, instead the iris sticks out slightly from the eyeball to form a bump in the area of the iris. As a result, the shape of the eye and the associated shape of the reflected structured light pattern is different depending on which direction the eye is pointed, when images of the eye are captured from a fixed position. Changes in the structured light pattern can subsequently be analyzed in captured eye images to determine the direction that the eye is looking.

The eye imaging system can also be used for the assessment of aspects of health of the user. In this case, information gained from analyzing captured images of the iris 5012 is different from information gained from analyzing captured images of the retina 5014. Where images of the retina 5014 are captured using light 5357 that illuminates the inner portions of the eye including the retina 5014. The light 5357 can be visible light, but in an embodiment, the light 5357 is infrared light (e.g. wavelength 1 to 5 microns) and the camera 3280 is an infrared light sensor (e.g. an InGaAs sensor) or a low resolution infrared image sensor that is used to determine the relative amount of light 5357 that is absorbed, reflected or scattered by the inner portions of the eye. Wherein the majority of the light that is absorbed, reflected or scattered can be attributed to materials in the inner portion of the eye including the retina where there are densely packed blood vessels with thin walls so that the absorption, reflection and scattering are caused by the material makeup of the blood. These measurements can be conducted automatically when the user is wearing the HWC, either at regular intervals, after identified events or when prompted by an external communication. In a preferred embodiment, the illuminating light is near infrared or mid infrared (e.g. 0.7 to 5 microns wavelength) to reduce the chance for thermal damage to the wearer's eye. In another embodiment, the polarizer 3285 is antireflection coated to reduce any reflections from this surface from the light 5357,

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the light 2969 or the light 3275 and thereby increase the sensitivity of the camera 3280. In a further embodiment, the light source 5355 and the camera 3280 together comprise a spectrometer wherein the relative intensity of the light reflected by the eye is analyzed over a series of narrow wavelengths within the range of wavelengths provided by the light source 5355 to determine a characteristic spectrum of the light that is absorbed, reflected or scattered by the eye. For example, the light source 5355 can provide a broad range of infrared light to illuminate the eye and the camera 3280 can include: a grating to laterally disperse the reflected light from the eye into a series of narrow wavelength bands that are captured by a linear photodetector so that the relative intensity by wavelength can be measured and a characteristic absorbance spectrum for the eye can be determined over the broad range of infrared. In a further example, the light source 5355 can provide a series of narrow wavelengths of light (ultraviolet, visible or infrared) to sequentially illuminate the eye and camera 3280 includes a photodetector that is selected to measure the relative intensity of the series of narrow wavelengths in a series of sequential measurements that together can be used to determine a characteristic spectrum of the eye. The determined characteristic spectrum is then compared to known characteristic spectra for different materials to determine the material makeup of the eye. In yet another embodiment, the illuminating light 5357 is focused on the retina 5014 and a characteristic spectrum of the retina 5014 is determined and the spectrum is compared to known spectra for materials that may be present in the user's blood. For example, in the visible wavelengths 540 nm is useful for detecting hemoglobin and 660 nm is useful for differentiating oxygenated hemoglobin. In a further example, in the infrared, a wide variety of materials can be identified as is known by those skilled in the art, including: glucose, urea, alcohol and controlled substances. FIG. 55 shows a series of example spectrum for a variety of controlled substances as measured using a form of infrared spectroscopy (ThermoScientific Application Note 51242, by C. Petty, B. Garland and the Mesa Police Department Forensic Laboratory, which is hereby incorporated by reference herein). FIG. 56 shows an infrared absorbance spectrum for glucose (Hewlett Packard Company 1999, G. Hopkins, G. Mauze; "In-vivo NIR Diffuse-reflectance Tissue Spectroscopy of Human Subjects," which is hereby incorporated by reference herein). U.S. Pat. No. 6,675,030, which is hereby incorporated by reference herein, provides a near infrared blood glucose monitoring system that includes infrared scans of a body part such as a foot. United States Patent publication 2006/0183986, which is hereby incorporated by reference herein, provides a blood glucose monitoring system including a light measurement of the retina. Embodiments of the present invention provide methods for automatic measurements of specific materials in the user's blood by illuminating at one or more narrow wavelengths into the iris of the wearer's eye and measuring the relative intensity of the light reflected by the eye to identify the relative absorbance spectrum and comparing the measured absorbance spectrum with known absorbance spectra for the specific material, such as illuminating at 540 and 660 nm to determine the level of hemoglobin present in the user's blood.

Another aspect of the present invention relates to collecting and using eye position and sight heading information. Head worn computing with motion heading, sight heading, and/or eye position prediction (sometimes referred to as "eye heading" herein) may be used to identify what a wearer of the HWC 102 is apparently interested in and the infor-

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mation may be captured and used. In embodiments, the information may be characterized as viewing information because the information apparently relates to what the wearer is looking at. The viewing information may be used to develop a personal profile for the wearer, which may indicate what the wearer tends to look at. The viewing information from several or many HWC's 102 may be captured such that group or crowd viewing trends may be established. For example, if the movement heading and sight heading are known, a prediction of what the wearer is looking at may be made and used to generate a personal profile or portion of a crowd profile. In another embodiment, if the eye heading and location, sight heading and/or movement heading are known, a prediction of what is being looked at may be predicted. The prediction may involve understanding what is in proximity of the wearer and this may be understood by establishing the position of the wearer (e.g. through GPS or other location technology) and establishing what mapped objects are known in the area. The prediction may involve interpreting images captured by the camera or other sensors associated with the HWC 102. For example, if the camera captures an image of a sign and the camera is in-line with the sight heading, the prediction may involve assessing the likelihood that the wearer is viewing the sign. The prediction may involve capturing an image or other sensory information and then performing object recognition analysis to determine what is being viewed. For example, the wearer may be walking down a street and the camera that is in the HWC 102 may capture an image and a processor, either on-board or remote from the HWC 102, may recognize a face, object, marker, image, etc. and it may be determined that the wearer may have been looking at it or towards it.

FIG. 57 illustrates a scene where a person is walking with a HWC 102 mounted on his head. In this scene, the person's geo-spatial location 5704 is known through a GPS sensor, which could be another location system, and his movement heading, sight heading 5714 and eye heading 5702 are known and can be recorded (e.g. through systems described herein). There are objects and a person in the scene. Person 5712 may be recognized by the wearer's HWC 102 system, the person may be mapped (e.g. the person's GPS location may be known or recognized), or otherwise known. The person may be wearing a garment or device that is recognizable. For example, the garment may be of a certain style and the HWC may recognize the style and record it's viewing. The scene also includes a mapped object 5718 and a recognized object 5720. As the wearer moves through the scene, the sight and/or eye headings may be recorded and communicated from the HWC 102. In embodiments, the time that the sight and/or eye heading maintains a particular position may be recorded. For example, if a person appears to look at an object or person for a predetermined period of time (e.g. 2 seconds or longer), the information may be communicated as gaze persistence information as an indication that the person may have been interested in the object.

In embodiments, sight headings may be used in conjunction with eye headings or eye and/or sight headings may be used alone. Sight headings can do a good job of predicting what direction a wearer is looking because many times the eyes are looking forward, in the same general direction as the sight heading. In other situations, eye headings may be a more desirable metric because the eye and sight headings are not always aligned. In embodiments herein examples may be provided with the term "eye/sight" heading, which indicates that either or both eye heading and sight heading may be used in the example.

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FIG. 58 illustrates a system for receiving, developing and using movement heading, sight heading, eye heading and/or persistence information from HWC(s) 102. The server 5804 may receive heading or gaze persistence information, which is noted as persistence information 5802, for processing and/or use. The heading and/or gaze persistence information may be used to generate a personal profile 5808 and/or a group profile 5810. The personal profile 5718 may reflect the wearer's general viewing tendencies and interests. The group profile 5810 may be an assemblage of different wearer's heading and persistence information to create impressions of general group viewing tendencies and interests. The group profile 5810 may be broken into different groups based on other information such as gender, likes, dislikes, biographical information, etc. such that certain groups can be distinguished from other groups. This may be useful in advertising because an advertiser may be interested in what a male adult sports goer is generally looking at as oppose to a younger female. The profiles 5808 and 5810 and raw heading and persistence information may be used by retailers 5814, advertisers 5818, trainers, etc. For example, an advertiser may have an advertisement posted in an environment and may be interested in knowing how many people look at the advertisement, how long they look at it and where they go after looking at it. This information may be used as conversion information to assess the value of the advertisement and thus the payment to be received for the advertisement.

In embodiments, the process involves collecting eye and/or sight heading information from a plurality of head-worn computers that come into proximity with an object in an environment. For example, a number of people may be walking through an area and each of the people may be wearing a head worn computer with the ability to track the position of the wearer's eye(s) as well as possibly the wearer's sight and movement headings. The various HWC wearing individuals may then walk, ride, or otherwise come into proximity with some object in the environment (e.g. a store, sign, person, vehicle, box, bag, etc.). When each person passes by or otherwise comes near the object, the eye imaging system may determine if the person is looking towards the object. All of the eye/sight heading information may be collected and used to form impressions of how the crowd reacted to the object. A store may be running a sale and so the store may put out a sign indicating such. The storeowners and managers may be very interested to know if anyone is looking at their sign. The sign may be set as the object of interest in the area and as people navigate near the sign, possibly determined by their GPS locations, the eye/sight heading determination system may record information relative to the environment and the sign. Once, or as, the eye/sight heading information is collected and associations between the eye headings and the sign are determined, feedback may be sent back to the storeowner, managers, advertiser, etc. as an indication of how well their sign is attracting people. In embodiments, the sign's effectiveness at attracting people's attention, as indicated through the eye/sight headings, may be considered a conversion metric and impact the economic value of the sign and/or the signs placement.

In embodiments, a map of the environment with the object may be generated by mapping the locations and movement paths of the people in the crowd as they navigate by the object (e.g. the sign). Layered on this map may be an indication of the various eye/sight headings. This may be useful in indicating where people were in relation to the object when they viewed the object. The map may also have

an indication of how long people looked at the object from the various positions in the environment and where they went after seeing the object.

In embodiments, the process involves collecting a plurality of eye/sight headings from a head-worn computer, wherein each of the plurality of eye/sight headings is associated with a different pre-determined object in an environment. This technology may be used to determine which of the different objects attracts more of the person's attention. For example, if there are three objects placed in an environment and a person enters the environment navigating his way through it, he may look at one or more of the objects and his eye/sight heading may persist on one or more objects longer than others. This may be used in making or refining the person's personal attention profile and/or it may be used in connection with other such people's data on the same or similar objects to determine an impression of how the population or crowd reacts to the objects. Testing advertisements in this way may provide good feedback of its effectiveness.

In embodiments, the process may involve capturing eye/sight headings once there is substantial alignment between the eye/sight heading and an object of interest. For example, the person with the HWC may be navigating through an environment and once the HWC detects substantial alignment or the projected occurrence of an upcoming substantial alignment between the eye/sight heading and the object of interest, the occurrence and/or persistence may be recorded for use.

In embodiments, the process may involve collecting eye/sight heading information from a head-worn computer and collecting a captured image from the head-worn computer that was taken at substantially the same time as the eye/sight heading information was captured. These two pieces of information may be used in conjunction to gain an understanding of what the wearer was looking at and possibly interested in. The process may further involve associating the eye/sight heading information with an object, person, or other thing found in the captured image. This may involve processing the captured image looking for objects or patterns. In embodiments, gaze time or persistence may be measured and used in conjunction with the image processing. The process may still involve object and/or pattern recognition, but it may also involve attempting to identify what the person gazed at for the period of time by more particularly identifying a portion of the image in conjunction with image processing.

In embodiments, the process may involve setting a pre-determined eye/sight heading from a pre-determined geospatial location and using them as triggers. In the event that a head worn computer enters the geospatial location and an eye/sight heading associated with the head worn computer aligns with the pre-determined eye/sight heading, the system may collect the fact that there was an apparent alignment and/or the system may record information identifying how long the eye/sight heading remains substantially aligned with the pre-determined eye/sight heading to form a persistence statistic. This may eliminate or reduce the need for image processing as the triggers can be used without having to image the area. In other embodiments, image capture and processing is performed in conjunction with the triggers. In embodiments, the triggers may be a series a geospatial locations with corresponding eye/sight headings such that many spots can be used as triggers that indicate when a person entered an area in proximity to an object of interest and/or when that person actually appeared to look at the object.

In embodiments, eye imaging may be used to capture images of both eyes of the wearer in order to determine the amount of convergence of the eyes (e.g. through technologies described herein elsewhere) to get an understanding of what focal plane is being concentrated on by the wearer. For example, if the convergence measurement suggests that the focal plane is within 15 feet of the wearer, than, even though the eye/sight headings may align with an object that is more than 15 feet away it may be determined that the wearer was not looking at the object. If the object were within the 15 foot suggested focal plane, the determination may be that the wearer was looking at the object. FIG. 59 illustrates an environmentally position locked digital content 5912 that is indicative of a person's location 5902. In this disclosure the term "BlueForce" is generally used to indicate team members or members for which geo-spatial locations are known and can be used. In embodiments, "Blue Force" is a term to indicate members of a tactical arms team (e.g. a police force, secret service force, security force, military force, national security force, intelligence force, etc.). In many embodiments herein one member may be referred to as the primary or first BlueForce member and it is this member, in many described embodiments, that is wearing the HWC. It should be understood that this terminology is to help the reader and make for clear presentations of the various situations and that other members of the Blueforce, or other people, may have HWC's 102 and have similar capabilities. In this embodiment, a first person is wearing a head-worn computer 102 that has a see through field of view ("FOV") 5914. The first person can see through the FOV to view the surrounding environment through the FOV and digital content can also be presented in the FOV such that the first person can view the actual surroundings, through the FOV, in a digitally augmented view. The other BlueForce person's location is known and is indicated at a position inside of a building at point 5902. This location is known in three dimensions, longitude, latitude and altitude, which may have been determined by GPS along with an altimeter associated with the other Blueforce person. Similarly, the location of the first person wearing the HWC 102 is also known, as indicated in FIG. 59 as point 5908. In this embodiment, the compass heading 5910 of the first person is also known. With the compass heading 5910 known, the angle in which the first person is viewing the surroundings can be estimated. A virtual target line between the location of the first person 5908 and the other person's location 5902 can be established in three dimensional space and emanating from the HWC 102 proximate the FOV 5914. The three dimensionally oriented virtual target line can then be used to present environmentally position locked digital content in the FOV 5914, which is indicative of the other person's location 5902. The environmentally position locked digital content 5902 can be positioned within the FOV 5914 such that the first person, who is wearing the HWC 102, perceives the content 5902 as locked in position within the environment and marking the location of the other person 5902.

The three dimensionally positioned virtual target line can be recalculated periodically (e.g. every millisecond, second, minute, etc.) to reposition the environmentally position locked content 5912 to remain in-line with the virtual target line. This can create the illusion that the content 5912 is staying positioned within the environment at a point that is associated with the other person's location 5902 independent of the location of the first person 5908 wearing the HWC 102 and independent of the compass heading of the HWC 102.

In embodiments, the environmentally locked digital content **5912** may be positioned with an object **5904** that is between the first person's location **5908** and the other person's location **5902**. The virtual target line may intersect the object **5904** before intersecting with the other person's location **5902**. In embodiments, the environmentally locked digital content **5912** may be associated with the object intersection point **5904**. In embodiments, the intersecting object **5904** may be identified by comparing the two person's locations **5902** and **5908** with obstructions identified on a map. In embodiments the intersecting object **5904** may be identified by processing images captured from a camera, or other sensor, associated with the HWC **102**. In embodiments, the digital content **5912** has an appearance that is indicative of being at the location of the other person **5902**, at the location of the intersecting object **5904** to provide a more clear indication of the position of the other person's position **5902** in the FOV **5914**.

FIG. **60** illustrates how and where digital content may be positioned within the FOV **6008** based on a virtual target line between the location of the first person **5908**, who's wearing the HWC **102**, and the other person **5902**. In addition to positioning the content in a position within the FOV **6008** that is in-line with the virtual target line, the digital content may be presented such that it comes into focus by the first person when the first person focuses at a certain plane or distance in the environment. Presented object **A 6018** is digitally generated content that is presented as an image at content position **A 6012**. The position **6012** is based on the virtual target line. The presented object **A 6018** is presented not only along the virtual target line but also at a focal plane **B 6014** such that the content at position **A 6012** in the FOV **6008** comes into focus by the first person when the first person's eye **6002** focuses at something in the surrounding environment at the focal plane **B 6014** distance. Setting the focal plane of the presented content provides content that does not come into focus until the eye **6002** focuses at the set focal plane. In embodiments, this allows the content at position **A** to be presented without when the HWC's compass is indicative of the first person looking in the direction of the other person **5902** but it will only come into focus when the first person focuses on in the direction of the other person **5902** and at the focal plane of the other person **5902**.

Presented object **B 6020** is aligned with a different virtual target line then presented object **A 6018**. Presented object **B 6020** is also presented at content position **B 6004** at a different focal plane than the content position **A 6012**. Presented content **B 6020** is presented at a further focal plane, which is indicative that the other person **5902** is physically located at a further distance. If the focal planes are sufficiently different, the content at position **A** will come into focus at a different time than the content at position **B** because the two focal planes require different focus from the eye **6002**.

FIG. **61** illustrates several BlueForce members at locations with various points of view from the first person's perspective. In embodiments, the relative positions, distances and obstacles may cause the digital content indicative of the other person's location to be altered. For example, if the other person can be seen by the first person through the first person's FOV, the digital content may be locked at the location of the other person and the digital content may be of a type that indicates the other person's position is being actively marked and tracked. If the other person is in relatively close proximity, but cannot be seen by the first person, the digital content may be locked to an intersecting

object or area and the digital content may indicate that the actual location of the other person cannot be seen but the mark is generally tracking the other persons general position. If the other person is not within a pre-determined proximity or is otherwise more significantly obscured from the first person's view, the digital content may generally indicate a direction or area where the other person is located and the digital content may indicate that the other person's location is not closely identified or tracked by the digital content, but that the other person is in the general area.

Continuing to refer to FIG. **61**, several BlueForce members are presented at various positions within an area where the first person is located. The primary BlueForce member **6102** (also referred to generally as the first person, or the person wherein the HWC with the FOV for example purposes) can directly see the BlueForce member in the open field **6104**. In embodiments, the digital content provided in the FOV of the primary BlueForce member may be based on a virtual target line and virtually locked in an environment position that is indicative of the open field position of the BlueForce member **6104**. The digital content may also indicate that the location of the open field BlueForce member is marked and is being tracked. The digital content may change forms if the Blue Force member becomes obscured from the vision of the primary BlueForce member or otherwise becomes unavailable for direct viewing.

BlueForce member **6108** is obscured from the primary BlueForce member's **6102** view by an obstacle that is in close proximity to the obscured member **6108**. As depicted, the obscured member **6108** is in a building but close to one of the front walls. In this situation, the digital content provided in the FOV of the primary member **6102** may be indicative of the general position of the obscured member **6108** and the digital content may indicate that, while the other person's location is fairly well marked, it is obscured so it is not as precise as if the person was in direct view. In addition, the digital content may be virtually positionally locked to some feature on the outside of the building that the obscured member is in. This may make the environmental locking more stable and also provide an indication that the location of the person is somewhat unknown.

BlueForce member **6110** is obscured by multiple obstacles. The member **6110** is in a building and there is another building **6112** in between the primary member **6102** and the obscured member **6110**. In this situation, the digital content in the FOV of the primary member will be spatially quite short of the actual obscured member and as such the digital content may need to be presented in a way that indicates that the obscured member **6110** is in a general direction but that the digital marker is not a reliable source of information for the particular location of obscured member **6110**.

FIG. **62** illustrates yet another method for positioning digital content within the FOV of a HWC where the digital content is intended to indicate a position of another person. This embodiment is similar to the embodiment described in connection with FIG. **62** herein. The main additional element in this embodiment is the additional step of verifying the distance between the first person **5908**, the one wearing the HWC with the FOV digital content presentation of location, and the other person at location **5902**. Here, the range finder may be included in the HWC and measure a distance at an angle that is represented by the virtual target line. In the event that the range finder finds an object obstructing the path of the virtual target line, the digital content presentation in the FOV may indicate such (e.g. as described herein elsewhere). In the event that the range

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finder confirms that there is a person or object at the end of the prescribed distance and angle defined by the virtual target line, the digital content may represent that the proper location has been marked, as described herein elsewhere.

Another aspect of the present invention relates to predicting the movement of BlueForce members to maintain proper virtual marking of the BlueForce member locations. FIG. 63 illustrates a situation where the primary BlueForce member 6302 is tracking the locations of the other BlueForce members through an augmented environment using a HWC 102, as described herein elsewhere (e.g. as described in connection with the above figures). The primary BlueForce member 6302 may have knowledge of the tactical movement plan 6308. The tactical movement plan may be maintained locally (e.g. on the HWCs 102 with sharing of the plan between the BlueForce members) or remotely (e.g. on a server and communicated to the HWC's 102, or communicated to a subset of HWC's 102 for HWC 102 sharing). In this case, the tactical plan involves the BlueForce group generally moving in the direction of the arrow 6308. The tactical plan may influence the presentations of digital content in the FOV of the HWC 102 of the primary BlueForce member. For example, the tactical plan may assist in the prediction of the location of the other BlueForce member and the virtual target line may be adjusted accordingly. In embodiments, the area in the tactical movement plan may be shaded or colored or otherwise marked with digital content in the FOV such that the primary BlueForce member can manage his activities with respect to the tactical plan. For example, he may be made aware that one or more BlueForce members are moving towards the tactical path 6308. He may also be made aware of movements in the tactical path that do not appear associated with BlueForce members.

FIG. 63 also illustrates that internal IMU sensors in the HWCs worn by the BlueForce members may provide guidance on the movement of the members 6304. This may be helpful in identifying when a GPS location should be updated and hence updating the position of the virtual marker in the FOV. This may also be helpful in assessing the validity of the GPS location. For example, if the GPS location has not updated but there is significant IMU sensor activity, the system may call into question the accuracy of the identified location. The IMU information may also be useful to help track the position of a member in the event the GPS information is unavailable. For example, dead reckoning may be used if the GPS signal is lost and the virtual marker in the FOV may indicate both indicated movements of the team member and indicate that the location identification is not ideal. The current tactical plan 6308 may be updated periodically and the updated plans may further refine what is presented in the FOV of the HWC 102.

FIG. 64 illustrates a BlueForce tracking system in accordance with the principles of the present invention. In embodiments, the BlueForce HWC's 102 may have directional antenna's that emit relatively low power directional RF signals such that other BlueForce members within the range of the relatively low power signal can receive and assess it's direction and/or distance based on the strength and varying strength of the signals. In embodiments, the tracking of such RF signals can be used to alter the presentation of the virtual markers of persons locations within the FOV of HWC 102.

Another aspect of the present invention relates to monitoring the health of BlueForce members. Each BlueForce member may be automatically monitored for health and stress events. For example, the members may have a watch-

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band as described herein elsewhere or other wearable biometric monitoring device and the device may continually monitor the biometric information and predict health concerns or stress events. As another example, the eye imaging systems described herein elsewhere may be used to monitor pupil dilatations as compared to normal conditions to predict head trauma. Each eye may be imaged to check for differences in pupil dilation for indications of head trauma. As another example, an IMU in the HWC 102 may monitor a person's walking gate looking for changes in pattern, which may be an indication of head or other trauma. Biometric feedback from a member indicative of a health or stress concern may be uploaded to a server for sharing with other members or the information may be shared with local members, for example. Once shared, the digital content in the FOV that indicates the location of the person having the health or stress event may include an indication of the health event.

FIG. 65 illustrates a situation where the primary BlueForce member 6502 is monitoring the location of the BlueForce member 6504 that has had a health event and caused a health alert to be transmitted from the HWC 102. As described herein elsewhere, the FOV of the HWC 102 of the primary BlueForce member may include an indication of the location of the BlueForce member with the health concern 6504. The digital content in the FOV may also include an indication of the health condition in association with the location indication. In embodiments, non-biometric sensors (e.g. IMU, camera, ranger finder, accelerometer, altimeter, etc.) may be used to provide health and/or situational conditions to the BlueForce team or other local or remote persons interested in the information. For example, if one of the BlueForce members is detected as quickly hitting the ground from a standing position an alter may be sent as an indication of a fall, the person is in trouble and had to drop down, was shot, etc.

Another aspect of the present invention relates to virtually marking various prior acts and events. For example, as depicted in FIG. 66, the techniques described herein elsewhere may be used to construct a virtual prior movement path 6604 of a BlueForce member. The virtual path may be displayed as digital content in the FOV of the primary BlueForce member 6602 using methods described herein elsewhere. As the BlueForce member moved along the path 6604 he may have virtually placed an event marker 6608 such that when another member views the location the mark can be displayed as digital content. For example, the BlueForce member may inspect and clear an area and then use an external user interface or gesture to indicate that the area has been cleared and then the location would be virtually marked and shared with BlueForce members. Then, when someone wants to understand if the location was inspected he can view the location's information. As indicated herein elsewhere, if the location is visible to the member, the digital content may be displayed in a way that indicates the specific location and if the location is not visible from the person's perspective, the digital content may be somewhat different in that it may not specifically mark the location.

Turning back to optical configurations, another aspect of the present invention relates to an optical configuration that provides digitally displayed content to an eye of a person wearing a head-worn display (e.g. as used in a HWC 102) and allows the person to see through the display such that the digital content is perceived by the person as augmenting the see through view of the surrounding environment. The optical configuration may have a variable transmission optical element that is in-line with the person's see-through view

such that the transmission of the see-through view can be increased and decreased. This may be helpful in situations where a person wants or would be better served with a high transmission see-through view and when, in the same HWC 102, the person wants or would be better served with less see-through transmission. The lower see-through transmission may be used in bright conditions and/or in conditions where higher contrast for the digitally presented content is desirable. The optical system may also have a camera that images the surrounding environment by receiving reflected light from the surrounding environment off of an optical element that is in-line with the person's see-through view of the surrounding. In embodiments, the camera may further be aligned in a dark light trap such that light reflected and/or transmitted in the direction of the camera that is not captured by the camera is trapped to reduce stray light.

In embodiments, a HWC 102 is provided that includes a camera that is coaxially aligned with the direction that the user is looking. FIG. 67 shows an illustration of an optical system 6715 that includes an absorptive polarizer 6737 and a camera 6739. The image source 6710 can include light sources, displays and reflective surfaces as well as one or more lenses 6720. Image light 6750 is provided by the image source 6710 wherein, a portion of the image light 6750 is reflected toward the user's eye 6730 by a partially reflective combiner 6735. At the same time, a portion of the image light 6750 may be transmitted by the combiner 6735 such that it is incident onto the absorptive polarizer 6737. In this embodiment, the image light 6750 is polarized light with the polarization state of the image light 6750 oriented relative to the transmission axis of the absorptive polarizer 6737 such that the incident image light 6750 is absorbed by the absorptive polarizer 6737. In this way, faceglow produced by escaping image light 6750 is reduced. In embodiments, the absorptive polarizer 6737 includes an antireflection coating to reduce reflections from the surface of the absorptive polarizer 6737.

FIG. 67 further shows a camera 6739 for capturing images of the environment in the direction that the user is looking. The camera 6739 is positioned behind the absorptive polarizer 6737 and below the combiner 6735 so that a portion of light from the environment 6770 is reflected by the combiner 6735 toward the camera 6739. Light from the environment 6770 can be unpolarized so that a portion of the light from the environment 6770 that is reflected by the combiner 6735 passes through the absorptive polarizer 6737 and it is this light that is captured by the camera 6739. As a result, the light captured by the camera will have a polarization state that is opposite that of the image light 6750. In addition, the camera 6739 is aligned relative to the combiner 6735 such that the field of view associated with the camera 6739 is coaxial to the display field of view provided by image light 6750. At the same time, a portion of scene light 6760 from the environment is transmitted by the combiner 6735 to provide a see-through view of the environment to the user's eye 6730. Where the display field of view associated with the image light 6750 is typically coincident to the see-through field of view associated with the scene light 6760 and thereby the see through field of view and the field of view of the camera 6739 are at least partially coaxial. By attaching the camera 6739 to the lower portion of the optical system 6715, the field of view of the camera 6739 as shown by the light from the environment 6770 moves as the user moves their head so that images captured by the camera 6739 correspond to the area of the environment that the user is looking at. By coaxially aligning the camera field of view with the displayed image and the user's view of the scene,

augmented reality images with improved alignment to objects in the scene can be provided. This is because the captured images from the camera 6739 provide an accurate representation of the user's perspective view of the scene. As an example, when the user sees an object in the scene as being located in the middle of the see-through view of the HWC, the object will be located in the middle of the image captured by the camera and any augmented reality imagery that is to be associated with the object can be located in the middle of the displayed image. As the user moves their head, the relative position of the object as seen in the see-through view of the scene will change and the position of the augmented reality imagery can be changed within the displayed image in a corresponding manner. When a camera 6739 is provided for each of the user's eyes, an accurate representation of the 3D view of the scene can be provided as well. This is an important advantage provided by the invention because images captured by a camera located in the frame of the HWC (e.g. between the eyes or at the corners) capture images that are laterally offset from the user's perspective of the scene and as a result it is difficult to align augmented reality images with objects in the scene as seen from the user's perspective.

In the optical system 6715 shown in FIG. 67, the absorptive polarizer 6737 simultaneously functions as a light trap for escaping image light 6750, a light blocker of the image light 6750 for the camera 6739 and a window for light from the environment 6770 to the camera 6739. This is possible because the polarization state of the image light 6750 is perpendicular to the transmission axis of the absorptive polarizer 6737 while the light from the environment 6770 is unpolarized so that a portion of the light from the environment 6770 that is the opposite polarization state to the image light is transmitted by the absorptive polarizer 6737. The combiner 6735 can be any partially reflective surface including a simple partial mirror, a notch mirror and a holographic mirror. The reflectivity of the combiner 6735 can be selected to be greater than 50% (e.g. 55% reflectivity and 45% transmission over the visible wavelength spectral band) whereby a majority of the image light 6750 will be reflected toward the user's eye 6730 and a majority of light from the environment 6770 will be reflected toward the camera 6739, this system will provide a brighter displayed image, a brighter captured image with a dimmer see-through view of the environment. Alternatively, the reflectivity of the combiner 6735 can be selected to be less than 50% (e.g. 20% reflectivity and 80% transmission over the visible wavelength spectral band) whereby the majority of the image light 6750 will be transmitted by the combiner 6735 and a majority of light from the environment 6770 will be transmitted to the user's eye 6730, this system will provide a brighter see-through view of the environment, while providing a dimmer displayed image and a dimmer captured image. As such, the system can be designed to favor the anticipated use by the user.

In embodiments, the combiner 6735 is planar with an optical flatness that is sufficient to enable a sharp displayed image and a sharp captured image, such as a flatness of less than 20 waves of light within the visible wavelengths. However, in embodiments, the combiner 6735 may be curved in which case the displayed image and the captured image will both be distorted and this distortion will have to be digitally corrected by the associated image processing system. In the case of the displayed image, the image is digitally distorted by the image processing system in a direction that is opposite to the distortion that is caused by the curved combiner so the two distortions cancel one

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another and as a result the user sees an undistorted displayed image. In the case of the captured image, the captured image is digitally distorted after capture to cancel out the distortion caused by the curved combiner so that the image appears to be undistorted after image processing.

In embodiments, the combiner **6735** is an adjustable partial mirror in which the reflectivity can be changed by the user or automatically to better function within different environmental conditions or different use cases. The adjustable partial mirror can be an electrically controllable mirror such as for example, the e-Transflector that can be obtained from Kent Optronics (<http://www.kentoptronics.com/mirror.html>) where the reflectivity can be adjusted based on an applied voltage. The adjustable partial mirror can also be a fast switchable mirror (e.g. a switching time of less than 0.03 seconds) wherein the perceived transparency is derived from the duty cycle of the mirror rapidly switching between a reflecting state and a transmitting state. In embodiments, the images captured by the camera **6739** can be synchronized to occur when the fast switchable mirror is in the reflecting state to provide an increased amount of light to the camera **6739** during image capture. As such, an adjustable partial mirror allows for the transmissivity of the partial mirror to be changed corresponding to the environmental conditions, e.g. the transmissivity can be low when the environment is bright and the transmissivity can be high when the environment is dim.

In a further embodiment, the combiner **6735** includes a hot mirror coating on the side facing the camera **6739** wherein visible wavelength light is substantially transmitted while a spectral wavelength band of infrared light is substantially reflected and the camera **6739** captures images that include at least a portion of the infrared wavelength light. In these embodiments, the image light **6750** includes visible wavelength light and a portion of the visible wavelength light is transmitted by the combiner **6735**, where it is then absorbed by the absorptive polarizer **6737**. A portion of the scene light **6760** is comprised of visible wavelength light and this is also transmitted by the combiner **6735**, to provide the user with a see-through view of the environment. The light from the environment **6770** is comprised of visible wavelength light and infrared wavelength light. A portion of the visible wavelength light along with substantially all of the infrared wavelength light within the spectral wavelength band associated with the hot mirror, is reflected by the combiner **6735** toward the camera **6739** thereby passing through the absorptive polarizer **6737**. In embodiments, the camera **6739** is selected to include an image sensor that is sensitive to infrared wavelengths of light and the absorptive polarizer **6737** is selected to substantially transmit infrared wavelengths of light of both polarization states (e.g. ITOS XP44 polarizer which transmits both polarization states of light with wavelengths above 750 nm: see <http://www.ito-s.de/english/polarisatoren/linear/linear.php>) so that an increased % of infrared light is captured by the camera **6739**. In these embodiments, the absorptive polarizer **6737** functions as a light trap for the escaping image light **6750** and thereby blocking the image light **6750** that is in the visible wavelengths from the camera **6739** while simultaneously acting as a window for infrared wavelength light from the environment **6770** for the camera **6739**.

By coaxially aligning the camera field of view with the displayed image and the user's view of the scene, augmented reality images with improved alignment to objects in the scene can be provided. This is because the captured images from the camera provide an accurate representation of the user's perspective view of the scene. In embodiments, the

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camera that is coaxially aligned with the user's view captures an image of the scene, the processor then identifies an object in the captured image and identifies a field of view position for the object, which can be compared to the displayed field of view correlated position so digital content is then displayed relative to the position of the object.

Another aspect of the present invention relates to an optical assembly that uses a reflective display where the reflective display is illuminated with a front light arranged to direct the illumination at angles around 90 degrees from the active reflective surface of the reflective display. In embodiments, the optical configuration is light weight, small and produces a high quality image in a head-worn see-through display.

FIG. **68** provides a cross sectional illustration of the compact optical display assembly for a HWC **102** according to principles of the present invention along with illustrative light rays to show how the light passes through the assembly. The display assembly is comprised of upper optics and lower optics. The upper optics include a reflective image source **6810**, a quarter wave film **6815**, a field lens **6820**, a reflective polarizer **6830** and a polarized light source **6850**. The upper optics convert illumination light **6837** into image light **6835**. The lower optics comprise a beam splitter plate **6870** and a rotationally curved partial mirror **6860**. The lower optics deliver the image light to a user who is wearing the HWC **102**. The compact optical display assembly provides the user with image light **6835** that conveys a displayed image along with scene light **6865** that provides a see-through view of the environment so that user sees the displayed image overlaid onto the view of the environment.

In the upper optics, linearly polarized light is provided by the polarized light source **6850**. Where the polarized light source **6850** can include one or more lights such as LEDs, QLEDs, laser diodes, fluorescent lights, etc. The polarized light source **6850** can also include a backlight assembly with light scattering surfaces or diffusers to spread the light uniformly across the output area of the polarized light source. Light control films or light control structures can be included as well to control the distribution of the light (also known as the cone angle) that is provided by the polarized light source **6850**. The light control films can include, for example, diffusers, elliptical diffusers, prism films and lenticular lens arrays. The light control structures can include prism arrays, lenticular lenses, cylindrical lenses, Fresnel lenses, refractive lenses, diffractive lenses or other structures that control the angular distribution of the illumination light **6837**. The output surface of the polarized light source **6850** is a polarizer film to ensure that the illumination light **6837** provided to the upper optics is linearly polarized.

The illumination light **6837** provided by the polarized light source **6850** is reflected by a reflective polarizer **6830**. Where the polarizer on the output surface of the polarized light source **6850** and the reflective polarizer **6830** are oriented so that their respective transmission axes are perpendicular to one another. As a result, the majority of the illumination light **6837** provided by the polarized light source **6850** is reflected by the reflective polarizer **6830**. In addition, the reflective polarizer **6830** is angled so that the illumination light **6837** is reflected toward the reflective image source **6810** thereby illuminating the reflective image source **6810** as shown in FIG. **68**.

The illumination light **6837** passes through a field lens **6820** and is then incident onto the reflective image source **6810**. The illumination light **6837** is then reflected by the reflective image source (otherwise referred to as a reflective display herein elsewhere) **6810**. Wherein the reflective

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image source **6810** can comprise a liquid crystal on silicon (LCOS) display, a ferroelectric liquid crystal on silicon (FLCSO) display, a reflective liquid crystal display, a cholesteric liquid crystal display, a bistable nematic liquid crystal display, or other such reflective display. The display can be a monochrome reflective display that is used with sequential red/green/blue illumination light **6837** or a full color display that is used with white illumination light **6837**. The reflective image source **6810** locally changes the polarization state of the illumination light **6837** in correspondence to the pixel by pixel image content that is displayed by the reflective image source **6810** thereby forming image light **6835**. Wherein if the reflective image source **6810** is a normally white display, the areas of the image light **6835** that correspond to bright areas of the image content end up with a polarization state that is opposite to the polarization state of the illumination light and dark areas of the image light **6835** end up with a polarization state that is the same as the illumination light **6837** (it should be noted that the invention can be used with normally black displays which provide an opposite effect on polarization in the image light). As such, the image light **6835** as initially reflected by the reflective image source **6810** has a mixed polarization state pixel by pixel. The image light **6835** then passes through the field lens **6820** which modifies the distribution of the image light **6835** while preserving the wavefront to match the requirements (such as for example, magnification and focus) of the lower optics. As the image light **6835** passes through the reflective polarizer **6830**, the bright areas of the image light **6835** that have a polarization state that is opposite to the illumination light **6837** are transmitted through the reflective polarizer **6830** and the dark areas of the image light **6835** that have the same polarization state as the illumination light **6837** are reflected back toward the polarized light source **6850**, as a result, the image light **6835** after passing through the reflective polarizer **6830** is linearly polarized with a single polarization state in all the pixels of the image but now with different intensities pixel by pixel. Thus the reflective polarizer **6830** acts first as a reflector for the illumination light **6837** and then second as an analyzer polarizer for the image light **6835**.

As such, the optical axis of the illumination light **6837** is coincident with the optical axis of the image light **6835** between the reflective polarizer **6830** and the reflective image source **6810**. The illumination light **6837** and the image light **6835** both pass through the field lens **6820**, but in opposite directions. Wherein the field lens acts to expand the illumination light **6837** so it illuminates the entire active area of the reflective image source **6810** and also to expand the image light **6835** so it fills the eyepiece **6882** after passing through the rest of the compact optical display system. By overlapping the portion of the compact optical display assembly associated with the illumination light **6837** with the portion of the compact optical display assembly associated with the image light **6835**, the overall size of the compact optical display assembly is reduced. Given that the focal length associated with the field lens **6820** requires some space in the compact optical display assembly, the reflective polarizer **6830** and the polarized light source **6850** are located in space that would otherwise be unused so the overall size of the display assembly is more compact.

The reflective polarizer **6830** can be a relatively thin film (e.g. 80 microns) or thin plate (e.g. 0.2 mm) as shown in FIG. **68**. The reflective polarizer **6830** can be a wiregrid polarizer such as is available from Asahi Kasei under the name WGF, or a multilayer dielectric film polarizer such as is available from 3M under the name DBEF. As previously

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described, the reflective polarizer **6830** has two functions. First, the reflective polarizer **6830** reflects the illumination light **6837** provided by the polarized light source **6850** and redirects the illumination light **6837** toward the reflective image source **6810**. Second, the reflective polarizer **6830** acts as an analyzer polarizer to the image light **6835** thereby converting the mixed polarization state of the image light **6835** above the reflective polarizer **6830** to linearly polarized light with a single polarization state below the reflective polarizer **6830**. While the illumination light **6837** incident on the reflective polarizer **6830** is incident on a relatively small portion of the reflective polarizer **6830**, the image light **6835** is incident on the majority of the area of the reflective polarizer **6830**. Consequently, the reflective polarizer **6830** extends at least across the entire area of the field lens **6820** and may extend across the entire area between the field lens **6820** and the beam splitter **6870** as shown in FIG. **68**. In addition, the reflective polarizer **6830** is angled at least in the portion where the illumination light **6837** is incident to redirect the illumination light **6837** toward the reflective image source **6810**. However, since reflective polarizers (such as a wiregrid polarizer) can be relatively insensitive to the incident angle, in a preferred embodiment, the reflective polarizer **6830** is a flat surface angled to redirect the illumination light **6837** toward the reflective image source **6810** wherein the flat surface extends substantially across the entire area between the field lens **6820** and the beam splitter **6870** in one continuously flat surface to make manufacturing easier. The thin film or thin plate of the reflective polarizer **6870** can be retained at the edges to position it at the desired angle and to make the surface flat.

The systems and methods described herein with respect to FIGS. **68** through **71** have a number of advantages. By avoiding grazing angles of the illumination light **6837** and the image light **6835** at all the surfaces in the compact optical display assembly, scattering of light in the assembly is reduced and as a result the contrast of the image presented to the user's eye **6880** is higher with blacker blacks. In addition, the reflective image source **6810** can include a compensating retarder film **6815** as is known to those skilled in the art, to enable the reflective image source **6810** to provide a higher contrast image with more uniform contrast over the area of the displayed image. Further, by providing an optical display assembly that is largely comprised of air, the weight of the compact optical display assembly is substantially reduced. By using coincident optical axes for the illumination light **6837** and the image light **6835** and overlapping the illumination light **6837** and image light **6835** for a substantial portion of the optical display assembly, the overall size of the compact optical display assembly is reduced. Where the coincident optical axes are provided by passing the illumination light **6837** and the image light **6835** in opposite directions through the field lens **6820**. To maintain a uniform polarization state for the illumination light **6837**, the field lens **6820** is made from a low birefringence material such as glass or a plastic such as OKP4 as available from Osaka Gas Chemicals. By positioning the polarized light source **6850** and the associated illumination light **6837** below the field lens **6820**, and by folding the optical path of both the illumination light **6837** at the reflective polarizer **6830** and the image light **6835** at the beam splitter **6870**, the overall height of the compact optical display assembly is greatly reduced. For example the overall height of the compact optical display assembly can be less than 24 mm as measured from the reflective image source **6810** to the bottom edge of the rotationally curved partial

mirror **6860** for a display that provides a 30 degree diagonal field of view with a 6×10 mm eyebox.

In a preferred case, the light control structure in the polarized light source **6850** includes a positive lens, such as for example a positive Fresnel lens, a positive diffractive lens or a positive refractive lens. Wherein a positive Fresnel lens or a positive diffractive lens is preferred because they can be very thin. The illumination light **6837** is thereby focused to form a smaller area or pupil at the reflective polarizer **6830** that has a direct relationship to the area of an eyebox **6882** at the other end of the optics wherein image light **6835** is provided to the user's eye **6880** as shown in FIG. **68**. Where the positive lens concentrates the illumination light **6837** from the polarized light source **6850** both in terms of intensity and angular distribution to match the extend of the optical system and thereby fills the eyebox with image light **6835**. By using the positive lens to converge the light from the polarized light source **6850** as provided to the reflective polarizer **6830** and then using the field lens **6820** to expand the illumination light **6837** to illuminate the active area of the reflective image source **6810**, efficiency is improved since illumination light **6837** is substantially delivered only where needed to form image light **6835**. Further, illumination light **6837** outside the pupil can be controlled by the positive lens and clipped by masked edges of the positive lens. By focusing the illumination light **6837** and clipping light outside the pupil, illumination light **6837** is prevented from impinging adjacent surfaces at grazing angles in the compact optical display assembly to reduce scattering of light and thereby increase contrast in the image provided to the user's eye **6880** by providing blacker blacks.

It should be noted that while FIGS. **68**, **69** and **70** show optical layouts wherein the illumination light **6837** is provided from behind the rotationally curved partial mirror **6860**, other optical layouts are possible within the invention. The location of the polarized light source **6850** can be changed for example to be at the side of the rotationally curved partial mirror **6860** wherein the reflective polarizer **6830** is oriented to receive the illumination light **6837** from the side. And reflect it toward the reflective image source **6810** (not shown).

In a further embodiment, the portion of the image light **6835** that is reflected back toward the polarized light source **6850** is recycled in the polarized light source **6850** to increase the efficiency of the polarized light source **6850**. In this case, a diffuser and a reflective surface is provided behind the polarized light source **6850** so the polarization of the light is scrambled and reflected back toward the reflective polarizer **6830**.

In yet another embodiment, another reflective polarizer is provided in the polarized light source **6850** and behind the linear polarizer previously disclosed. Wherein the respective transmission axes of the reflective polarizer and the linear polarizer are parallel to one another. The other reflective polarizer then reflects the light back into the backlight that has the polarization state that would not be transmitted by the linear polarizer. The light that is reflected back into the backlight passes through diffusers associated with the polarized light source **6850** where the polarization state is scrambled and reemitted thereby recycling the light and increasing efficiency.

In another embodiment, the system according to the principles of the present invention includes an eye imaging system. FIG. **69** is an illustration of a compact optical display assembly, which includes an eye imaging camera **6992** that captures an image of the user's eye **6880** that is

coaxial with the displayed image provided to the user so that a full image of the user's iris can be reliably captured. The eye imaging camera **6992** is reflected into the lower optics by a reflective polarizer **6930** that includes a notch mirror coating, facing the eye imaging camera **6992**, that reflects the wavelengths of light that are captured by the eye imaging camera **6992** (e.g. near infrared wavelengths) while transmitting wavelengths associated with the image light **6835** (e.g. visible wavelengths). Eye light rays **6995** shown in FIG. **69** illustrate how the field of view associated with the eye imaging camera **6992** is a relatively narrow field of view because it is multiply reflected through the lower optics to capture an image of the user's eye **6880**. However, to enable the eye imaging camera **6992** to focus onto the user's eye **6880**, the eye imaging camera **6992** needs to have a very near focus distance (e.g. 35 mm). In addition, the field of view and focus distance of the eye imaging camera must take into account the reducing effect of the optical power provided by the rotationally curved partial mirror **6860**. To increase the efficiency of capturing the light reflected from the user's eye **6880** and thereby enable a brighter image of the eye, the rotationally curved partial mirror **6860** can be coated with a partial mirror coating that acts as a full mirror in the wavelengths being captured by the eye imaging camera **6992**, for example the coating can reflect 50% of visible light associated with the image light and 90% of near infrared light associated with the eye light **6995**. Where the reflections and associated changes in polarization state are similar to those associated with the image light **6835** but in the opposite order since the eye light rays **6995** are coming from the user's eye **6880**. LEDs or other miniature lights are provided adjacent to the user's eye **6880** to illuminate the user's eye **6880** wherein the wavelengths associated with the LED's or other miniature lights are different than the wavelengths associated with the image light **6835** such as for example near infrared wavelengths (e.g. 850 nm, 940 nm or 1050 nm). Alternatively, the image light **6835** is used to illuminate the user's eye **6880** and a reflective polarizer **6930** with a low extinction ratio in reflection (e.g. reflective extinction ratio <15) is used so that some of the eye light rays are reflected toward the eye imaging camera **6992**.

In an alternative embodiment, the reflective and partially reflective surfaces can extend laterally to the sides of the areas used for displaying an image to the user. In this case, the eye imaging camera can be located adjacent to the field lens and pointed in a direction to image the user's eye after reflecting from the beam splitter and the rotationally curved partial mirror as shown in FIG. **70**. Where FIG. **70** is an illustration that shows an eye imaging camera **7092** positioned to the side of the field lens **6820** and reflective polarizer **6830**. The eye imaging camera **7092** is pointed such that the field of view captured by the eye imaging camera **7092** includes the user's eye **6880** as illustrated by the eye light rays **7095**. The quarter wave film **6890** is also extended laterally to change the polarization state of the eye light **7095** in the same way that the polarization state of the image light is changed so that the eye light passes through the beam splitter **6870** and quarter wave **6890**, is partially reflected by the rotationally curved partial mirror **6860** and is then reflected by the beam splitter **6870** and is then captured by the eye imaging camera **7092**. By positioning the eye imaging camera **7092** to the side of the field lens **6820** and reflective polarizer **6830**, the complexity of the optics associated with displaying an image to the user is reduced. In addition, the space available for the eye imaging camera **7092** is increased since interferences with the display optics are reduced. By positioning the eye imaging

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camera 7092 adjacent to the display optics, the eye image is captured nearly coaxially with the displayed image.

In a yet another embodiment, the systems according to the principles of the present invention include a field lens with an internal reflective polarizer and one or more surfaces with optical power. FIG. 71 is an illustration of the upper optics including a field lens 7121 comprised of upper prism 7122 and lower prism 7123. The upper prism 7122 and the lower prism 7123 can be molded to shape or grind and polished. A reflective polarizer 7124 is interposed on the flat surface between the upper prism 7122 and the lower prism 7123. The reflective polarizer 7124 can be a wiregrid polarizer film or a multilayer dielectric polarizer as previously mentioned. The reflective polarizer 7124 can be bonded into place with a transparent UV curable adhesive that has the same refractive index as the upper prism 7122 or the lower prism 7123. Typically the upper prism 7122 and the lower prism 7123 would have the same refractive index. Wherein upper prism 7122 includes an angled surface for illumination light 6837 to be provided to illuminate the reflective image source 6810. The illumination light is provided by a light source that includes lights such as LEDs, a backlight 7151, a diffuser 7152 and a polarizer 7153 as has been previously described. The lower prism 7123 includes a curved surface on the exit surface for controlling the wavefront of the image light 6835 as supplied to the lower optics. The upper prism may also include a curved surface on the upper surface next to the reflective image source 6810 as shown in FIG. 71 for manipulating the chief ray angles of the light at the surface of the reflective image source 6810. Illumination light 6837 is polarized by the polarizer 7153 prior to entering the upper prism 7122. The transmission axes of the polarizer 7153 and the reflective polarizer 7124 are perpendicular to one another so that the illumination light 6837 is reflected by the reflective polarizer 7124 so that the illumination light is redirected toward the reflective image source 6810. The polarization state of the illumination light 6837 is then changed by the reflective image source 6810 in correspondence with the image content to be displayed as previously described and the resulting image light 6835 then passes through the reflective polarizer 7124 to form the bright and dark areas associated with the image that is displayed to the user's eye 6880.

In another embodiment, the field lens 7121 of FIG. 71 comprises a polarizing beam splitter cube including two prisms, upper prism 7122 and lower prism 7123. In this case, the reflective polarizer 7124 is replaced by a coating that is polarization sensitive so that light of one polarization state (typically S polarized light for example) is reflected and light of the other polarization state is transmitted. The illumination light 6837 is then provided with the polarization state that is reflected by the coating and the image light is provided with the polarization state that is transmitted by the coating. As shown in FIG. 71, the beam splitter cube includes one or more curved surfaces in the upper prism 7122 or the lower prism 7123. The beam splitter cube can also include one or more angled surfaces where the illumination light is supplied. The angled surface can include light control structures such as a microlens array to improve the uniformity of the illumination light 6837, or a lenticular array to collimate the illumination light 6837.

In yet another embodiment, the curved surface(s) or the angled surface(s) illustrated in FIG. 71 can be molded onto a rectangularly shaped beam splitter cube by casting a UV curable material (e.g. UV curable acrylic) onto a flat surface of a beam splitter cube, placing a transparent mold with a cavity that has the desired curve onto the flat surface to force

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the UV curable material into the desired curve and applying UV light to cure the UV curable material. The beam splitter cube can be made of a material that has the same or different refractive index than the UV curable material.

In a further embodiment, polarization sensitive reflective coatings such as dielectric partial mirror coatings, can be used in place of reflective polarizers or beam splitters as shown in FIG. 68. In this case, the reflective films and plates that comprise the reflective polarizers 6830 and beam splitters 6870 include polarization sensitive coatings that substantially reflect light with one polarization state (e.g. S polarization) while substantially transmitting light with the other polarization state (e.g. P polarization). Since the illumination light source includes a polarizer 7153, the illumination light 6837 is one polarization state and it is not important that the reflective polarizer 7124 be sensitive to the polarization state in reflection, the polarization state just needs to be maintained and presented uniformly over the surface of the reflective image source 6810. However, it is important that the reflective polarizer 7124 be highly sensitive to polarization state in transmission (e.g. extinction ratio >200) to be an effective polarizer analyzer and to provide a high contrast image (e.g. contrast ratio >200) to the user's eye 6880.

In a further embodiment, the field lens 7121 shown in FIG. 71 can comprise a reflective polarizer 7124 with a curved surface (not shown) instead of a flat surface and wherein the reflective polarizer 7124 is not a film and instead is a polarization sensitive coating, a printed wiregrid polarizer or a molded wiregrid pattern that is then metallized. In this case, the upper prism 7122 and the lower prism 7123 are made as a matched pair with mating curved surfaces that together form the surface of the reflective polarizer. Wherein the polarization sensitive coating, the printed wiregrid or the molded wiregrid pattern are applied to the mating curved surface associated either the upper prism 7122 or the lower prism 7123 and a transparent adhesive is applied to the other mating surface to bond the upper prism 7122 and lower prism 7123 together to form the field lens 7121 with an internal curved reflective polarizer 7121.

Another aspect of the present invention relates to manufacturing and providing an optical element for use in a see-through computer display system. In embodiments, a lightweight low-cost and high optical quality optical element.

In a head mounted display, a beam splitter can be used to direct illuminating light from a light source toward a reflective image source such as an LCOS or a DLP. Where it is desirable to have a low weight beam splitter with a flat partially reflective surface to provide good image quality. The flat partially reflective surface is particularly important when an eye camera is provided for eye imaging that utilizes the flat partially reflective surface for directing the field of view of the eye camera toward the user's eye.

Systems and methods provide for a lightweight beam splitter comprised of molded plastic elements and an internal plate element to provide a flat partially reflective surface. Together the pieces form a triplet beam splitter optic including two molded elements and a plate element. By providing the plate element internal to the beam splitter, the matching surfaces of the molded elements do not have to be optically flat, instead the plate element provides the flat surface and an index matching material is used to join the plate element to the molded elements. All three elements can be plastic elements to reduce the weight and cost of the lightweight beam splitter. To provide a more uniform refractive effect,

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the molded elements and the plate element are preferentially made from plastic materials with similar refractive index and have low birefringence.

FIG. 72 shows an illustration of the two molded elements **7210** and **7220**. These molded elements are molded with a relatively uniform thickness to provide uniform flow of the plastic material during molding (either injection molding, compression molding or casting) and thereby enable a low birefringence in the elements as molded. To further reduce birefringence in the molded elements as molded, materials with low viscosity and low stress optic coefficients are preferred including: OKP4 from Osaka Gas Company; Zeonex F52R, K26R or 350R from Zeon Chemical; PanLite SP3810 from Teijin.

The molded elements **7210** and **7220** can include flat surfaces and surfaces with optical power, where the surfaces with optical power can include spherical or aspheric curved surfaces, diffractive surfaces or Fresnel surfaces. Flat surfaces, diffractive surfaces or Fresnel surfaces are preferred on the surfaces associated with light that illuminates the image source and flat surfaces, spherical surfaces or aspheric surfaces are preferred on the surfaces associated with image light. Molded element **7210** is shown with a spherical or aspheric surface **7215** and molded element **7220** is shown with a flat surface **7225**, however, any of the surfaces shown can be molded as flat surfaces or surfaces with optical power.

After molding the molded elements **7210** and **7220** are machined to provide matching angled surfaces. Molded element **7210** is shown in FIG. 73 where a milling cutter **7328** is shown machining angled surface **7329**. FIG. 74 shows an illustration of molded elements **7210** and **7220** after they have been machined to respectively provide beam splitter elements **7430** and **7440** that are prisms. The angled surface of beam splitter elements **7430** and **7440** are machined to have matching angles. Alternatively, beam splitter elements **7430** and **7440** can be machined from sheet material or molded pucks. In either case of using machined angled surfaces or molded angled surface in the beam splitter elements, the surfaces will not be optically flat.

FIG. 75 shows an illustration of the assembled triplet beam splitter optic, wherein the beam splitter elements **7430** and **7440** have been assembled with a partially reflecting plate element **7560** to form a beam splitter cube. Wherein the beam splitter elements **7430** and **7440** are made from either the same material or different materials that have a very similar refractive index (e.g. within 0.05 of each other). An index matching material is used at the interfaces between the beam splitter elements and the plate element. The index matching material can be a fluid, a light curing adhesive, a moisture curing adhesive or a thermally curing adhesive. The index matching material should have a refractive index that is very similar to that of the beam splitter elements (e.g. within 0.1).

The partially reflective plate element **7560** can be a transparent plate with a partially reflective layer that is either a partially reflective coating or a laminated partially reflective film. The transparent plate is preferably a cast sheet such as cell cast acrylic that has low birefringence, or a molded plaque of a low birefringence material such as OKP4, Zeonex F52R, Zeonex K26R, Zeonex 350R or PanLite SP3810. In addition, the transparent plate should be optically flat (e.g. within 20 microns over the surface and with a surface finish of less than 15 nanometers), however optically flat surfaces are easily obtained in sheet stock. By using an index matching material at the interfaces between the beam splitter elements **7430** and **7440** and the partially

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reflective plate element **7560**, the lack of optical flatness of the surface of the beam splitter elements **7430** and **7440** can be filled by the index matching material so that the flatness of the reflective surface is determined by the flatness of the more easily obtained partially reflective plate element **7560** thereby providing a manufacturing advantage. The partially reflective layer can be a partial mirror, a reflective polarizer or a wiregrid polarizer where the reflective polarizer can be a coating or a film and the wiregrid polarizer can be a film or a molded structure that is partially coated with a conductive layer. Where a suitable reflective polarizer film can be obtained from 3M available under the trade name of DBEFQ and a wiregrid polarizer film can be obtained from Asahi-Kasei available under the trade name of WGF. In a preferred embodiment, the transparent plate of the partially reflective plate element **7560** has a refractive index that is very similar (e.g. within 0.1) to the refractive indices of the beam splitter elements **7430** and **7440**.

FIG. 76 shows an illustration of an optical system for a head mounted display system. The system includes a reflective display as an image source **7667**, a light source **7665** that can be a white light source or a sequential color light source as appropriate for the image source **7665**. Wherein the light source **7665** provides illumination light **7674** that can be polarized light provided that a quarter wave layer is associated with the image source **7667** or the partially reflecting plate element **7560** so that the polarization of the illumination light **7674** is changed before becoming image light **7672**. The illumination light **7674** is reflected by a surface of the partially reflecting plate element **7560**, and then reflected by the image source **7667**, whereupon it passes through the partially reflective plate element **7560** thereby becoming image light **7672**. The image light **7672** is then reflected by a partially reflective combiner **7682** so that the image light is directed toward the user's eye **7680** to display an image to the user while simultaneously providing a see-through view of the environment. In the optical system, an index matching material can be used at the interface between the image source **7665** and the beam splitter element **7440** so that the surface of the beam splitter element **7440** does not have to be flat. It is contemplated by the current inventions that the optical system may include additional lenses and other optical structures that are not shown to improve the image quality or change the form factor of the optical system.

In another embodiment, beam splitter elements **7430** and **7440** are molded directly to shape using injection molding or casting. The molded beam splitter elements are then assembled as shown in FIG. 75 as described previously herein.

In further embodiments, surfaces of the beam splitter elements are molded or machined to have additional structures to provide further features. FIG. 77 shows an illustration of lightweight beam splitter **7750** that includes an extended partially reflective plate element **7760** and an extended beam splitter element **7740**, wherein the partially reflective surface is extended to provide additional area for the illumination light **7674** to be reflected toward the image source **7665**. Where having an extended partially reflective surface is particularly useful when the image source **7665** is a DLP and the illumination light **7665** must be delivered at an oblique angle. FIG. 78 shows a lightweight beam splitter **7850** that includes an entrance surface **7840** for the illumination light **7674** that is angled so the illumination light **7674** passes substantially perpendicularly through the entrance surface **7840**.

In yet further embodiments, beam splitter element elements **7430** and **7440** are machined from a single molded element. Where the single molded element is designed to provide the desired optical surfaces. For example, the molded element **7210** as shown in FIG. **72** has surfaces that could be used for both surface **7215** and **7225**. A series of molded elements **7210** could then be molded and some would be used to make machined beam splitter elements **7430** and some for beam splitter elements **7440**. A partially reflective plate element **7560** would then be bonded with the beam splitter element **7430** and **7440** using index-matched adhesive as previously described herein. Alternatively, the single molded element **7210** could be designed with extra thickness across the dimension where the partially reflective plate element **7560** will be added, so that a single molded element **7210** could be sawn, machined or laser cut into beam splitter elements **7430** and **7440**.

In another embodiment, a first molded optical element is molded in a geometry that enables improved optical characteristics including: low birefringence; more accurate replication of the optical surfaces of the mold (reduced power and irregularity deviation). The first molded optical element is then cut to a different shape wherein the cutting process leaves an optically rough surface finish. A second optical element with an optically smooth surface is then bonded to the optically rough surface of the first molded optical element using an index-matched adhesive to provide a combined optical element. The index-matched adhesive fills in the optically rough surface on the first molded optical element so that the optically rough surface is no longer visible and an optically smooth surface is provided in the combined optical element by the second optical element. The optical characteristics of the combined optical element are improved as compared to a directly molded optical element that has the geometry of the combined optical element. The cut surface can be flat or curved, as long as the cut surface of the first molded optical element is substantially similar to the bonding surface of the second optical element. In addition, both the first molded optical element and the second optical element can provide optical surface with independent optical features such as optical power, wedge, diffraction, grating, dispersion, filtering and reflection. For example, optically flat surfaces can be difficult to mold on plastic lenses. A lens can be molded to provide a spherically curved surface and another surface that is subsequently milled off to provide a flat surface with a rough surface finish. An optically flat sheet can then be bonded to the milled surface using an index matched adhesive to provide a combined optical element with an optically flat surface.

In yet further embodiments, surfaces of the beam splitter elements include molded or machined structures to collimate, converge, diverge, diffuse, partially absorb, redirect or polarize the illumination light **7674** or the image light **7672**. In this way, the number of parts in the lightweight beam splitter is reduced and the cost and manufacturing complexity is reduced.

The multi-piece lightweight solid optic has been described in connection with certain embodiments; it should be understood that the multi-piece lightweight solid optic may be used in connection with other optical arrangements (e.g. other see-through head-worn display optical configuration described herein elsewhere).

In embodiments, the invention provides methods for aligning images, along with methods and apparatus for controlling light within the optics of the display assembly

associated with a HMD to prevent scattering and also to trap excess light to thereby improve the image quality provided to the user.

FIG. **79a** is a schematic illustration of a cross section of a display assembly for a HMD. Wherein, the display assembly includes upper optics **795** and lower optics **797** that operate together to display an image to a user while simultaneously providing a see-through view of the environment. Aspects of the upper optics **795** will be discussed in more detail herein. The lower optics **797** can comprise optical elements to manipulate image light **7940** from the upper optics **795** and thereby present the image light **7940** to the user's eye **799**. Lower optics **797** can comprise one or more lenses **7950** and a combiner **793**. The combiner **793** presents the image light **7940** to the user's eye **799** while simultaneously allowing light from the environment **791** to pass through to the user's eye **799** so that the user sees the displayed image overlaid onto a view of the environment.

FIG. **79** is a schematic drawing of a cross section of the upper optics **795** for a HMD. Included are a light source **7910**, a partially reflective layer **7930**, a reflective image source **7935** and a lens **7950**. The light source **7910** provides illumination light **7920** to the HMD. The illumination light **7920** is redirected by the partially reflective layer **7930** to illuminate the reflective image source **7935**. The illumination light **7920** is then reflected by the reflective image source **7935** in correspondence with the image content in the displayed image so that it passes through the partially reflective layer **7930** and thereby forms image light **7940**. The image light **7940** is optically manipulated by the lens **7950** and other optical elements (not shown) in the lower optics **797** so that a displayed image is provided to a user's eye **799**. Together, the light source **7910**, the partially reflective layer **7930** and the reflective image source **7935** form a frontlighted image source. Where, the reflective image source **7935** can comprise a LCOS, a FLCOS, DLP or other reflective display. FIGS. **79**, **80**, **82** and **83** are shown with the illumination light **7920** provided so that it is incident on the reflective image source **7935** at an oblique angle as is required for a DLP. FIGS. **84c**, **84d**, **85**, **86**, **87**, **88** and **89** are shown with the illumination light **7920** provided perpendicular to the reflective image source **8535** as is required for an LCOS or FLCOS. The principles of the invention described herein apply to any type of reflective image source where stray reduction is needed. The light source **7910** can include light sources such as LEDs, laser diodes or other light sources (e.g. as described herein) along with various light control elements including: diffusers, prismatic films, lenticular films, Fresnel lenses, refractive lenses and polarizers. Polarizers included in the light source **7910** polarize the light exiting the light source **7910** so that the illumination light **7920** is polarized. The partially reflective layer **7930** can be a partial mirror coating on a substrate or it can be a reflective polarizer film such as a wire grid film supplied by Asahi-Kasei under the name WGF or a multi-layer polarizer film supplied by 3M under the name DBEF. When the partially reflective layer **7930** is a reflective polarizer, the illumination light **7920** is supplied as polarized light wherein the polarization axis of the illumination light **7920** is oriented relative to the polarization axis of the reflective polarizer so that the illumination light **7920** is substantially reflected. The reflective image source **7935** then includes a quarter wave retarder (e.g. a quarter wave film) so that the polarization state of the illumination light **7920** is reversed in the process of being reflected by the reflective image source **79345**. This enables the reflected illumination light **7920** to then be substantially transmitted

by the reflective polarizer. After passing through the partially reflective layer **7930**, the light becomes image light **7940**. The image light **7940** then passes into a lens **7950** which is part of the lower optics **797** or display optics which manipulates the light to provide a displayed image to the user's eye. While the partially reflective layer **7930** is illustrated as a flat surface, the inventors have contemplated that the surface may be curved, shaped, have simple or complex angles, etc. and such surface shapes are encompassed by the principles of the present invention.

In HMDs that provide images to both eyes of the user, it is desirable to provide the images so that they are aligned to one another. This is particularly important when the images are viewed as stereo images where the perceived alignment of the images seen with each eye is critical to achieving the perception of depth. To provide an accurate alignment of the images, an active alignment of the optics can be performed after the optics have been assembled into a rigid frame of the HMD. Where active alignment includes aligning the images for each eye to one another by moving portions of the display assembly and affixing the portions into position relative to one another. To this end, FIG. **79** shows space **7952** that extends around the reflective image source **7935** so that the reflective image source **7935** can be moved laterally and rotationally. The light source **7910** and partially reflective layer **7930** are arranged to illuminate the area that includes the reflective image source **7935** and a portion of the adjacent space **7952**. As a result, the reflective image source **7935** can be moved within the space **7952** during the active alignment process without losing illumination or degrading the brightness of the displayed image. Where movements of the reflective image source **7935** during the active alignment can include movements that correspond to horizontal, vertical and rotational movements of the image provided to one eye relative to the image provided to the other eye of the user. The movements can be 0.5 mm in size for example when the reflective image source **7935** is approximately 5x8.5 mm in size (this equates to approximately 10% of the reflective image source dimension) and as such the space **7952** can be 0.5 mm wide or wider.

However, by including the space **7952**, in the illuminated area, visible artifacts can occur due to light scattering or reflecting from the edges of the reflective image source **7935** or from structures adjacent to the space **7952**. Consequently, a mask **8055** is provided that extends from the edge of the active area of the reflective image source **7935** across the space **7952** to cover the edges of the reflective image source **7935** and structures adjacent to the space **7952** as shown in FIG. **80**. The mask **8055** is black and non-reflecting so that incident illumination light **7920** is absorbed. In addition the mask **8055** is designed to not interfere with the movements of the reflective image source **7935** that occur during active alignment. To this end, the mask **8055** can be stiff (e.g. a black plastic or a black coated metal) and designed to slide under the adjacent structures such as the light source **7910**, the edge of the partially reflective layer **7930** and the sides of the housing that contain the frontlight. Alternatively, the mask **8055** can be flexible (e.g. a black plastic film or a black rubber film or tape) so that it deforms when it contacts the adjacent structures. FIG. **81a** shows an illustration of the reflective image source **7935**, the light source **7910** and the space **7952** as viewed from above. As is typically found with image source of all kinds, there is a mask **8168** applied to the surface of the image source that surrounds the active area **8165**, however this mask **8168** does not cover the space **7952**. FIG. **81b** shows a further illustration of the system shown in FIG. **81a** wherein the mask **8055** is applied to the

reflective image source **7935** so that it attaches within the mask **8168** in a way that covers the space **7952** and does not block the active area **8165**.

In another embodiment, the image produced by the image source does not use all of the active display area of the image source so there is room to shift the image in an x and/or y perspective within the active display area for alignment of the content. For example, if a misalignment is observed (as indicated above) rather than physically moving the image source, or in addition to moving the image source, the image is digitally shifted in the x and/or y direction to create better combined content alignment. The originally inactive display area around the content may be referred to as a content shift buffer zone.

In a further embodiment for aligning images in a HMD with see-through, a first image containing features is provided to one eye of the user using a display assembly similar to that shown in FIG. **79a** or FIG. **85**. A second image containing features in the same locations is provided to the other eye of the user. The position of at least one of the image sources is then moved within the space provided for adjustment to align the first image to the second image as seen by the user's eyes. This image alignment can also be done using cameras in place of the user's eyes.

In the case where the first and second images are smaller in size than the active area of the reflective image source, thereby leaving a digital space adjacent to the images that can be used for digital shifting of the images for further alignment adjustment. This adjustment can be used in combination with physical movements of the reflective image sources to align the first image to the second image.

FIG. **82** is an illustration of upper optics **825** that includes the elements of upper optics **795** with the addition of a trim polarizer **8260**. Where the polarization axis of the trim polarizer **8260** is oriented so the image light **7940** is transmitted to the lower optics (not shown). Light that has the opposite polarization state compared to the image light **7940** is absorbed by the trim polarizer **8260**. As such, light that is scattered from surfaces such as the walls of the housing **8262** that typically has a mixed polarization state will be partially absorbed by the trim polarizer **8260**. The trim polarizer **8260** can also absorb a portion of colored light caused by birefringence in the lens **7950** provided the trim polarizer **8260** is located after the lens **7950**. In this case, the trim polarizer **8260** absorbs the light that has the opposite polarization state caused by the birefringence and transmits the light that has the polarization state of the image light **7940** prior to the lens **7950**. In some cases, it is advantageous to change the polarization state of the image light **7940** to improve the reflection of the image light **7940** from the combiner **793** so that a half wave retarder is needed in addition to the trim polarizer **8260**. For proper operation, the half wave retarder is positioned with its fast axis oriented at 45 degrees to the transmission axis of the trim polarizer **8260**. In this case, it is advantageous to position the half wave retarder (not shown) below the trim polarizer **8260** so that the trim polarizer can absorb any elliptical polarization that may be present due to birefringence in the lens **7950** before the image light is acted upon by the half wave retarder. In this way, any variation in retardation with wavelength that may be present in the half wave retarder will not act to increase the elliptical polarization or act to increase color artifacts in the image light **7940** caused by birefringence in the lens **7950**. In an example, the trim polarizer can be a polarizer film that is laminated to a half wave retarder film and antireflection coatings can be applied to the outer surfaces.

In FIG. 83, the partially reflective layer 8330 is a laminated multiple polarizer film comprised of a reflective polarizer film 8332 laminated to an absorptive polarizer film 8331. Where, the reflective polarizer film 8332 is only big enough to reflect the illumination light 7920 that illuminates the active area 8165 of the reflective image source 7935. The absorptive polarizer film 8331 is larger than the reflective polarizer film 8332 and extends across the entire aperture between the reflective image source 7935 and the lens 7950, so that no edges of the absorptive polarizer film 8331 are visible and all the light reflected from the reflective image source 7935 passes through the absorptive polarizer 8331. For the case when the reflective image source 7935 is an LCOS, the absorptive polarizer 8331 acts as an analyzer polarizer to only allow the polarization state of the image light to be transmitted. As such, the reflective polarizer film 8332 only covers a portion of the absorptive polarizer film 8331. The polarization axes of the reflective polarizer film 8332 and the absorptive polarizer film 8331 are aligned so that polarized light that is transmitted by the reflective polarizer film 8332 is also transmitted by the absorptive polarizer film 8331. In contrast, polarized light that is reflected by the reflective polarizer film 8332 is absorbed by the absorptive polarizer film 8331. Thereby, illumination light 7920 that is incident onto the reflective polarizer film 8332 is reflected toward the reflective image source 7935 where the polarization state is reversed so that it is transmitted by the reflective polarizer film 8332 and the absorptive polarizer film 8331 as it becomes image light 7940. At the same time, illumination light 7920 that is incident onto the absorptive polarizer film 8331 in the area surrounding the reflective polarizer film 8332 is absorbed by the absorptive polarizer film 8331. By absorbing this excess illumination light 7920, that would not illuminate the active area 8165 of the reflective image source 7935, stray light is reduced within the display assembly and the contrast in the image presented to the user's eye is increased as a result. By aligning the polarization axes of the reflective polarizer film 8332 and the absorptive polarizer film 8331, the transmission is only reduced by approximately 12%, in the regions that include both reflective polarizer film 8332 and absorptive polarizer film 8331 compared to the regions that include just absorptive polarizer film 8331. Given the location of the partially reflective layer 8330 in the optical system and the fact that it is remote from the reflective image source 7935, having local differences in transmission on the partially reflective layer 8330 comprised of a laminated multiple polarizer will have a very small effect on the brightness uniformity in the image provided to the user's eye. In addition, the fact that the partially reflective layer 8330 is remote from the reflective image source 8330 makes the edges of the reflective polarizer film 8332 indistinct as seen by the user.

FIGS. 84a and 84b show illustrations of examples of partially reflective layers 8330, comprised of a reflective polarizer film 8430 and 8431 laminated to an absorptive polarizer film 8432. The reflective polarizer films 8430 and 8431 are cut to a shape that covers only the area where illumination light 7920 will be reflected to illuminate the active area 8165 of the reflective image source 7935. The shape required for the reflective polarizer film will vary depending on the type of frontlight. For the frontlight shown in FIG. 83 where the partially reflective layer 8330 is located adjacent to the reflective image source 7935, the shape of the reflective polarizer film 8431 will be rectangular or oval such as shown in FIG. 84b. For the frontlight included in the display assembly shown in FIG. 85 where the lens 8550 is

located between the partially reflective layer 8530 and the reflective image source 8535, the influence of the illumination light 8520 passing through the lens 8550 changes the distribution of illumination light 8520 needed from the light source 8510. As a result, the illumination light 8520 can cover only a portion of the partially reflective layer 8530 and the use of a laminated multiple polarizer is advantageous. In embodiments, the reflective polarizer film can cover less than 80% of the area of the absorptive polarizer film in the laminated partially reflective layer. In further embodiments, the reflective polarizer film can cover less than 50% of the area of the absorptive polarizer film in the laminated partially reflective layer. In this case, the partially reflective layer 8530 can include a reflective polarizer film 8430 with a shape similar to that shown in FIG. 84a. In any case, the shape of the reflective polarizer film is selected in concert with the optical elements in the frontlight and display optics associated with the display assembly of the HMD.

FIG. 84c shows an example illustration of a frontlight for a display assembly similar to that shown in FIG. 85 wherein a laminated multiple polarizer film 8436 is shown with a complex curved shape that resembles an S with a central flat portion and curved ends. The laminated multiple polarizer 8436 includes a reflective polarizer film 8438 and an absorptive polarizer film 8437. Illumination light 8520 includes rays 8522 that are incident on the reflective polarizer film 8438 and rays 8521 that are incident on the absorptive polarizer film 8437. Due to the alignment of the polarization of the illumination light 8520 to the polarization axes of the reflective polarizer film 8438 and the absorptive polarizer film 8437 as previously described herein, rays 8522 are reflected by the reflective polarizer film 8438 and rays 8521 are absorbed by the absorptive polarizer film 8437. In this way, rays 8521 are prevented from contributing to stray light. It is beneficial to absorb rays 8521 since they cannot contribute to image light 8540 because if they were reflected by the laminated multiple polarizer 8436 they would be incident on the reflective image source 8535 outside of the active area 8165, and if they were transmitted by the laminated multiple polarizer 8436, they would be incident on the housing sidewalls 8262. Consequently, by absorbing rays 8521, the laminated multiple polarizer 8436 reduces stray light and thereby increases the contrast in the image displayed to the user.

FIG. 84d shows a further example illustration of a frontlight for a display assembly similar to that shown in FIG. 79 wherein the partially reflective layer 7930 comprises a laminated multiple polarizer film with a curved surface. The laminated polarizer includes an absorptive polarizer film 8442 with a laminated reflective polarizer film 8441. The reflective polarizer film 8441 is positioned in the central portion of the absorptive polarizer film 8442 where the illumination light 7920 is reflected toward the reflective image source 7935. The polarization axes of the reflective polarizer film 8441 and the absorptive polarizer film 8442 are aligned in parallel to each other and perpendicular to the polarization axis of the illumination light 7920 as provided by the polarized light source 7910. The rays 8421 of the illumination light 7920 that are incident on the partially reflective layer 7930 outside of the reflective polarizer film 8441 are absorbed by the absorptive polarizer film 8442. The reflective light source 8535 includes a quarter wave layer 8443 so that the polarization axis of the illuminating light 7920 is changed during the process of being reflected from the reflective image source 8535. As a result, the reflected illumination light 7920 is transmitted by the reflective polarizer film 8441 and the absorptive polarizer film 8442,

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thereby becoming image light **7940**. By absorbing the rays **8421**, before they are incident on external surfaces such as housing walls or other optical surfaces, stray light is reduced and as a result the contrast in the image provided to the user's eye is increased. It should be noted that while FIGS. **84c** and **84d** show the reflective polarizer film being positioned to reduce stray light from the left and right sides as shown in the figure, the reflective polarizer can similarly be positioned to reduce stray light in the direction in and out of the paper as shown in the figure. FIGS. **84a** and **84b** show reflective polarizer films **8430** and **8431** positioned in a center portion of the absorptive polarizer **8432** so that stray light can be reduced in all directions. An important aspect of the invention is that this stray light reduction is obtained without a reduction in the brightness of the image provided to the user's eye since the reflective polarizer films **8430** and **8431** reflect illumination light over the entire area that is needed to fully illuminate the reflective image source.

FIG. **85** shows a schematic illustration of a display assembly for a HMD wherein the optical elements of the frontlight are overlapped with the display optics, as the lens **8550** is located between the partially reflective layer **8530** and the reflective image source **8535**. The display assembly is then comprised of upper optics and lower optics. The upper optics include a reflective image source **8535**, a lens **8550**, a partially reflective layer **8530** and a light source **8510**. The upper optics convert illumination light **8520** into image light **8540**. As shown, the lower optics comprise a beam splitter plate **8580**, a quarter wave film **8575** and a rotationally curved partial mirror **8570** (lower optics similar to those shown in FIG. **79a** are also possible). The lower optics deliver the image light **8540** to a user's eye **8582**. As previously stated herein, the display assembly provides the user with image light **8540** that conveys a displayed image along with scene light **8583** that provides a see-through view of the environment so that the user sees the displayed image overlaid onto a view of the environment.

FIG. **85** shows a display assembly wherein the partially reflective layer **8530** is a single flat film. However, it can be advantageous to use a segmented partially reflective layer **8630** such as is shown in FIG. **86**. In this way, the angle of the central portion **8631** of the partially reflective layer **8630** can be selected to position the light source **8610** differently to reduce the clipping of illumination light **8620** by the lens **8550** or other portions of the supporting structure associated with the display assembly and thereby improve brightness uniformity in the displayed image seen by the user's eye **8582**. To this end, a comparison of FIG. **85** to FIG. **86** shows that by changing the angle of the central portion of the partially reflective film, the position of the light source **8610** is moved downward and the clearance of the illumination light **8620** is increased relative to the lens **8550**.

Segmented partially reflective layers can be used which a variety of geometries and makeups. FIG. **86** shows a segmented partially reflective layer **8630** that includes a folded Z shape with three flat sections. FIG. **87** shows a segmented partially reflective layer that includes an S shape with a central flat section **8731** and ends that are curved similar to that shown in FIG. **84c**. The segmented partially reflective layer can comprise a single partially reflective layer such as a reflective polarizer film or a partial mirror film. In addition, illumination light **8620** can be reflected from just the central flat section or it can be reflected from the central flat section plus one or more of the other segments of the segmented partially reflective layer. Alternatively, the partially reflective layer **8630** can comprise a multiple polarizer film to selectively provide a partially reflective layer over just the

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portions of the partially reflective layer that are actually needed to reflect illumination light to uniformly illuminate the reflective image source **7935** as previously described herein. FIG. **88** shows a display assembly wherein the partially reflective layer **8830** is comprised of a laminated multiple polarizer film with a central portion **8831** that includes a reflective polarizer film and the remainder of which is an absorptive polarizer as previously described herein. Where the segmented shape of the partially reflective layer **8830** is similar to that shown in FIG. **86**, FIG. **89** shows a display assembly wherein the partially reflective layer **8930** is comprised of a laminated multiple polarizer film with a central portion **8931** that includes a reflective polarizer film and the remainder of which is an absorptive polarizer as previously described herein. Where the segmented shape of the partially reflective layer **8930** is similar to that shown in FIG. **87**. While FIGS. **88** and **89** show the reflective polarizer film as just occupying the flat central segment of the segmented partially reflective layers **8830** and **8930** respectively, the reflective polarizer can extend into the adjacent segments as needed to reflect the illumination light **8620** in the pattern needed to uniformly illuminate the reflective image source **8535**. Alternatively the segments associated with the segmented partially reflective layers **8830** and **8930** can have three dimensional shapes when the reflective polarizer portion is shaped like that shown in FIG. **84a** to keep the reflective polarizer **8430** portion flat.

In a further embodiment, the reflective polarizer film is laminated to a flexible transparent carrier film to increase the flexibility and the absorptive polarizer film is a separate layer. FIG. **90** shows a partially reflective layer **9030** comprised of a reflective polarizer film **8441** laminated to a flexible transparent carrier film **9043**. Where the flexible transparent carrier film **9043** does not reflect the illumination light **7920** or change polarization state of the illumination light **7920** and as a result rays **8421** pass through the flexible transparent carrier film **9043**. The purpose of the flexible transparent carrier film is to support the reflective polarizer film **8441** while allowing the partially reflective layer **9030** to be substantially as flexible as the reflective polarizer film **8441** alone. Absorptive polarizer film **9042** is then provided as a separate layer positioned adjacent to the partially reflective layer **9030**. While the absorptive polarizer film **9042** can be flat or curved as needed to fit within the available space, in a preferred embodiment, the absorptive polarizer film **9042** is curved to be better positioned to absorb rays **8421** that are incident on the partially reflective layer **9030** outside of the reflective polarizer film **8441** as shown in FIG. **90**.

In yet another embodiment, the reflective polarizer film is modified to make the portions transparent and non-reflective where illumination light is incident that is not needed to illuminate the active area of the reflective image source and a separate absorptive polarizer is provided to absorb light that is transmitted through the non-reflective portions. FIG. **91** is an illustration of a partially reflective layer **9130** comprised of a reflective polarizer film wherein portions **9143** are modified to be transparent and non-reflective while the portion **9141** is a reflective polarizer. As such, polarized illumination light **7920** is reflected by the reflective polarizer portion **9141** and is transmitted by the modified portions **9143**. An absorptive polarizer **9042** is provided as a separate layer adjacent to the partially reflective layer **9130** so that rays **8421** of the illumination light **7920** are transmitted by the modified portions **9143** and absorbed by absorptive polarizer. Wherein the transmission axis of the reflective

polarizer portion **9141** is parallel aligned to the transmission axis of the absorptive polarizer **9042**. The modification of the reflective polarizer film can be accomplished by etching the reflective polarizer film, when the reflective polarizer film is a wiregrid polarizer, and thereby removing the metal wires of the wiregrid in the modified portions. Alternatively the wiregrid polarizer can be masked during the metal deposition step to provide shaped portions of wiregrid polarizer during manufacturing. An advantage provided by modifying the reflective polarizer film is that the flexibility of the partially reflective layer **9130** is substantially unchanged by the modification and as a result the partially reflective layer **9130** remains uniformly flexible in both the modified portions **9143** and the reflective polarizer portion **9141**. Another advantage provided by using a modified reflective polarizer film is that the transition from the modified portion **9143** to the reflective polarizer portion **9141** does not include a sharp edge that can cause visible artifacts in the image provided to the user's eye due to scattering by the edge or a change in optical density from a thickness change. This embodiment can also be applied to other types of display assemblies such as for example that shown in FIG. **85**.

In a yet further embodiment, the partially reflective layer comprises a reflective polarizer film laminated to an absorptive polarizer and the partially reflective layer includes a flat portion and a curved portion. FIG. **92** is an illustration of a frontlight for a display assembly similar to that shown in FIG. **79a** with the addition of a laminated partially reflective layer **9230** that has a portion that is a reflective polarizer laminated to an absorptive polarizer **9230**. Where the partially reflective layer **9230** is segmented with a flat segment and a curved segment. By including a flat segment in the portion of the partially reflective layer **9230** that is a reflective polarizer **9241**, the uniformity of illumination light **7920** that is reflected onto the reflective image source **7935** is improved because a larger portion of the light source **7910** is mapped to the image as can be seen in FIG. **92**. Wherein when using a small scale light source and associated light control films such as diffusers, it is important to map a large portion of the light source area to avoid darker or brighter lines across the image produced by a dark or bright spot on the light source. Including a flat segment in the partially reflective layer **9230** also reduces local distortions in the image provided to the user's eye that are caused by local changes in optical path length or localized refraction due to changes in the surface angles that the light is exposed to. This embodiment can also be applied to other types of display assemblies such as for example that shown in FIG. **85**.

In head mounted displays that provide a displayed image overlaid onto a see-through view of the environment, it is advantageous to have high see-through transmission both so the user can better interact with the environment and so that people in the environment can see the user's eyes so they feel more engaged with the user. It is also advantageous to have a thin optics module with low height to make the head mounted display more compact and thereby more attractive.

FIG. **93** shows an illustration of an optics module that provides the user with a displayed image while simultaneously providing high see-thru transmission. In this way, the user is provided with a displayed image overlaid onto a clear view of the environment. The optics module includes a combiner **9320** that can have a partial mirror coating that transmits a majority (greater than 50% transmission of visible light) of the available light from the environment, with transmission higher than 70% preferred. For example,

the combiner **9320** can have a broadband partial mirror that reflects less than 30% and transmits over 70% of the entire visible wavelength band. Alternatively, the combiner **9320** can have a notch mirror coating where the reflectivity band of the notch mirror coating is matched to the wavelength bands provided by the light source **9340**, where the light source **9340** can include one or more LEDs, QLEDs, diode lasers or other light source, each with narrow wavelength bands (e.g. 50 nm wide bands or less, full width half max). The notch mirror coating can provide for example, greater than 20% reflectivity (e.g. 50% reflectivity) in the wavelength bands provided by the light source **9340** while providing greater than 80% transmission in the remaining wavelength bands in the visible. For full color images to be provided by the optics module, at least three LEDs with complimentary colors are required such as red, green and blue light or, cyan, magenta and yellow light. In a preferred embodiment, the combiner **9320** has a tristimulus notch mirror that reflects over 50% of the light within the wavelength bands provided by the light source **9340** and transmits an average of over 80% across the entire visible wavelength band. In this way, the tristimulus notch mirror coating provides improved efficiency compared to the partial mirror coating previously described. In an example, if the combiner is to provide 75% transmission of visible light from the environment **9362**, the partial mirror coating will reflect only 25% of image light **9360** so that 75% of the image light will be transmitted through the combiner and will not contribute to the brightness of the image provided to the user's eye **9310**. In contrast, a tristimulus notch mirror coating can be used to reflect over 50% of the image light **9360** over the wavelengths of light provided by the LEDs in the light source **9340** while transmitting over 90% of the remaining wavelengths of visible light that are not provided by the LEDs so that the average transmission over the entire range of visible light is over 75%. Consequently, the tristimulus notch mirror is twice as efficient as the partial mirror in terms of the ability to reflect image light **9360** toward the user's eye **9310**.

To enable the optics module to operate with a combiner **9320** as shown in FIG. **93**, image light **9360** is provided to a lens **9330** which focuses the image light **9360** at the user's eye **9310**. Wherein lens **9330** is shown as a single lens element for simplicity, but multiple lens elements are also possible. The image light **9360** is provided from illumination light **9364** that comes from the light source **9340**. Where, the illumination light **9364** is reflected by a beam splitter **9352** toward a reflective image source **9350**. The image source **9350** can be a liquid crystal on silicon display (LCOS), a ferroelectric liquid crystal display (FLCOS) or other such reflective display. A polarizer **9342** can be associated with the light source **9340** to provide polarized illumination light **9364**. The beam splitter **9352** can then be a reflective polarizer that is oriented to substantially reflect the polarized illumination light **9364**. The image source **9350** changes the polarization state of the illumination light **9364** when the light is reflected by the image source **9350** to form image light **9360** that has a polarization state that is opposite to that of the illumination light **9364**. By changing the polarization state of the illumination light **9364** to the polarization state of the image light **9360**, the image light **9360** can then be transmitted by the reflective polarizer of the beam splitter **9352**. It is important to note that the image light **9360** is polarized to enable a folded illumination system and not because polarized light is required by the combiner **9320**. In

fact, to provide a transmission of light from the environment **9362** that is greater than 50%, the combiner **9320** cannot include a polarizer.

FIG. **94** is an illustration of an optics module than includes multiply folded optics to reduce the overall height of the optics module. In this case, illumination light **9464** is transmitted by the beam splitter **9452** so that it passes directly toward the image source **9450** wherein the beam splitter **9452** is a reflective polarizer and the light source **9340** includes a polarizer **9342** that is oriented so the transmission axis of the polarizer **9342** is parallel to the transmission axis of the beam splitter **9452**. The illumination light **9464** is then reflected and changed in polarization state by the image source **9450** so that the image light **9360** with its changed polarization state is reflected by beam splitter **9452** toward the lens **9330**. As can be seen by comparing FIG. **93** to FIG. **94**, the overall height of the optics module shown in FIG. **94** is substantially reduced.

However, the orientation of the additional fold in the optical path of the image light **9360** in the optics module of FIG. **94** increases the thickness of the optics module, where thickness is defined as the distance from the closest back surface of the optics module that is nearest to the user's eye to the farthest front surface of the optics module that is farthest from the user's eye. FIGS. **95** and **96** show illustrations of an optical module where the added fold in the optical path of the image light **9360** is oriented perpendicular to the fold shown in FIG. **94**. In this case, the optics module in FIGS. **95** and **96** is wider but thinner than that shown in FIG. **94**. FIG. **95** shows the optics module from the side and FIG. **96** shows the optics module from the position of the user's eye **9310**. As such, in the multiply folded optics shown in FIGS. **95** and **96**, optical axis **935** associated with the illumination light **9464** is perpendicular to both the optical axis **934** associated with the image light **9360** as it passes through the lens **9330** and the optical axis **933** associated with the image light **9360** as it proceeds toward the user's eye **9310** in the eyepiece. In the case of a head mounted display, it can be very important to have a thin optics module because a thick optics module can cause the head mounted display to stick outward from the user's forehead, which can be uncomfortable and unattractive. Thus, the multiply folded optics module shown in FIGS. **95** and **96** are shorter and thinner than the optic module shown in FIG. **93**. The optics module shown in FIGS. **95** and **96** is wider than the optics module shown in FIG. **93**, but in a glasses configuration of the head mounted display, wider optics modules can be better fit into the glasses frames than taller or thicker optics modules.

A further advantage that is provided by an optics module that includes multiply folded optics is that twists can be introduced at the fold surfaces to modify the orientation of different portions of the optics module relative to each other. This can be important when the optics module needs to fit into a thin curved glasses frame, a visor or a helmet where the increased width associated with the upper portion of the multiply folded optics module can make it more difficult to fit into structures that are not parallel to the combiner. In this case, the upper portion including for example (based on FIG. **96**), the light source **9340**, the polarizer **9342**, the beam splitter **9452** and the image source **9450**, can be twisted relative to the lower portion including the lens **9330** and the combiner **9320**. Where to avoid distortion of the image due to the compound angles between the fold surfaces, a twist of the upper portion about the axis **934** must be combined with a corresponding twist of the lower portion about the axis **933**. In this way, the effects of the increased width of the

upper portion of the multiply folded optics can be reduced when fitting the optics module into a curved structure such as glasses frames, a visor frame or a helmet structure.

FIG. **99** shows a further embodiment wherein the lens **9930** includes a diffractive surface **9931** to enable a more compact and shorter optical design with reduced chromatic aberration. Where the diffractive surface **9931** can be comprised of a series of small annular sections of a refractive lens curve such as for example in a Fresnel lens. The diffractive surface **9931** can be flat as shown in FIG. **99** or it can have a base curve to provide additional optical power. The diffractive surface **9931** can be a single order diffractive or a multiple order diffractive. To reduce scattering of wide angle illumination light **9964** that could be incident on the diffractive surface **9931**, an absorptive polarizer **9932** is provided and is oriented with its transmission axis perpendicular to the transmission axis of the reflective polarizer of the beam splitter **9452**. In this way, illumination light **9964** that is transmitted by the beam splitter **9452** in the direction that would cause it to be incident on the diffractive surface **9931** is absorbed by the absorptive polarizer **9932** before it can be scattered by the diffractive surface **9931**. At the same time, image light **9360** has a polarization state that is opposite to that of the illumination light **9964** so that it is reflected by the beam splitter **9452** and transmitted by the absorptive polarizer **9932** as it passes into the lens **9930**.

FIG. **100** shows an illustration of an optics module that includes a reduced angle between the beam splitter **9452** and the lens **9930** to reduce the overall height of the optics module. The fold angle of the image light **9360** (the deflection angle between **934** and **1005**) is then more than 90 degrees and as a result, the upper edge of the beam splitter is closer to the lens **9330** thereby providing a reduced overall height of the optics module.

FIG. **100** also shows a compact planar light source **10040** comprised of a thin edge-lit backlight similar to what is provided in displays used in displays for mobile devices like cellphones. The compact planar light source **10040** is positioned directly behind the beam splitter **9452** to reduce the overall size of the optics module. The compact planar light source can include a light guide film or light guide plate with an edge lit light such as one or more LEDs and a reflector on the side opposite the beam splitter **9452**. The compact planar light source can include a polarizer so the illumination light **10064** is polarized as previously described herein. To direct the illumination light **10064** toward the image source **9450** for improved efficiency, a turning film **10043** is positioned between the compact planar light source **10040** and the beam splitter **9452**. A 20 degree prismatic turning film can be obtained for example, from Luminit **103C** (Torrance, CA) under the name DTF. To obtain greater degrees of turning, such as 40 degrees, multiple layers of turning film **10043** can be stacked together provided they are oriented such that the turning effect is additive. A diffuser layer (not shown) can be used in addition to the turning film **10043** to reduce artifacts such as linear shadows that can be associated with prismatic structures that are typically associated with turning films **10043**. FIG. **101** shows an illustration of an optics module as seen from the position of the user's eye, which is similar to that shown in FIG. **100** but with a perpendicular orientation of the added fold in the image light **10164** to reduce the thickness of the optics module as previously described herein. As in the optics module shown in FIGS. **95** and **96**, the multiply folded optics shown in FIG. **101** have an optical axis **1005** associated with the illumination light **10164** that is perpendicular to both the optical axis **934** associated with the image light

9360 as it passes through the lens 9330 and the optical axis 933 associated with the image light 9360 as it proceeds toward the user's eye 9310 in the eyebox. As a result, the optics module of FIG. 101 is thinner and shorter than the optics module of FIG. 93. FIG. 101 also includes a field lens 10130 to improve the optical performance of the optics module. The addition of this second lens element is possible because of the change in fold orientation so that the field lens 10130 does not increase the thickness of the optics module, instead the added length of the optical path from the field lens 10130 occurs in the width of the optics module where space is more readily available in the head mounted display.

FIG. 102 shows an illustration of an optics module similar to that shown in FIG. 99 but with a different orientation of the upper portion of the optics module relative to the combiner so that the combiner 10220 can be more vertical. This rearrangement of the elements within the optics module can be important to achieve a good fit of the head mounted display onto the user's face. By making the combiner 10220 more vertical, the optics module can be made to have less interference with the user's cheekbones.

FIGS. 103, 103a and 103b show illustrations of optics modules as seen from the position of the user's eye, that include multiply folded optics and digital light projector (DLP) image sources 10350. In this case, the illumination light 10364 is provided at an oblique angle to the image source 10350 as required by the micromirrors in the DLP, to reflect image light 9360 along the optical axis 934 of the lens 9930. Where, in the case of a DLP image source 10350, image light 9360 is comprised of on-state light reflected by on-state micromirrors in the DLP image source 10350 along optical axis 934, in correspondence to the brightness of pixels in the image to be displayed to the user's eye 9310 in the eyebox. The micromirrors in the DLP image source 10350 also reflect off-state light 10371 to the side of the optics module in correspondence to the dark image content and as a result, a light trap 10372 is provided in the optics module to absorb light 10371. The light trap 10372 can be a black absorptive surface or a textured black surface. The purpose of the light trap 10372 is to absorb incident light 10371 and thereby reduce stray light and subsequently improve the contrast of the image displayed to the user's eye 9310. As previously described in other embodiments herein, the light source 10340 is provided to the side of the optics module with a multiply folded optical path to reduce the overall thickness and height of the optics module. FIG. 103 provides the DLP image source 10350 at the top of the optics module so that the image light 9360 proceeds straight along the optical axis 934, through the lens 9930 and down to the combiner 9320 where the image light is reflected toward the user's eye 9310 located in the eyebox. A polarizer 10341 is provided with the light source 10340 so that polarized illumination light 10364 is reflected by the beam splitter 9452 to illuminate the DLP image source 10350. Where, the beam splitter 9452 in this case, is a reflective polarizer that is aligned with the polarizer 10341 so that the polarized illumination light 10364 is reflected by the beam splitter 9452 and image light 9360 is transmitted by the beam splitter 9452. A quarter wave film 10351 is located adjacent to the surface of the DLP image source 10350 so that the polarization state of the image light 9360 is opposite to that of the illumination light 10364 after being reflected by the DLP image source 10350. The light source 10340 and the reflective polarizer 9452 are angularly arranged so that the illumination light 10364 is incident onto the DLP image source 10350 at the oblique angle required so that the image light 9360 when reflected by the on-state pixels in the DLP

image source 10350 proceeds along the optical axis 934 of the lens 9930. A field lens (similar to 10130 as shown in FIG. 101) or other lens elements may be included in the optics of FIG. 103 but is not shown, in which case, the illumination light 10364 and the image light 9360 may pass thru the field lens or other lens elements in opposite directions.

FIG. 103a is an illustration of another optics module with a multiply folded optical path that includes a DLP image source 10350 and is shown from the position of the user's eye. The light source 10340 is again provided to the side of the optics module to reduce the thickness of the optics module. In this case, the light source 10340 is provided on the same side of the lens 9930 and combiner 9320, as the DLP image source 10350. Lens 9930 can optionally include one or more diffractive surfaces 9931. The light source 10340 directly illuminates the DLP image source 10350 where the illumination light 10364 is incident on the DLP image source 10350 at an oblique angle so that the image light 9360, after being reflected by the on-state micromirrors in the DLP image source 10350, proceeds along the folded optical axis 934. At least one light trap 10372 is also provided to absorb light 10371 that is reflected from off-state micromirrors in the DLP and thereby improve the contrast of the displayed image as seen by the user. A field lens 10332 is provided between the DLP image source 10350 and the fold mirror 10352. The illumination light 10364 in this case can be unpolarized light whereupon the fold mirror 10352 can be comprised of a full mirror coating (e.g. a coating that reflects the entire visible light spectrum) on a substrate. The field lens 10332 can be a single lens element as shown in FIG. 103a or it can include multiple lens elements as needed. The field lens 10332 is designed to provide a large air gap between the field lens 10352 and the DLP image source 10350, so that the illumination light 10364 can be introduced to the optics module to directly illuminate the active area associated with the DLP image source 10350. By using unpolarized illumination light 10364, the optics module shown in FIG. 103a has improved efficiency over the optics module with DLP image sources 10350 shown in FIGS. 103 and 103b.

FIG. 103b is an illustration of another optics module with multiply folded optical path that includes a DLP image source 10350 and is shown from the position of the user's eye 9310 in the eyebox. As in the optics modules shown in FIGS. 103 and 103a, the optics module of FIG. 103b has the light source 10340 positioned at the side of the optics module to reduce the height and thickness of the optics module. The DLP image source 10350 is positioned opposite the light source 10340 however in this embodiment they do not share an optical axis. The illumination light 10364 passes through the beam splitter 10352, which in this case can be a first reflective polarizer. A second reflective polarizer 10332 is positioned adjacent to the lens 9930 so that the illumination light 10364 is reflected toward the DLP image source 10350. To reflect the illumination light 10364, the first reflective polarizer (beam splitter 10352) and the second reflective polarizer 10332 are oriented with perpendicular transmission axes. A quarter wave film 10351 (or quarter wave coating on the DLP cover glass) is provided adjacent to the DLP image source 10350 so that the polarization state of the illumination light 10364 is changed upon reflection from the DLP image source 10350 as it becomes image light 9360. As a result, the polarization of the illumination light 10364 is opposite to that of the image light 9360. Consequently, the illumination light 10364 is transmitted by the beam splitter 10352 and reflected by the second reflective polarizer 10332, while the image light 9360 is reflected by

the beam splitter 10352 and transmitted by the second reflective polarizer 10332. The light source 10340 is oriented relative to the second reflective polarizer 10332 so that it is reflected at an oblique angle relative to the DLP image source 10350 as required to provide image light 9360 reflected from on-state micromirrors in the DLP image source 10350 along the folded optical axis 934. The second reflective polarizer 10332 can be extended beyond the lens 9930 to provide the required oblique angle to fully illuminate the DLP image source 10350 as shown in FIG. 103b. Because the light source 10340 is located behind the beam splitter 10352, which is a reflective polarizer, the light source 10340 does not affect the image light 9360 and as a result, the light source 10340 can be a different size and orientation than the beam splitter 10352. One or more light traps 10372 are provided to absorb light 10371 that is reflected from off-state micromirrors in the DLP image source 10350 and thereby improve the contrast of the displayed image. In this case, the light trap 10372 can be positioned under the second reflective polarizer 10332 because the polarization state of the light 10371 is such that it is reflected by the beam splitter 10352 and transmitted by the second reflective polarizer 10332. The combined orientation of the light source 10340, the beam splitter 10352 and the DLP image source 10350 provides an optics module that is relatively thin and relatively short compared to optics modules where the image source or the light source are positioned above the fold mirror or beam splitter (e.g. such as the optics module shown in FIG. 103).

FIGS. 97 and 98 show illustrations of optics modules similar to those shown in FIG. 94 but with the addition of an eye imaging camera 979 for capturing images of the user's eye 9310 during use. In these cases, the light source 9340 and image source 9450 are positioned opposite one another so that the eye imaging camera 979 can be positioned directly above the lens 9340 so that the optical axis 934 is shared between the optics module and the eye imaging camera 979. By sharing a common optical axis, the eye imaging camera 979 can capture an image of the user's eye 9310 that has a perspective from directly in front of the user's eye 9310. Image light 9360 can then be used to illuminate the user's eye 9310 during image capture. A portion of the light reflected from the user's eye 9310, which can be unpolarized, passes through the beam splitter 9452 before being captured by the eye imaging camera 979. Because the eye imaging camera 979 is located above the beam splitter 9452, if the beam splitter 9452 is a reflective polarizer, the polarization state of the image light 9360 will be opposite to that of the light 978 captured by the eye imaging camera 979. The eye imaging camera 979 can be used to capture still images or video. Where video images can be used to track movements of the user's eye when looking at displayed images or when looking at a see-through view of the environment. Still images can be used to capture images of the user's eye 9310 for the purpose of identifying the user based on patterns on the iris. Given the small size of available camera modules, an eye imaging camera 979 can be added to the optics module with little impact on the overall size of the optics module. Additional lighting can be provided adjacent to the combiner 9320 to illuminate the user's eye. The additional lighting can be infrared, so the user can simultaneously view images displayed with visible light. If the additional lighting is infrared, the eye camera 979 must be capable of capturing images at matching infrared wavelengths. By capturing images of the user's eye from the perspective of directly in front of the

user's eye, undistorted images of the user's eye can be obtained over a wide range of eye movement.

FIG. 120 shows an illustration of another embodiment of an eye imaging camera associated with the optics module shown in FIG. 101, however the eye imaging camera can be similarly included in optics modules such as those shown in FIGS. 99, 100, 103, 103b. These optics modules include absorptive polarizers 9932 to reduce stray light as previously disclosed herein. These optics modules can also include a diffractive surface 9931, but the diffractive surface 9931 is not required for the operation of the eye imaging camera 979. In this embodiment, the polarization state of the image light 9360 is the same as that of the light that is reflected by the user's eye and captured by the eye imaging camera 979 since they both pass through the absorptive polarizer 9932. In this embodiment, the eye imaging camera 979 is positioned adjacent to the beam splitter 9452 and the compact planar light source 10040 and between the beam splitter and the field lens 10130. The optical axis 12034 of the light reflected by the eye is then angled somewhat relative to the optical axis 934 of the image light 9360, so that the center of the user's eye 9310 and the associated eyebox are within the field of view of the eye imaging camera 979. In this way, the eye imaging camera 979 captures images of the user's eye from nearly directly in front and only slightly to the side of the user's eye 9310 as shown in FIG. 120. While FIG. 120 shows the eye imaging camera 979 positioned adjacent to an end of the beam splitter 9452, it is also possible to position the eye imaging camera 979 adjacent to a side of the beam splitter 9452. The advantage of this embodiment is that the eye imaging camera 979 is provided with a simple optical path so that high image quality is possible in the captured images of the user's eye 9310. It should be noted that the optics associated with the eye imaging camera must take into account the effect of the lens 9930 since the light reflected by the user's eye 9310 that is captured by the eye imaging camera passes through the lens 9930. Also, the addition of the eye imaging camera 979 does not substantially increase the volume of the optics module as can be seen by comparing FIG. 120 to FIG. 101.

FIG. 121 shows an illustration of a further embodiment of an optics module that includes an eye imaging camera 979. Similar to the embodiment shown in FIG. 120, this optics module also includes an absorptive polarizer 9932 to reduce stray light and a diffractive surface 9931 may be included, but is not required. In this embodiment, the eye imaging camera 979 is positioned between the beam splitter 9452 and the field lens 10130 and pointed towards the beam splitter 9452. In this way, light reflected by the user's eye 9310 is reflected upwards by the combiner 9320, passes through the lens 9930 and the absorptive polarizer 9932 and then is reflected laterally toward the eye imaging camera 979 by the beam splitter 9452. The light captured by the eye imaging camera 979 is thereby the same polarization state as the image light 9360, so that it is reflected by the beam splitter 9452 and transmitted by the absorptive polarizer 9932. The light reflected by the user's eye 9310 can be unpolarized as initially reflected by the user's eye 9310, however, after passing through the absorptive polarizer 9932, the light becomes polarized with the same polarization state as the image light 9360. An advantage of this embodiment is that it is even more compact than the embodiment shown in FIG. 120. This arrangement of the eye imaging camera 979 is also possible in the optics modules shown in FIGS. 99, 100, 103, 103a and 103b.

In the embodiments shown in FIGS. 120 and 121, the user's eye 9310 and the associated eyebox can be illumi-

nated by image light **9360** or an additional light source can be provided for example, by an LED positioned adjacent to the combiner **9320**. Where the LED can provide visible light or infrared light, provided the eye imaging camera can capture at least a portion of the wavelengths of light provided by the LED.

In an alternative embodiment for the optics module shown in FIG. **103a**, the light source **10340** provides polarized illumination light **10364** and the fold mirror **10352** is a reflective polarizer plate so that an eye camera (not shown) can be positioned above the fold mirror **10352** and along the optical axis **934** for capturing images of the user's eye **9310** similar to that shown in FIGS. **97** and **98**. The eye camera and the optics module then share a common optical axis **934** so that images of the user's eye **9310** are captured from directly in front of the eye. In this arrangement, the polarization state of the image light **9360** is opposite to that of the light captured by the eye camera because the image light **9360** is reflected by the fold mirror **10352** and the light captured by the eye camera is transmitted by the fold mirror **10352**.

FIG. **104** shows an illustration of the optics module of FIG. **95** with the additional element of a controllable light blocking element to improve contrast in portions of the displayed image and also to improve the appearance of opacity in displayed objects such as augmented reality objects. Where the controllable light blocking element can operate by absorbing the incident light or scattering the incident light as provided, for example, by an electrochromic element, a polymer stabilized liquid crystal or a ferroelectric liquid crystal. Examples of suitable light blocking elements includes: 3G Switchable Film from Scienstry (Richardson, TX); Switchable Mirror or Switchable Glass from Kent Optronics (Hopewell Junction, NY). The controllable light blocking element **10420** is shown in FIG. **104** as being attached to the lower surface of the combiner **9320** so that it doesn't interfere with the displayed image while blocking see-thru light from the environment **9362**. Provided the combiner **9320** is flat, the addition of controllable light blocking elements **10420** adjacent to the combiner **9320** is easily done either by attaching directly to the combiner or attaching to the sidewalls of the optics module housing. The controllable light blocking element **10420** can have a single area that can be used to block a selectable portion of the see-through light from the environment over the entire combiner **9320** area thereby enabling a selectable optical density. Alternatively the controllable light blocking element **10420** can provide an array of areas **10520**, as shown in FIG. **105**, that can be separately selectively controlled to block portions of the combiner **9320** area that correspond to areas in the displayed image where high contrast areas of the image are located. FIG. **105** shows an illustration of an array of separately controllable light blocking elements **10520**. FIGS. **106a**, **106b** and **106c** are illustrations of how the array of separately controllable light blocking elements **10520** can be used. FIG. **106a** shows how the array of separately controllable light blocking elements **10520** can be put into blocking modes in areas **10622** and non-blocking modes in areas **10623**. Where the blocking mode areas **10622** correspond to areas where information or objects are to be displayed such as is shown in the corresponding areas in the illustration of FIG. **106b**. FIG. **106c** shows what the user sees when the image of FIG. **106b** is displayed with the array of controllable light blocking elements **10520** used in light blocking modes **10622** and non-blocking modes **10623**. The user then sees the displayed information or objects overlaid onto a see-through view of

the environment, but in the areas where information of objects are displayed, the see-through view is blocked to improve the contrast of the displayed information or object and provide a sense of solidness to the displayed information or objects.

In addition, FIG. **104** shows a rear optical element **10490** that can be a protective plate or a corrective optic. The protective plate can be connected to sidewalls and other structural elements to stiffen the positioning of the combiner **9320** and to prevent dust and dirt from getting onto the inner surface of the combiner **9320**. The corrective optics can include a prescriptive optic, which includes the ophthalmic prescription (optical power and astigmatism for example) of the user to improve the viewing experience.

Head mounted displays provide the user with freedom to move their head while watching displayed information. See-through head mounted displays also provide the user with a see-through view of the environment whereupon the displayed information is overlaid. While head mounted displays can include various types of image sources, image sources that provide sequential color display typically provide higher perceived resolution relative to the number of pixels in the displayed images because each pixel provides image content for each of the colors and the image perceived by the user as a displayed full color image frame is actually the sum of a series of rapidly displayed sequential color subframes. For example, the image source can sequentially provide subframe images comprised of a red image, a green image and then a blue image that are all derived from a single full color frame image. In this case, full color images are displayed at an image frame rate that includes a series of at least three sequentially colored subframes that are displayed at a subframe rate which is at least 3× the image frame rate. Sequential color images sources include reflective image sources such as LCOS and DLP.

The color breakup that occurs with a sequential color display occurs because the different color subframe images that together provide the user with a full color frame image are displayed at different times. The inventors realized that with sequential color display in a head mounted display, when there is movement of the head-mounted display or movement of the user's eyes, such that the user's eyes do not move in synch with the displayed image that under such movement conditions the perceived locations of each of the sequential color image subframes are different within the user's field of view. This can happen when the user moves his head and the user's eyes do not follow the same trajectory as the head mounted display, which can be due to the user's eyes moving in a jerky trajectory as the eyes pause to look at an object in the see-through view of the environment. Another way this can happen is if an object passes through the see-through view of the environment and the user's eyes follow the movement of the object. Due to this difference in perceived locations within the user's field of view, the user sees the sequential color images slightly separated at the edges of objects. This separation of colors at the edge of objects is referred to as color breakup. Color breakup may be easily perceived during certain movements because the sequential colors are vividly colored in areas where they do not overlap one another. The faster the user moves their head or the faster the user's eyes move across the display field of view, the more noticeable the color breakup becomes, because the different color subframe images are separated by a greater distance within the field of view. Color breakup is particularly noticeable with see-through head mounted displays, because the user can see the environment and the user's eyes tend to linger on objects

seen in the environment as the user turns his head. So even though the user may turn his head at a steady rotational rate, the user's eye movement tends to be jerky and this creates the conditions where color breakup is observed. As such there are two different conditions that tend to be associated with color breakup: rapid head movement and rapid eye movement.

It is important to note that when the user is not moving his head and the head mounted display is not moving on the user's head, color breakup will not be observed because the subframe images are provided at the same positions within the field of view of the user's eyes. Also, if the user were to move his head and the user moves his eyes in synch with the head movement, color breakup will not be observed. So movement of the head mounted display is indicative of conditions that can lead to color breakup and is also indicative of the degree of color breakup that can occur if the user moves his eyes relative to the movement of the head mounted display. Color breakup is less of an issue with head mounted displays that do not have see-through to the environment, because only the displayed image content is visible to the user and it moves in synch with the movement of the head mounted display. Color breakup is also not an issue if a monochrome image is displayed with a monochrome light source (i.e. there are no sequential color subframes, instead there are only single color frames) since all the displayed images are comprised of the same color. Thus, color breakup is an issue that is most noticeable with head mounted displays that provide a see-through view of the environment.

Systems and methods according to the principles of the present invention reduce color breakup and thereby improve the viewing experience provided by a head-mounted display with see-through when the user is moving through the environment.

In embodiments, systems and methods are provided where the head-mounted display detects the speed of movement of the head-mounted display and in response, the resolution of the image is reduced or the bit depth of the image is reduced, while the image frame rate at which the image is displayed and the associated subframe rate are correspondingly increased. In this way, the bandwidth associated with the display of the image can be maintained constant, in spite of the frame rate being increased. Where, by increasing the frame rate associated with the display of images, the time between the display of each sequential color subframe image is reduced and as a result the visually perceived separation between the sequential color images is reduced. Similarly the image frame rate can be reduced while the subframe rate is increased by increasing the number of subframes displayed for each image frame.

In further embodiments, systems and methods are provided where the sequential color subframe images are shifted laterally or vertically relative to one another by a number of pixels that corresponds to the detected movement of the head mounted display. In this way, the color sequential subframe images are displayed to the user such that they are visually overlaid on top of each other within the displayed field of view. This compensates for separation between subframes and thereby reduces color breakup.

In yet another embodiment, systems and methods are provided where an eye-imaging camera in the head-mounted display is used to track the movement of the user's eyes. The movement of the head-mounted display may be simultaneously measured. An accommodation in the presentation may then be made to reduce color breakup. For example, the resolution of the images and the frame rate may be changed or the image frame rate can be reduced while increasing the

subframe rate, in correspondence to the difference in movement of the user's eyes and the movement of the head mounted display. As another example, the subframes may be shifted to align the subframes in correspondence to the determined difference in movement between the user's eyes and the head mounted display. As a further example, the color saturation of the content may be reduced to reduce the perception of color breakup due to the fact that the colors, while positionally separated as perceived by the user, are not as separated in color space. In yet a further example, the content could be converted to monochrome imagery which is displayed as a single color image (e.g. white) during the detected movement so that color breakup is not visible.

FIG. 107 shows an example of a full color image 10700 that includes an array of pixels, including portions of red, green and blue pixels. For sequential color display, three subframe images are created that are each comprised of only one color, such as only red or only green or only blue. Those skilled in the art will recognize that sequential color images that together provide a perceived full color image can also be comprised of subframes of cyan, magenta and yellow. These subframe images are rapidly displayed in sequence to the user on the head-mounted display so that the user perceives a full color image that combines all three colors. With a reflective display such as an LCOS or a DLP, the subframe images are displayed by changing the reflective display to provide the respective image content associated with the particular subframe image and then illuminating the reflective display with the associated color light, so the light is reflected to provide the subframe image to the optics of the head-mounted display and from there to the user's eye.

If the subframe images are accurately aligned with each other, then the full color image perceived by the user will be full color out to the edges of the image and there will be no color breakup. This is what is typically seen by the user of a head-mounted display when the head-mounted display is stationary on the user's head and the user is not moving his eyes. However, if the user moves his head or the head-mounted display moves on the user's head (such as due to vibration) and the user's eyes are not moved in unison with the displayed image, the user will perceive the subframe images to be laterally (or vertically) offset relative to one another as shown by illustrations 10802 and 10804 in FIGS. 108A and 108B. The perceived amount of lateral offset between the displayed subframe images is related to the speed of movement of the head-mounted display and the time between the display of the sequential subframe images, which is also known as subframe time or 1/subframe rate. The lateral shifting between subframe images, that is perceived by the user, is the color breakup and color breakup is perceived as fringes of color at the edges of objects. When the user moves his head (or eyes) quickly and the subframe rate is slow, color breakup can be substantial as illustrated in FIG. 108A. If the user moves his head slowly or the subframe rate is higher, the color breakup is less as illustrated in FIG. 108B. If the color breakup is less than one pixel, in digital imaging, in lateral shifting, the user will perceive there to be no color breakup.

Display frame rate in a head-mounted display is typically limited by either the bandwidth of the processor and associated electronics or by the power required to drive the processor and associated electronics, which translates into battery life. The bandwidth required to display images at a given frame rate is related to the number of frames displayed in a period of time and the number of pixels in each frame image. As such, simply increasing the frame rate to reduce color breakup is not always a good solution as it requires a

higher bandwidth which the processor or associated electronics may not be able to support and power usage will be increased thereby reducing battery life. Instead, systems and methods in accordance with the principles of the present invention provide a method of display wherein the number of pixels in each subframe image is reduced thereby reducing the bandwidth required to display each subframe image while simultaneously increasing the subframe rate by a corresponding amount to maintain bandwidth while reducing color breakup. This embodiment is suitable for situations wherein subframe images can be provided with different numbers of pixels and different frame rates. For example, it would be suitable in camera and display systems where the capture conditions can be changed to provide images with a lower resolution that can then be displayed with a faster subframe rate. Static images such as text or illustrations can be displayed with a lower frame rate and a faster subframe rate to reduce color breakup since the image content doesn't change quickly. Alternatively, images can be modified to be displayed at lower resolution (fewer pixels) with a faster frame rate or subframe rate to reduce color breakup.

FIG. 109 shows an illustration of the timing of a sequential color image comprised of sequential display of a red subframe image 10902 followed by a green subframe image 10904 followed by a blue subframe image 10908 in a repeating process. As long as the subframes together are displayed at a full color image frame rate that is greater than approximately 24 frames/sec, such that the sequential color subframes are displayed at a subframe rate of greater than 72 subframes/sec, the human eye will perceive full color moving images without flicker. This condition is suitable for displaying a video image without color breakup when the head-mounted display is stationary or moving relatively slowly. However, if the user moves his head such that the head-mounted display moves rapidly, color breakup will occur. This color breakup occurs because rapid head movements are typically a reaction of the user to something occurring in the environment (e.g. a loud noise) so that the user's eyes are searching the environment during the rapid head movement, which leads to jerky eye movements and substantial color breakup.

Movement of the head-mounted display can be detected by an inertial measurement unit, which can include accelerometers, gyro sensors, magnetometers, tilt sensors, vibration sensors, etc. Where only the movements within the plane of the display field of view (e.g. x and y movements and not z movement) are important for detecting conditions where color breakup may occur. If the head-mounted display is detected to be moving above a predetermined threshold where color breakup is predicted to occur (e.g. greater than 9 degrees/sec), in embodiments, the resolution of the images may be reduced (thereby reducing the number of pixels in the images and effectively making each pixel larger within the display field of view) and the subframe rate may be correspondingly increased. Note that the subframe rate can be increased without changing the image frame rate by increasing the number of subframes that are displayed sequentially, for example six subframes could be displayed for each image frame wherein the sequential color subframe images are each displayed twice. By increasing the number of subframes displayed for each image frame, the subframe rate can be increased without having to increase the image frame rate, which can be more difficult to change because the image frame rate is typically provided by the source of the image content such as in a movie. FIG. 110 shows an illustration of a faster subframe rate, wherein the display time for each subframe, 11002, 11004, and 11008 is reduced

and the time between display of each sequential subframe is also reduced. FIG. 110 shows a subframe rate that is approximately twice as fast as that shown in FIG. 109. The associated image frame rate can be twice as fast in FIG. 110 as compared to FIG. 109, where both the image frame rate and the subframe rate are doubled. Alternatively, as previously described, the image frame rate can be unchanged between FIGS. 109 and 110, where only the subframe rate is doubled to reduce color breakup. To enable the bandwidth associated with the display of the images shown in FIG. 110 to be approximately the same as the bandwidth associated with the display of subframe images shown in FIG. 109, the resolution (number of pixels in each subframe image) is reduced by approximately a factor of two.

While reducing the resolution of the displayed subframe images in correspondence to an increase in the subframe rate may seem to degrade the image quality perceived by the user, the human eye is not capable of perceiving high resolution when there is substantial movement. As such, color breakup is more visible than a reduction in the resolution of the image when the eye is moving. Consequently, the systems and methods of the present invention trade reduced image resolution for increased image frame rate to reduce color breakup without a perceptible loss in resolution, and bandwidth is thereby maintained. This technique can be used, for example, to reduce color breakup by a factor of up to 16, where the resolution of the displayed image is reduced to 1/16th the original resolution and the frame rate of the displayed image is increased by 16x.

In another embodiment of the invention, when movement of the head-mounted display is detected, the subframe images associated with a full color frame image are digitally shifted relative to one another in a direction counter to the detected direction of movement and with an amount that corresponds to the detected speed of movement. This effectively compensates for the perceived offset between the displayed subframe images that causes color breakup. The digital shifting is applied only to the subframes that together comprise a full color frame image. This is different from typical digital image stabilization wherein full color frame images are digitally shifted relative to one another to compensate for movement as described, for example, in United States patent publication 2008/0165280. By applying the digital shifting to the subframes that constitute a single full color frame image, the amount of digital shifting required to reduce color breakup is typically only a few pixels even when the detected movement speed is high, this is in contrast to typical digital image stabilization where fast movements result in accumulating shifts of the frame image so that the image effectively moves outside of the display field of view or the amount of digital stabilization that can be applied is limited. FIGS. 111a and 111b illustrate this embodiment. FIG. 111a shows how sequentially displayed subframe images, 11102, 11104, and 11108 would be perceived by the user when there is substantial movement, wherein the different colors associated with the subframes are separately visible along the edges of objects, evenly spaced across the field of view in the direction of movement. In contrast, FIG. 111b shows how the visibility of the subframes is changed when the subframes are digitally shifted to compensate for the detected movement and thereby reduce the separation between the subframes across the field of view, and as a result the user perceives a series of full color frame images 11120 with reduced color breakup. As shown in FIG. 111b, the full color frame images are not image stabilized or digitally shifted in response to the detected movement.

In embodiments, movement direction and speed of the head-mounted display is detected by the IMU sensor immediately prior to the display of each full color frame image. If the movement speed is above a predetermined threshold, the sequentially displayed color subframes associated with

each full color frame are digitally shifted relative to one another so that they are displayed in an aligned position within the display field of view. The magnitude of the shift corresponds to the speed of the detected movement and the direction of the shift is counter to the detected direction of movement.

In an example, the movement of the head-mounted display is detected immediately prior to display of a first subframe associated with a full color frame image. The first subframe associated with the full color frame image can then be displayed without a shift. The second subframe can be shifted by an amount and direction that compensates for the movement that occurs between the display of the first and second subframes and then is displayed. The third subframe can be shifted by an amount and direction that compensates for the movement that occurs between the display of the first subframe and the third subframe and is then displayed. The movement of the head-mounted display is then detected again to determine the shifts to be applied to the subframes associated with the next full color frame image. Alternatively, the subframes can be shifted by an amount that compensates for a portion of the movement that occurs between the subframes.

In a further example, the direction and speed of movement of the head-mounted display is detected immediately prior to the display of a reference subframe. Subsequent subframes are then shifted to compensate for movement that occurs between the time the reference subframe is displayed and the time that the subsequent subframe is displayed. Wherein the time that the reference subframe is displayed and the time that the subsequent subframe is displayed may be up to 5 frame times.

An advantage of this embodiment is illustrated by examining the effective frame rates associated with the color breakup and the blur of the image. If the full color image is displayed with an image frame rate of 60 frames/sec, the subframes would typically be displayed at a subframe rate of 180 frames/sec to provide three subframes for each image frame. The described system and method effectively shifts the subframes so that they are positioned on top of one another, so the color breakup is reduced to an amount that corresponds to 180 frames/sec. At the same time, the blur perceived by the user between image frames corresponds to 60 frames/sec since each of the subframes is derived from the same full color frame image.

In further embodiments, the digital shifting of the subframes that is based on detected movement immediately prior to the display of each full color frame image can be combined with digital image stabilization that is applied between the full color frame images.

In yet further embodiments, the method of digital shifting of subframes is combined with the method of increasing frame rate with a simultaneous reduction in image resolution. These two methods of reducing color breakup operate on different aspects of the image processing associated with displaying an image in a head mounted display, as such they can be independently applied in either order in the image processing system associated with the processor.

In yet another embodiment, the head mounted display includes a camera for detecting the eye movements of the user (e.g. as described herein) relative to the movement of the head mounted display. The eye camera can be used to

measure the speed of eye movement and the direction of eye movement. In embodiments, the resolution of eye cameras can be relatively low (e.g. QVGA or VGA) so that the frame rate can be relatively high (e.g. 120 frames/sec) without introducing bandwidth limitations. The detected eye movements relative to the head-mounted display can be used to determine when to apply methods to reduce color breakup including, for example, increasing the frame rate and digitally shifting the subframes as has been previously described herein. For example, if the detected eye movement is above a predetermined angular speed, the resolution of the displayed images can be reduced and the subframe rate can be increased. In another example, the detected eye movement can be used to determine the amount and direction of digital shifting applied to subframes within an image frame prior to display of the subframes. In yet another example, measured eye movements can be used in combination with detected movements of the head-mounted display to determine the amount and direction of digital shifting applied to subframes within an image frame prior to display of the subframes. The amount and direction of digital shifting applied to the subframes can be in correspondence to the difference between the detected movements of the head mounted display and the detected eye movements of the user. Where the detection of a condition where the user's eye is moving one direction and the head mounted display is moving in an opposing direction represents a situation where particularly bad color breakup can occur. In this case, combined methods for reducing color breakup are advantageous.

In another yet further embodiment, when movement of the head-mounted display or eye movement is detected above a predetermined threshold, the images are changed from color sequentially displayed full color images to monochrome images. The monochrome images can be comprised of combined image content from each of the color sequential subframes associated with each full color image frame. Where the monochrome images can be grey scale or luma images wherein the luma code values (Y) for each pixel can be calculated for example as given in Equation 1 below as taken from <http://en.wikipedia.org/wiki/Grayscale> and as referenced to the CIE 1931 standard for digital photography:

$$Y=0.2126R+0.7152G+0.0722B \quad \text{Equation 1}$$

where R is the red code value for the pixel, G is the green code value for the pixel and B is the blue code value for the pixel. Alternatively, monochrome images can be comprised of single color images such as the green subframe image, and this image can be displayed either with a single color or preferably with simultaneous application of all the sequential colors (e.g. red, green and blue) so that the applied illumination onto the reflective image source is white light and as a result, the displayed image appears as a grey scale image.

Several more specific examples are provided below.

Example 1

For a 26 deg display field of view and a 1280 pixel horizontally wide image, a pixel occupies 0.020 deg within the display field of view. If the frame rate of the full color images is 60 Hz, with three color sequential subframes images, the subframe time is 0.006 sec. The rotational speed of the head mounted display needed to produce one pixel of color breakup is then 3.6 deg/sec. If the number of horizontal pixels in the display field of view is reduced to 640 pixels and simultaneously the frame rate of the full color images is increased to 120 Hz, with three color sequential subframes

images, the subframe time is reduced to 0.003, the size of a pixel is increased to 0.041 deg and the rotational speed to produce one pixel of color breakup is 14.6 deg/sec.

Example 2

For a 26 deg display field of view and a 1280 pixel horizontally wide image, a pixel is 0.020 deg within the display field of view. If the smallest size that the user can detect for color breakup is one pixel wide, then a rotational speed of over 3.6 deg/sec is required if the subframe rate is 180 Hz, before color breakup is detected by the user. Even though the color breakup is an analog effect, the user's eye does not have the resolution to detect the color fringes that are present during movement below this speed. So below this rotational speed, color breakup management is not required.

Example 3

For a 26 deg display field of view and a 1280 pixel horizontally wide image, a pixel is 0.020 deg within the display field of view. If the user can detect color breakup as small as one pixel wide, then a rotational speed of 3.6 deg/sec will require a shift of the subframes relative to each other of one pixel if the subframe rate is 180 Hz, to align the subframes so that color breakup is not visible to the user. If the user rotates their head at 15 deg/sec, then the subframes will require a shift of 4 pixels relative to one another to align the subframes so that color breakup is not visible. If the image frame begins with the display of the red subframe image, then no digital shifting is required for the red subframe image. A 4 pixel shift is required for the green subframe image. And, an 8 pixel shift is required for the blue subframe image. The next red subframe associated with the next image frame would then be effectively shifted 12 pixels relative to the previous red subframe within the field of view.

Each of the color breakup reduction technologies described herein may be used in combination with each of the other color breakup reduction technologies.

The inventors appreciated that fitting see-through computer displays into certain head-worn form factors is a challenge, even when reduced in size as described herein. A further advantage that is provided by an optics module that includes multiply folded optics is that twists can be introduced at the fold surfaces to modify the orientation of different portions of the optics module relative to each other. This can be important when the optics module needs to fit into a thin curved glasses frame, a visor or a helmet where the increased width associated with the upper portion of the multiply folded optics module can make it more difficult to fit into structures that are not parallel to the combiner. As such, another aspect of the present invention relates to twisting certain optical components within the see-through computer display such that the optical components better fit certain form factors (e.g. glasses) yet continue to perform as high quality image displays. In embodiments, optics systems with dual mirror systems to fold the optical path (e.g. optical systems described herein with respect to FIGS. 6, and 93 through 106) are provided such that the image production module (e.g. upper module), which includes a first image light reflective surface, is turned about a first optical axis leading from the upper module to the lower module and in a direction to fit the upper module more compactly into a frame of a head-worn computer. At the same time, to avoid distorting the image provided to the eye of the user, the image delivery optics (e.g. lower module), which includes a

second image light reflective surface, is turned about a second optical axis that leads to the user's eye and in the opposite direction relative to the image, thereby introducing a compound angle between the first image light reflective surface and the second image light reflective surface. Provided that the first and second optical axes are perpendicular to one another in the non-twisted state, the distortion in the image associated with the twist about the first axis is compensated by a twist of the same angular magnitude about the second axis so that the image presented to the eye of the user is undistorted by the twisting.

FIG. 112 illustrates a head-worn computer with see-through displays in accordance with the principles of the present invention. The head-worn computer has a frame 11202 that houses/holds the optics modules in position in front of the users eyes. As illustrated in FIG. 112, the frame 11202 holds two sets of optical modules 11204 and 11208 each of which have upper and lower optics modules. Optics module 11204 is non-twisted and is presented to illustrate the difficulty in fitting the non-twisted version into the frame. One will note that the dotted box, which represents the outer bounds of the optics module 11204 doesn't fit within the bounds of the frame 11202. Fitting optics module 11204 into the frame 11202 would normally require that the frame 11202 become thicker, from front to back, which would lead to more offset of the glasses form factor from the face of the user, which is less desirable and is less compact. In contrast, optics module 11208 is a twisted optics module, where the upper module is twisted (or rotated) to better fit into the confines of the frame 11202 as shown in FIG. 112.

FIG. 113 shows a more detailed illustration of the twists imparted within multiply folded optics in optics module 11208. Upper module 11214 is twisted relative to the lower module 11218 along optical axis 934 to better fit into the frame 11202. It is this twist which enables optics module 11208 to better fit within the frame 11202 as shown in FIG. 112 and as a result frame 11202 can be thinner and more compact than if non-twisted optics modules were used. To avoid distorting the image provided to the user, a second twist is required to introduce a compound angle between the first reflecting surface 11225 in the upper module 11214 and second reflecting surface 11226 in the lower optics module 11218. The second twist is imparted to the second reflecting surface about the optical axis 933 and in an opposite direction relative to the image from the twist in the upper module 11214. In this way, the effects of the increased width of the upper portion of the multiply folded optics can be reduced when fitting the optics module into a curved structure such as glasses frames, a visor frame or a helmet structure. Where it is preferred, but not required that the optical axis 934 be perpendicular to the optical axis 933 so that the magnitude of the angular twist imparted to the first reflecting surface 11225 can be the same as the twist imparted to the second reflecting surface 11226 to provide an image to the user's eye that is not distorted due to the twisting.

Another aspect of the present invention relates to the configuration of the optics and electronics in a head-worn frame such that the frame maintains a minimal form factor to resemble standard glasses. In embodiments, a see through optical display with multiply folded optics to provide a reduced thickness (e.g. as described herein) may be mounted in the frame. In embodiments, the multiply folded optical configuration may be twisted at the fold surfaces (e.g. as described herein) to better fit the optics into the frame. In embodiments, the electronics that operate the displays, processor, memory, sensors, etc. are positioned between, above,

below, on a side, etc. of the optical modules and oriented to provide a reduced thickness in the frame to match the thickness of the optics. Orienting the board can be particularly important when the board includes large components that limit the width of the board, such as for example the processor chip. For example, an electronics board or components on the electronics board may be mounted in a vertical orientation between and/or above the optical modules to reduce the thickness of the electronics board as mounted into the frame. In another configuration the board may be mounted between the optical modules at a height near the top of the optical modules to minimize the height of the glasses frame. In yet another configuration the board may be mounted such that it extends over the optical modules to minimize the thickness of the frame. In further embodiments, the board may be mounted in an angled configuration to enable the thickness and height of the frame to be reduced simultaneously. In embodiments, the electronics may be divided between multiple boards. For example, a longer board over a shorter board where the space between the optical modules is used for the lower board. This configuration uses some of the space between the eyes for some of the electronics.

FIG. 114 illustrates a top view and front view of a configuration including optical modules 11208, electronics board 11402 and a heat sink 11404. The board 11402 is mounted in a vertical orientation to maintain a thin frame portion that sits across the user's brow. As illustrated, the optical modules 11208 include upper modules 11214 and a second reflecting surface 11226 in front of the user's eye. The upper module may have a flat reflecting surface and the upper 11214 may be turned or twisted with respect to the second reflecting surface 11226 as described herein. The second reflecting surface 11226 may be a partial mirror, notch filter, holographic filter, etc. to reflect at least a portion of the image light to the eye of the user while allowing scene light to transmit through to the eye.

FIG. 115 illustrates a front view of a configuration that includes optics illustrated in FIG. 114; however, the electronics board 11402 is mounted in the space between the optical modules at a height that is similar to the height of the optical modules. This configuration reduces the overall height of the frame.

FIG. 116 illustrates a front view of a configuration that includes optics illustrated in FIGS. 114 and 115. The electronics layout in this configuration is done with multiple boards, 11402, 11602 and 11604. The multiple board configuration allows the boards to be thinner from front to back thereby enabling the brow section of the frame to be thinner. A heat sink 11404 (not shown in FIG. 116) may be mounted on the front face between the optical modules. This configuration also causes the heat to be drawn in a direction away from the user's head. In embodiments, the processor, which is a main heat generator in the electronics, is mounted vertically (e.g. on board 11604) and the heat sink 11404 may be mounted in front such that it contacts the processor. In this configuration, the heat sink 11404 causes heat to spread to the front of the device, away from the user's head. In other embodiments, the processor is mounted horizontally (e.g. on board 11602 or 11402). In embodiments, the board(s) may be tilted (e.g. 20 degrees) from front to back to create an even thinner brow section.

Another aspect of the present invention relates to concealing the optical modules such that a person viewing the user does not clearly see the optical modules, electronics or boards. For example, in configurations described herein, the optical modules include lenses that hang below the top of the

brow section of the head-worn device frame and the electronics board(s) hang down as well so that the see-through view is partially blocked. To conceal these features and thereby provide the head worn computer with the appearance of conventional glasses, an outer lens may be included in the glasses frame so that it covers a portion of the frame that contain the optical modules or electronics, and the outer lens may include a progressive tint from top to bottom. In embodiments, the tint may have less transmission at the top for concealment of a portion of the frame that includes the optical modules or electronics board while having higher transmission below the concealment point such that a high see-through transmission is maintained.

Aspects of the present invention provide multiply folded optics to reduce the thickness of the optics modules along with vertically oriented or angled electronics to reduce the mounted thickness of the electronics and progressively tinted outer lenses to conceal a portion of the optics or electronics. In this way, a head worn computer is provided with a thinner form factor and an appearance of conventional glasses.

Another aspect of the present invention relates to an intuitive user interface mounted on the HWC 102 where the user interface includes tactile feedback to the user to provide the user an indication of engagement and change. In embodiments, the user interface is a rotating element on a temple section of a glasses form factor of the HWC 102. The rotating element may include segments such that it positively engages at certain predetermined angles. This facilitates a tactile feedback to the user. As the user turns the rotating element it 'clicks' through its predetermined steps or angles and each step causes a displayed user interface content to be changed. For example, the user may cycle through a set of menu items or selectable applications. In embodiments, the rotating element also includes a selection element, such as a pressure-induced section where the user can push to make a selection.

FIG. 117 illustrates a human head wearing a head-worn computer in a glasses form factor. The glasses have a temple section 11702 and a rotating user interface element 11704. The user can rotate the rotating element 11704 to cycle through options presented as content in the see-through display of the glasses. FIG. 118 illustrates several examples of different rotating user interface elements 11704a, 11704b and 11704c. Rotating element 11704a is mounted at the front end of the temple and has significant side and top exposure for user interaction. Rotating element 11704b is mounted further back and also has significant exposure (e.g. 270 degrees of touch). Rotating element 11704c has less exposure and is exposed for interaction on the top of the temple. Other embodiments may have a side or bottom exposure.

As discussed above, a specially designed lens may be used to conceal portions of the optics modules and/or electronics modules. FIG. 119 illustrates an embodiment of one such lens 11902. Two lenses, 11902 are illustrated with Base 6 and 1.3 mm thickness but other geometries with, for example, different curvatures and thicknesses can be used. The lenses 11902 are shaped to look like conventional glasses lenses with features including magnetic mounting attachment and special tinting in portions of the lenses 11902 where opaque structures such as electronics are located behind the lenses.

The lenses 11902 includes blind holes 11904 for the mounting of a magnetic attachment system (not shown). The magnetic attachment system may include magnets, magnetic material, dual magnets, opposite polarization magnets, etc. such that the lenses 11902 can be removed and remounted to

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the head-worn computer (e.g. HWC 102). In the magnetic attachment system, the lenses 11902 are held by magnetic force into the frame of the HWC. The magnets can be inserted into the blind holes 11904 or inserted into the frame of the HWC in corresponding matching positions. As long as either the lens 11902 or the matching position on the frame of the HWC includes a magnet and the other position has a similar sized piece of magnetic material or another magnet oriented to attract the lens 11902 and hold it in the frame of the HWC. To this end, the frame of the HWC can provide guidance features to position the lens 11902 in front of the optics modules in the HWC. Where the guidance features can be a ridge or flange that the lens is seated in so the lens 11902 cannot move laterally when held in place by the magnetic attachment system. In this way, the function of the magnetic attachment system is simply to hold the lenses 11902 in place, while the guidance features position the lenses 11902. The guidance features can be robustly made to hold the lenses 11902 in place when dropped or subjected to impact even when the force provided by the magnetic attachment system is relatively low, so that the lenses 11902 can be easily removed by the user for cleaning or replacement. Where easy replacement enables a variety of lenses with different optical features (e.g. polarized, photochromic, different optical density) or different appearance (e.g. colors, level of tinting, mirror coating) to be changed out by the user as desired.

FIG. 119 also illustrates an example of how the lens 11902 may be tinted to conceal or at least partially conceal certain optical components (e.g. the non-see-through components or opaque components) such as, electronics, electronics boards, auxiliary sensors such as an infrared camera and/or other components. As illustrated, the blind holes 11904 may also be concealed or at least partially concealed by the tinting. As illustrated in FIG. 119, a top portion 11908, approximately 15 mm as illustrated, may be more heavily tinted (e.g. 0 to 30% transmission) or mirrored to better conceal the non-see-through portions of the optics and other components. Below the top portion 11908, the lens 11902 may have a gradient zone 11909 where the tinting level gradually changes from top to bottom and leads into the lower zone 11910. The lower zone 11910 includes the area where the user primarily views the see-through surrounding and this zone may be tinted to suit the viewing application. For example, if the application requires a high see through, the lower zone 11910 may be tinted, between 90% and 100% transmissive. If the application requires some see-through tint, than the lower area may be more heavily tinted or mirrored (e.g. 20% to 90%). In embodiments, the lower area 11910 may be a photochromic layer, an electrochromic layer, a controllable mirror or other variable transmission layer. In embodiments, the entire lens or portions thereof may have a variable transmission layer such as a photochromic layer, electrochromic layer, controllable mirror, etc. In embodiments, any of the areas or whole lens 11902 may include polarization.

Another aspect of the present invention relates to cooling the internal component through the use of micro-holes sized such they are large enough to allow gas to escape but small enough to not allow water to pass through (e.g. 25 μ m, 0.2 mm, 0.3 mm, etc.). The micro-holes may be included in a heat sink, for example. The heat sink, or other area, may be populated with hundreds or thousands of such micro-holes. The micro-holes may be laser cut or CNC holes, for example, that are small enough to keep large droplets of water out of the device but allow air to exchange through the heat sink. Besides increasing surface area of the heat sink, they also have matching holes on the underside of the frame

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to enable convective cooling where cool air is pulled in from the bottom as the heat raises from the top, like a chimney and as such, the heat sink with the micro-holes is preferably located on the top or side of the frame of the HWC. In embodiments, the micro-holes are aligned in the troughs formed by the fins on the top of the heat sink. This causes the exiting air to flow through the troughs thereby increasing the heat transfer from the fins. In embodiments, the micro-holes may be angled such that the length of the hole in the heat sink material is increased and the air flow can be directed away from the head of the user. In addition, the micro-holes may be of a size to cause turbulence in the air flow as it passes through the micro-holes. Where, turbulence substantially increases the heat transfer rate associated with the air flow through the heat sink. In embodiments, the heat management system of the HWC 102 is passive, including no active cooling systems such as fans or other energized mechanical cooling systems to force air flow through the micro-holes. In other embodiments, the heat management system includes energized mechanical cooling, such as a fan or multiple fans or other systems to force air movement through the HWC and the micro-holes.

Another aspect of the present invention relates to finding items in the surrounding environment based on similarity to items identified. Augmented reality is often rigidly defined in terms of what is included and how it is used, it would be advantageous to provide a more flexible interface so people can use augmented reality to do whatever they want it to do. An example is to use the HWC camera, image analysis and display to designate items to be found. FIG. 122 shows an illustration of an image 12210 of a scene containing an object that the user would like the HWC to assist in looking for the object as the user moves through the environment. In this example, the user has circled the object 12220 that is being looked for, where in this case the object is a cat. The HWC then analyses the circled region of the image for shapes, patterns and colors to identify the target to be searched for. The HWC then uses the camera to capture images of the scene as the user moves about. The HWC analyses the images and compares the shapes, patterns and colors in the captured images of the scene and compares them to the shapes, patterns and colors of the target. When there is a match, the HWC alerts the user to a potential find. The alert can be a vibration, a sound or a visual cue in an image displayed in the HWC such as a pointer, a flash or a circle that corresponds to the location of the potential find in the scene. This method provides a versatile and flexible augmented reality system wherein an item is described visually and a command of "find something like this" is given to the HWC. Examples of ways to identify an object to be searched for include: circle an item in a previously captured image that is stored on the HWC (as shown in FIG. 122); point to an item in a physical image held in front of the camera in the HWC; point to an item in the live image provided by the camera in the HWC and viewed in the see-through display of the HWC, etc. Alternately, text can be input to the HWC with a command of "find wording like this", e.g. a street sign or an item in a store and the HWC can then search for the text as the user moves through the environment. In another example, the user can indicate a color with a command of "find a color like this". The camera used to search for the item can even be a hyperspectral camera in HWC to search for the item using infrared or ultraviolet light to thereby augment the visual search that the user is conducting. This method can be extended to any pattern that the user can identify for the HWC such as sounds, vibrations, movements, etc. and the HWC can then

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use any of the sensors included in the HWC to search for the identified pattern as the target. As such the finding system provided by the invention is very flexible and can react to any pattern that can be identified by the sensors in the HWC, all the user has to do is provide an example of the pattern to look for as a target. In this way the finding system assists the user and the user can do other things while the HWC looks for the target. The finding system can be provided as an operating mode in the HWC where the user selects the mode and then inputs the pattern to be used as the search target by the HWC. Examples of items that can be searched for include: household objects, animals, plants, street signs, weather activity (e.g. cloud formations), people, voices, songs, bird calls, specific sounds, spoken words, temperatures, wind direction shifts as identified by wind sound relative to the compass heading, vibrations, objects to be purchased, brand names in stores, labels on items in a warehouse, bar codes or numbers on objects and colors of objects to be matched. In a further embodiment, the rate of searching (e.g. how often an analysis is conducted) can be selected by the user or the rate can be automatically selected by the HWC in response to the rate of change of the conditions related to the target. In a yet further embodiment, the sensors in the HWC include a rangefinder or a camera capable of generating a depth map to measure the distance to an object in an image captured by the camera. The HWC can then analyze the image along with the distance to determine the size of the object. The user can then input the size of the object to the finding system as a characteristic of the target pattern to enable the HWC to more accurately identify potential finds.

Another aspect of the present invention relates to assisting a person in reading text that is presented in a physical form, such as a book, magazine, on a computer screen or phone screen, etc. In embodiments, the camera on the HWC can image the page and the processor in the HWC can recognize the words on the page. Lines, boxes, or other indicators may be presented in the HWC to indicate which words are being captured and recognized. The user would then be viewing the page of words through the see-through display with an indication of which words have been recognized. The recognized words can then be translated or converted from text that is then presented to the user in the see-through display. Alternately, the recognized words can be converted from text to speech, which is then presented to the user through the head worn speakers, headphones, visual displays, etc. This gives the user a better understanding of the accuracy associated with the text recognition relative to the translated text or converted speech.

In a further aspect of the invention, a magnetic attachment structure is provided for the combiner to enable the combiner to be removable. In the optics associated with a HWC 102 such as for example the optics shown in FIG. 6, it is important that the combiner 602 be accurately positioned and rigidly held below the frame of the HWC and the upper optical module 202 located inside the frame. At the same time, the combiner 602 can become damaged so that it needs to be replaced, or it may need to be cleaned periodically so that it is advantageous for the combiner to be removable. FIG. 123 shows an illustration of a cross section of a single combiner 12360 with the magnetic attachment structure as shown from the side to show the angle of the combiner 12360. FIG. 124 shows an illustration of two combiners 12360 with magnetic attachment structures attaching the combiners 12360 to the frame of the HWC 12350 as shown from the front of the HWC. The combiner 12360 has two or more pins 12365 that are attached to the combiner 12360

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such that the pins have parallel axes. The pins 12365 are shown as being inserted into holes drilled through the combiner 12365 and attached in place with adhesive such as UV cured adhesive. The pins 12365 are made of a magnetic material such as for example 420 stainless steel. The pins 12365 extend into parallel bores in the frame of the HWC 12350 so that the combiner 12360 is fixedly held in place relative to the frame of the HWC 12350. The attachment and bend of the pins 12365 establish the angle between the combiner 12360 and the optics in the frame of the HWC 12350. A magnet 12370 is bonded into the frame of the HWC 12350 such that the pin 12365 attracted by the magnet 12370 and thereby the pin 12365 and the attached combiner 12360 are held in place relative to the frame of the HWC 12350. The magnet 12370 is selected so that the force exerted by the magnet 12370 onto the pin 12365 is strong enough to hold the combiner 12360 in place during normal use, but weak enough that removal of the combiner 12350 is possible by the user. By having the pins 12365 and associated bores parallel, the combiner 12350 can be easily removed for cleaning, or replaced if damaged. To provide a more rigid and repeatable connection between the combiner 12360 and the frame of the HWC 12350, the pins can fit into an extended tight bore in the frame of the HWC 12350. In addition, the pins 12365 can include a flange as shown that seats onto an associated flat surface of the frame of the frame 12350 or a flat surface of the magnet 12370 to further establish the angle of the combiner 12360 and the vertical position of the combiner 12360. In a preferred embodiment, the magnet 12370 is a ring magnet and the pin 12365 extends through the center of the ring magnet. The magnet 12370 can also be included in an insert (not shown) that further includes a precision bore to precisely align and guide the pin 12365. The insert can be made of a hardened material such as a ceramic to provide a bore for the pin 12365 that is resistant to wear during repeated removal and reinstallation of the combiner 12360. The pins can be accurately positioned within the combiner through the use of a jig that holds the pins and the combiner. The holes for the pins in the combiner are then made larger than the pins so there is a clearance to allow the combiner and pins to be fully positioned by the jig. An adhesive such as a UV curing adhesive is then introduced to the holes and cured in place to fasten the pins to the combiner in a position that is established by the jig. In a further embodiment, the combined structure of the pins 12365 and the combiner 12350 are designed to break if subjected to a high impact force, to thereby protect the user from injury. Where the pin 12365 or the combiner are designed to break at a previously selected impact force that is less than the impact force required to break the frame of the HWC 12350 so that the combiner 12350 with the attached pins 12365 can be simply replaced when damaged. In yet a further embodiment, by providing a method for easily replacing the combiners 12360, different types of combiners can also be provided to the user such as: polarized combiners, combiners with different tints, combiners with different spectral properties, combiners with different levels of physical properties, combiners with different shapes or sizes, combiners that are partial mirrors or combiners that are notch mirrors, combiners with features to block faceglow as previously described herein.

Although embodiments of HWC have been described in language specific to features, systems, computer processes and/or methods, the appended claims are not necessarily limited to the specific features, systems, computer processes and/or methods described. Rather, the specific features, systems, computer processes and/or methods are dis-

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closed as non-limited example implementations of HWC. All documents referenced herein are hereby incorporated by reference.

What is claimed is:

1. A wearable head device comprising:
 - an optics module comprising:
 - a polarized light source configured to provide an illumination light;
 - a reflective image source configured to form an image light based on the illumination light; and
 - a partial reflector configured to transmit the illumination light toward the reflective image source, receive the image light from the reflective image source, and transmit the image light toward a first lens configured to converge the image light;
 - a combiner configured to transmit the converged image light, along with an environment light from a surrounding environment, toward a user's eye concurrently with the user wearing the wearable head device; and
 - a frame configured to hold a transparent element, wherein:
 - the transparent element comprises a first region having a first transmissivity configured to occlude viewing of the optics module through the first region of the transparent element,
 - the transparent element further comprises a second region having a second transmissivity configured to transmit the environment light toward the user's eye concurrently with the user wearing the wearable head device, and
 - the second transmissivity is greater than the first transmissivity.
2. The wearable head device of claim 1, wherein the transparent element comprises a corrective lens.
3. The wearable head device of claim 1, wherein the first transmissivity is less than 30 percent.
4. The wearable head device of claim 1, wherein the second transmissivity is greater than 90 percent.
5. The wearable head device of claim 1, wherein the transparent element further comprises a gradient zone, the gradient zone comprising a tint gradient from the first transmissivity to the second transmissivity.
6. The wearable head device of claim 1, wherein the frame is configured to hold the transparent element via a magnetic attachment system.
7. The wearable head device of claim 1, wherein the first region corresponds to an upper region of the transparent element and the second region corresponds to a lower region of the transparent element.
8. The wearable head device of claim 1, wherein:
 - the transparent element further comprises an electrochromic element;
 - the wearable head device further comprises one or more processors in communication with the electrochromic element, the one or more processors configured to vary the second transmissivity via the electrochromic element.
9. The wearable head device of claim 8, wherein varying the second transmissivity via the electrochromic element comprises varying the second transmissivity based on a signal received via a software application executed by the one or more processors.

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10. The wearable head device of claim 1, further comprising an eye imaging camera positioned adjacent to the partial reflector and configured to capture light reflected by the user's eye.

11. A method comprising:
 - providing, via a polarized light source of an optics module of a wearable head device, an illumination light;
 - transmitting, via a partial reflector of the optics module, the illumination light toward a reflective image source of the optics module;
 - receiving an image light from the reflective image source, the image light based on the illumination light;
 - transmitting, via the partial reflector, the image light toward a first lens of the wearable head device;
 - converging the image light via the first lens; and
 - concurrently transmitting, toward a user's eye via a combiner of the wearable head device, the converged image light and an environment light from a surrounding environment, wherein:
 - the wearable head device comprises a frame configured to hold a transparent element,
 - the transparent element comprises a first region having a first transmissivity configured to occlude viewing of the optics module through the first region of the transparent element,
 - the transparent element further comprises a second region having a second transmissivity configured to transmit the environment light toward the user's eye, and
 - the second transmissivity is greater than the first transmissivity.

12. The method of claim 11, wherein the transparent element comprises a corrective lens.

13. The method of claim 11, wherein the first transmissivity is less than 30 percent.

14. The method of claim 11, wherein the second transmissivity is greater than 90 percent.

15. The method of claim 11, wherein the transparent element further comprises a gradient zone, the gradient zone comprising a tint gradient from the first transmissivity to the second transmissivity.

16. The method of claim 11, wherein the frame is configured to hold the transparent element via a magnetic attachment system.

17. The method of claim 11, wherein the first region corresponds to an upper region of the transparent element and the second region corresponds to a lower region of the transparent element.

18. The method of claim 11, wherein:

- the transparent element further comprises an electrochromic element;
- the method further comprises varying the second transmissivity via the electrochromic element.

19. The method of claim 18, wherein varying the second transmissivity via the electrochromic element comprises varying the second transmissivity based on a signal received via a software application executed by one or more processors in communication with the wearable head device.

20. The method of claim 11, wherein:

- the wearable head device further comprises an eye imaging camera positioned adjacent to the partial reflector, and
- the method further comprises capturing, via the eye imaging camera, light reflected by the user's eye.

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